

Organic Chemistry



Previous
Year
Question



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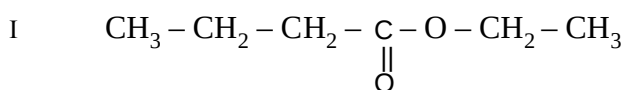


Let's Crack it !

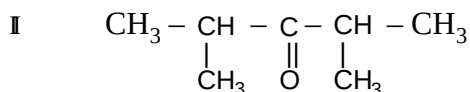
1 NOMENCLATURE OF ORGANIC COMPOUND

[JEE-MAIN / AIEEE]

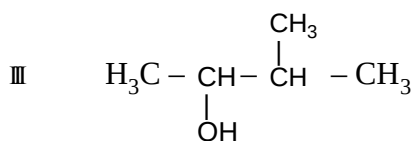
1. Of the following compounds which has a wrong IUPAC name - [AIEEE-2002]



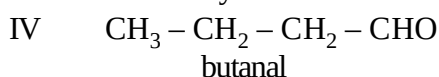
ethyl butanoate



2,4- dimethyl -3- pentanone



2-methyl -3- butanol

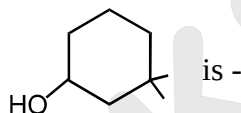


- (A) I (B) II (C) III (D) IV

2. The IUPAC name of $\text{CH}_3\text{COCH}(\text{CH}_3)_2$ is - [AIEEE-2003]

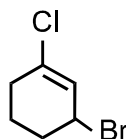
- (A) 4-methylisopropyl ketone (B) 3-methyl-2-butanone
(C) Isopropylmethyl ketone (D) 2-methyl-3-butanone

3. The IUPAC name of the compound [AIEEE-2004]



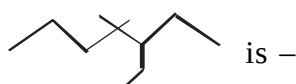
- (A) 3,3 -dimethyl-1-hydroxy cyclohexane (B) 1, 1 - dimethyl-3- hydroxy cyclohexane
(C) 3,3-dimethyl-1-cyclohexanol (D) 1,1 - dimethyl - 3- cyclohexanol

4. The IUPAC name of the compound shown below is - [AIEEE-2006]



- (A) 6-bromo-2-chlorocyclohexene (B) 3-bromo-1-chlorocyclohexene
(C) 1-bromo-3-chlorocyclohexene (D) 2-bromo-6-chlorocyclohex-1-ene

5. The IUPAC name of [AIEEE-2007]

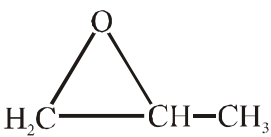


- (A) 1, 1-diethyl-2-dimethylpentane (B) 4, 4-dimethyl-5, 5-diethylpentane
(C) 5, 5-diethyl-4, 4-dimethylpentane (D) 3-ethyl-4, 4 dimethylheptane

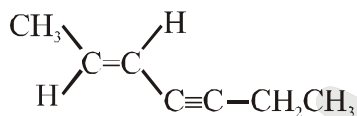
6. The correct decreasing order of priority for the functional groups of organic compounds in the IUPAC system of nomenclature is [AIEEE 2008]

- (A) $-\text{SO}_3\text{H}$, $-\text{COOH}$, $-\text{CONH}_2$, $-\text{CHO}$ (B) $-\text{CHO}$, $-\text{COOH}$, $-\text{SO}_3\text{H}$, $-\text{CONH}_2$
(C) $-\text{CONH}_2$, $-\text{CHO}$, $-\text{SO}_3\text{H}$, $-\text{COOH}$ (D) $-\text{COOH}$, $-\text{SO}_3\text{H}$, $-\text{CONH}_2$, $-\text{CHO}$

7. The IUPAC name of neopentane is : [AIEEE -2009]
(A) 2-methylpropane (B) 2, 2-dimethylbutane
(C) 2-methylbutane (D) 2, 2-dimethylpropane

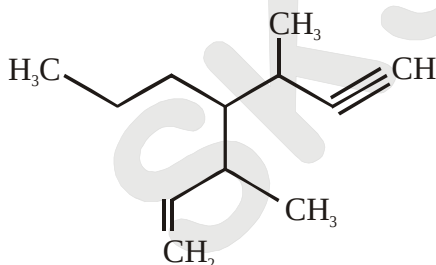
8. The IUPAC name of compound  is :- [AIEEE -2012 (Online)]
(A) 1, 2-Epoxy propane (B) Propylene oxide
(C) 1, 2-Oxo propane (D) 1, 2-Propoxide

9. The IUPAC name of the following compounds is : [AIEEE -2012 (Online)]

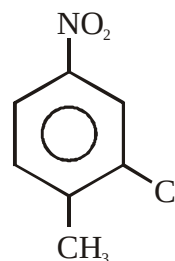


- (A) (Z) – 5– hepten – 3 – yne (B) (Z) – 2– hepten – 4 – yne
(C) (E) – 5– hepten – 3 – yne (D) (E) – 2– hepten – 4 – yne
10. Aspirin is known as : [AIEEE -2012]
(A) Acetyl salicylic acid (B) Phenyl salicylate
(C) Acetyl salicylate (D) Methyl salicylic acid

11. The IUPAC name for the following compound is [JEE Main 2019]



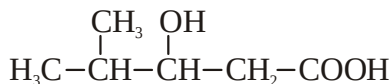
- (A) 3-methyl-4-(3-methylprop-1-enyl)-1-heptyne
(B) 3, 5-dimethyl-4-propylhept-6-en-1-yne
(C) 3-methyl-4-(1-methylprop-2-ynyl)-1-heptene
(D) 3, 5-dimethyl-4-propylhept1-en-6-yne
12. The correct IUPAC name of the following compound is [JEE Main 2019]
(A) 2-methyl-5-nitro-1-chlorobenzene
(B) 3-chloro-1-methyl-nitrobenzene
(C) 2-chloro-1-methyl 1-4-nitrobenzene
(D) 5-chloro-4-methyl 1-1-nitrobenzene



13. Which of the following compounds will show the maximum 'enol' content ? [JEE Main 2019]

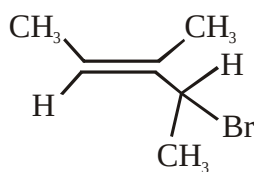


14. The IUPAC name of the following compound is [JEE Main 2019]



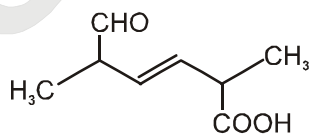
- (A) 4,4-dimethyl-3-hydroxybutanoic acid (B) 2-methyl-3-hydroxypentan-5-oic acid
(C) 3-hydroxy-4-methylpentanoic acid (D) 4-methyl-3-hydroxypentanoic acid

15. What is the IUPAC name of the following compound ? [JEE Main 2019]



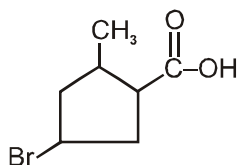
- (A) 3-bromo-3-methyl-1,2-dimethylprop-1-ene
(B) 3-bromo-1,2-dimethylbut-1-ene
(C) 2-bromo-3-methylpent-3-ene
(D) 4-bromo-methylpent-2-ene

16. The IUPAC name for the following compound is : [JEE-Main-2020]



- (A) 2,5-dimethyl-6-carboxy-hex-3-enal (B) 2,5-dimethyl-5-carboxy-hex-3-enal
(C) 6-formyl-2-methyl-hex-3-enoic (D) 2,5-dimethyl-6-oxo-hex-3-enoic acid

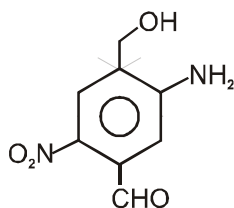
17. The IUPAC name of the following compound is : [JEE-Main-2020]



- (A) 3-bromo-5-methylcyclopentane (B) 3-bromo-5-methylcyclopentanecarboxylic acid
(C) 5-bromo-5-methylcyclopentanecarboxylic acid (D) 4-bromo-2-methylcyclopentanecarboxylic acid

18. The IUPAC name of the following compound is :

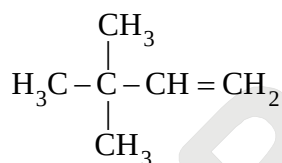
[JEE-Main-2020]



- (A) 2-nitro-4-hydroxymethyl-5-aminobenzaldehyde
 (B) 3-amino-4-hydroxymethyl-5-nitrobenzaldehyde
 (C) 4-amino-2-formyl-5-hydroxymethylnitrobenzene
 (D) 5-amino-4-hydroxymethyl-2-nitrobenzaldehyde

[JEE-ADVANCED]

1. The IUPAC name of the compound having the formula is:



- (A) 3,3,3-trimethyl-1-propene
 (B) 1,1,1-trimethyl-2-propene
 (C) 3,3-dimethyl-1-butene
 (D) 2,2-dimethyl-3-butene

[JEE 1984]

2. Write the IUPAC name of $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCOOH}$

[JEE 1986]

3. The IUPAC name of the compound $\text{CH}_2=\text{CH}-\text{CH}(\text{CH}_3)_2$ is:

- (A) 1,1-dimethyl-2-propene
 (B) 3-methyl-1-butene
 (C) 2-vinyl propane
 (D) None of the above

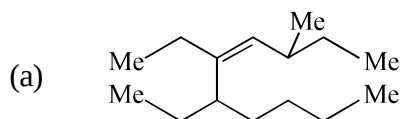
[JEE 1987]

4. The number of sigma and pi-bonds in 1-butene 3-yne are:

[JEE 1989]

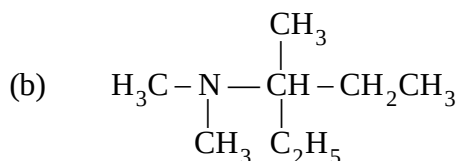
- (A) 5 sigma and 5 pi
 (B) 7 sigma and 3 pi
 (C) 8 sigma and 2 pi
 (D) 6 sigma and 4 pi

5. Write I.U.P.A.C name of following:



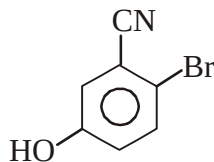
Me = methyl group

[JEE 1990]

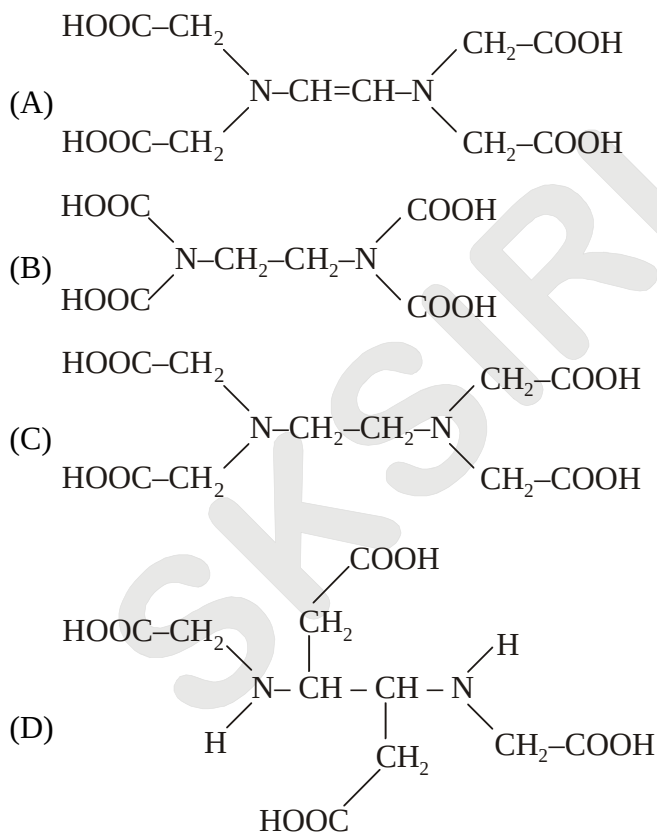


[JEE 1991]

6. Write IUPAC name of succinic acid. [JEE 1994]
7. The IUPAC name of C_6H_5COCl is
(A) Benzoyl chloride (B) Benzene chloro ketone
(C) Benzene carbonyl chloride (D) Chloro phenyl ketone [JEE 2006]
8. The IUPAC name of the following compound is [JEE 2009]



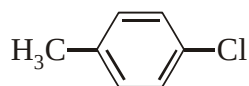
- (A) 4-Bromo-3-cyanophenol (B) 2-Bromo-5-hydroxybenzonitrile
(C) 2-Cyano-4-hydroxybromobenzene (D) 6-Bromo-3-hydroxybenzonitrile
9. The correct structure of ethylenediaminetetraacetic acid (EDTA) is [IIT-JEE 2010]



10. The total number of lone-pairs of electrons in melamine is [JEE Advance 2013]
11. The correct combination of names for isomeric alcohols with molecular formula $C_4H_{10}O$ is/are
(A) tert-butanol and 2-methylpropan-2-ol
(B) tert-butanol and 1,1-dimethylethan-1-ol
(C) n-butanol and butan-1-ol
(D) isobutyl alcohol and 2-methylpropan-1-ol [JEE Advance 2014]

12. The IUPAC name(s) of the following compound is(are)

[JEE Advance 2017]



(A) 1-chloro-4-methylbenzene

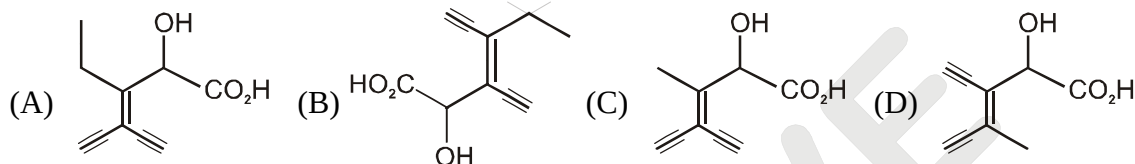
(B) 4-chlorotoluene

(C) 1-methyl-4-chlorobenzene

(D) 4-methylchlorobenzene

13. Which one of the following structures has the IUPAC name
3-ethynyl-2-hydroxy-4-methylhex-3-en-5-ynoic acid ?

[JEE Advance 2020]

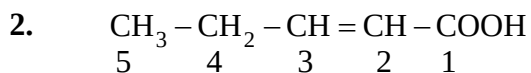


ANSWER KEY**[JEE-MAIN/AIEEE]**

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. C | 2. B | 3. C | 4. B | 5. D | 6. D | 7. D |
| 8. A | 9. D | 10. A | 11. D | 12. C | 13. B | 14. C |
| 15. D | 16. D | 17. D | 18. D | | | |

[JEE-ADVANCED]

1. C



Pent-2-en-1-oic acid or Pent-2-enoic acid

3. B 4. B

5. (a) 5,6-diethyl-3-methyl-dec-4-ene (b) N,N,3-trimethyl-3-pentanamine

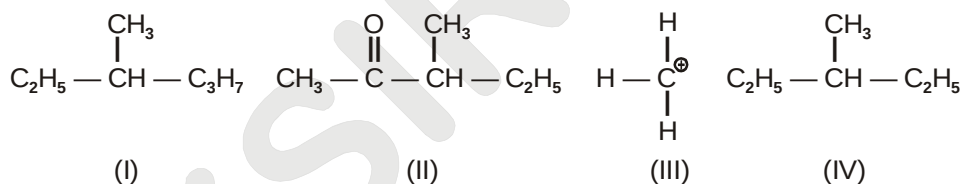
6. Butane-1,4-dioic acid 7. C 8. B 9. C 10. 6

11. ACD 12. AB 13. D

[JEE-MAIN / AIEEE]

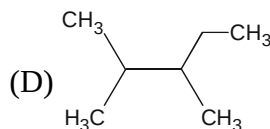
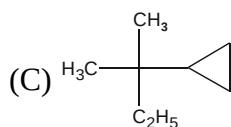
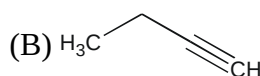
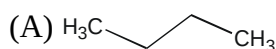
Single Choice Questions :

1. A similarity between optical and geometrical isomerism is that [AIEEE-2002]
 (A) each forms equal number of isomers for a given compound
 (B) If in a compound one is present then so is the other
 (C) both are included in stereoisomerism
 (D) they have no similarity
2. Which of the following does not show geometrical isomerism ? [AIEEE-2002]
 (A) 1, 2-dichloro-1-pentene (B) 1, 3-dichloro-2-pentene
 (C) 1, 1-dichloro-1-pentene (D) 1,4-dichloro-2-pentene
3. Racemic mixture is formed by mixing two [AIEEE-2002]
 (A) isomeric compounds (B) chiral compounds
 (C) meso compounds (D) optical isomers
4. Among the following four structures I to IV [AIEEE-2003]

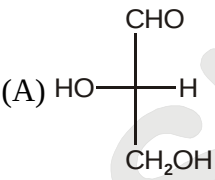


it is true that :

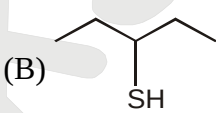
- (A) all four are chiral compounds (B) only I and II are chiral compounds
 (C) only III is a chiral compound (D) only II and IV are chiral compounds
5. On mixing a certain alkane with chlorine and irradiating it with ultraviolet light, it forms only one monochloroalkane. This alkane could be [AIEEE-2003]
 (A) propane (B) pentane (C) isopentane (D) neopentane
6. Which of the following will have meso-isomer also ? [AIEEE-2004]
 (A) 2-chlorobutane (B) 2-hydroxy propanoic acid
 (C) 2,3-dichloropentane (D) 2,3-dichlorobutane
7. Amongst the following compound, the optically active alkane having lowest molecular mass is [AIEEE-2004]



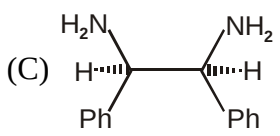
8. Which of the following compound is not chiral ? [AIEEE-2004]
 (A) 1-chloropentane (B) 3-chloro-2-methylpentane
 (C) 1-chloro-2-methylpentane (D) 2-chloropentane
9. Due to the presence of an unpaired electron, free radicals are: [AIEEE-2005]
 (A) Chemically reactive (B) Chemically inactive
 (C) Anions (D) Cations
10. 2-methylbutane on reacting with bromine in the presence of sunlight gives mainly [AIEEE-2005]
 (A) 1-bromo-2-methylbutane (B) 2-bromo-2-methylbutane
 (C) 2-bromo-3-methylbutane (D) 1-bromo-3-methylbutane
11. Of the five isomeric hexanes, the isomer which can give two monochlorinated compounds is [AIEEE-2005]
 (A) n-hexane (B) 2,3-dimethylbutane
 (C) 2,2-dimethylbutane (D) 2-methylpentane
12. Which types of isomerism is shown by 2,3-dichlorobutane? [AIEEE-2005]
 (A) Tautomerism (B) Optical (C) Geometric (D) Structural
13. Increasing order of stability among the three main conformations (i.e. Eclipse, Anti, Gauche) of 2-fluoroethanol is [AIEEE-2006]
 (A) Eclipse, Gauche, Anti (B) Gauche, Eclipse, Anti
 (C) Eclipse, Anti, Gauche (D) Anti, Gauche, Eclipse
14. Which of the following molecules is expected to rotate the plane of plane polarized light ? [AIEEE-2007]
- (A)



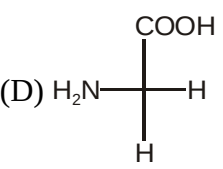
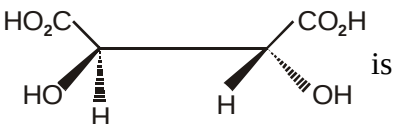
(B)



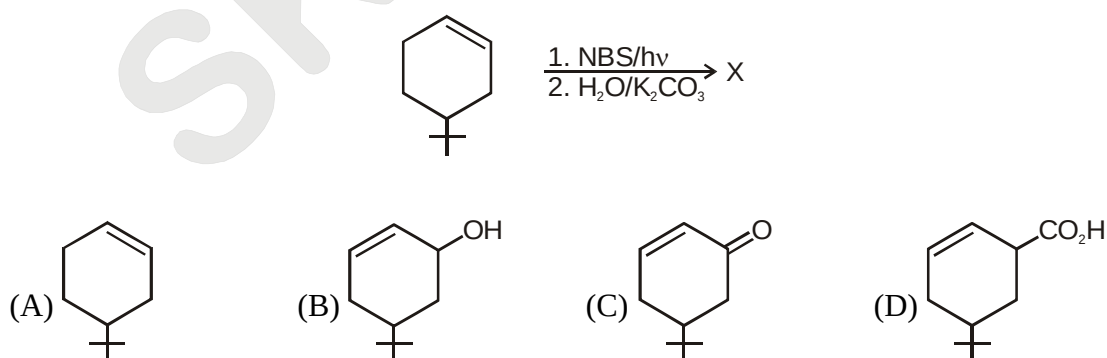
(C)



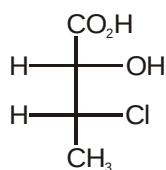
(D)


15. Which one of the following conformation of cyclohexane is chiral? [AIEEE-2007]
 (A) Twist boat (B) Rigid (C) Chair (D) Boat
16. The absolute configuration of  is [AIEEE-2008]
 (A) S, S (B) R, R (C) R, S (D) S, R
17. The alkene that exhibits geometrical isomerism is : [AIEEE-2009]
 (A) 2-butene (B) 2-methyl-2-butene (C) propene (D) 2-methyl propene
18. The number of stereoisomers possible for a compound of the molecular formula [AIEEE-2009]
 $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}(\text{OH}) - \text{Me}$ is :
 (A) 4 (B) 6 (C) 3 (D) 2

19. The IUPAC name of neopentane is : [AIEEE-2009]
(A) 2-methylpropane (B) 2,2-dimethylbutane
(C) 2-methylbutane (D) 2,2-dimethylpropane
20. One mole of a symmetrical alkene on ozonolysis gives two moles of an aldehyde having a molecular mass of 44 u. The alkene is [AIEEE-2010]
(A) 2-butene (B) ethene (C) propene (D) 1-butene
21. Identify the compound that exhibits tautomerism. [AIEEE-2011]
(A) 2-Butene (B) Lactic acid (C) 2-Pentanone (D) Phenol
22. Ozonolysis of an organic compound gives formaldehyde as one of the products. This confirms the presence of : [AIEEE-2011]
(A) two ethylenic double bonds (B) a vinyl group
(C) an isopropyl group (D) an acetylenic triple bond
23. How many chiral compounds are possible on monochlorination of 2-methyl butane? [AIEEE-2012]
(A) 8 (B) 2 (C) 4 (D) 6
24. Which branched chain isomer of the hydrocarbon with molecular mass 72u gives only one isomer of mono substituted alkyl halide ? [AIEEE-2012]
(A) Tertiary butyl chloride (B) Neopentane
(C) Isohexane (D) Neohexane
25. The product of the reaction given below is : [JEE Mains-2016]



26. The absolute configuration of [JEE Mains-2016]

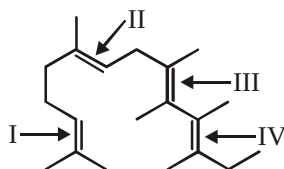


is:

- (A) (2R, 3S) (B) (2S, 3R) (C) (2S, 3S) (D) (2R, 3R)

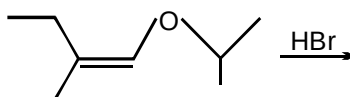
27. In the following structure, the double bonds are marked as I, II, III and IV

[JEE Mains Online-2017]



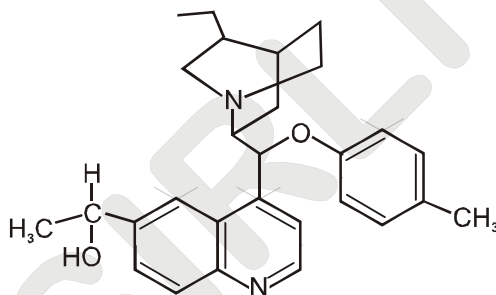
28. The total number of optically active compounds formed in the following reaction is :

[JEE Mains Online, 2018]



- (A) Two (B) Four (C) Six (D) Zero
29. The number of chiral carbons present in the molecule given below is

[JEE Mains Online, 2020]

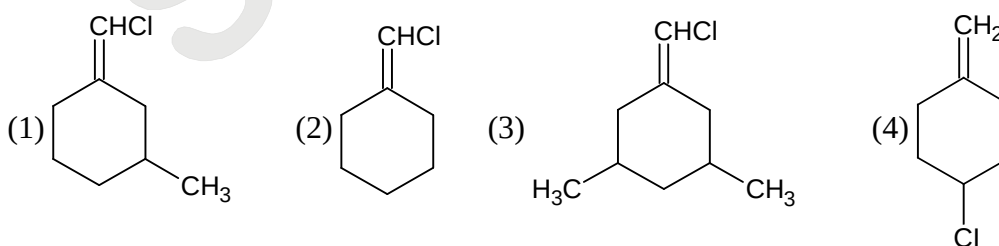


30. The number of chiral centres present in threonine is

[JEE Mains Online, 2020]

31. Among the following compounds, geometrical isomerism is exhibited by :

[JEE Mains Online, 2020]



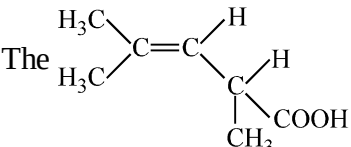
31. Which of the following compounds shows geometrical isomerism ?

[JEE Mains Online, 2020]

- (1) 4-methylpent-1-ene (2) 2-methylpent-2-ene
(3) 4-methylpent-2-ene (4) 2-methylpent-1-ene

[JEE-ADVANCED]

1. Isomers which can be interconverted through rotation around a single bond are [JEE 1992]
 (A) Conformers (B) Diastereomers (C) Enantiomers (D) Positional isomers

2. The  shows: [JEE 1995(Scr.)]

- (A) geometrical isomerism (B) optical isomerism
 (C) geometrical & optical isomerism (D) tautomerism

3. How many optically active stereoisomers are possible for butane-2,3-diol? [JEE 1997]
 (A) 1 (B) 2 (C) 3 (D) 4

4. When cyclohexane is poured on water, it floats, because: [JEE 1997]
 (A) cyclohexane is in 'boat' form (B) cyclohexane is in 'chair' form
 (C) cyclohexane is in 'crown' form (D) cyclohexane is less dense than water

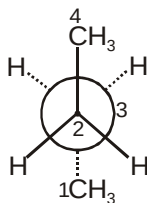
5. Which of the following compounds will show geometrical isomerism? [JEE 1998]
 (A) 2-butene (B) propene (C) 1-phenylpropene (D) 2-methyl-2-butene

6. Which of the following compounds will exhibit geometrical isomerism? [JEE 2000 (Scr.)]
 (A) 1-Phenyl-2-butene (B) 3-Phenyl-1-butene
 (C) 2-Phenyl-1-butene (D) 1,1-Diphenyl-1-propene

7. The number of isomers for the compound with molecular formula $C_2BrClFI$ is [JEE 2001 (Scr.)]
 (A) 3 (B) 4 (C) 5 (D) 6

8. Which of the following compounds exhibits stereoisomerism? [JEE 2002 (Scr.)]
 (A) 2-methylbutene-1 (B) 3-methylbutyne-1
 (C) 3-methylbutanoic acid (D) 2-methylbutanoic acid

9. In the given conformation, if C_2 is rotated about C_2-C_3 bond anticlockwise by an angle of 120° then the conformation obtained is [JEE 2004 (Scr.)]



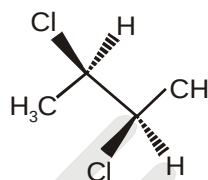
- (A) fully eclipsed conformation (B) partially eclipsed conformation
 (C) gauche conformation (D) staggered conformation

10. The number of structural isomers for C_6H_{14} is [JEE 2007]
 (A) 3 (B) 4 (C) 5 (D) 6

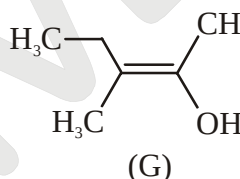
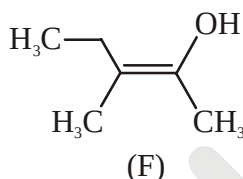
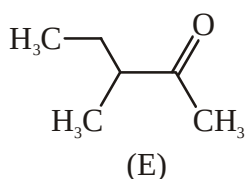
11. **Statement-1** : Molecules that are not superimposable on their mirror images are chiral. [JEE 2007]
Statement-2 : All chiral molecules have chiral centres.
 (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.

12. The correct statement(s) about the compound given below is (are) [JEE 2008]

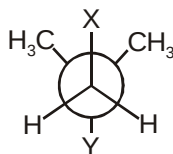
- (A) The compound is optically active
 (B) The compound possesses centre of symmetry
 (C) The compound possesses plane of symmetry
 (D) The compound possesses axis of symmetry



13. The correct statement(s) concerning the structures E, F and G is (are) [JEE 2008]

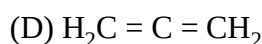
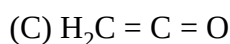
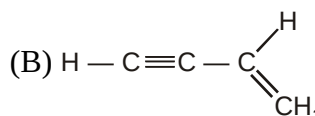
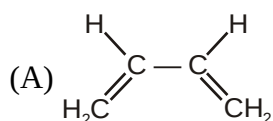


- (A) E, F and G are resonance structures
 (B) E, F and E, G are tautomers
 (C) F and G are geometrical isomers
 (D) F and G are diastereomers
14. The correct statement(s) about the compound $\text{H}_3\text{C}(\text{HO})\text{HC} - \text{CH} = \text{CH} - \text{CH}(\text{OH})\text{CH}_3(\text{X})$ is (are) [JEE 2009]
 (A) The total number of stereoisomers possible for X is 6
 (B) The total number of diastereomers possible for X is 3
 (C) If the stereochemistry about the double bond in X is trans, the number of enantiomers possible for X is 4
 (D) If the stereochemistry about the double bond in X is cis, the number of enantiomers possible for X is 2
15. In the Newman projection for 2,2-dimethylbutane [JEE 2010]

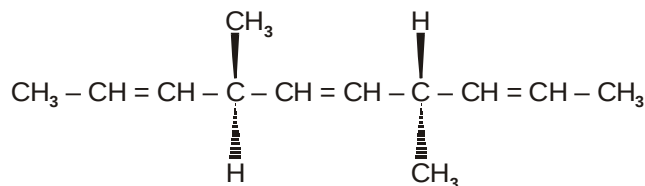


X and Y can respectively be

- (A) H and H
 (B) H and C_2H_5
 (C) C_2H_5 and H
 (D) CH_3 and CH_3
16. Amongst the given options, the compound(s) in which all the atoms are in one plane in all the possible conformations (if any), is (are) [JEE 2011]



17. In allene (C_3H_4), the type(s) of hybridisation of the carbon atoms is (are) : [JEE 2012]
 (A) sp and sp^3 (B) sp and sp^2 (C) only sp^2 (D) sp^2 and sp^3
18. The number of optically active products obtained from the complete ozonolysis of the given compound is: [JEE 2012]

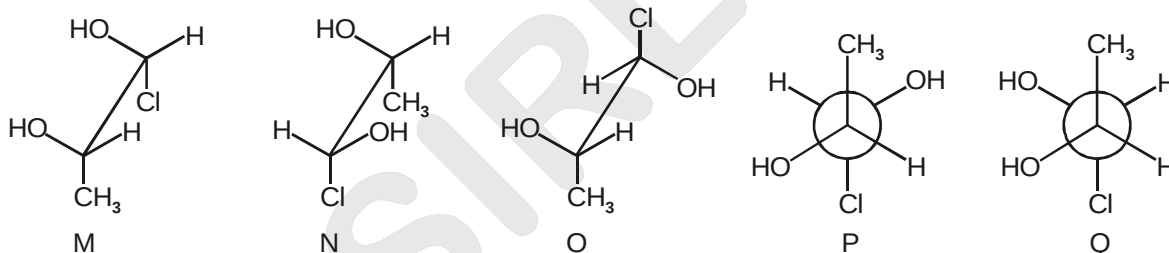


- (A) 0 (B) 1 (C) 2 (D) 4
19. Which of the following molecules, in pure form, is (are) **unstable** at room temperature ? [JEE 2012]



20. Which of the given statement(s) about N, O, P and Q with respect to M is (are) correct ?

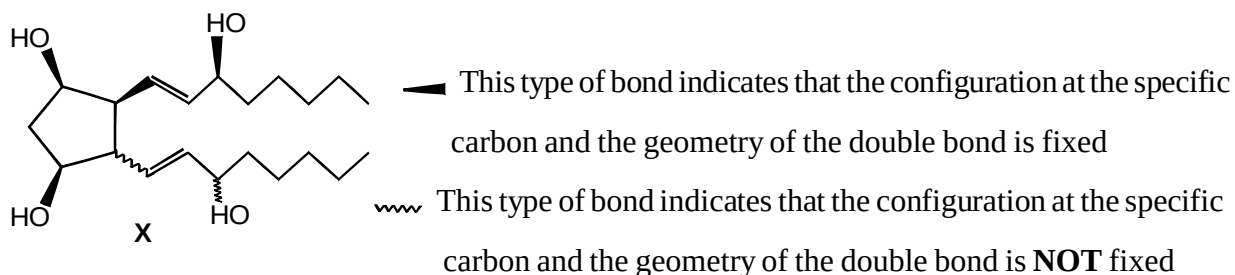
[JEE 2012]



- (A) M and N are non-mirror image stereoisomers
 (B) M and O are identical
 (C) M and P are enantiomers
 (D) M and Q are identical
21. For 'invert sugar', the correct statement(s) is(are) [JEE Advance 2016]
 (Given : specific rotations of (+)-sucrose, (+)-maltose, L-(−)-glucose and L-(+)-fructose in aqueous solution are $+66^\circ$, $+140^\circ$, -52° and $+92^\circ$, respectively)
 (A) 'invert sugar' is prepared by acid catalyzed hydrolysis of maltose
 (B) 'invert sugar' is an equimolar mixture of D-(+)-glucose and D-(−)-fructose
 (C) specific rotation of 'invert sugar' is -20°
 (D) on reaction with Br_2 water, 'invert sugar' forms saccharic acid as one of the products

22. For the given compound X, the total number of optically active stereoisomers is_____.

[JEE Advance 2018]

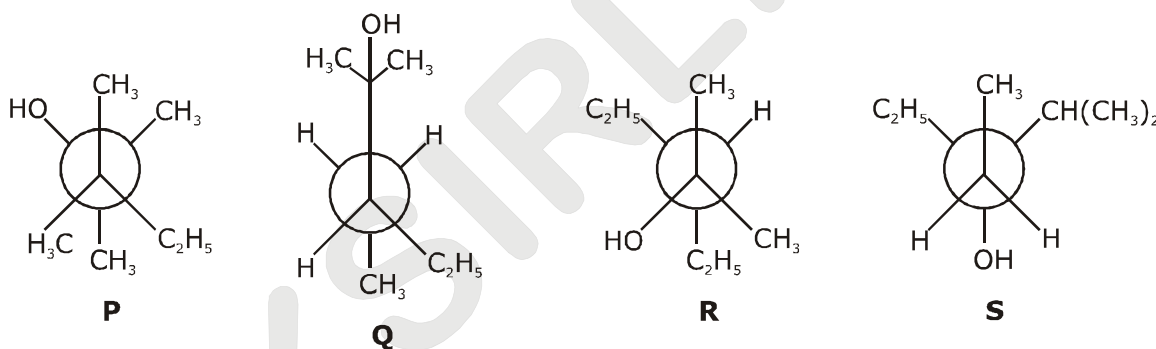


23. Total number of isomers, considering both structural and stereoisomers, of cyclic ethers with the molecular formula C_4H_8O is _____.

[JEE Advance 2019]

24. Newman projections P, Q, R and S are shown below :

[JEE Advance 2020]

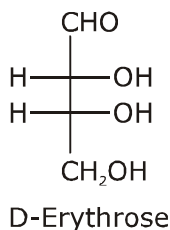


Which one of the following options represents identical molecules ?

- (A) P and Q (B) Q and S (C) Q and R (D) R and S

25. The Fischer projection of D-erythrose is shown below :

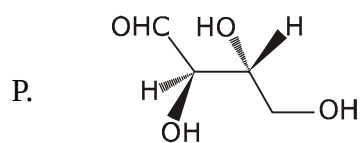
[JEE Advance 2020]



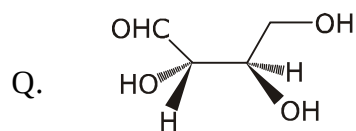
D-Erythrose and its isomers are listed as P, Q, R, and S in Column-I. Choose the correct relationship of P, Q, R, and S with D-erythrose from Column II.

Column – I

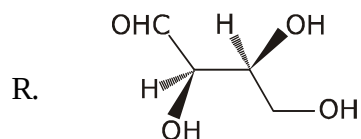
Column – II



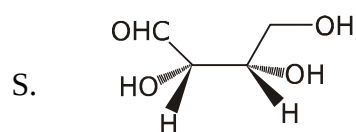
1. Diastereomer



2. Identical



3. Enantiomer



(A) P → 2, Q → 3, R → 2, S → 2

(B) P → 3, Q → 1, R → 1, S → 2

(C) P → 2, Q → 1, R → 1, S → 3

(D) P → 2, Q → 3, R → 3, S → 1

ANSWER KEY**[JEE-MAIN/AIEEE]**

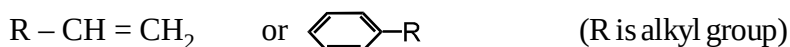
1.	C	2.	C	3.	D	4.	B	5.	D	6.	D	7.	D
8.	A	9.	A	10.	B	11.	B	12.	B	13.	C	14.	A
15.	A	16.	B	17.	A	18.	A	19.	D	20.	A	21.	C
22.	B	23.	C	24.	B	25.	B	26.	B	27.	B	28.	B
29.	5	30.	2	31.	1 (Answer should be 1 and 3 both)					31.	3		

[JEE-ADVANCED]

1.	A	2.	B	3.	B	4.	D	5.	A, C	6.	A	7.	D
8.	D	9.	C	10.	C	11.	C	12.	A, D	13.	B, C, D		
14.	A, D	15.	B, D	16.	B, C	17.	B	18.	A	19.	BC		
20.	ABC	21.	B, C	22.	7	23.	10	24.	C	25.	C		

[JEE-MAIN / AIEEE]

1. In the following benzyl/alkyl system



increasing order of inductive effect is –

[AIEEE-2002]

- (A) $(CH_3)_3C - > (CH_3)_2CH - > CH_3CH_2 -$
 (B) $(CH_3CH_2 - > (CH_3)_2CH - > (CH_3)_3C -$
 (C) $(CH_3)_2CH - > CH_3CH_2 - > (CH_3)_3C -$
 (D) $(CH_3)C - > CH_3CH_2 - > (CH_3)_2CH -$

2. When $-CH_3$, $CH_3 - CH -$ & $CH_3 - \underset{\begin{array}{c} | \\ CH_3 \end{array}}{\overset{\begin{array}{c} CH_3 \\ | \end{array}}{C}} -$ groups are introduced on benzene ring then correct order

of their inductive effect is –

[AIEEE-2002]

- (A) $CH_3 - > CH_3 - \underset{\begin{array}{c} | \\ CH_3 \end{array}}{\overset{\begin{array}{c} CH_3 \\ | \end{array}}{C}} - > CH_3 - CH -$ (B) $CH_3 - \underset{\begin{array}{c} | \\ CH_3 \end{array}}{\overset{\begin{array}{c} CH_3 \\ | \end{array}}{C}} - > CH_3 - CH - > CH_3 -$
 (C) $CH_3 - CH - > CH_3 - > CH_3 - \underset{\begin{array}{c} | \\ CH_3 \end{array}}{\overset{\begin{array}{c} CH_3 \\ | \end{array}}{C}} -$ (D) $CH_3 - \underset{\begin{array}{c} | \\ CH_3 \end{array}}{\overset{\begin{array}{c} CH_3 \\ | \end{array}}{C}} - > CH_3 - > CH_3 - CH -$

3. The correct order of increasing basic no. of the bases NH_3 , CH_3NH_2 and $(CH_3)_2NH$ is –

[AIEEE-2003]

- (A) $NH_3 < CH_3NH_2 < (CH_3)_2NH$ (B) $CH_3NH_2 < (CH_3)_2NH < NH_3$
 (C) $CH_3NH_2 < NH_3 < (CH_3)_2NH$ (D) $(CH_3)_2NH < NH_3 < CH_3NH_2$

4. Consider the acidity of the carboxylic acids :

[AIEEE-2004]

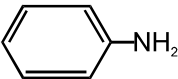
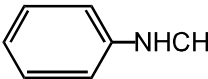
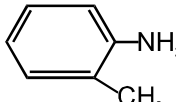
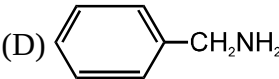
- (a) $PhCOOH$ (b) $o - NO_2C_6H_4COOH$
 (c) $p - NO_2C_6H_4COOH$ (d) $m - NO_2C_6H_4COOH$

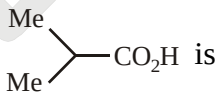
Which of the following order is correct ?

- (A) $a > b > c > d$ (B) $b > d > c > a$ (C) $b > d > a > c$ (D) $b > c > d > a$

5. Which of the following is the strongest base –

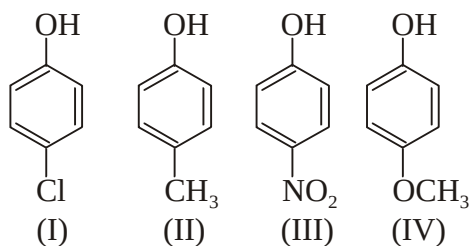
[AIEEE-2004]

- (A)  (B)  (C)  (D) 

6. Amongst the following the most basic compound is— [AIEEE-2005]
 (A) aniline (B) benzylamine (C) p-nitroaniline (D) acetanilide
7. The increasing order of stability of the following free radicals is— [AIEEE 2006]
 (A) $(\text{C}_6\text{H}_5)_3\dot{\text{C}} < (\text{C}_6\text{H}_5)_2\dot{\text{C}}\text{H} < (\text{CH}_3)_3\dot{\text{C}} < (\text{CH}_3)_2\dot{\text{C}}\text{H}$
 (B) $(\text{C}_6\text{H}_5)_2\dot{\text{C}}\text{H} < (\text{C}_6\text{H}_5)_3\dot{\text{C}} < (\text{CH}_3)_3\dot{\text{C}} < (\text{CH}_3)_2\dot{\text{C}}\text{H}$
 (C) $(\text{CH}_3)_2\dot{\text{C}}\text{H} < (\text{CH}_3)_3\dot{\text{C}} < (\text{C}_6\text{H}_5)_3\dot{\text{C}} < (\text{C}_6\text{H}_5)_2\dot{\text{C}}\text{H}$
 (D) $(\text{CH}_3)_2\dot{\text{C}}\text{H} < (\text{CH}_3)_3\dot{\text{C}} < (\text{C}_6\text{H}_5)_2\dot{\text{C}}\text{H} < (\text{C}_6\text{H}_5)_3\dot{\text{C}}$
8. The correct order of increasing acid strength of the compounds [AIEEE 2006]
 (a) $\text{CH}_3\text{CO}_2\text{H}$ (b) $\text{MeOCH}_2\text{CO}_2\text{H}$ (c) $\text{CF}_3\text{CO}_2\text{H}$ (d)  is
 (A) $d < a < c < b$ (B) $d < a < b < c$ (C) $a < d < c < b$ (D) $b < d < a < c$
9. Which one of the following is the strongest base in aqueous solution? [AIEEE-2007]
 (A) Trimethylamine (B) Aniline (C) Dimethylamine (D) Methylamine
10. Arrange the carbanions, $(\text{CH}_3)_3\dot{\text{C}}$, $\dot{\text{C}}\text{Cl}_3$, $(\text{CH}_3)_2\dot{\text{C}}\text{H}$, $\text{C}_6\text{H}_5\dot{\text{C}}\text{H}_2$, in order of their decreasing stability- [AIEEE 2009]
 (A) $(\text{CH}_3)_2\dot{\text{C}}\text{H} > \dot{\text{C}}\text{Cl}_3 > \text{C}_6\text{H}_5\dot{\text{C}}\text{H}_2 > (\text{CH}_3)_3\dot{\text{C}}$
 (B) $\dot{\text{C}}\text{Cl}_3 > \text{C}_6\text{H}_5\dot{\text{C}}\text{H}_2 > (\text{CH}_3)_2\dot{\text{C}}\text{H} > (\text{CH}_3)_3\dot{\text{C}}$
 (C) $(\text{CH}_3)_3\dot{\text{C}} > (\text{CH}_3)_2\dot{\text{C}}\text{H} > \text{C}_6\text{H}_5\dot{\text{C}}\text{H}_2 > \dot{\text{C}}\text{Cl}_3$
 (D) $\text{C}_6\text{H}_5\dot{\text{C}}\text{H}_2 > \dot{\text{C}}\text{Cl}_3 > (\text{CH}_3)_3\dot{\text{C}} > (\text{CH}_3)_2\dot{\text{C}}\text{H}$
11. The correct order of increasing basicity of the given conjugate bases ($\text{R} = \text{CH}_3$) is : [AIEEE 2010]
 (A) $\text{RCOO}^- < \text{NH}_2^- < \text{HC} \equiv \text{C}^- < \text{R}^-$ (B) $\text{RCOO}^- < \text{HC} \equiv \text{C}^- < \text{NH}_2^- < \text{R}^-$
 (C) $\text{RCOO}^- < \text{HC} \equiv \text{C}^- < \text{R}^- < \text{NH}_2^-$ (D) $\text{R}^- < \text{HC} \equiv \text{C}^- < \text{RCOO}^- < \text{NH}_2^-$
12. The strongest acid amongst the following compounds is [AIEEE 2011]
 (A) CH_3COOH (B) HCOOH
 (C) $\text{CH}_3\text{CH}_2\text{CH}(\text{Cl})\text{CO}_2\text{H}$ (D) $\text{ClCH}_2\text{CH}_2\text{CH}_2\text{COOH}$

13. Arrange the following compounds in order of decreasing acidity :

[JEE Main 2013]



(A) $\text{I} > \text{II} > \text{III} > \text{IV}$

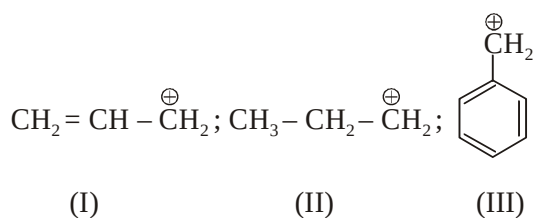
(B) $\text{III} > \text{I} > \text{II} > \text{IV}$

(C) $\text{IV} > \text{III} > \text{I} > \text{II}$

(D) $\text{II} > \text{IV} > \text{I} > \text{III}$

14. The order of stability of the following carbocations:

[JEE Main 2013]



is

(A) $\text{II} > \text{III} > \text{I}$

(B) $\text{I} > \text{II} > \text{III}$

(C) $\text{III} > \text{I} > \text{II}$

(D) $\text{III} > \text{II} > \text{I}$

15. Considering the basic strength of amines in aqueous solution, which one has the smallest pK_b value?

[JEE Main 2014]

(A) CH_3NH_2

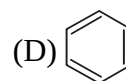
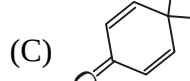
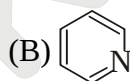
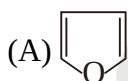
(B) $(\text{CH}_3)_3\text{N}$

(C) $\text{C}_6\text{H}_5\text{NH}_2$

(D) $(\text{CH}_3)_2\text{NH}$

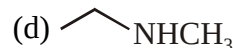
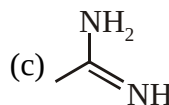
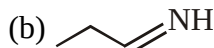
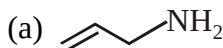
16. Which of the following molecules is least resonance stabilized ?

[JEE Main 2017]



17. The increasing order of basicity of the following compounds is :

[JEE Main 2018]



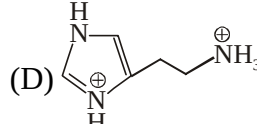
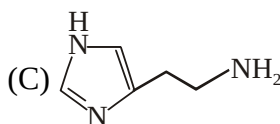
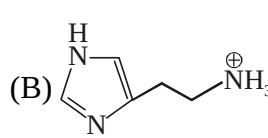
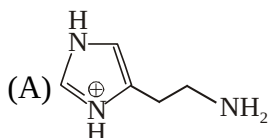
(A) $(b) < (a) < (d) < (c)$

(B) $(d) < (b) < (a) < (c)$

(C) $(a) < (b) < (c) < (d)$

(D) $(b) < (a) < (c) < (d)$

18. The predominant form of histamine present in human blood is (pK_a , Histidine = 6.0) [JEE Main 2018]



19. Which of the following compounds will show the maximum 'enol' content ? [JEE Main 2019]

- (A) CH_3COCH_3 (B) $\text{CH}_3\text{COCH}_2\text{COCH}_3$
(C) $\text{CH}_3\text{COCH}_2\text{COOC}_2\text{H}_5$ (D) $\text{CH}_3\text{COCH}_2\text{CONH}_2$

20. Which amongst the following is the strongest acid ? [JEE Main 2019]

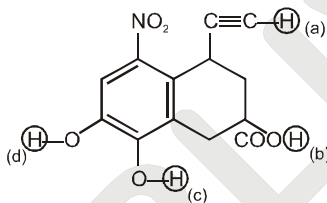
- (A) CHBr_3 (B) CHI_3 (C) CHCl_3 (D) $\text{CH}(\text{CN})_3$

21. The correct order for acid strength of compounds : [JEE Main 2019]

$\text{CH} \equiv \text{CH}$, $\text{CH}_3 - \text{C} \equiv \text{CH}$ and $\text{CH}_2 = \text{CH}_2$ is as follows :

- (A) $\text{CH}_3 - \text{C} \equiv \text{CH} > \text{CH}_2 = \text{CH}_2 > \text{HC} \equiv \text{CH}$
(B) $\text{CH}_3 - \text{C} \equiv \text{CH} > \text{CH} \equiv \text{CH} > \text{CH}_2 = \text{CH}_2$
(C) $\text{HC} \equiv \text{CH} > \text{CH}_3 - \text{C} \equiv \text{CH} > \text{CH}_2 = \text{CH}_2$
(D) $\text{CH} \equiv \text{CH} > \text{CH}_2 = \text{CH}_2 > \text{CH}_3 - \text{C} \equiv \text{CH}$

22. Arrange the following labelled hydrogens in decreasing order of acidity : [JEE Main 2020]



- (A) $b > c > d > a$ (B) $c > b > a > d$ (C) $b > a > c > d$ (D) $c > b > d > a$

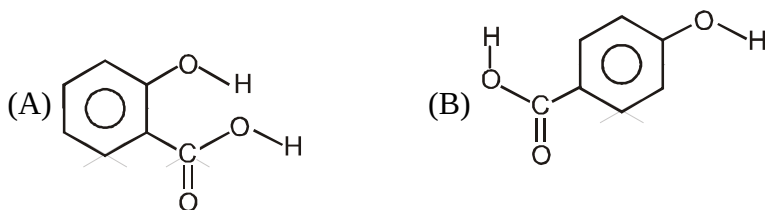
23. If you spill a chemical toilet cleaning liquid on your hand, your first aid would be: [JEE Main 2020]

- (A) vinegar (B) aqueous NaOH (C) aqueous NaHCO_3 (D) aqueous NH_3

24. Which one of the following compounds possesses the most acidic hydrogen ? [JEE Main 2020]

- (A) $\text{H}_3\text{C} - \text{C} \equiv \text{C} - \text{H}$ (B) $\text{N} \equiv \text{C} - \text{C}(\text{H})_2 - \text{C} \equiv \text{N}$
(C) (D)

25. Consider the following molecules and statements related to them : [JEE Main 2020]

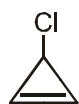


- (a) (B) is more likely to be crystalline than (A) (b) (B) has higher boiling point than (A)
(c) (B) dissolves more readily than (A) in water

Identify the correct option from below :

- (A) only (a) is true (B) (a) and (b) are true (C) (b) and (c) are true (D) (a) and (c) are true

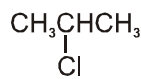
26. The decreasing order of reactivity of the following organic molecules towards AgNO_3 solution is :



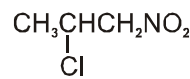
(A)



(B)



(C)



(D)

(A) (C) > (D) > (A) > (B)

(B) (A) > (B) > (C) > (D)

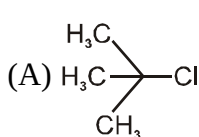
[JEE Main 2020]

(C) (B) > (A) > (C) > (D)

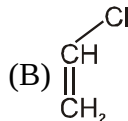
(D) (A) > (B) > (D) > (C)

27. Among the following compounds, which one has the shortest C-Cl bond ?

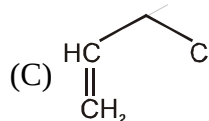
[JEE Main 2020]



(A)



(B)

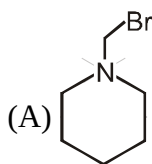


(C)

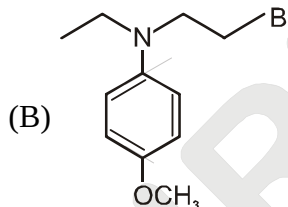
(D)

28. Which of the following compounds will form the precipitate with aq. AgNO_3 solution most readily ?

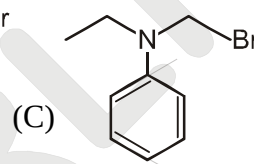
[JEE Main 2020]



(A)



(B)

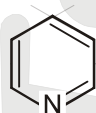


(C)

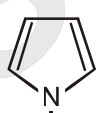
(D)

29. The increasing order of basicity of the following compounds is :

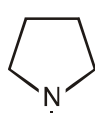
[JEE Main 2020]



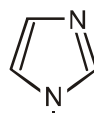
(A)



(B)



(C)



(D)

(A) (B) < (A) < (C) < (D)

(B) (A) < (B) < (C) < (D)

(C) (D) < (A) < (B) < (C)

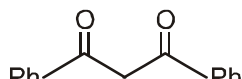
(D) (B) < (A) < (D) < (C)

30. The increasing order of the acidity of the α -hydrogen of the following compounds is :

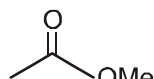
[JEE Main 2020]



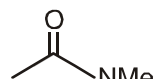
(A)



(B)



(C)



(D)

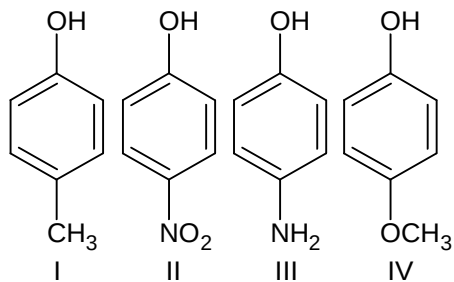
(A) (C) < (A) < (B) < (D)

(B) (B) < (C) < (A) < (D)

(C) (A) < (C) < (D) < (B)

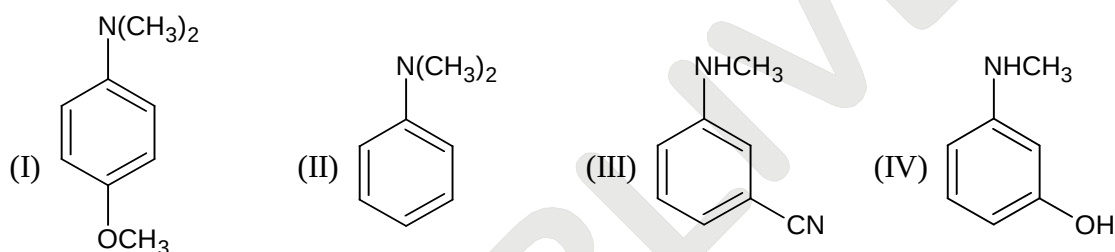
(D) (D) < (C) < (A) < (B)

31. The increasing order of boiling points of the following compounds is : [JEE Main 2020]



- (A) $\text{I} < \text{IV} < \text{III} < \text{II}$ (B) $\text{IV} < \text{I} < \text{II} < \text{III}$ (C) $\text{I} < \text{III} < \text{IV} < \text{II}$ (D) $\text{III} < \text{I} < \text{II} < \text{IV}$

32. The increasing order of pK_b values of the following compounds is : [JEE Main 2020]



- (A) $\text{II} < \text{I} < \text{III} < \text{IV}$ (B) $\text{II} < \text{IV} < \text{III} < \text{I}$ (C) $\text{I} < \text{II} < \text{IV} < \text{III}$ (D) $\text{I} < \text{II} < \text{III} < \text{IV}$

[JEE-ADVANCED]

1. Phenol is less acidic than [JEE 1986]
 (A) Acetic acid (B) p-methoxy phenol (C) p-nitrophenol (D) Ethanol

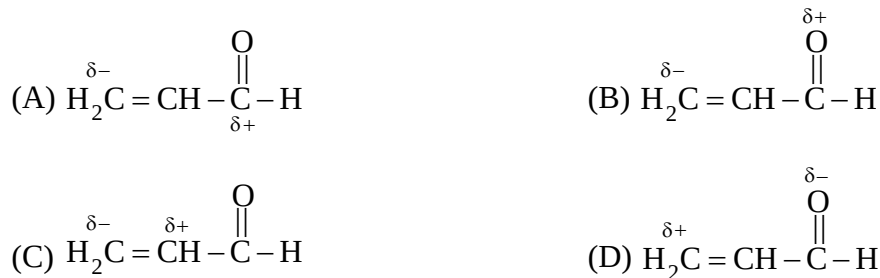
2. The bond between carbon atom (1) and carbon atom (2) in compound [JEE 1987]

$$\text{N} \equiv \underset{\text{1}}{\text{C}} - \underset{\text{2}}{\text{CH}} = \text{CH}_2$$

involves the hybridisations as

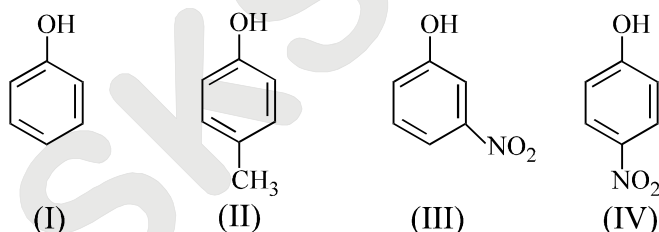
- (A) sp^2 and sp^2 (B) sp^3 and sp (C) sp and sp^2 (D) sp and sp

3. Polarisation of electrons in acrolein may be written as [JEE 1988]



4. The number of sigma and pi-bonds in 1-butene 3-yne are [JEE 1989]
 (A) 5 sigma and 5 pi (B) 7 sigma and 3 pi
 (C) 8 sigma and 2 pi (D) 6 sigma and 4 pi

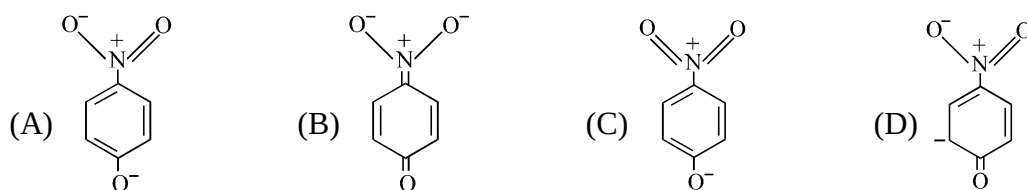
5. Amongst the following, the most basic compound is [JEE 1990]
 (A) benzylamine (B) aniline (C) acetaniline (D) p-nitroaniline
6. The shape of $(\text{CH}_3)^+$ is [JEE 1990]
7. The hybridisation of carbon atoms in C–C single bond $\text{H}-\text{C}\equiv\text{C}-\text{CH}=\text{CH}_2$ is [JEE 1991]
 (A) sp^3-sp^3 (B) sp^2-sp^3 (C) $\text{sp}-\text{sp}^3$ (D) $\text{sp}-\text{sp}^2$
8. What is the decreasing order of strength of the base? [JEE 1992]
 OH^- , NH_2^- , $\text{H}-\text{C}\equiv\text{C}^-$ and $\text{CH}_3-\text{CH}_2^-$
 (A) $\text{CH}_3-\text{CH}_2^- > \text{NH}_2^- > \text{H}-\text{C}\equiv\text{C}^- > \text{OH}^-$
 (B) $\text{H}-\text{C}\equiv\text{C}^- > \text{CH}_3-\text{CH}_2^- > \text{NH}_2^- > \text{OH}^-$
 (C) $\text{OH}^- > \text{NH}_2^- > \text{H}-\text{C}\equiv\text{C}^- > \text{CH}_3-\text{CH}_2^-$
 (D) $\text{NH}_2^- > \text{H}-\text{C}\equiv\text{C}^- > \text{OH}^- > \text{CH}_3-\text{CH}_2^-$
9. The bond dissociation energy needed to form the benzyl radical from toluene is _____ than the formation of the methyl radical from methane [JEE 1994]
10. The kind of delocalisation involving sigma bond orbitals is called [JEE 1994]
11. Arrange the following in the order of their increasing basicity : p-toluidine, N,N-dimethyl-p-toluidine, p-nitroaniline, aniline. [JEE 1995]
12. In the following compounds: [JEE 1996]



The order of acidity is:

- (A) $\text{III} > \text{IV} > \text{I} > \text{II}$ (B) $\text{I} > \text{IV} > \text{III} > \text{II}$ (C) $\text{II} > \text{I} > \text{III} > \text{IV}$ (D) $\text{IV} > \text{III} > \text{I} > \text{II}$

13. Although phenoxide ion has more number of resonating structures than benzoate ion, benzoic acid is a stronger acid than phenol. Why? [JEE 1997]
14. Among the following compounds, the strongest acid is [JEE 1998]
 (A) $\text{HC}\equiv\text{CH}$ (B) C_6H_6 (C) C_2H_6 (D) CH_3OH
15. The most unlikely representation of resonance structure of p-nitrophenoxide ion is: [JEE 1998]



16. Give reasons for the following in one or two sentences.
The central carbon - carbon bond in 1,3 butadiene is shorter than that of n-butane. [JEE 1998]
17. Explain, why o-hydroxybenzaldehyde is a liquid at room temperature while p-hydroxybenzaldehyde is a high melting solid [JEE 1999]
18. Give reasons for the following [JEE 2000]
 $\text{CH}_2 = \text{CH}^-$ is more basic than $\text{HC} \equiv \text{C}^-$
19. The compound in which C uses only its sp^3 -hybrid orbitals for bond formation is : [JEE 2000]
(A) HCOOH (B) $(\text{H}_2\text{N})_2\text{CO}$ (C) $(\text{CH}_3)_3\text{COH}$ (D) CH_3CHO
20. Which of the following has the most acidic hydrogen? [JEE 2000]
(A) 3-hexanone (B) 2,4-hexanedione (C) 2,5-hexanedione (D) 2,3-hexanedione
21. Amongst the following the strongest base is: [JEE 2000]
(A) $\text{C}_6\text{H}_5\text{NH}_2$ (B) $\text{p-O}_2\text{NC}_6\text{H}_4\text{NH}_2$ (C) $\text{m-O}_2\text{NC}_6\text{H}_4\text{NH}_2$ (D) $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$
22. The correct order of basicities of the following compounds is: [JEE 2001]
- $\begin{array}{c} \text{NH} \\ \diagup \\ \text{CH}_3-\text{C} \\ \diagdown \\ \text{NH}_2 \end{array}$
(1)

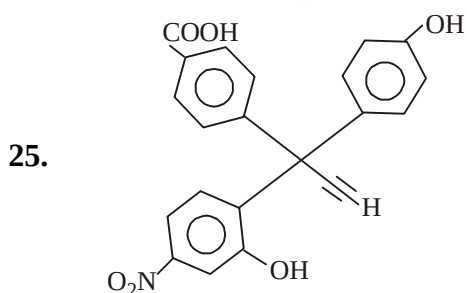
$\text{CH}_3\text{CH}_2\text{NH}_2$
(2)

$(\text{CH}_3)_2\text{NH}$
(3)

$\begin{array}{c} \text{O} \\ \parallel \\ \text{CH}_3\text{CNH}_2 \end{array}$
(4)
- (A) $2 > 1 > 3 > 4$ (B) $1 > 3 > 2 > 4$ (C) $3 > 1 > 2 > 4$ (D) $1 > 2 > 3 > 4$
23. Identify the correct order of boiling points of the following compounds. [JEE 2002]
- (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$

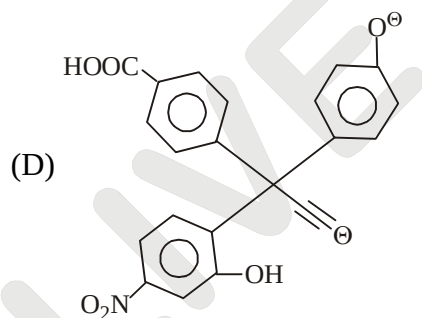
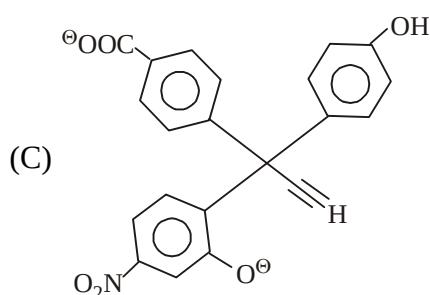
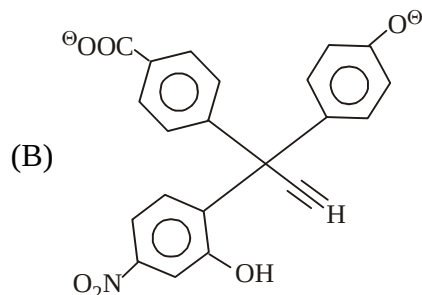
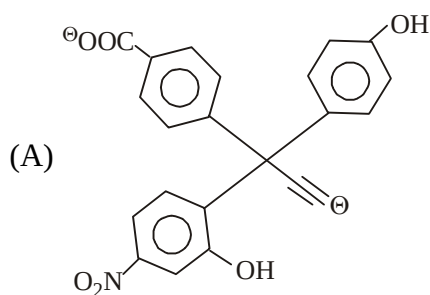
(3) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$

(2) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$
- (A) $1 > 2 > 3$ (B) $3 > 1 > 2$ (C) $1 > 3 > 2$ (D) $3 > 2 > 1$
24. Which of the following acids has the smallest dissociation constant? [JEE 2002]
(A) $\text{CH}_3\text{CHF}\text{COOH}$ (B) $\text{FCH}_2\text{CH}_2\text{COOH}$
(C) $\text{BrCH}_2\text{CH}_2\text{COOH}$ (D) $\text{CH}_3\text{CHBr}\text{COOH}$

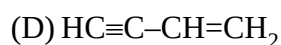
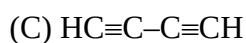
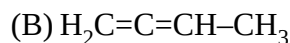
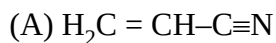


when X is made to react with 2 eq. of NaNH_2 the product formed will be:

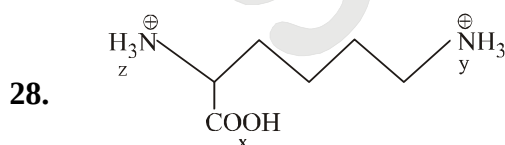
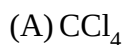
[JEE 2003]



26. Which of the following represent the given mode of hybridisation sp^2 , sp^2 , sp , sp from left to right ? [JEE 2003]

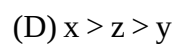
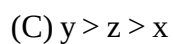
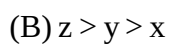
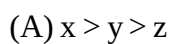


27. Maximum dipole moment will be of : [JEE 2003]

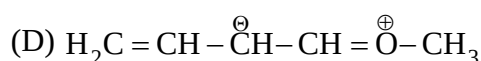
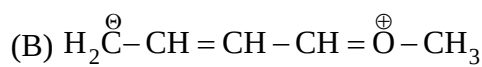
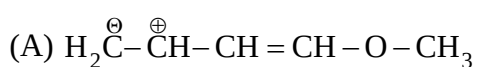


[JEE 2004]

Correct order of acidic strength is:



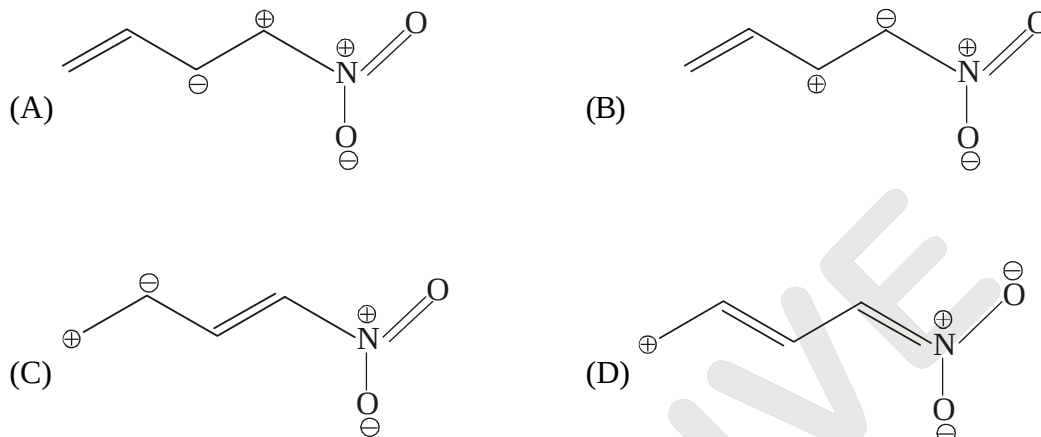
29. For 1-methoxy-1,3-butadiene, which of the following resonating structure is the least stable? [JEE 2005]



30. When benzene sulfonic acid and p-nitrophenol are treated with NaHCO_3 , the gases released respectively are [JEE 2006]

(A) SO_2 , NO_2 (B) SO_2 , NO (C) SO_2 , CO_2 (D) CO_2 , CO_2

31. Among the following, the least stable resonance structure is [JEE 2007]



32. The correct stability order for the following species is [JEE 2008]

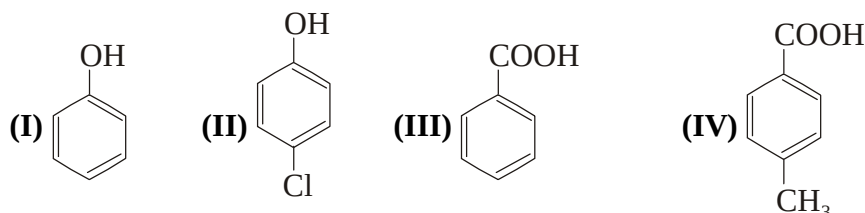


(A) (II) > (IV) > (I) > (III) (B) (I) > (II) > (III) > (IV)
(C) (II) > (I) > (IV) > (III) (D) (I) > (III) > (II) > (IV)

33. Hyperconjugation involves overlap of the following orbitals [JEE 2008]

(A) $\sigma - \sigma$ (B) $\sigma - p$ (C) $p - p$ (D) $\pi - \pi$

34. The **correct** acidity order of the following is : [JEE 2008]



(A) (III) > (IV) > (II) > (I) (B) (IV) > (III) > (I) > (II)
(C) (III) > (II) > (I) > (IV) (D) (II) > (III) > (IV) > (I)

35. The correct stability order of the following resonance structure is : [JEE 2009]



(A) (I) > (II) > (IV) > (III)

(B) (I) > (III) > (II) > (IV)

(C) (II) > (I) > (III) > (IV)

(D) (III) > (I) > (IV) > (II)

36. Among the following compounds, the most acidic is

[JEE 2011]

(A) p-nitrophenol

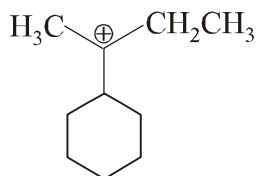
(B) p-hydroxybenzoic acid

(C) o-hydroxybenzoic acid

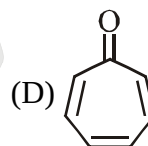
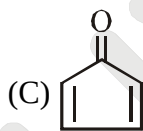
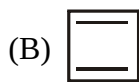
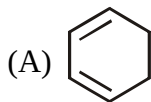
(D) p-toluic acid

37. The total number of contributing structures showing hyperconjugation (involving C-H bonds) for the following carbocation is

[JEE 2011]



38. Which of the following molecules, in pure form, is (are) unstable at room temperature ? [JEE 2012]

39. The compound that does **NOT** liberate CO₂, on treatment with aqueous sodium bicarbonate solution, is:

[JEE Advance 2013]

(A) Benzoic acid

(B) Benzenesulphonic acid

(C) Salicylic acid

(D) Carboic acid (Phenol)

40. The hyperconjugative stabilities of tert-butyl cation and 2-butene, respectively, are due to

[JEE Advance 2013]

(A) $\sigma \rightarrow p$ (empty) and $\sigma \rightarrow \pi^*$ electron delocalisations.(B) $\sigma \rightarrow \sigma^*$ and $\sigma \rightarrow \pi$ electron delocalisations(C) $\sigma \rightarrow p$ (filled) and $\sigma \rightarrow \pi$ electron delocalisations(D) p (filled) $\rightarrow \sigma^*$ and $\sigma \rightarrow \pi^*$ electron delocalisations

41. The total number of lone-pairs of electrons in melamine is

[JEE Advance 2013]

42. Hydrogen bonding plays a central role in the following phenomena :

[JEE Advance 2014]

(A) Ice floats in water.

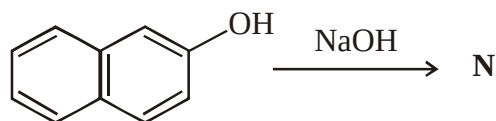
(B) Higher Lewis basicity of primary amines than tertiary amines in aqueous solutions

(C) Formic acid is more acidic than acetic acid.

(D) Dimerisation of acetic acid in benzene.

43. The number of resonance structures for **N** is :

[JEE Advance 2015]

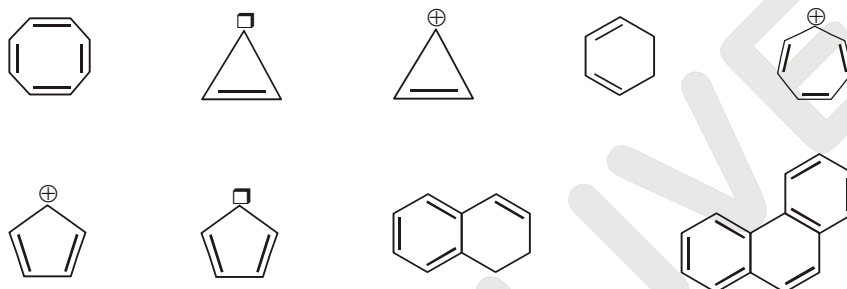


44. The correct order of acidity for the following compound is : [JEE Advance 2016]

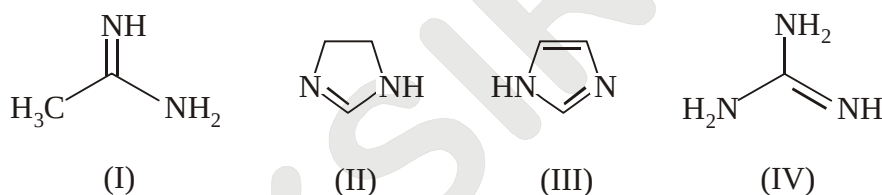


- (A) I > II > III > IV (B) III > I > II > IV (C) III > IV > II > I (D) I > III > IV > II

45. Among the following, the number of aromatic compound(s) is [JEE Advance 2017]

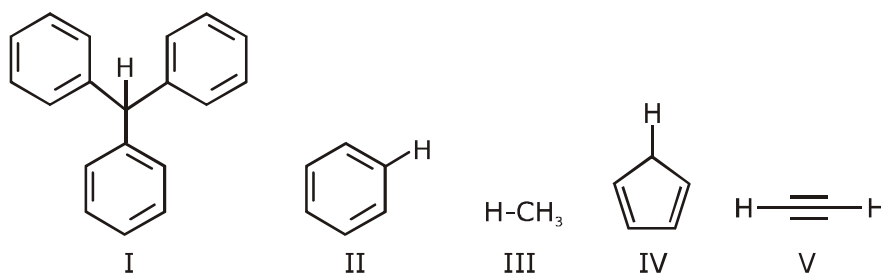


46. The order of basicity among the following compounds is [JEE Advance 2017]



- (A) I > IV > III > II (B) II > I > IV > III (C) IV > I > II > III (D) IV > II > III > I

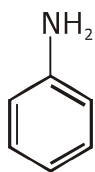
47. With respect to the compounds I–V, choose the correct statement(s). [JEE Advance 2020]



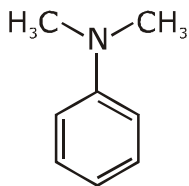
- (A) The acidity of compound I is due to delocalization in the conjugate base.
 (B) The conjugate base of compound IV is aromatic.
 (C) Compound II becomes more acidic, when it has a $-\text{NO}_2$ substituent.
 (D) The acidity of compounds follows the order I > IV > V > II > III.

48. Consider the following four compounds I, II, III, and IV.

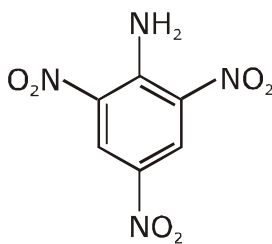
[JEE Advance 2020]



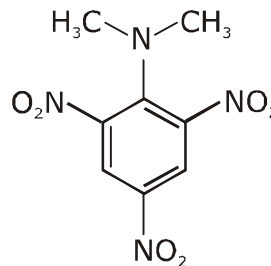
I



II



III



IV

Choose the correct statement(s).

- (A) The order of basicity is $\text{II} > \text{I} > \text{III} > \text{IV}$.
(B) The magnitude of pK_b difference between I and II is more than that between III and IV.
(C) Resonance effect is more in III than in IV
(D) Steric effect makes compound IV more basic than III.

ANSWER KEY**[JEE-MAIN/AIEEE]**

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. A | 2. B | 3. A | 4. D | 5. D | 6. B | 7. D |
| 8. B | 9. C | 10. B | 11. B | 12. C | 13. B | 14. C |
| 15. D | 16. C | 17. A | 18. B | 19. B | 20. D | 21. C |
| 22. A | 23. C | 24. C | 25. B | 26. C | 27. B | 28. A |
| 29. D | 30. D | 31. A | 32. C | | | |

[JEE-ADVANCED]

- | | | | | | |
|--|---|---------|----------------------|---------------------|----------------------|
| 1. AC | 2. C | 3. D | 4. B | 5. A | 6. Triangular planar |
| 7. D | 8. A | 9. less | 10. hyperconjugation | 11. $2 > 1 > 4 > 3$ | |
| 12. D | 13. Better resonance occurs in Benzoic acid | 14. D | 15. C | | |
| 16. Because of Resonance | | | | | |
| 17. Becoz of intramolecular H-bonding is o-hydroxybenzaldehyde | | | | | |
| 18. Acidic strength : $\text{CH} \equiv \text{CH} > \text{CH}_2=\text{CH}_2$ | | | | | |
| Basic Strength : $\text{CH} \equiv \text{C}^- < \text{CH}_2=\text{CH}^-$ | 19. C | 20. B | 21. D | 22. B | |
| 23. B | 24. C | 25. C | 26. A | 27. D | 28. D |
| 30. D | 31. A | 32. D | 33. B | 34. A | 35. B |
| 37. 6 | 38. BC | 39. D | 40. A | 41. 6 | 42. ABD |
| 44. A | 45. 5 | 46. C | 47. ABC | 48. CD | 43. 9 |

[JEE-MAIN / AIEEE]

1. The compound

$$\begin{array}{c} \text{H}_3\text{C} - \text{C} = \text{CH} - \text{CH}_2 - \text{CH}_3 \\ | \\ \text{H}_3\text{C} \end{array} \xrightarrow{\text{Vigorous oxidation}}$$
 product, here product is [AIEEE-2002]

- (A) CH_3COOH & $\text{CH}_3 - \overset{\text{O}}{\underset{\text{O}}{\text{C}}} - \text{CH}_3$ (B) $\text{CH}_3 - \text{CH}_2 - \text{COOH}$ & $\text{CH}_3 - \overset{\text{O}}{\underset{\text{CH}_3}{\text{C}}} = \text{O}$
- (C) $\text{CH}_3 - \text{CH}_2 - \text{COOH}$ only (D) HCOOH & $\text{CH}_3 - \overset{\text{O}}{\underset{\text{O}}{\text{C}}} - \text{CH}_3$

2. Reaction

$\text{H} - \text{C} \equiv \text{C} - \text{H} + \text{HOCl} \longrightarrow$ product, here product will be - [AIEEE-2002]

- (A) $\text{CHCl}_2 - \text{CHO}$ (B) $\text{CHO} - \text{CHO}$ (C) $\text{CH} - \text{Cl} = \text{CHCl}$ (D) $\text{CHCl}_2 - \text{CHCl}_2$

3. Acetylene does not react with -

[AIEEE-2002]

- (A) NaNH_2 (B) NaOH (C) Na metal (D) Ammonical AgNO_3

4. Butene-1 may be converted to butane by reaction with -

[AIEEE-2003]

- (A) $\text{Zn} - \text{Hg}$ (B) Pd / H_2 (C) $\text{Zn} - \text{HCl}$ (D) $\text{Sn} - \text{HCl}$

5. During dehydration of alcohols to alkenes by heating with conc.
- H_2SO_4
- the initiation step is -

[AIEEE-2003]

- (A) Elimination of water
(B) Formation of an ester
(C) Protonation of alcohol molecule
(D) Formation of carbocation

6. On mixing a certain alkane with chlorine and irradiating it with ultraviolet light, it forms only one monochloroalkane. This alkane could be -

[AIEEE-2003]

- (A) Isopentane (B) Neopentane (C) Propane (D) Pentane

7. Which one of the following has the minimum boiling point ?

[AIEEE-2004]

- (A) n - Butane (B) 1- Butyne (C) 1- Butene (D) isobutane

8. Amongst the following compounds, the optically active alkane having lowest molecular mass is-

[AIEEE-2004]

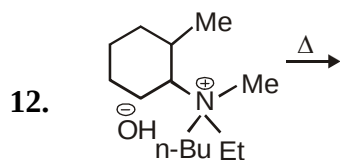
- (A) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$ (B) $\text{CH}_3 - \text{CH}_2 - \overset{\text{CH}_3}{\underset{|}{\text{CH}}} - \text{CH}_3$

- (C) $\text{CH}_3 - \overset{\text{H}}{\underset{\text{C}_2\text{H}_5}{\text{C}}} - \triangle$ (D) $\text{CH}_3 - \text{CH}_2 - \text{C} \equiv \text{CH}$

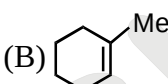
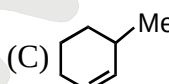
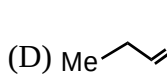
9. Reaction of one molecule of HBr with one molecule of 1,3-butadiene at 40°C given predominantly
 (A) 1-bromo-2-butene under thermodynamically controlled conditions [AIEEE-2005]
 (B) 3-bromobutene under kinetically controlled conditions
 (C) 1-bromo-2-butene under kinetically controlled conditions
 (D) 3-bromobutene under thermodynamically controlled conditions

10. Acid catalyzed hydration of alkenes except ethene leads to the formation of – [AIEEE-2005]
 (A) secondary or tertiary alcohol (B) primary alcohol
 (C) mixture of secondary and tertiary alcohols (D) mixture of primary and secondary alcohols

11. Reaction of trans 2-phenyl-1 bromocyclopentane on reaction with alcoholic KOH produces
 (A) 2-phenylcyclopentene (B) 1-phenylcyclopentene [AIEEE 2006]
 (C) 3-phenylcyclopentene (D) 4-phenylcyclopentene



The alkene formed as a major product in the above elimination reaction is [AIEEE 2006]

- (A) $\text{CH}_2 = \text{CH}_2$ (B)  (C)  (D) 

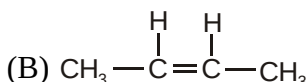
13. Which of the following reactions will yield 2, 2-dibromopropane? [AIEEE 2007]
 (A) $\text{CH}_3 - \text{C} \equiv \text{CH} + 2\text{HBr} \rightarrow$ (B) $\text{CH}_3\text{CH} = \text{CHBr} + \text{HBr} \rightarrow$
 (C) $\text{CH} \equiv \text{CH} + 2\text{HBr} \rightarrow$ (D) $\text{CH}_3 - \text{CH} = \text{CH}_2 + \text{HBr} \rightarrow$

14. In the following sequence of reactions, the alkene affords the compound 'B'

$\text{CH}_3\text{CH} = \text{CHCH}_3 \xrightarrow{\text{O}_3} \text{A} \xrightarrow[\text{Zn}]{\text{H}_2\text{O}} \text{B}$, The compound B is [AIEEE 2008]

- (A) CH_3COCH_3 (B) $\text{CH}_3\text{CH}_2\text{COCH}_3$
 (C) CH_3CHO (D) $\text{CH}_3\text{CH}_2\text{CHO}$

15. The treatment of CH_3MgX with $\text{CH}_3\text{C} \equiv \text{C} - \text{H}$ produces [AIEEE 2008]

- (A) $\text{CH}_3\text{C} \equiv \text{C} - \text{CH}_3$ (B) 
 (C) CH_4 (D) $\text{CH}_3 - \text{CH} = \text{CH}_2$

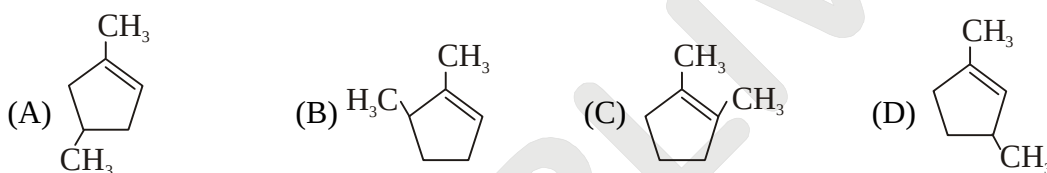
16. The hydrocarbon which can not react with sodium in liquid ammonia to give H_2 is [AIEEE 2008]

- (A) $\text{CH}_3\text{CH}_2\text{C} \equiv \text{CH}$ (B) $\text{CH}_3\text{CH} = \text{CHCH}_3$
 (C) $\text{CH}_3\text{CH}_2\text{C} \equiv \text{CCH}_2\text{CH}_3$ (D) $\text{CH}_3\text{CH}_2\text{CH}_2\text{C} \equiv \text{CCH}_2\text{CH}_2\text{CH}_3$

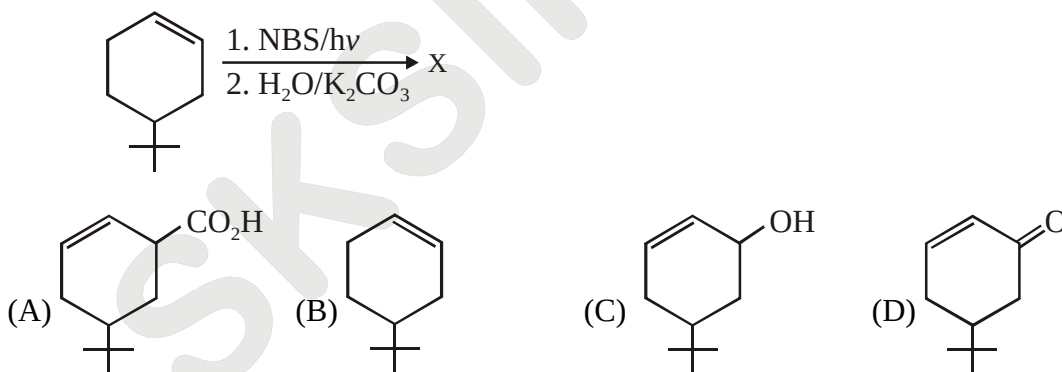
17. Out of the following, the alkene that exhibits optical isomerism is : [AIEEE 2010]

- (A) 3-methyl-1-pentene (B) 2-methyl-2-pentene
 (C) 3-methyl-2-pentene (D) 4-methyl-1-penten

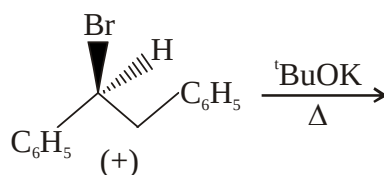
18. Which branched chain isomer of the hydrocarbon with molecular mass 72u gives only one isomer of mono substituted alkyl halide? [AIEEE 2012]
 (A) Isohexane (B) Neohexane
 (C) Tertiary butyl chloride (D) Neopentane.
19. 2-Hexyne gives trans-2-Hexene on treatment with : [AIEEE 2012]
 (A) Pd/BaSO₄ (B) LiAlH₄
 (C) Pt/H₂ (D) Li/NH₃
20. A gaseous hydrocarbon gives upon combustion 0.72 g of water and 3.08 g of CO₂. The empirical formula of the hydrocarbon is: [JEE Main 2013]
 (A) C₃H₄ (B) C₆H₅ (C) C₇H₈ (D) C₂H₄
21. Which compound would give 5-keto-2-methyl hexanal upon ozonolysis? [JEE Main 2015]



22. The product of the reaction given below is: [JEE Main 2016]

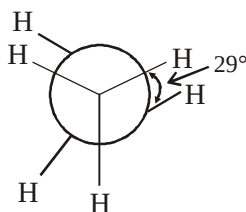


23. The reaction of propene with HOCl (Cl₂ + H₂O) proceeds through the intermediate: [JEE Main 2016]
 (A) CH₃-CHCl-CH₂⁺ (B) CH₃-CH⁺-CH₂-OH
 (C) CH₃-CH⁺-CH₂-Cl (D) CH₃-CH(OH)-CH₂⁺
24. The major product obtained in the following reaction is : [JEE Main 2017]

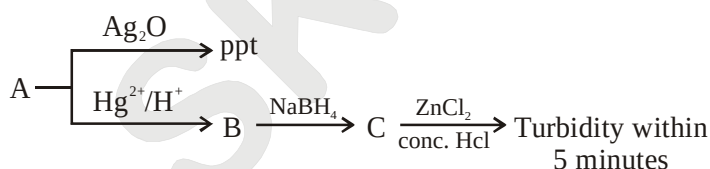


- (A) C₆H₅CH = CHC₆H₅ (B) (+) C₆H₅CH(O^tBu)CH₂C₆H₅
 (C) (-) C₆H₅CH(O^tBu)CH₂C₆H₅ (D) (±) C₆H₅CH(O^tBu)CH₂C₆H₅

25. 3-Methyl-pent-2-ene on reaction with HBr in presence of peroxide forms an addition product. The number of possible stereoisomers for the product is : **[JEE Main 2017]**
 (A) Zero (B) Two (C) Four (D) Six
26. The trans-alkenes are formed by the reduction of alkynes with: **[JEE Main 2018]**
 (A) Na/liq. NH_3 (B) Sn-HCl (C) H_2 -Pd/C, BaSO_4 (D) NaBH_4
27. In the following skew conformation of ethane, $\text{H}' - \text{C} - \text{C} - \text{H}''$ dihedral angle is **[JEE MAIN 2019]**



- (A) 58° (B) 149° (C) 151° (D) 120°
28. Which of these factors does not govern the stability of a conformation in acyclic compounds ? **[JEE MAIN 2019]**
 (A) Electrostatic forces of interaction
 (B) Torsional strain
 (C) Angle strain
 (D) Steric interactions
29. Consider the following reactions, **[JEE MAIN 2019]**

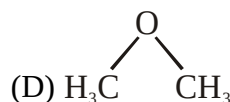
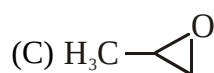
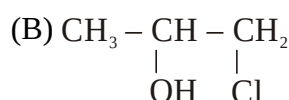
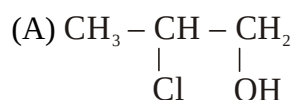


A is

- (A) $\text{CH} \equiv \text{CH}$ (B) $\text{CH}_3 - \text{C} \equiv \text{C} - \text{CH}_3$
 (C) $\text{CH}_3 - \text{C} \equiv \text{CH}$ (D) $\text{CH}_2 = \text{CH}_2$
30. But-2-ene on reaction with alkaline KMnO_4 at elevated temperature followed by acidification will give **[JEE MAIN 2019]**
 (A) $\text{CH}_3 - \underset{\text{OH}}{\text{CH}} - \underset{\text{OH}}{\text{CH}} - \text{CH}_3$
 (B) one molecule of CH_3CHO and molecule of CH_3COOH
 (C) 2 molecules of CH_3COOH
 (D) 2 molecules of CH_3CHO

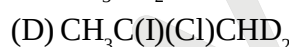
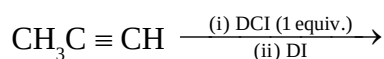
31. The major product of the following addition reaction is

[JEE MAIN 2019]



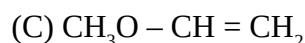
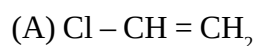
32. The major product of the following reaction is

[JEE MAIN 2019]



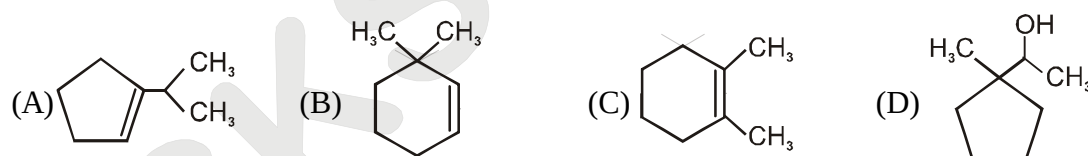
33. Which one of the following alkenes when treated with HCl yields majorly an anti Markownikov product ?

[JEE MAIN 2019]



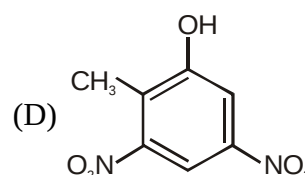
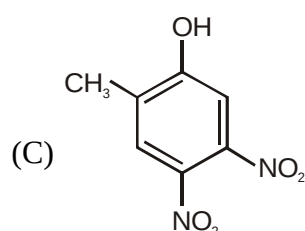
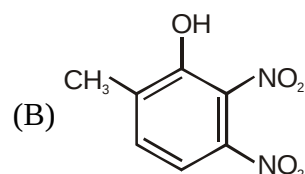
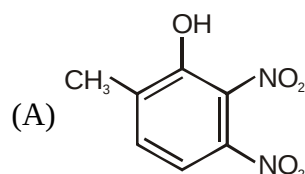
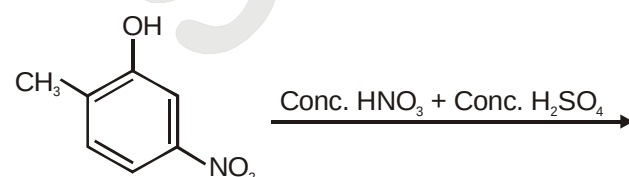
34. The major product in the following reaction is :

[JEE MAIN 2020]



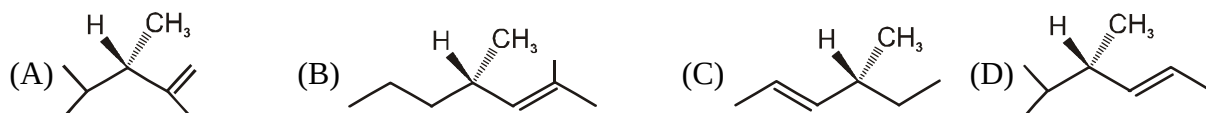
35. The major product of the following reaction is:

[JEE MAIN 2020]



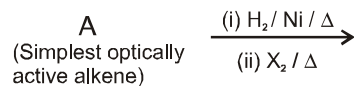
36. Which of the following compounds produces an optically inactive compound on hydrogenation ?

[JEE MAIN 2020]



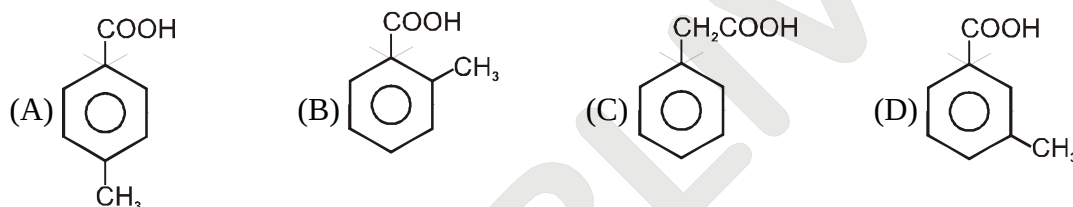
37. The total number of monohalogenated organic products in the following (including stereoisomers) reaction is _____.

[JEE MAIN 2020]



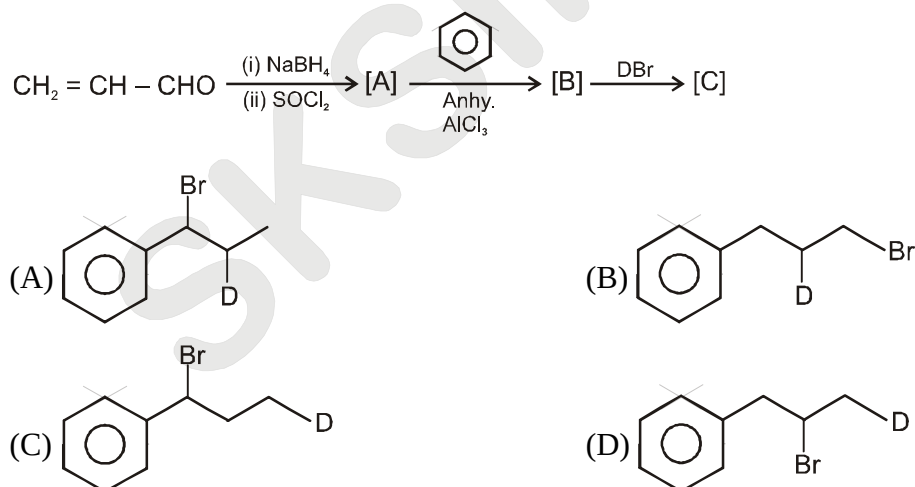
38. [P] on treatment with $Br_2/FeBr_3$ in CCl_4 produced a single isomer $C_8H_7O_2Br$ while heating [P] with sodalime gas toluene. The compound [P] is :

[JEE MAIN 2020]



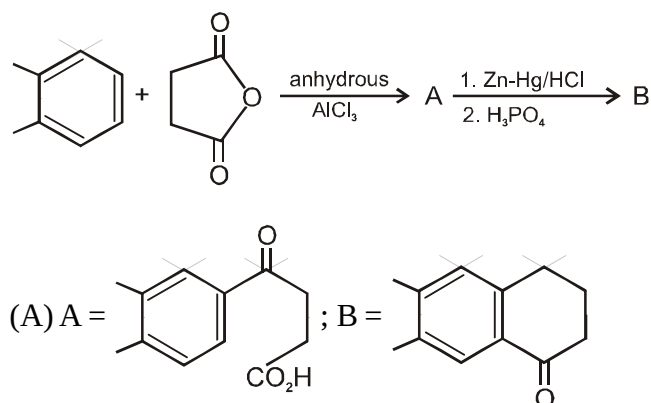
39. The major product [C] of the following reaction sequence will be :

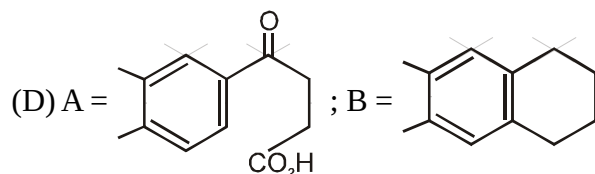
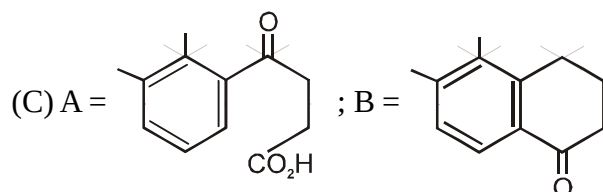
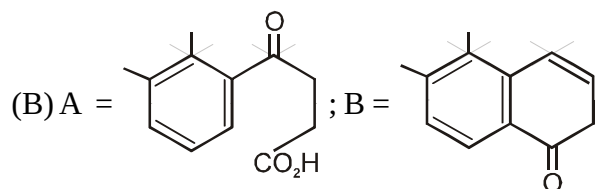
[JEE MAIN 2020]



40. In the following reaction sequence the major product A and B are :

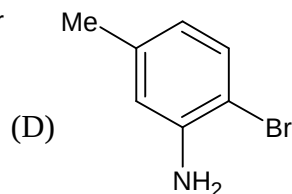
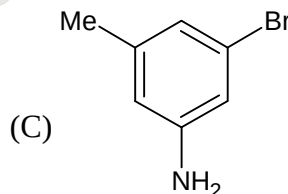
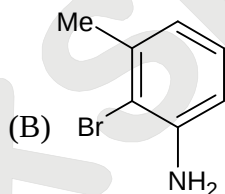
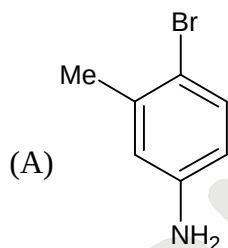
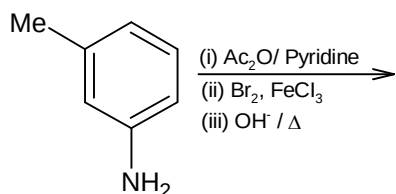
[JEE MAIN 2020]





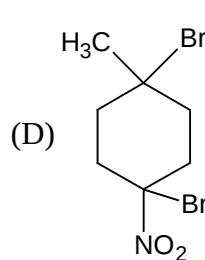
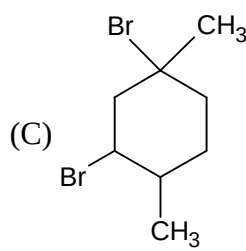
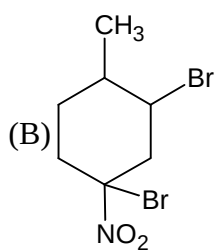
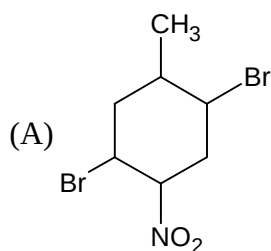
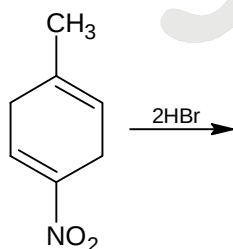
41. The final major product of the following reaction is :

[JEE MAIN 2020]



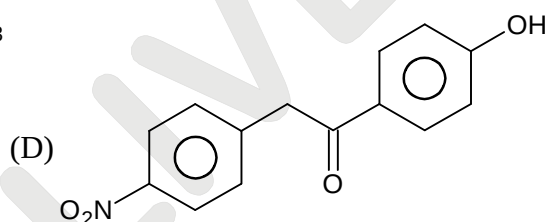
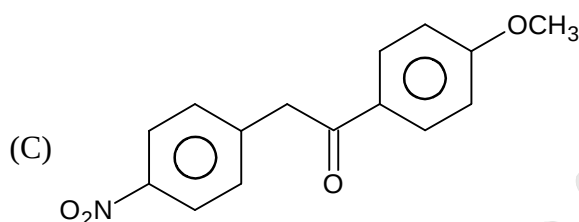
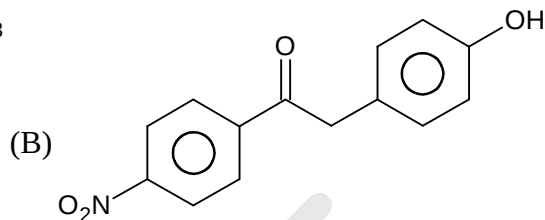
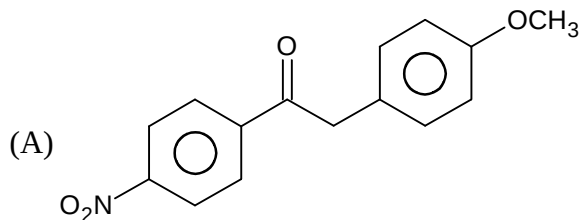
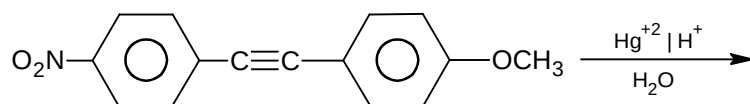
42. The major product of the following reaction is :

[JEE MAIN 2020]



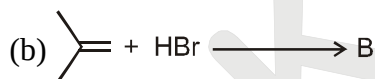
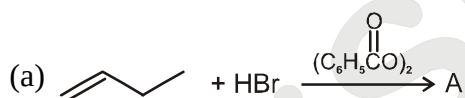
43. The major product obtained from the following reaction is :

[JEE MAIN 2020]



44. The increasing order of the boiling points of the major product A, B and C of the following reactions will be

[JEE MAIN 2020]



(A) $A < B < C$

(B) $A < C < B$

(C) $B < C < A$

(D) $C < A < B$

[JEE-ADVANCED]

1. Alcoholic solution of KOH is a specific reagent for –

[IIT 1990]

(A) Dehydration

(B) Dehydrogenation

(C) Dehydro halogenation

(D) Dehalogenation

2. Of the following, unsaturated hydrocarbons are –

[IIT 1990]

(A) ethyne

(B) cyclohexane

(C) n-propane

(D) ethene

3. 1-chlorobutane on reaction with alcoholic potash gives –

[IIT 1991]

(A) 1-butene

(B) 1-butanol

(C) 2-butene

(D) 2-butanol

4. The hybridisation of carbon atoms in C–C single bond of $\text{HC}\equiv\text{C}-\text{CH}=\text{CH}_2$ is –

[IIT 1991]

(A) sp^3-sp^3

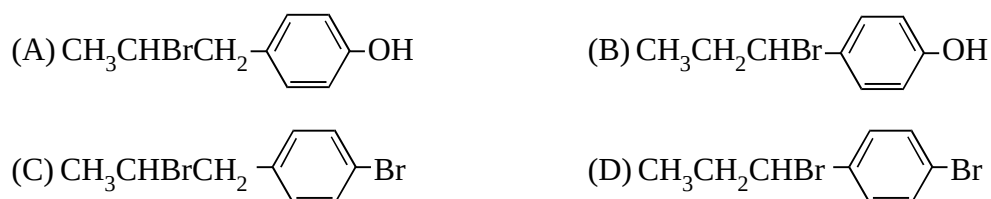
(B) sp^2-sp^3

(C) $\text{sp}-\text{sp}^2$

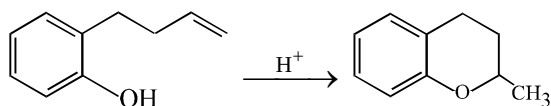
(D) sp^2-sp^2

5. $(\text{CH}_3)_2\overset{\text{Cl}}{\underset{|}{\text{C}}}-\text{CH}_2\text{CH}_3 \xrightarrow{\text{alc.KOH}} ?$ [IIT 1992]
6. $\text{C}_6\text{H}_5\text{CH}_2\underset{\text{Br}}{\underset{|}{\text{CH}}}\text{CH}_3 \xrightarrow[\text{heat}]{\text{alc.KOH}} ? \xrightarrow{\text{HBr}} ?$ [IIT 1993]
7. $\text{C}(\text{C}_6\text{H}_{12})_4$, an optically active hydrocarbon which on catalytic hydrogenation gives an optically inactive compound, C_6H_{14} . [IIT 1993]
8. Draw the stereochemical structure of the product in the following reactions. [IIT 1994]
- $\text{R}-\text{C}\equiv\text{C}-\text{R} \xrightarrow[\text{Lindlar catalyst}]{\text{H}_2}$
9. Write down the structures of the stereoisomers formed when cis-2-butene is reacted with bromine. [IIT 1995]
10. An organic compound E (C_5H_8) on hydrogenation gives compound F (C_5H_{12}). Compound E on ozonolysis gives formaldehyde and 2-ketopropanal. Deduce the structure of compound E. [IIT 1995]
11. Give the structures of the major organic products from 3-ethyl-2-pentene under each of the following reaction conditions. [IIT 1996]
- (a) HBr in the presence of peroxide (b) $\text{Br}_2/\text{H}_2\text{O}$
(c) $\text{Hg}(\text{OAc})_2/\text{H}_2\text{O}$; NaBH_4
12. An alkyl halide, (X) of formula $\text{C}_6\text{H}_{13}\text{Cl}$ on treatment with potassium tertiary butoxide gives two isomeric alkenes (Y) and (Z) (C_6H_{12}). Both alkenes on hydrogenation give 2, 3-dimethylbutane. Predict the structures of (X), (Y) and (Z). [IIT 1996]
13. 3,3-Dimethylbutan-2-ol loses a molecule of water in the presence of concentrated sulphuric acid to give tetramethylethylene as a major product. Suggest a suitable mechanism. [IIT 1996]
14. One mole of the compound A (molecular formula C_8H_{12}), incapable of showing stereoisomerism, reacts with only one mole of H_2 on hydrogenation over Pd. A undergoes ozonolysis to give a symmetrical diketone B ($\text{C}_8\text{H}_{12}\text{O}_2$). What are the structure of A and B? [IIT 1997]
15. Compound (A) C_6H_{12} gives a positive test with bromine in carbon tetrachloride. Reaction of (A) with alkaline KMnO_4 yields only (B) which is the potassium salt of an acid. Write structure formulae and IUPAC name of (A) and (B). [IIT 1997]
16. The central carbon-carbon bond in 1,3-butadiene is shorter than that of n-butane. Why?
17. Which of the following compounds will show geometrical isomerism? [IIT 1998]
- (A) 2-butene (B) Propene (C) 1-phenylpropene (D) 2-methyl-2-butene

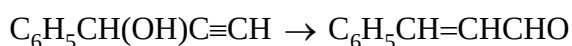
18. The reaction of $\text{CH}_3-\text{CH}=\text{CH}-\text{C}_6\text{H}_4-\text{OH}$ with HBr gives: [JEE 1998]



19. Write the intermediate steps for each of the following reaction. [IIT 1998]



20. Write the intermediate steps for each of the following reaction [IIT 1998]



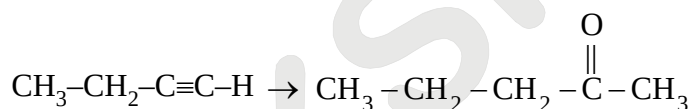
21. Give reasons for the following in one or two sentences. [JEE 1998]

"Acid catalysed dehydration of t-butanol is faster than that of n-butanol."

22. Benzyl chloride ($\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$) can be prepared from toluene by chlorination with:

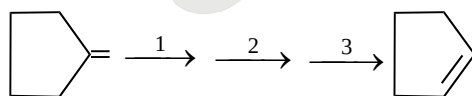


23. Carry out the following transformation in not more than three steps. [IIT 1999]

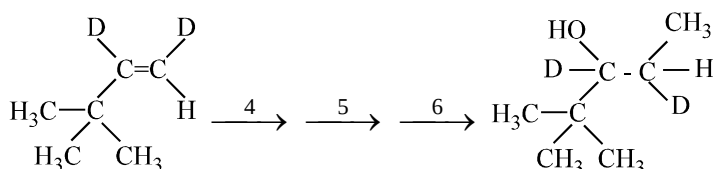


24. Discuss the hybridisation of carbon atoms in allene (C_3H_4) and show the π -orbital overlaps. [IIT 1999]

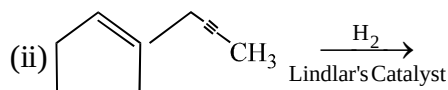
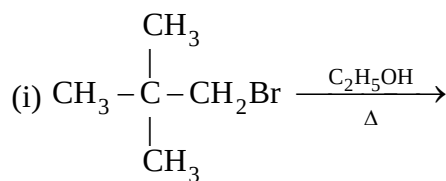
25. Complete the following – [IIT 1999]



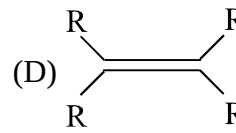
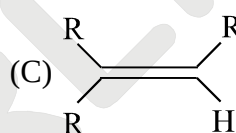
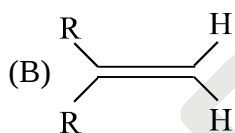
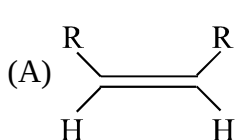
26. Complete the following – [IIT 1999]



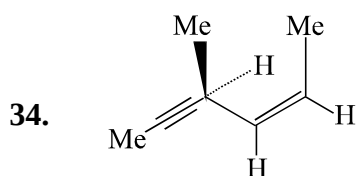
27. In the compound $\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}_2-\text{C}\equiv\text{CH}$, the C_2-C_3 bond is of the type – [IIT 1999]
 (A) $\text{sp}-\text{sp}^2$ (B) sp^3-sp^3 (C) $\text{sp}-\text{sp}^3$ (D) sp^2-sp^3
28. What would be the major product in each of the following reactions? [IIT 2000]



29. $\text{CH}_2=\text{CH}^-$ is more basic than $\text{HC}\equiv\text{C}^-$ [IIT 2000]
30. Propyne and propene can be distinguished by – [IIT 2000]
 (A) conc. H_2SO_4 (B) Br_2 in CCl_4 (C) dil. KMnO_4 (D) AgNO_3 in ammonia
31. Which one of the following alkenes will react fastest with H_2 under catalytic hydrogenation condition – [IIT 2000]



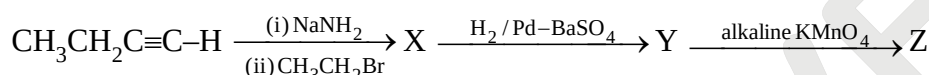
32. In the presence of peroxide, hydrogen chloride and hydrogen iodide do not give anti-Markovnikov addition to alkene because – [IIT 2001]
 (A) both are highly ionic
 (B) one is oxidising and the other is reducing
 (C) one of the step is endothermic in both the cases
 (D) All the steps are exothermic in both cases
33. The reaction of propene with HOCl proceeds via the addition of – [IIT 2001]
 (A) H^+ in first step (B) Cl^+ in first step (C) OH^- in first step (D) Cl^+ and OH^- in single step



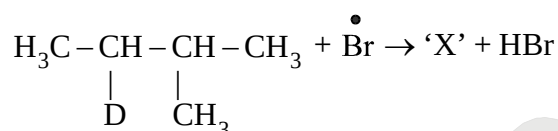
Hydrogenation of the above compound in the presence of poisoned palladium catalyst gives –

- (A) An optically active compound (B) An optically inactive compound [IIT 2001]
 (C) A racemic mixture (D) A diastereomeric mixture
35. On reaction with 4N alcoholic KOH at 175°C 1-pentyne is slowly converted into equilibrium mixture of 1.3% 1-pentyne (A), 95.2% 2-pentyne (B) and 3.5% 1,2-pentadiene (C). Give the suitable mechanism of formation of A, B and C with all intermediates. [IIT 2001]

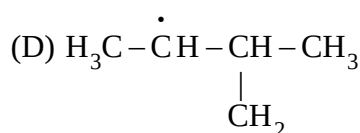
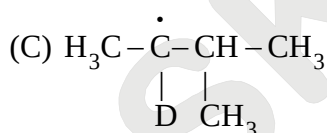
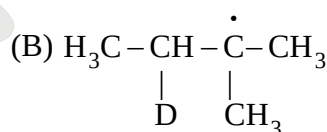
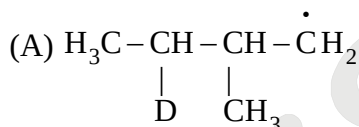
36. A biologically active compound, Bombykol ($C_{16}H_{30}O$) is obtained from a natural source. The structure of the compound is determined by the following reactions.
- (a) On hydrogenation, Bombykol gives a compound A, $C_{16}H_{34}O$, which reacts with acetic anhydride to give an ester.
- (b) Bombykol also reacts with acetic anhydride to give another ester, which on oxidative ozonolysis (O_3/H_2O_2) gives a mixture of butanoic acid, oxalic acid and 10-acetoxy decanoic acid.
- Determine the number of double bonds in Bombykol. Write the structures of compound A and Bombykol. How many geometrical isomers are possible for Bombykol? **[IIT 2002]**
37. Identify X, Y and Z in the following synthetic scheme and write their structures. Is the compound Z optically active? Justify your answer. **[IIT 2002]**



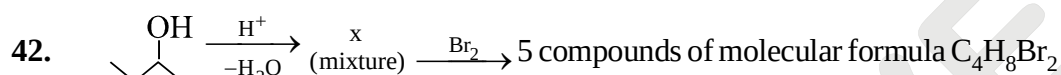
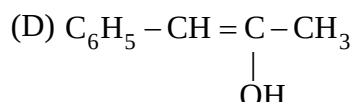
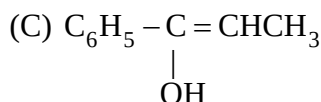
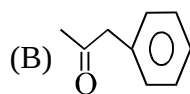
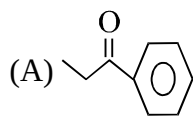
38. Consider the following reactions – **[IIT 2002]**



Identify the structure of the major product 'X'



39. The nodal plane in the π -bond of ethene is located in – **[IIT '2002]**
- (A) the molecular plane
 (B) a plane parallel to the molecular plane
 (C) a plane perpendicular to the molecular plane which contains the carbon-carbon σ -bond at right angle
 (D) a plane perpendicular to the molecular plane which contains the carbon-carbon σ -bond
40. Identify a reagent from the following list which can easily distinguish between 1-butyne and 2-butyne- **[IIT 2002]**
- (A) bromine, CCl_4 (B) H_2 , Lindlar catalyst
 (C) dilute H_2SO_4 , $HgSO_4$ (D) ammonical Cu_2Cl_2 solution



Number of compounds in X will be:

- (A) 2 (B) 3 (C) 4 (D) 5

[IIT 2003]

43. 2-hexyne can be converted into trans-2-hexene by the action of :

[IIT 2004]

- (A) $\text{H}_2-\text{Pd}-\text{BaSO}_4$ (B) Li in liq. NH_3 (C) H_2-PtO_2 (D) NaBH_4

44. Cyclohexene is best prepared from cyclohexanol by which of the following :

[IIT 2005]

- (A) conc. H_3PO_4 (B) conc. HCl/ZnCl_2 (C) conc. HCl (D) conc. HBr

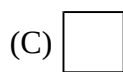
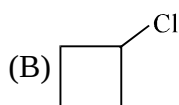
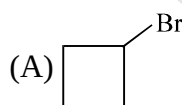
45. When Phenyl Magnesium Bromide reacts with tert. butanol, which of the following is formed?

- (A) Tert. butyl methyl ether (B) Benzene
(C) Tert. butyl benzene (D) Phenol

[IIT 2005]

46. 1-bromo-3-chlorocyclobutane when treated with two equivalents of Na, in the presence of ether which of the following will be formed?

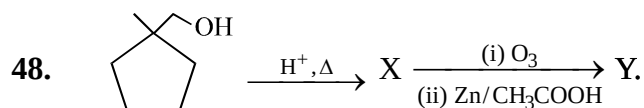
[IIT 2005]



47. If after complete ozonolysis of one mole of monomer of natural polymer gives two moles of CH_2O and

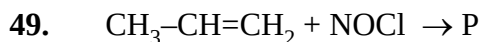
one mole of $\text{O}=\text{C}(\text{CH}_3)-\text{CH}=\text{O}$. Identify the monomer and draw the all-cis structure of natural polymer.

[IIT 2005]



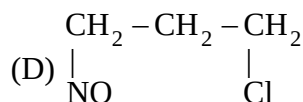
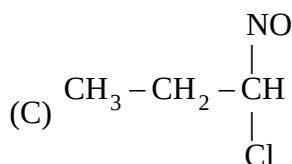
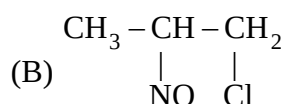
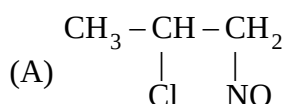
Identify X and Y.

[IIT 2005]



[IIT 2006]

Identify the adduct.



50. The number of stereoisomers obtained by bromination of trans-2-butene is

[IIT 2007]

- (A) 1 (B) 2 (C) 3 (D) 4

51. The number of structural isomers for C_6H_{14} is

[IIT 2007]

- (A) 3 (B) 4 (C) 5 (D) 6

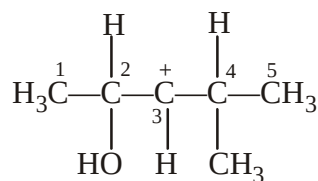
52. The reagent(s) for the following conversion,

[IIT 2007]



- (A) alcoholic KOH (B) alcoholic KOH followed by NaNH_2
(C) aqueous KOH followed by NaNH_2 (D) $\text{Zn} / \text{CH}_3\text{OH}$

53. In the following carbocation, H/ CH_3 that is most likely to migrate to the positively charged carbon is



[JEE 2009]

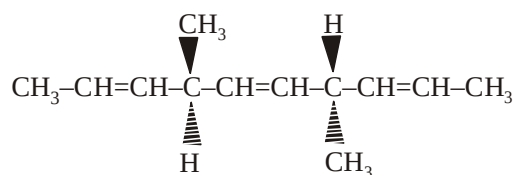
- (A) CH_3 at C-4 (B) H at C-4 (C) CH_3 at C-2 (D) H at C-2

54. The synthesis of 3-octyne is achieved by adding a bromoalkane into a mixture of sodium amide and an alkyne. The bromoalkane and alkyne respectively are

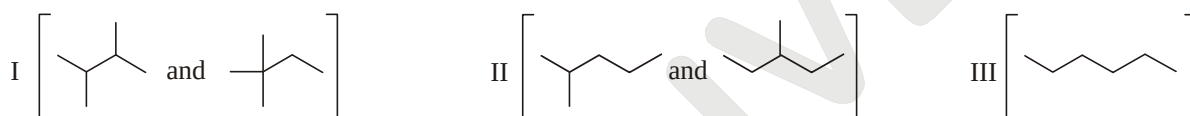
[IIT 2010]

- (A) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH}$
(B) $\text{BrCH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{C}\equiv\text{CH}$
(C) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{C}\equiv\text{CH}$
(D) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH}$

55. The number of optically active products obtained from the complete ozonolysis of the given compound is [JEE 2012]

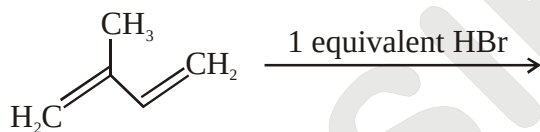


- (A) 0 (B) 1 (C) 2 (D) 4
56. In allene (C_3H_4), the type(s) of hybridisation of the carbon atoms is (are) [JEE 2012]
 (A) sp and sp^3 (B) sp and sp^2 (C) only sp^2 (D) sp^2 and sp^3
57. Isomers of hexane, based on their branching, can be divided into three distinct classes as shown in the figure. [JEE Advance 2014]



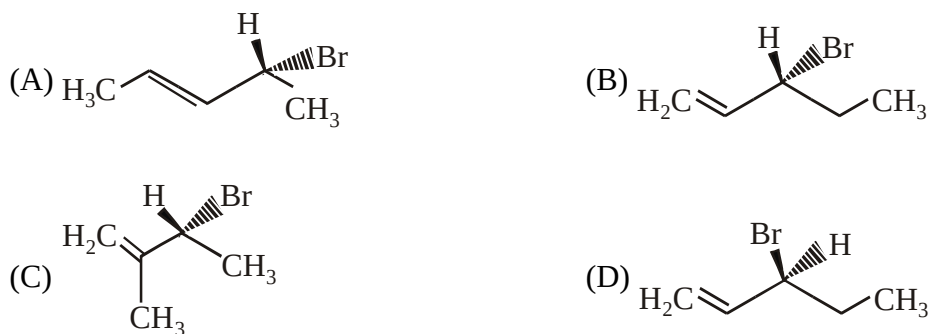
The correct order of their boiling point is

- (A) $\text{I} > \text{II} > \text{III}$ (B) $\text{III} > \text{II} > \text{I}$ (C) $\text{II} > \text{III} > \text{I}$ (D) $\text{III} > \text{I} > \text{II}$
58. In the following reaction, the major product is : [JEE Advance 2015]



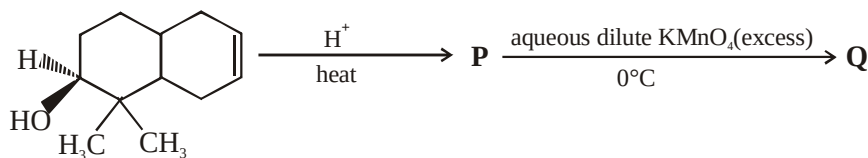
- (A) $\text{H}_2\text{C}=\underset{\text{Br}}{\overset{\text{CH}_3}{\text{C}}}-\text{CH}_3$ (B) $\text{H}_3\text{C}-\underset{\text{Br}}{\overset{\text{CH}_3}{\text{C}}}=\text{CH}_2$
 (C) $\text{H}_2\text{C}=\underset{\text{CH}_3}{\text{C}}-\text{CH}_2-\text{CH}_2-\text{Br}$ (D) $\text{H}_3\text{C}-\underset{\text{CH}_3}{\text{C}}=\text{CH}-\text{CH}_2-\text{Br}$

59. Compound(s) that on hydrogenation produce(s) optically inactive compound(s) is(are) [JEE Advance 2015]



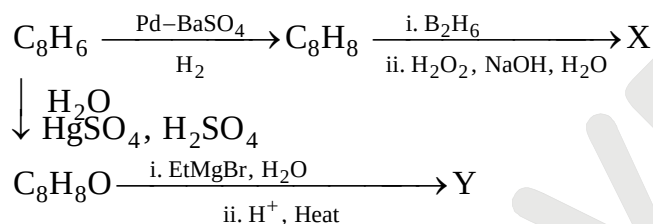
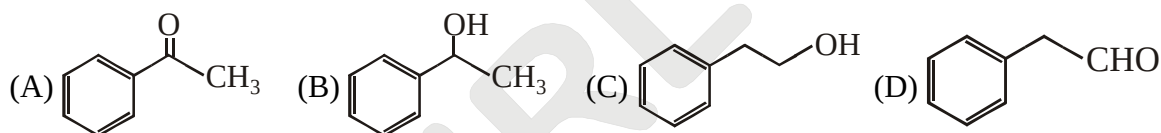
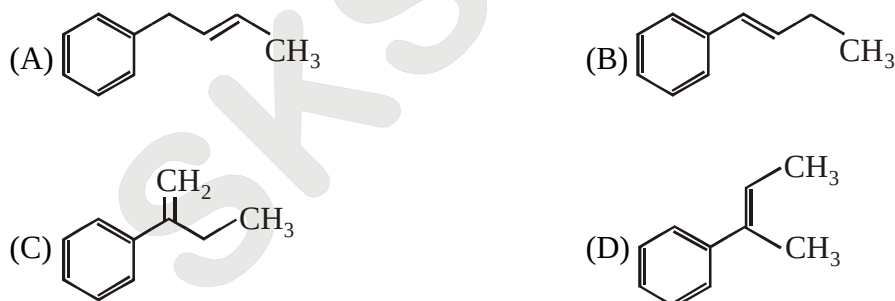
60. The number of hydroxyl group(s) in **Q** is

[JEE Advance 2015]

**Paragraph for question nos. 61 & 62**

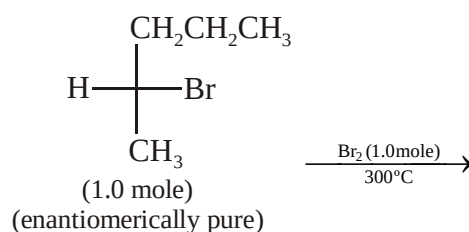
In the following reactions :

[JEE Advance 2015]

61. Compound **X** is62. The major compound **Y** is

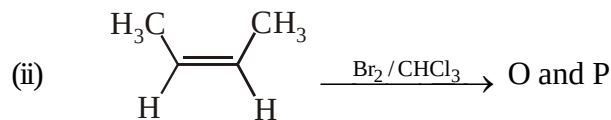
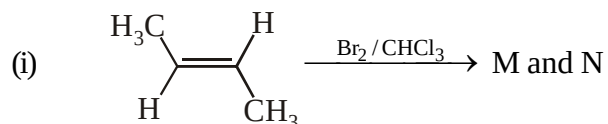
63. In the following monobromination reaction, the number of possible chiral products is

[JEE Advance 2016]



64. The correct statement(s) for the following addition reactions is(are)

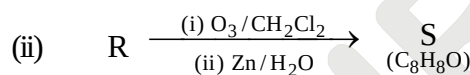
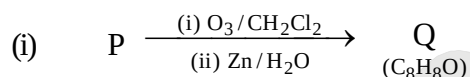
[JEE Advance 2017]



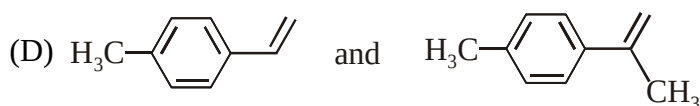
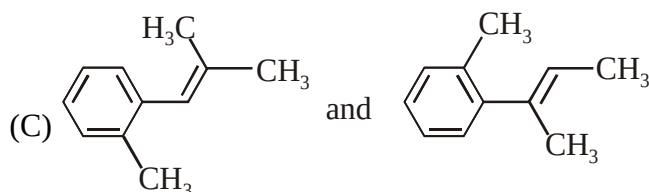
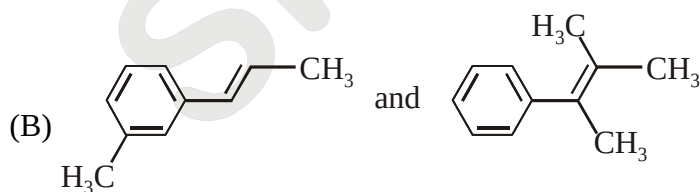
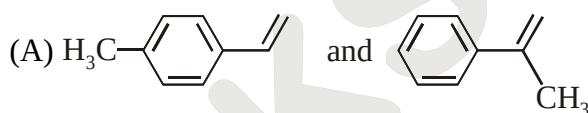
- (A) O and P are identical molecules
 (B) Bromination proceeds through trans-addition in both the reactions
 (C) (M and O) and (N and P) are two pairs of enantiomers
 (D) (M and O) and (N and P) are two pairs of diastereomers

65. Compounds **P** and **R** upon ozonolysis produce **Q** and **S**, respectively. The molecular formula of **Q** and **S** is $\text{C}_8\text{H}_8\text{O}$. **Q** undergoes Cannizzaro reaction but not haloform reaction, whereas **S** undergoes haloform reaction but not Cannizzaro reaction.

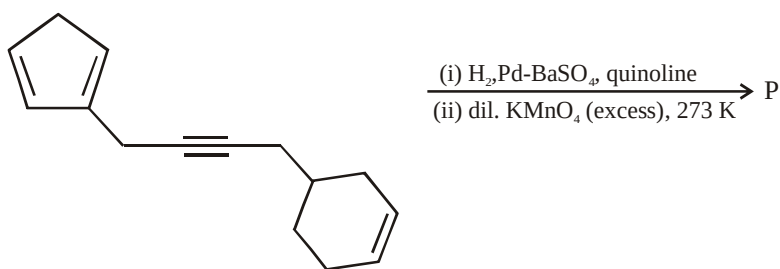
[JEE Advance 2017]



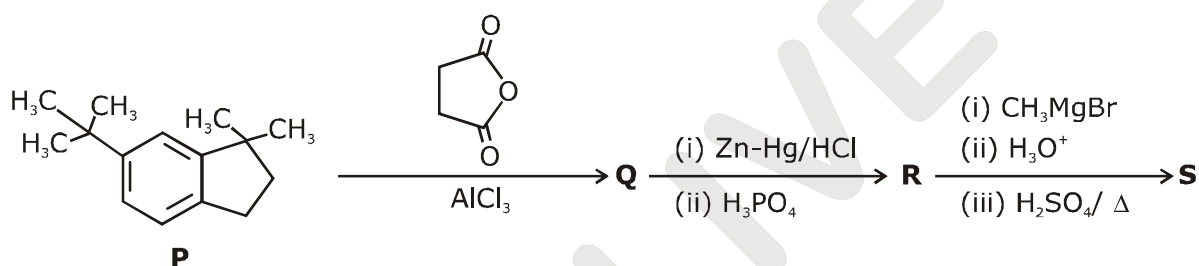
The option(s) with suitable combination of **P** and **R**, respectively, is(are)



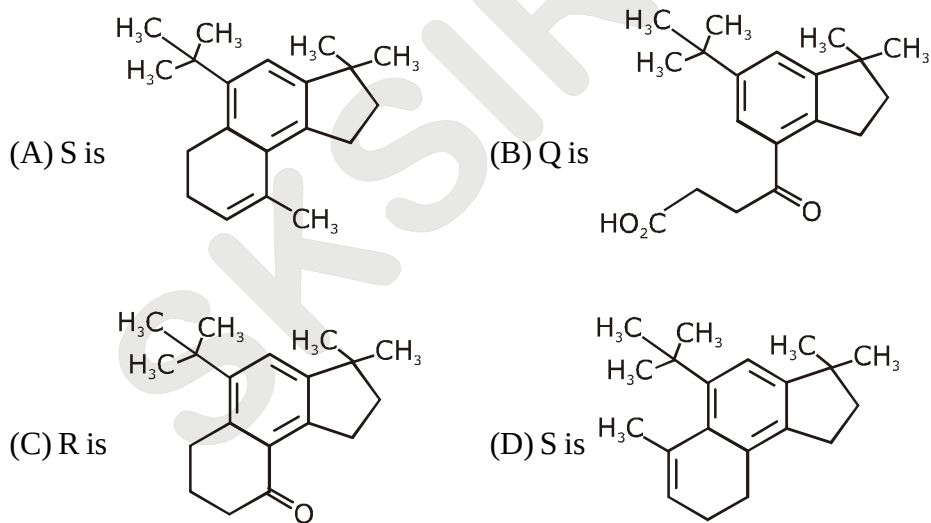
66. Total number of hydroxyl groups present in a molecule of the major product P is _____
[JEE Adv. 2019]



67. In the reaction scheme shown below, Q, R and S are the major products. [JEE Adv. 2020]



The correct structure of

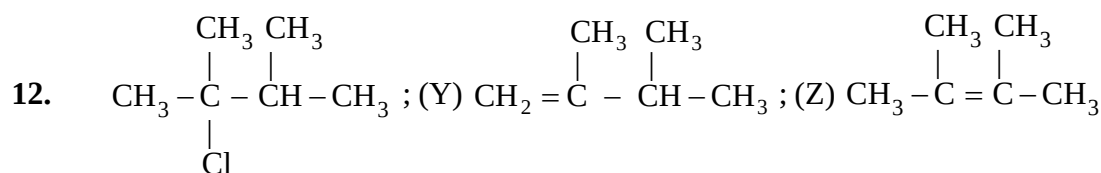
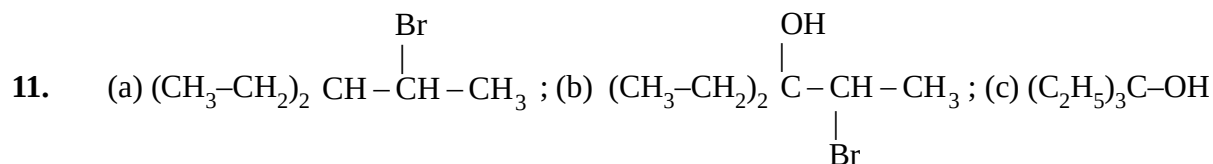
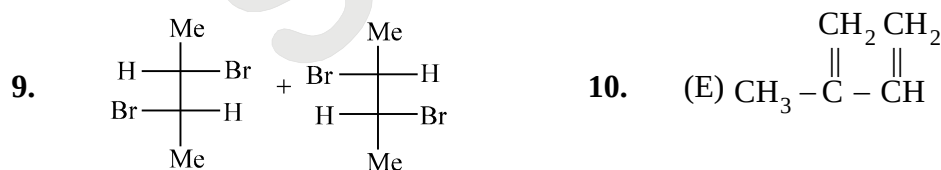
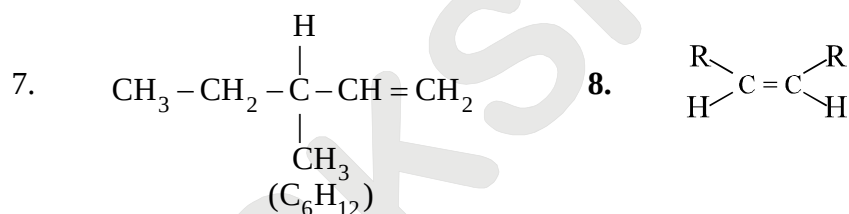
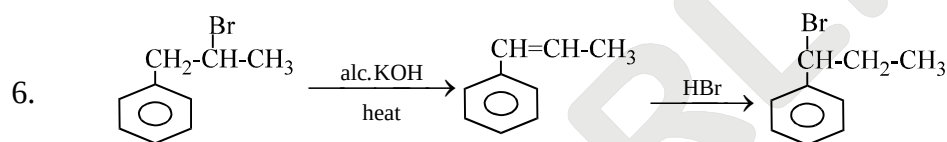


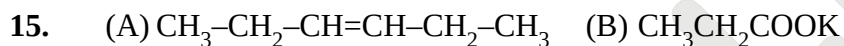
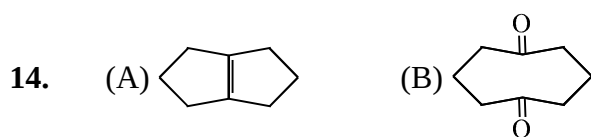
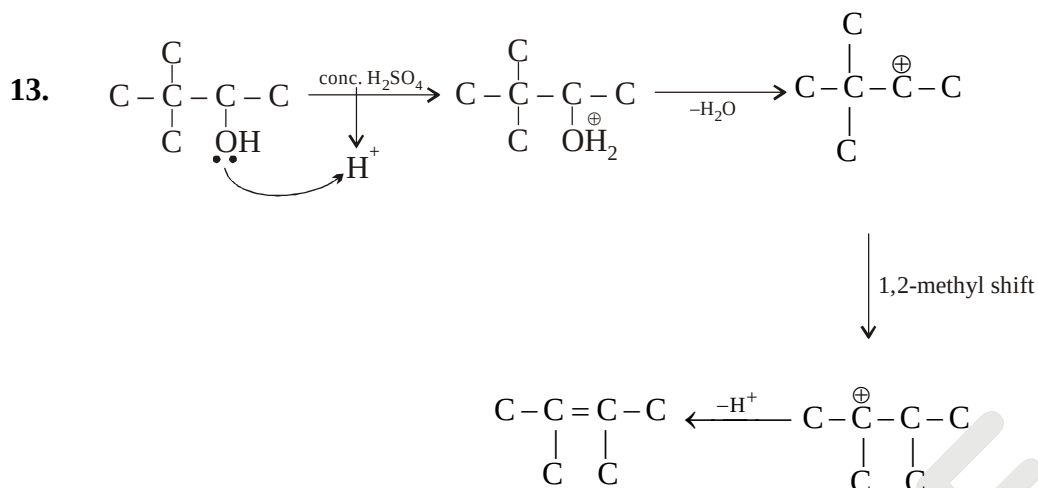
ANSWER KEY**[JEE-MAIN/AIEEE]**

1.	B	2.	A	3.	B	4.	B	5.	C	6.	B	7.	D
8.	C	9.	A	10.	A	11.	C	12.	A	13.	A	14.	C
15.	C	16.	B	17.	A	18.	D	19.	D	20.	C	21.	D
22.	C	23.	C	24.	A	25.	C	26.	A	27.	A	28.	C
29.	C	30.	C	31.	B	32.	D	33.	D	34.	C	35.	C
36.	A	37.	8	38.	A	39.	C	40.	A	41.	A	42.	C
43.	C	44.	C										

[JEE-ADVANCED]

1. C 2. AD 3. A 4. C 5. $\text{CH}_3 - \underset{\text{CH}_3}{\text{C}} = \text{CH} - \text{CH}_3$

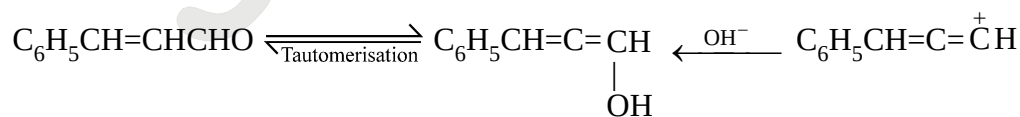
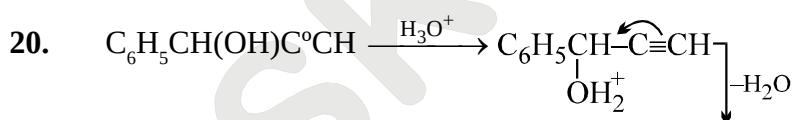
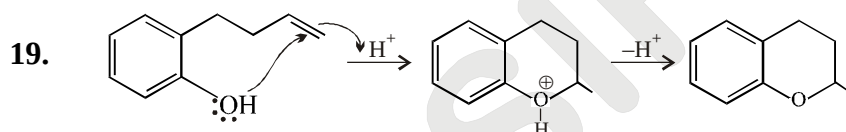




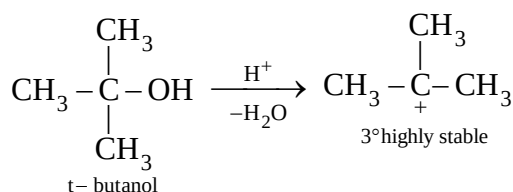
16. Presence of partial double bond character due to resonance.

17. AC

18. B



21. Acid catalysed dehydration proceeds via carbocation intermediate. Also, greater the stability of reactive intermediate, faster the reaction:

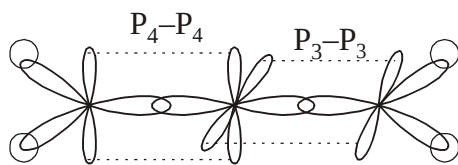


n-butanol forms less stable (1°) carbocation.

22. AC

23. (1) NaNH_2 , (2) Me-I , (3) HgSO_4 dil H_2SO_4

24. $\text{H}_2\text{C} = \text{C} = \text{CH}_2$



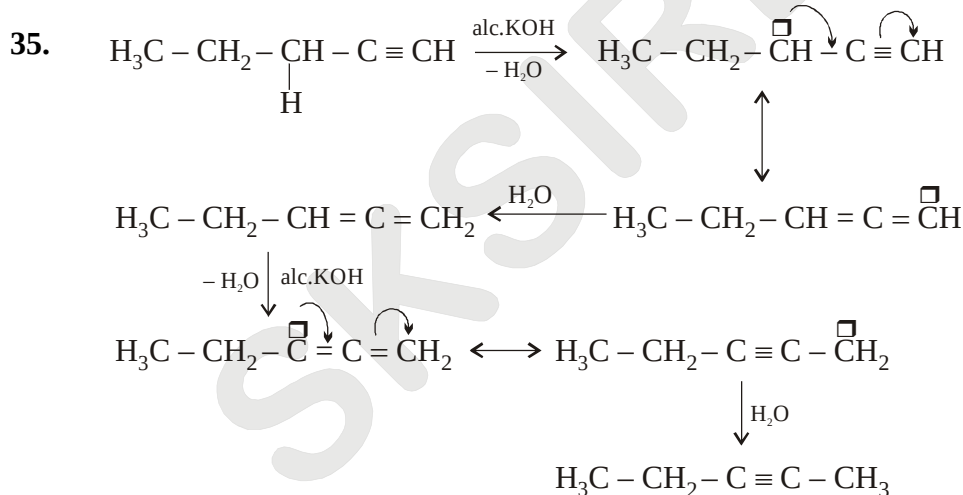
25. 1 $\rightarrow$ ozonolysis ; 2 $\rightarrow \text{LiAlH}_4$; 3 $\rightarrow \text{H}_2\text{SO}_4$

26. (4) $\rightarrow \text{HO-Cl}$; (5) $\rightarrow \text{CH}_3\text{MgCl}$; (6) $\rightarrow \text{H}_2\text{O}/\text{H}^+$

27. D 28. (i) $\text{CH}_3 - \overset{\text{CH}_3}{\underset{|}{\text{C}}} = \text{CH} - \text{CH}_3$ (ii)

29. higher electronegativity of sp carbon

30. D 31. A 32. C 33. B 34. B



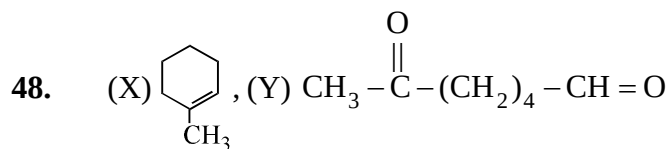
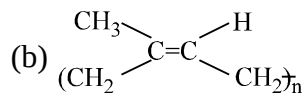
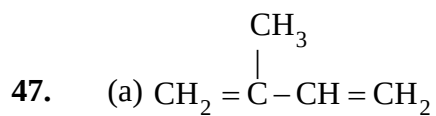
36. Bombykol :- $\text{HO-C-C-C-C-C-C-C-C-C-C}=\text{C-C}=\text{C-C-C-C}$

(A) :- $\text{HO}(\text{CH}_3)_{15}\text{CH}_3$ 4 geometrical isomers are possible

37. (X) $\rightarrow \text{Et-C}\equiv\text{C-Et}$ (Y) $\rightarrow \text{Et} \diagup \text{C}=\text{C} \diagdown \text{Et}$ (Z) $\rightarrow \begin{array}{c} \text{Et} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{Et} \end{array}$ Z is meso so optically inactive.

38. B 39. A 40. D 41. A 42. B 43. B 44. A

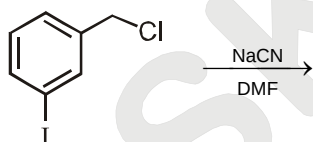
45. B 46. D



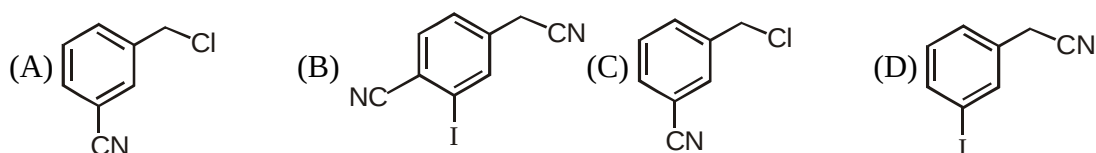
- | | | | | | | | | | | | | | |
|-----|---|-----|----|-----|----|-----|----|-----|----|-----|---|-----|---|
| 49. | A | 50. | A | 51. | C | 52. | B | 53. | D | 54. | D | 55. | A |
| 56. | B | 57. | B | 58. | D | 59. | BD | 60. | 4 | 61. | C | 62. | D |
| 63. | 5 | 64. | BD | 65. | AB | 66. | 6 | 67. | BD | | | | |

[JEE-MAIN / AIEEE]

- The reaction : $(\text{CH}_3)_3\text{C}-\text{Br} \xrightarrow{\text{H}_2\text{O}} (\text{CH}_3)_3\text{C}-\text{OH}$ [AIEEE-2002]
 (A) elimination reaction (B) substitution reaction
 (C) free radical reaction (D) displacement reaction
- The correct order of the thermal stability of hydrogen halides ($\text{H}-\text{X}$) is [AIEEE-2005]
 (A) $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$ (B) $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$
 (C) $\text{HCl} < \text{HF} > \text{HBr} < \text{HI}$ (D) $\text{HI} > \text{HCl} < \text{HF} < \text{HBr}$
- Tertiary alkyl halides are practically inert to substitution by S_N^2 mechanism because of [AIEEE-2005]
 (A) insolubility (B) instability (C) inductive effect (D) steric hindrance
- Elimination of bromine from 2-bromobutane results in the formation of- [AIEEE-2005]
 (A) equimolar mixture of 1 and 2-butene (B) predominantly 2-butene
 (C) predominantly 1-butene (D) predominantly 2-butyne
- $\text{CH}_3\text{Br} + \text{Nu}^- \longrightarrow \text{CH}_3-\text{Nu} + \text{Br}^-$ [AIEEE-2006]
 The decreasing order of the rate of the above reaction with nucleophiles (Nu^-) A to D is
 $[\text{Nu}^- = (\text{A}) \text{PhO}^-, (\text{B}) \text{AcO}^-, (\text{C}) \text{HO}^-, (\text{D}) \text{CH}_3\text{O}^-]$
 (A) $\text{D} > \text{C} > \text{A} > \text{B}$ (B) $\text{D} > \text{C} > \text{B} > \text{A}$ (C) $\text{A} > \text{B} > \text{C} > \text{D}$ (D) $\text{B} > \text{D} > \text{C} > \text{A}$
- The structure of the major product formed in the following reaction [AIEEE-2006]

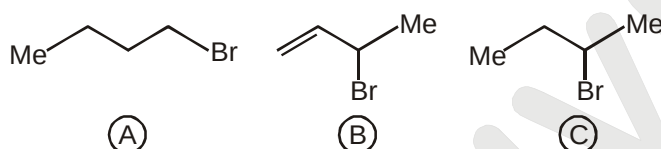


is



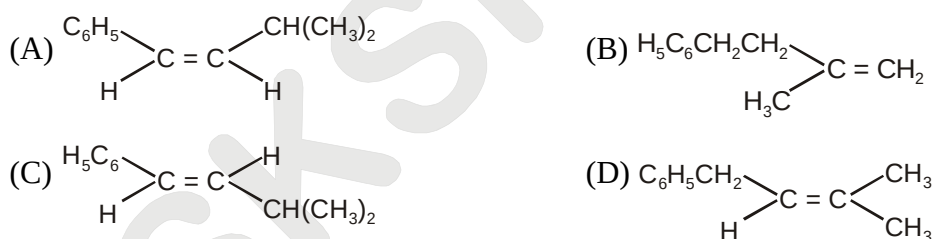
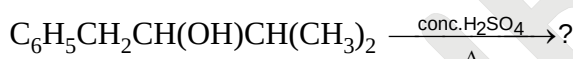
- Reaction of trans-2-phenyl-1-bromocyclopentane on reaction with alcoholic KOH produces [AIEEE-2006]
 (A) 4-phenylcyclopentene (B) 2-phenylcyclopentene
 (C) 1-phenylcyclopentene (D) 3-phenylcyclopentene
- Which of the following is the correct order of decreasing S_N^2 reactivity? [AIEEE-2007]
 (A) $\text{RCH}_2\text{X} > \text{R}_3\text{CX} > \text{R}_2\text{CHX}$ (B) $\text{RCH}_2\text{X} > \text{R}_2\text{CHX} > \text{R}_3\text{CX}$
 (C) $\text{R}_3\text{CX} > \text{R}_2\text{CHX} > \text{RCH}_2\text{X}$ (D) $\text{R}_2\text{CHX} > \text{R}_3\text{CX} > \text{RCH}_2\text{X}$

9. The organic chloro compound, which shows complete stereochemical inversion during a S_N2 reaction, is [AIEEE-2008]
 (A) $(C_2H_5)_2CHCl$ (B) $(CH_3)_3CCl$ (C) $(CH_3)_2CHCl$ (D) CH_3Cl
10. Which of the following on heating with aqueous KOH , produces acetaldehyde ? [AIEEE-2009]
 (A) CH_2ClCH_2Cl (B) CH_3CHCl_2 (C) CH_3COCl (D) CH_3CH_2Cl
11. From amongst the following alcohols the one that would react fastest with conc. HCl and anhydrous $ZnCl_2$, is [AIEEE-2010]
 (A) 2-Methylpropanol (B) 1-Butanol (C) 2-Butanol (D) 2-Methylpropan-2-ol
12. Consider the following bromides : [AIEEE-2010]

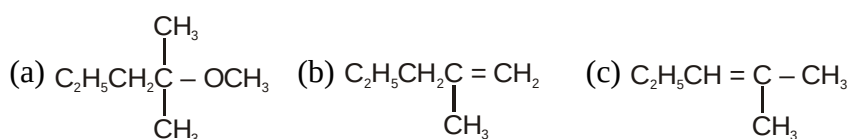


The correct order of S_N1 reactivity is

- (A) $C > B > A$ (B) $A > B > C$ (C) $B > C > A$ (D) $B > A > C$
13. The main product of the following reaction is [AIEEE-2010]



14. An unknown alcohol is treated with the "Lucas reagent" to determine whether the alcohol is primary, secondary or tertiary. Which alcohol reacts fastest and by what mechanism : [JEE-Main-2013]
 (A) tertiary alcohol by S_N2 (B) secondary alcohol by S_N1
 (C) tertiary alcohol by S_N1 (D) secondary alcohol by S_N2
15. A solution of (–) – 1 – chloro – 1 – phenylethane is toluene racemises slowly in the presence of a small amount of $SbCl_5$, due to the formation of : [JEE-Main-2013]
 (A) free radical (B) carbanion (C) carbene (D) carbocation
16. 2-chloro-2methylpentane on reaction with sodium methoxide in methanol yields :



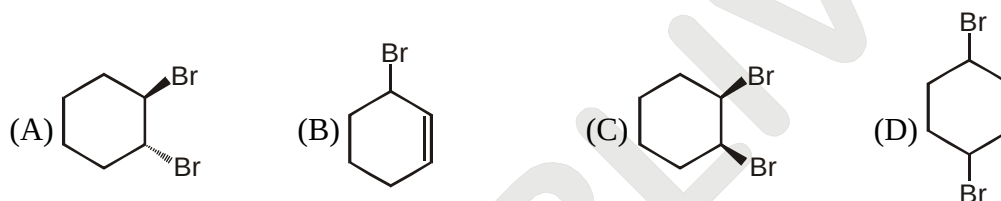
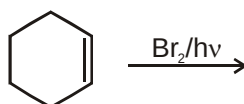
[JEE-Main-2016]

- (A) All of these (B) (a) and (c) (C) (c) only (D) (a) and (b)

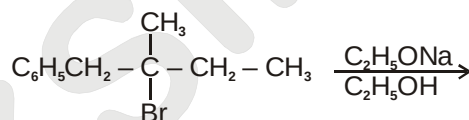
17. The reaction of propene with HOCl ($\text{Cl}_2 + \text{H}_2\text{O}$) proceeds through the intermediate :
 (A) $\text{CH}_3 - \text{CH}^+ - \text{CH}_2 - \text{OH}$ (B) $\text{CH}_3 - \text{CH}^+ - \text{CH}_2 - \text{Cl}$ [JEE-Mains-2016]
 (C) $\text{CH}_3 - \text{CH}(\text{OH}) - \text{CH}_2^+$ (D) $\text{CH}_3 - \text{CHCl} - \text{CH}_2^+$
18. Which one of the following reagents is not suitable for the elimination reaction ?
 [JEE-Mains-2016 (online)]



- (A) NaI (B) NaOH/ H_2O -EtOH (C) NaOH/ H_2O (D) NaOEt/EtOH
19. Bromination of cyclohexene under conditions given below yields : [JEE-Mains-2016 (online)]

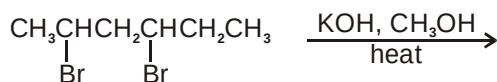


20. The major product of the following reaction is : [JEE-Mains Online -2017]

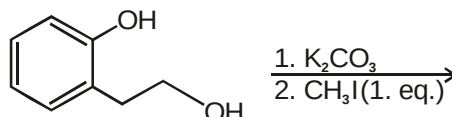


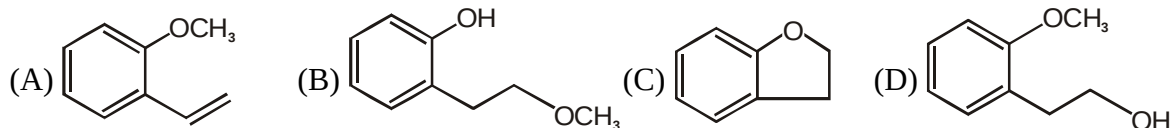
- (A) $\text{C}_6\text{H}_5\text{CH}=\underset{\text{CH}_3}{\text{C}}-\text{CH}_2\text{CH}_3$ (B) $\text{C}_6\text{H}_5\text{CH}_2-\underset{\text{CH}_2\text{CH}_3}{\text{C}}=\text{CH}_2$
 (C) $\text{C}_6\text{H}_5\text{CH}_2-\underset{\text{OC}_2\text{H}_5}{\overset{\text{CH}_3}{\text{C}}}-\text{CH}_2\text{CH}_3$ (D) $\text{C}_6\text{H}_5\text{CH}_2-\underset{\text{CH}_3}{\text{C}}=\text{CHCH}_3$

21. The major product of the following reaction is : [JEE-Mains Online -2017]

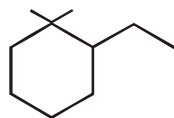


- (A) $\text{CH}_2=\text{CHCH}=\text{CHCH}_2\text{CH}_3$ (B) $\text{CH}_3\text{CH}=\text{CH}-\text{CH}=\text{CHCH}_3$
 (C) $\text{CH}_2=\text{CHCH}_2\text{CH}=\text{CHCH}_3$ (D) $\text{CH}_3\text{CH}=\text{C}=\text{CHCH}_2\text{CH}_3$
22. The major product of the following reaction is : [JEE-Mains Online -2017]

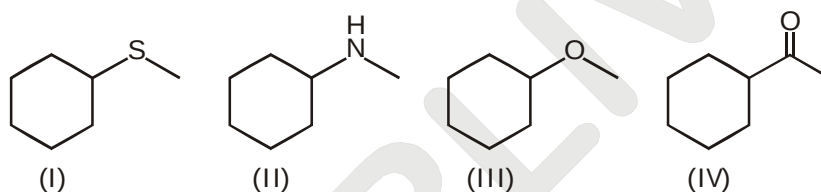




23. The IUPAC name of the following compound is : [JEE-Mains Online -2017]



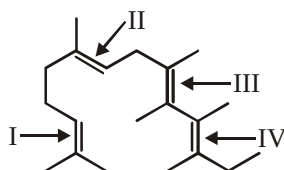
- (A) 2-Ethyl-1,1-dimethylcyclohexane (B) 1-Ethyl-2,2-dimethylcyclohexane
(C) 1,1-Dimethyl-2-ethylcyclohexane (D) 2,2-Dimethyl-1-ethylcyclohexane
24. A mixture containing the following four compounds is extracted with 1M HCl. The compound that goes to aqueous layer is : [JEE-Mains Online -2017]



- (A) (IV) (B) (II) (C) (I) (D) (III)
25. Which of the following compounds is most reactive to an aqueous solution of sodium carbonate? [JEE-Mains Online -2017]

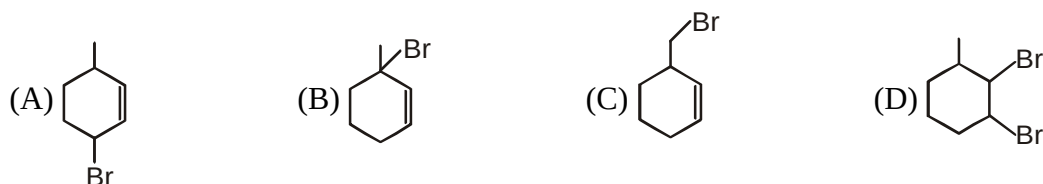
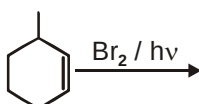


26. In the following structure, the double bonds are marked as I, II, III and IV



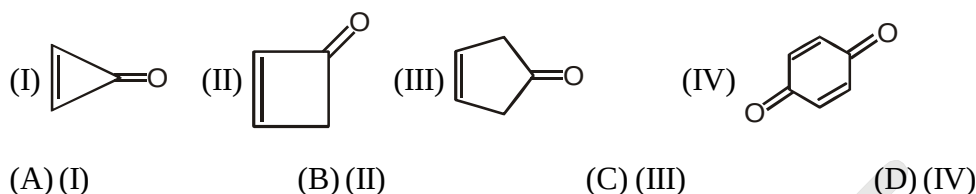
- Geometrical isomerism is not possible at site (s): [JEE-Mains Online -2017]
(A) III (B) I (C) I and III (D) III and IV

27. The major product of the following reaction is : [JEE-Mains Online -2017]

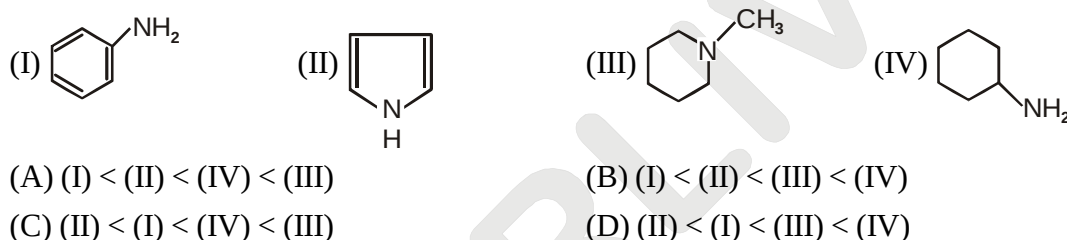


28. The increasing order of the boiling points for the following compounds is :
 (I) C_2H_5OH (II) C_2H_5Cl (III) $C_2H_5CH_3$ (IV) $C_2H_5OCH_3$ [JEE-Mains Online -2017]
 (A) (III) < (IV) < (II) < (I) (B) (IV) < (III) < (I) < (II)
 (C) (II) < (III) < (IV) < (I) (D) (III) < (II) < (I) < (IV)

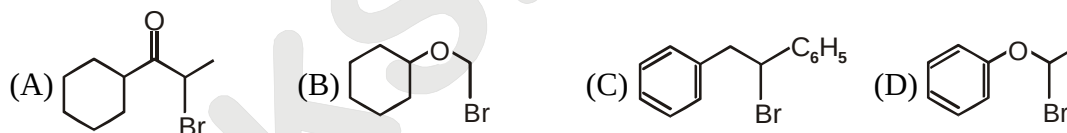
29. Which of the following compounds will show highest dipole moment? [JEE-Mains Online -2017]



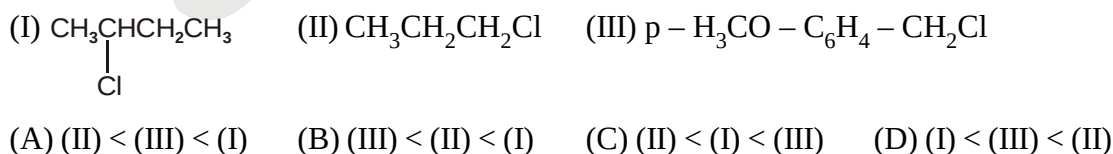
30. Among the following compounds, the increasing order of their basic strength is :
 [JEE-Mains Online -2017]



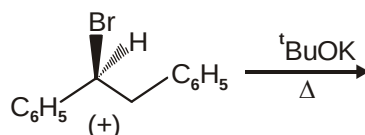
31. Which of the following, upon treatment with $tert\text{-BuONa}$ followed by addition of bromine water, fails to decolourize the colour of bromine? [JEE-Mains-2017]



32. The increasing order of the reactivity of the following halides for the S_N1 reaction is :
 [JEE-Mains-2017]



33. The major product obtained in the following reaction is [JEE-Mains-2017]



- (A) $(-)\text{-C}_6\text{H}_5\text{CH(O}^t\text{Bu)CH}_2\text{C}_6\text{H}_5$ (B) $(\pm)\text{-C}_6\text{H}_5\text{CH(O}^t\text{Bu)CH}_2\text{C}_6\text{H}_5$
 (C) $\text{C}_6\text{H}_5\text{CH=CHC}_6\text{H}_5$ (D) $(+)\text{-C}_6\text{H}_5\text{CH(O}^t\text{Bu)CH}_2\text{C}_6\text{H}_5$

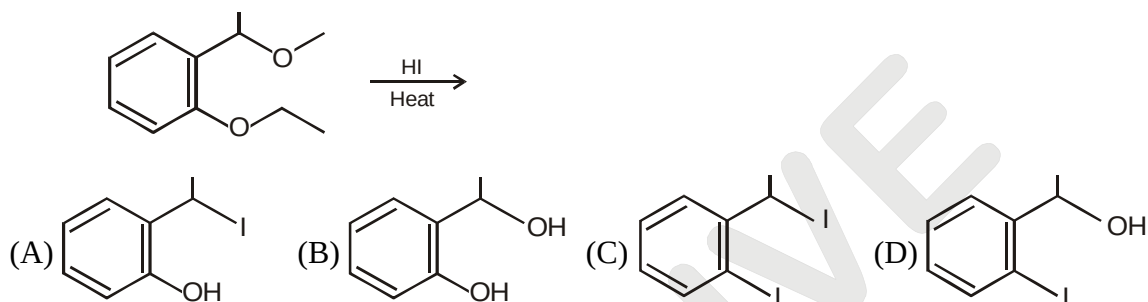
34. Which of the following molecules is least resonance stabilized? [JEE-Mains-2017]



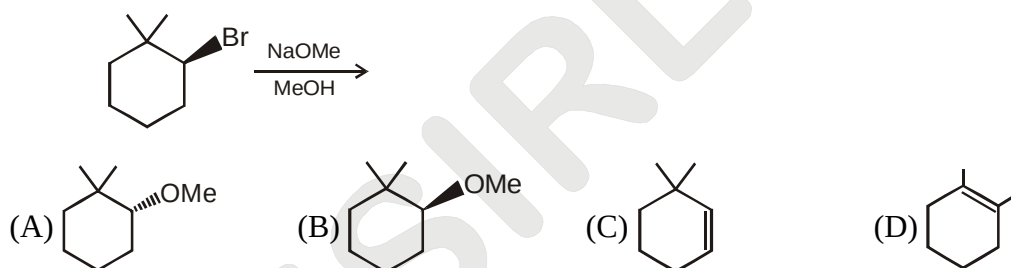
35. 3-Methyl-pent-2-ene on reaction with HBr in presence of peroxide forms an addition product. The number of possible stereoisomers for the product is [JEE-Mains-2017]



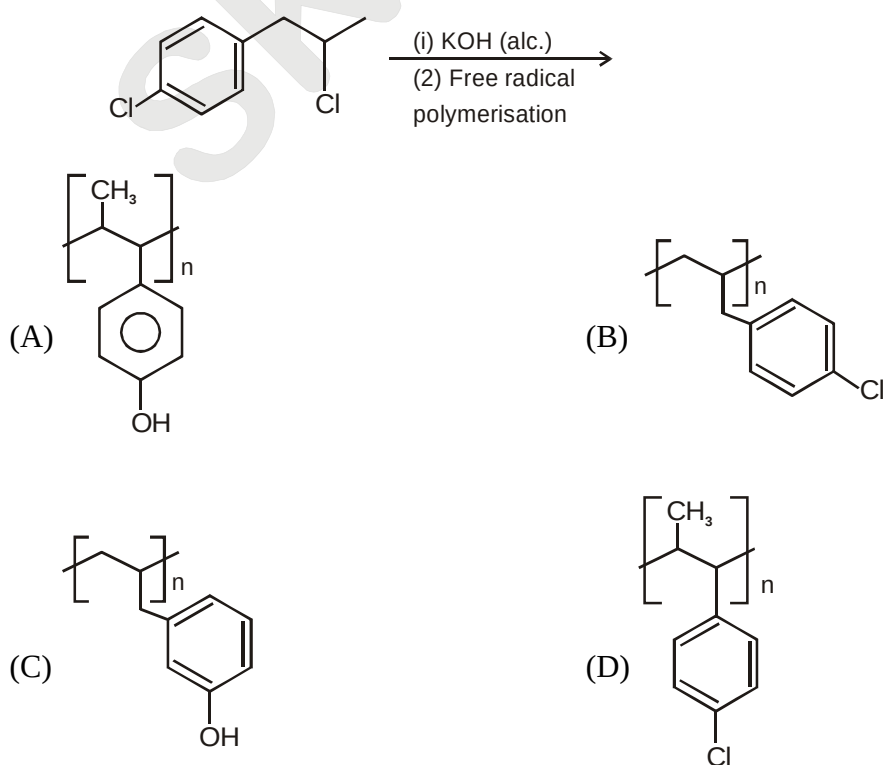
36. The major product formed in the following reaction is : [JEE-Mains-2018]



37. The major product of the following reaction is : [JEE-Mains-2018]

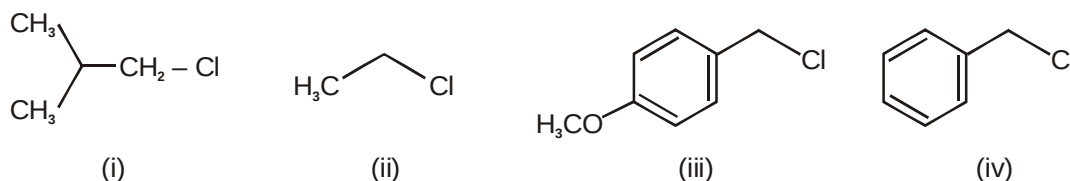


38. The major product of the following reaction is : [JEE-Mains Online -2019]



39. Increasing order of reactivity of the following compounds for S_N1 substitution is :

[JEE-Mains Online -2019]



(A) (ii) < (i) < (iv) < (iii)

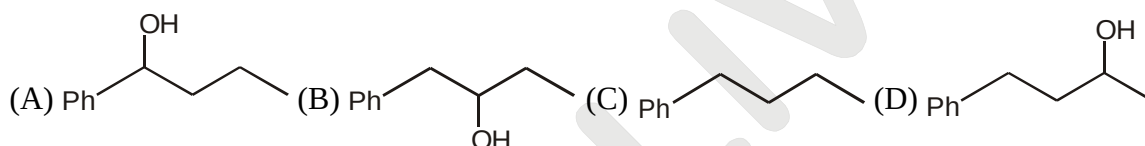
(B) (ii) < (iii) < (i) < (iv)

(C) (i) < (ii) < (iv) < (iii)

(D) (ii) < (iii) < (iv) < (i)

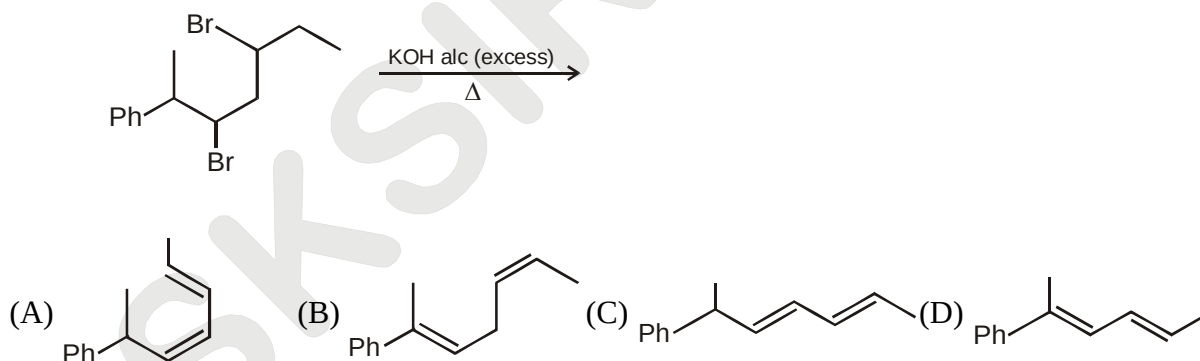
40. Heating of 2-chloro-1 phenylbutane with EtOK/EtOH gives X as the major product. Reaction of X with $Hg(OAc)_2/H_2O$ followed by $NaBH_4$ gives Y as the major product . Y is :

[JEE-Mains Online -2019]



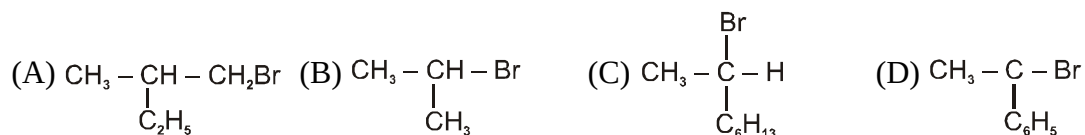
41. The major product of the following reaction is :

[JEE-Mains Online -2019]



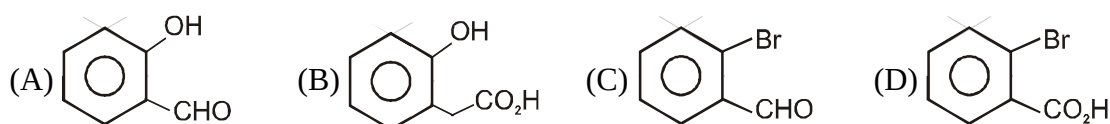
42. Which of the following compounds will show retention in configuration on nucleophile substitution by OH^- ion ?

[JEE-Mains Online -2020]



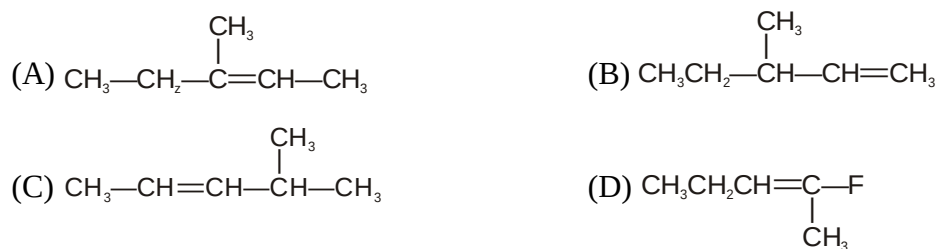
43. The major aromatic product C in the following reaction sequence will be :

[JEE-Mains Online -2020]



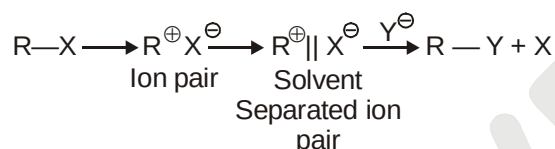
44. The major product obtained from E_2 -elimination of 3-bromo-2-fluoropentane is

[JEE-Mains Online -2020]



45. The mechanism of S_N1 reaction is given as:

[JEE-Mains Online -2020]



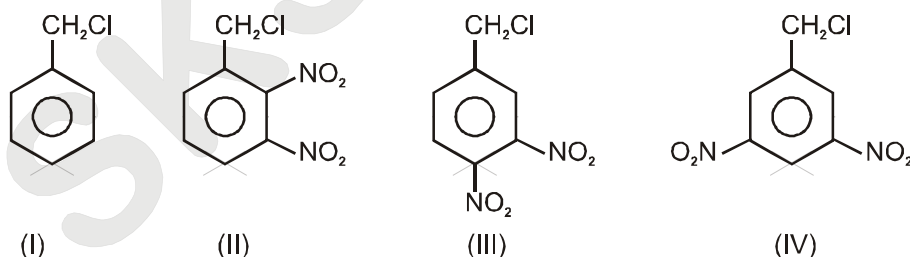
A student writes general characteristics based on the given mechanism as :

- (a) The reaction is favoured by weak nucleophiles.
 (b) $\text{R}^{\oplus}$ would be easily formed if the substituents are bulky
 (c) The reaction is accompanied by racemization
 (d) The reaction is favoured by non-polar solvents.

Which observations are correct ?

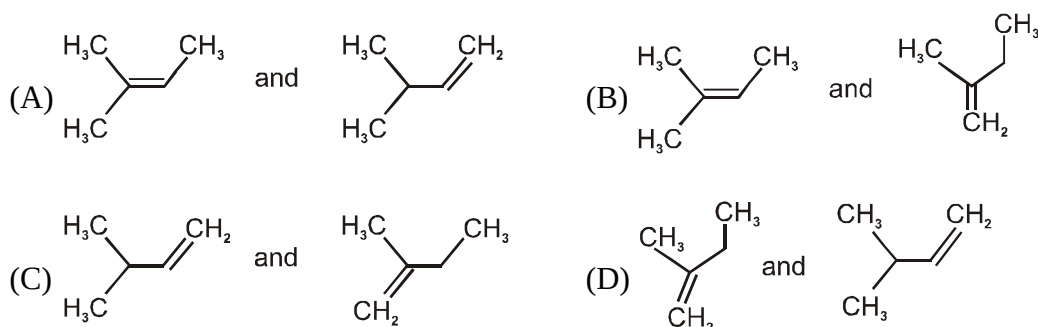
- (A) (b) and (d) (B) (a) and (c) (C) (a) and (b) (D) (a), (b) and (c)
46. The decreasing order of reactivity of the following compounds towards nucleophilic substitution (S_N2) is:

[JEE-Mains Online -2020]



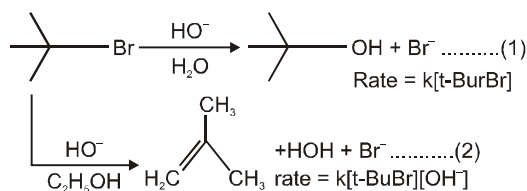
- (A) (II) > (III) > (I) > (IV) (B) (II) > (III) > (IV) > (I)
 (C) (III) > (II) > (IV) > (I) (D) (IV) > (II) > (III) > (I)
47. When neopentyl alcohol is heated with an acid, it slowly converted into an 85 : 15 mixture of alkenes A and B, respectively. What are these alkenes ?

[JEE-Mains Online -2020]



48. Consider the reaction sequence given below :

[JEE-Mains Online -2020]

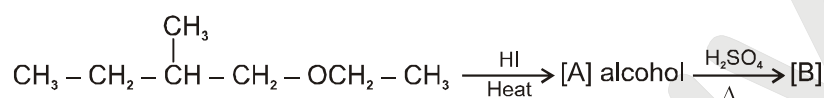


Which of the following statements is true:

- (A) Changing the concentration of base will have no effect on reaction (A)
 (B) Doubling the concentration of base will double the rate of both the reactions
 (C) Changing the concentration of base will have no effect on reaction (B)
 (D) $^\ominus\text{OH}$ will have no effect on reaction (B)

49. The major product [B] in the following reaction is :

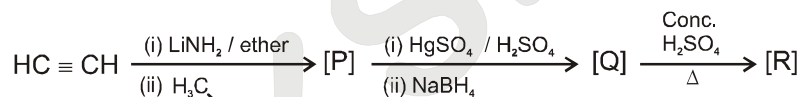
[JEE-Mains Online -2020]



- (A) $\text{CH}_2 = \text{CH}_2$ (B) $\text{CH}_3 - \text{CH} = \overset{\text{CH}_3}{\underset{|}{\text{C}}} - \text{CH}_3$
 (C) $\text{CH}_3 - \text{CH}_2 - \overset{\text{CH}_3}{\underset{|}{\text{C}}} = \text{CH}_2$ (D) $\text{CH}_3 - \text{CH}_2\text{CH} = \text{CH} - \text{CH}_3$

50. The major product [R] in the following sequence of reaction is :

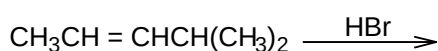
[JEE-Mains Online -2020]



- (A) $\text{H}_3\text{C}-\overset{\text{CH}_3}{\underset{|}{\text{C}}} = \text{CH}-\text{CH}_3$ (B) $\text{H}_3\text{C}-\text{CH}(\text{CH}_3)-\text{CH}=\text{CH}_2$
 (C) $\text{H}_2\text{C}=\text{C}(\text{CH}_3)-\text{CH}_2-\text{CH}_3$ (D) $\text{H}_3\text{C}-\overset{\text{H}_3\text{C}}{\underset{|}{\text{C}}} = \text{C}(\text{CH}_3)_2$

51. The major product in the following reaction is :

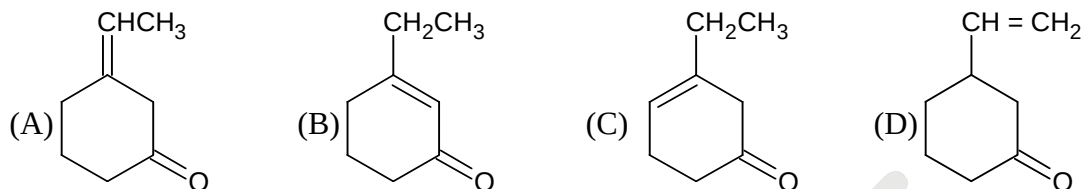
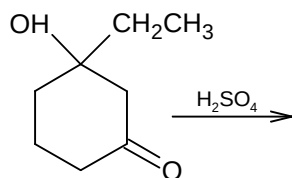
[JEE-Mains Online -2020]



- (A) $\text{CH}_3\text{CH}(\text{Br})\text{CH}_2\text{CH}(\text{CH}_3)_2$ (B) $\text{Br}(\text{CH}_2)_3\text{CH}(\text{CH}_3)_2$
 (C) $\text{CH}_3\text{CH}_2\text{CH}_2\text{C}(\text{Br})(\text{CH}_3)_2$ (D) $\text{CH}_3\text{CH}_2\text{CH}(\text{Br})\text{CH}(\text{CH}_3)_2$

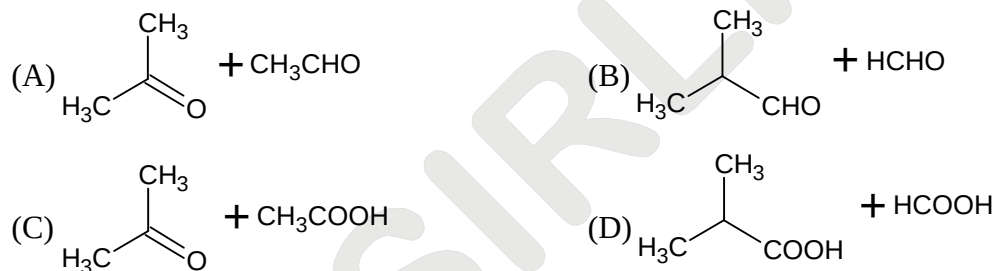
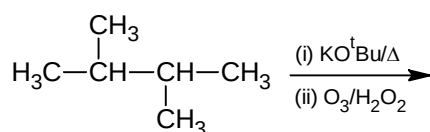
52. The major product of the following reaction is :

[JEE-Mains Online -2020]



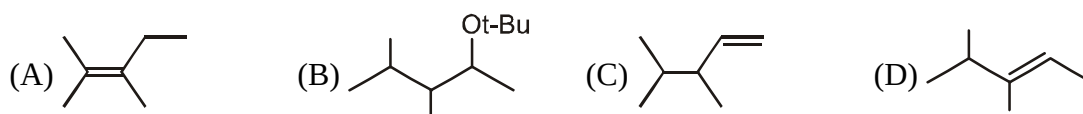
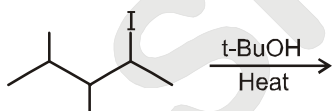
53. The major products of the following reaction are :

[JEE-Mains Online -2020]



54. The major product in the following reaction is:

[JEE-Mains Online -2020]

**[JEE-ADVANCED]**

1. Chloroform is stored in dark coloured bottles. Explain in not more than two sentences. [JEE 1980]

2. The compound which reacts fastest with Lucas reagent at room temperature is [JEE 1981]

(A) butan-2-ol

(B) butan-1-ol

(C) 2-methyl propan-1-ol

(D) 2-methyl propan-2-ol

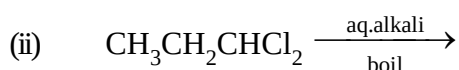
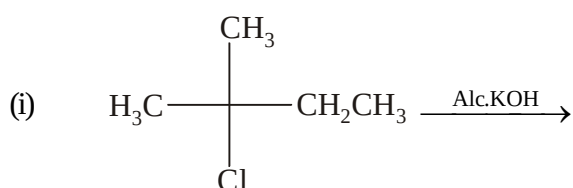
3. The halogen which is most reactive in the halogenation of alkanes under sunlight is _____. [JEE 1981]

4. Carbon tetrachloride burns in air to give phosgene. [IIT 1983]
5. Diethyl ether on heating with concentration HI gives two moles of [JEE 1983]
(A) ethanol (B) iodoform (C) ethyl iodide (D) methyl iodide
6. Carbon tetrachloride is inflammable. [IIT 1985]
7. Iodide is better nucleophile than bromide. [IIT 1985]
8. Write down the main product of the following reaction: [JEE 1985]
Ethanol $\xrightarrow{I_2 / NaOH}$
9. HBr reacts fastest with [JEE 1986]
(A) 2-methyl propan-2-ol (B) propan-1-ol
(C) propan-2-ol (D) 2-methyl propan-1-ol
10. The reaction condition leading to the best yield of C_2H_5Cl are [JEE 1986]
(A) C_2H_6 (excess) + $Cl_2 \xrightarrow{UV \text{ light}}$ (B) $C_2H_6 + Cl_2$ (excess) $\xrightarrow[\text{room temp.}]{\text{dark}}$
(C) $C_2H_6 + Cl_2$ (excess) $\xrightarrow{UV \text{ light}}$ (D) $C_2H_6 + Cl_2 \xrightarrow{UV \text{ light}}$
11. HBr reacts fastest with [JEE 1986]
(A) 2-methyl propan-2-ol (B) propan-1-ol
(C) Propan-2-ol (D) 2-methyl propan-1-ol
12. How may be the following transformation be carried out (in not more than six steps)? [JEE 1986]
"Ethyl alcohol to vinyl acetate".
13. n-propyl bromide on treatment with ethanolic potassium hydroxide produces. [IIT 1987]
(A) Propane (B) Propene (C) Propyne (D) Propanol
14. The reaction of vinyl chloride with hydrogen iodide to give 1-chloro-1-iodo ethane is an example of anti-Markownikoff's rule. [JEE 1989]
15. During S_N1 reactions, the leaving group leaves the molecule before the incoming group is attached to the molecule. [IIT 1990]
16. Aryl halides are less reactive towards nucleophilic substitution reaction as compared to alkyl halides due to [IIT 1990]
(A) The formation of less stable carbonium ion
(B) Resonance stabilization
(C) The inductive effect
(D) sp^2 hybridised carbon attached to the halogen
17. The compounds used as refrigerant are [JEE 1990]
(A) NH_3 (B) CCl_4 (C) CF_4 (D) CF_2Cl_2
(E) CH_2F_2
18. 1-Chlorobutane on reaction with alcoholic potash gives: [ITT 1991]
(A) 1-butene (B) 1-butanol (C) 2-butene (D) 2-butanol

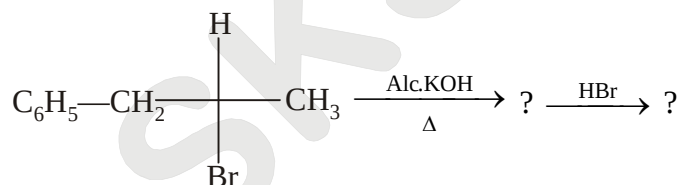
19. The products of reaction of alcoholic AgNO_2 with ethyl bromide are [IIT 1991]
(A) Ethane (B) Ethyl nitrite (C) Nitroethane (D) Ethyl alcohol
20. Fill in the blanks:
(a) Butan nitrile can be prepared by heating _____ with alcoholic KCN. [IIT 1992]
(b) Amongst three isomers of nitrophenol, the one that is least soluble in water is _____ [IIT 1992]

21. Arrange the following in order of their
(i) Increasing basicity
 H_2O , OH^- , CH_3OH , CH_3O^-
(ii) Increasing reactivity in nucleophilic substitution reactions
 CH_3F , CH_3I , CH_3Br , CH_3Cl [IIT 1992]

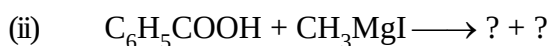
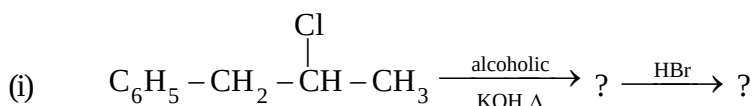
22. Write the structures of the major organic product expected from each of the following reactions: [IIT 1992]



23. Photobromination of 2-methylpropane gives a mixture of 1-bromo-2-methyl propane and 2-bromo-2-methyl propane in the ratio 9 : 1. [JEE 1993]
24. Identify the major product in the following reaction. [IIT 1993]

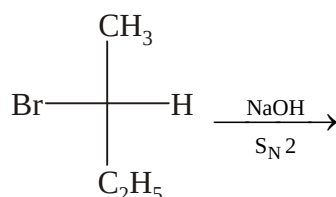


25. Identify the major product in the following reactions : [IIT 1993]



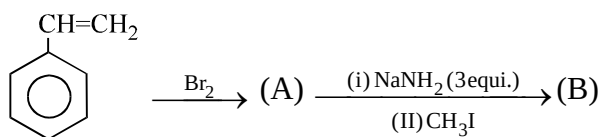
26. Aryl halides are less reactive than alkyl halides towards nucleophilic reagents. Give reason. [IIT 1994]

27. Draw the stereochemical structure of product in the following reaction. [IIT 1994]



28. In the following reactions, identify the compounds A, B, C and D. [JEE 1994]
- (i) $\text{PCl}_5 + \text{SO}_2 \longrightarrow \text{A} + \text{B}$
- (ii) $\text{A} + \text{CH}_3\text{COOH} \longrightarrow \text{C} + \text{SO}_2 + \text{HCl}$
- (iii) $2\text{C} + (\text{CH}_3)_2\text{Cd} \longrightarrow 2\text{D} + \text{CdCl}_2$
29. Optically active 2-iodobutane on treatment with NaI in acetone gives a product which does not show optical activity. Explain briefly. [IIT 1995]
30. Write down the structure of the stereoisomers formed when cis-2-butene is reacted with bromine. [JEE 1995]
31. Arrange the following compounds in order of increasing dipole moment [IIT 1996]
- | | | | |
|---------|-------------------|-------------------|-------------------|
| Toluene | m-dichlorobenzene | o-dichlorobenzene | p-dichlorobenzene |
| I | II | III | IV |
- (A) $\text{I} < \text{IV} < \text{II} < \text{III}$ (B) $\text{IV} < \text{I} < \text{II} < \text{III}$ (C) $\text{IV} < \text{I} < \text{III} < \text{II}$ (D) $\text{IV} < \text{II} < \text{I} < \text{III}$
32. An alkyl halide X of formula $\text{C}_6\text{H}_{13}\text{Cl}$ on treatment with potassium tertiary butoxide gives two isomeric alkenes Y and Z (C_6H_{12}). Both alkenes on hydrogenation give 2,3-dimethylbutane. Predict the structures of X, Y and Z. [IIT 1996]
33. Predict the structure of the intermediates/products in the following reaction sequence – [IIT 1996]
-
34. A compound D ($\text{C}_8\text{H}_8\text{O}$) upon treatment with alkaline solution of iodine gives a yellow precipitate. The filtrate on acidification gives a white solid E ($\text{C}_7\text{H}_6\text{O}_2$). Write the structures of D, E and explain the formation of E. [JEE 1996]
35. Which of the following is the correct method for synthesising methyl-t-butyl ether and why?
- $(\text{CH}_3)_3\text{CBr} + \text{NaOMe} \longrightarrow$
- or $\text{CH}_3\text{Br} + \text{NaO-t-Bu} \longrightarrow$ [IIT 1997]
36. **Statement-1 :** Addition of Br_2 to 1-butene gives two optical isomers. [JEE 1998]
- Statement-2 :** The product contains one asymmetric carbon.
- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
- (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
- (C) Statement-1 is true, statement-2 is false.
- (D) Statement-1 is false, statement-2 is true.
37. $(\text{CH}_3)_2\text{CHOCH}_3 \xrightarrow[\text{Heat}]{\text{HI(excess)}} \text{A} + \text{B}$ Write structures of A and B. [IIT 1998]

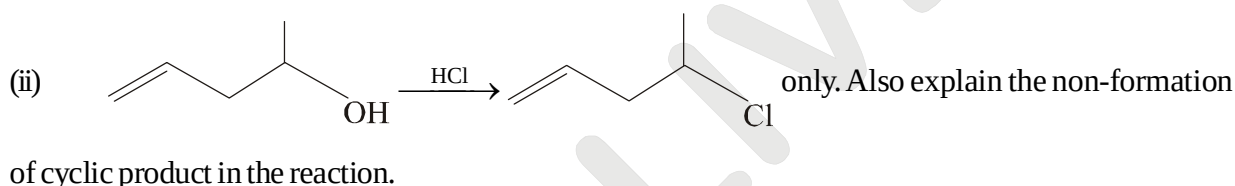
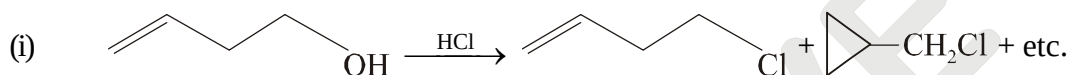
38. Complete the following reaction with appropriate structures of products/reagents. [IIT 1998]



39. A solution of (+) 2-chloro-2-phenylethane in toluene racemises slowly in the presence of small amount of SbCl_5 , due to the formation of [IIT 1999]

(A) carbanion (B) carbene (C) free radical (D) carbocation

40. Explain briefly the formation of products giving the structures of the intermediates. [JEE 1999]



41. The order of reactivity of the following alkyl halides for a $\text{S}_\text{N}2$ reaction is: [IIT 2000]

(A) $\text{RF} > \text{RC} > \text{R-Br} > \text{R-I}$ (B) $\text{R-F} > \text{R-Br} > \text{R-Cl} > \text{R-I}$
 (C) $\text{R-Cl} > \text{R-Br} > \text{RF} > \text{RI}$ (D) $\text{R-I} > \text{RBr} > \text{R-Cl} > \text{R-F}$

42. Which of the following has the highest nucleophilicity?

(A) F^- (B) OH^- (C) CH_3^- (D) NH_2^- [IIT 2000]

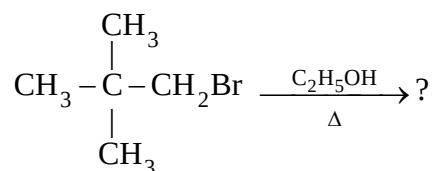
43. **Statement-1 :** 1-butene on reaction with HBr in the presence of a peroxide produces 1-bromobutane.

Statement-2 : It involves the formation of a primary radical.

[JEE 2000]

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

44. What would be major product? [IIT 2000]



45. An $\text{S}_\text{N}2$ reaction at an asymmetric carbon of a compound always gives.

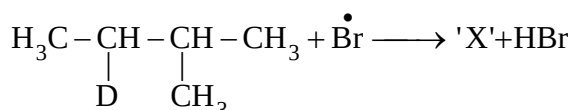
(A) an enantiomer of the substance (B) a product with opposite optical rotation
 (C) a mixture of diastereomers (D) a single stereoisomer [IIT 2001]

46. The compound that will react most readily with NaOH to form methanol is

(A) $(\text{CH}_3)_4\text{N}^+ \text{I}^-$ (B) CH_3OCH_3 (C) $(\text{CH}_3)_3\text{S}^+ \text{I}^-$ (D) $(\text{CH}_3)_3\text{Cl}$ [IIT 2001]

47. **Statement-1 :** Addition of bromine to trans-2-butene yields meso-2,3-dibromo butane.
Statement-2 : Bromine addition to an alkene is an electrophilic addition. [JEE 2001]
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

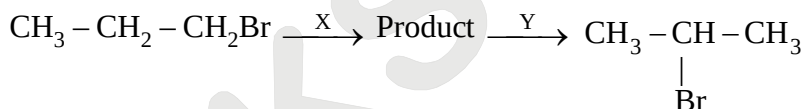
48. Consider the following reaction. [JEE 2002]



Identify the structure of the major product X.

- (A) $\text{H}_3\text{C}-\underset{\text{D}}{\text{CH}}-\underset{\text{CH}_3}{\text{CH}}-\dot{\text{C}}\text{H}_2$ (B) $\text{H}_3\text{C}-\underset{\text{D}}{\text{CH}}-\dot{\text{C}}(\text{CH}_3)-\text{CH}_3$
 (C) $\text{H}_3\text{C}-\dot{\text{C}}(\text{D})-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_3$ (D) $\text{H}_3\text{C}-\dot{\text{C}}\text{H}-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_3$

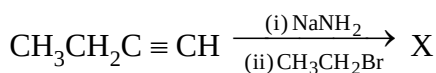
49. Identify the set of reagents / reaction conditions 'X' and 'Y' in the following set of transformation:



- (A) X = dilute aqueous NaOH, 20°C; Y = HBr / acetic acid, 20°C
 (B) X = concentrated alcoholic NaOH, 80°C; Y = HBr / acetic acid 20°C
 (C) X = dilute aqueous NaOH, 20°C; Y = Br₂ / CHCl₃, 0°C
 (D) X = concentrated alcoholic NaOH, 80°C; Y = Br₂/CHCl₃, 0°C

[IIT 2002]

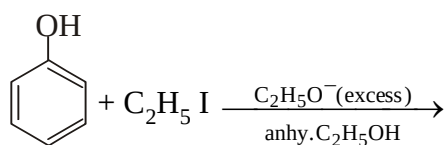
50. Identify X, Y and Z in the following synthetic scheme and write their structures



[IIT 2002]

51. The product of following reaction is

[IIT 2003]

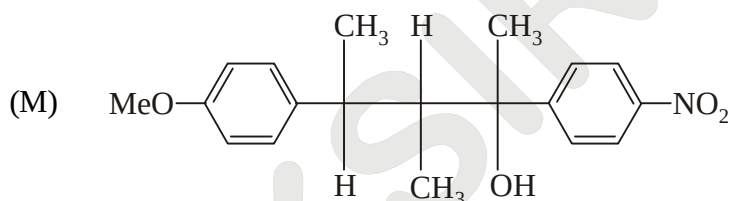
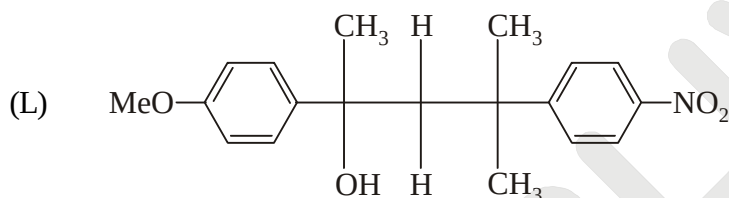
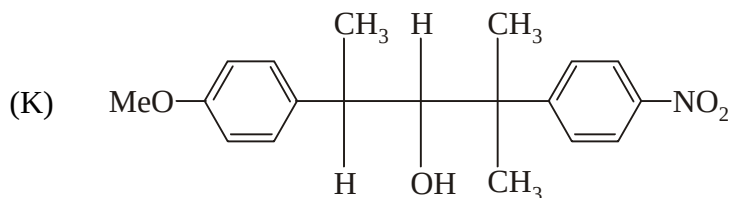
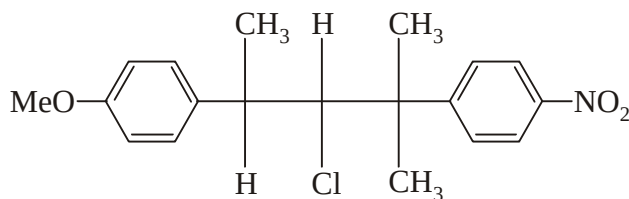


- (A) C₆H₅OC₂H₅ (B) C₂H₅OC₂H₅ (C) C₆H₅OC₆H₅ (D) C₆H₅I

52. How many chiral compounds are possible on monochlorination of 2-methyl butane? [JEE 2004]
 (A) 2 (B) 4 (C) 6 (D) 8

53. The following compound on hydrolysis in aqueous acetone will give:

[IIT 2005]



It mainly gives

(A) K and L

(B) Only K

(C) L and M

(D) Only M

54. Match the following:

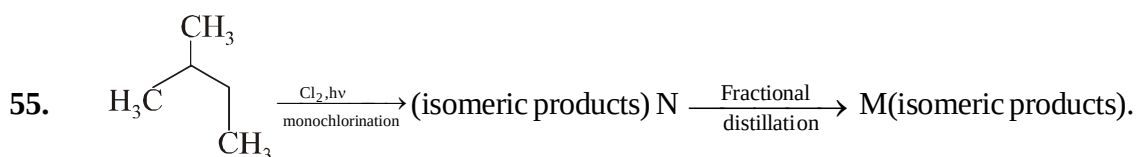
[IIT 2006]

Column I

- (A) $\text{CH}_3\text{-CHBr-CD}_3$ on treatment with alc. KOH gives $\text{CH}_2=\text{CH-CD}_3$ as a major product.
 (B) Ph-CHBr-CH_3 reacts faster than Ph-CHBr-CD_3 .
 (C) $\text{Ph-CD}_2\text{-CH}_2\text{Br}$ on treatment with $\text{C}_2\text{H}_5\text{OD/C}_2\text{H}_5\text{O}^-$ gives Ph-CD=CH_2 as the major product.
 (D) $\text{PhCH}_2\text{CH}_2\text{Br}$ and $\text{PhCD}_2\text{CH}_2\text{Br}$ react with same rate.

Column II

- (P) E1 reaction
 (Q) E2 reaction
 (R) E1cb reaction
 (S) First order reaction



What are N and M?

(A) 6, 6

(B) 6, 4

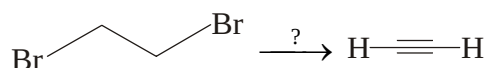
(C) 4, 4

(D) 3, 3

[JEE 2006]

56. The reagent(s) for the following conversion:

[JEE 2007]

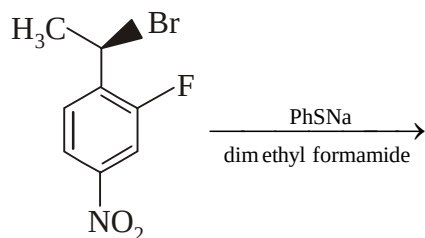


is / are :

- (A) alcoholic KOH (B) alcoholic KOH followed by NaNH_2
 (C) aqueous KOH followed by NaNH_2 (D) $\text{Zn} / \text{CH}_3\text{OH}$

57. The major product of the following reaction is

[IIT 2008]



- (A) (B) (C) (D)

58. In the reaction $\xrightarrow{\text{HBr}}$ the products are

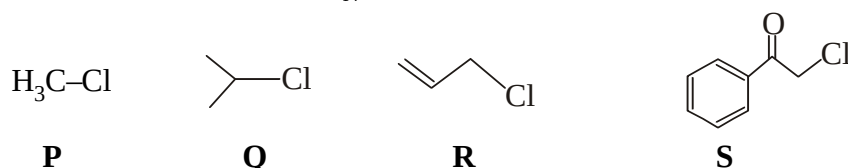
[IIT 2010]

- (A) and H_2 (B) and CH_3Br
 (C) and CH_3OH (D) and CH_3Br

59. The total number of alkenes possible by dehydrobromination of 3-bromo-3-cyclopentylhexane using alcoholic KOH is

[IIT 2011]

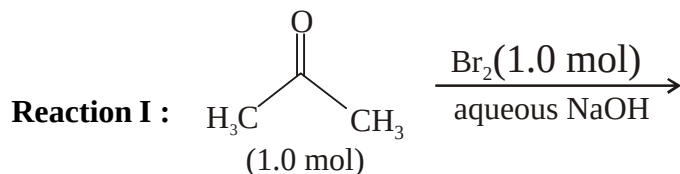
60. KI in acetone, undergoes $\text{S}_{\text{N}}2$ reaction with each of P, Q, R and S. The rates of the reaction vary as



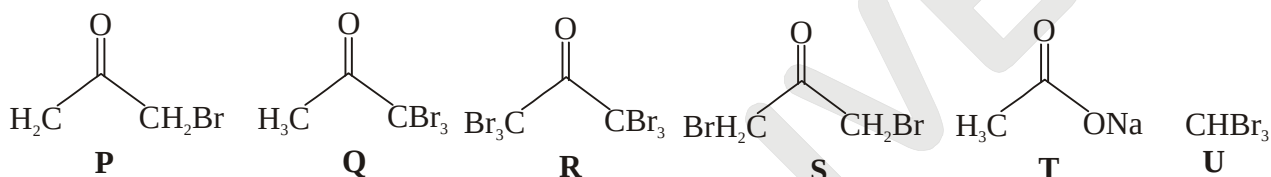
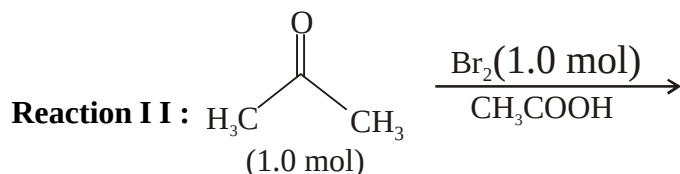
[JEE Advance 2013]

- (A) $\text{P} > \text{Q} > \text{R} > \text{S}$ (B) $\text{S} > \text{P} > \text{R} > \text{Q}$ (C) $\text{P} > \text{R} > \text{Q} > \text{S}$ (D) $\text{R} > \text{P} > \text{S} > \text{Q}$

61. After completion of the reactions (I and II), the organic compound(s) in the reaction mixtures is(are):



[JEE Advance 2013]

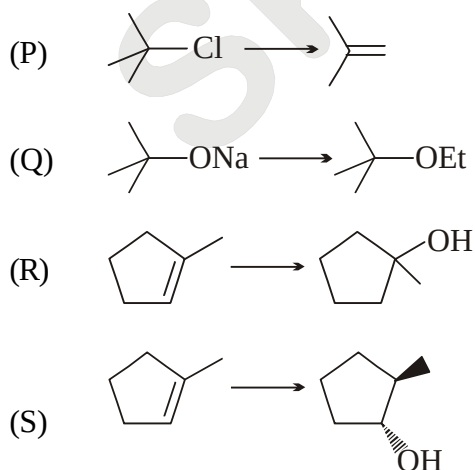


- (A) Reaction I : **P** and Reaction II : **P**
 (B) Reaction I : **U**, acetone and Reaction II : **Q**, acetone
 (C) Reaction I : **T**, **U**, acetone and Reaction II : **P**
 (D) Reaction I : **R**, acetone and Reaction II : **S**, acetone

62. Match the chemical conversions in List I with the appropriate reagents in List II and select the correct answer using the code given below the lists: [JEE Advance 2013]

List I

List II



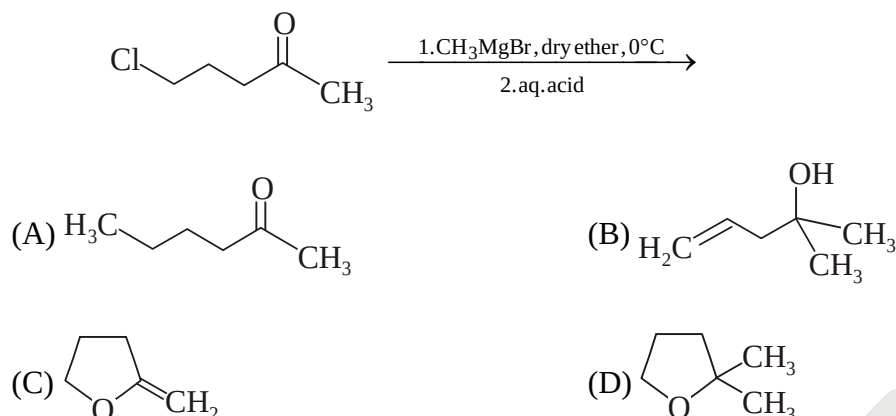
- (1) (i) $\text{Hg}(\text{OAc})_2$; (ii) NaBH_4
 (2) NaOEt
 (3) Et-Br
 (4) (i) BH_3 ; (ii) $\text{H}_2\text{O}_2/\text{NaOH}$

Codes :

	P	Q	R	S
(A)	2	3	1	4
(B)	3	2	1	4
(C)	2	3	4	1
(D)	3	2	4	1

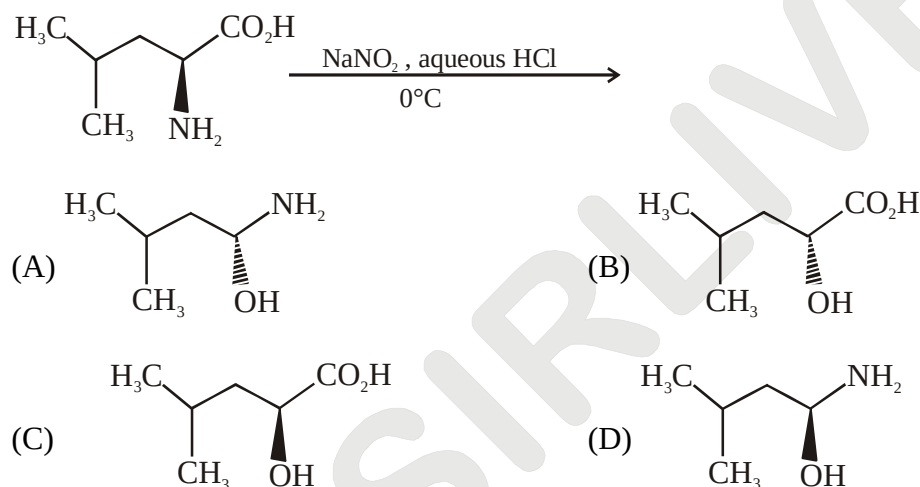
63. The major product in the following reaction is

[JEE Advance 2014]



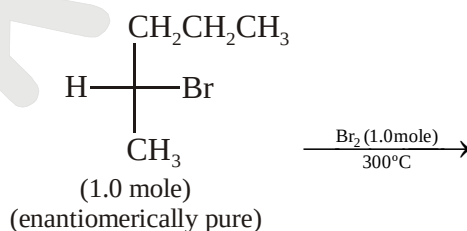
64. The major product of the reaction is :

[JEE Advance 2015]



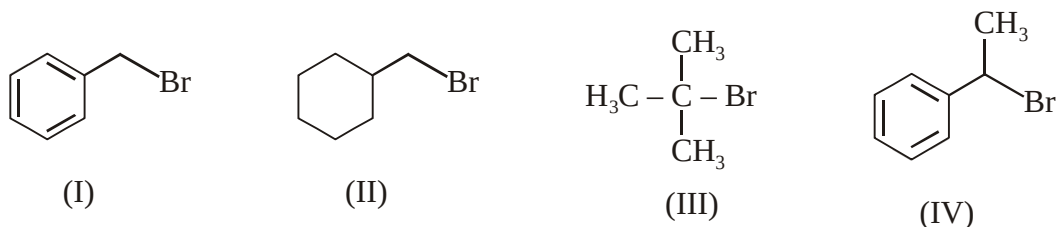
65. In the following monobromination reaction, the number of possible chiral products is

[JEE Advance 2016]



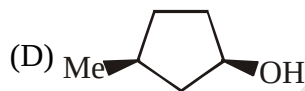
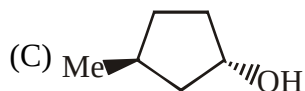
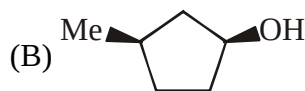
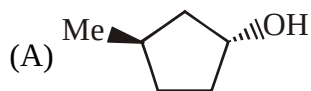
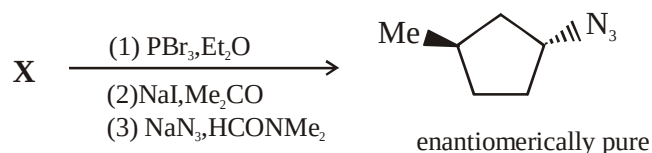
66. For the following compounds, the correct statement(s) with respect to nucleophilic substitution reactions is(are)

[JEE Advance 2017]



- (A) I and III follow S_N1 mechanism
 (B) Compound IV undergoes inversion of configuration
 (C) I and II follow S_N2 mechanism
 (D) The order of reactivity for I, III and IV is : IV > I > III

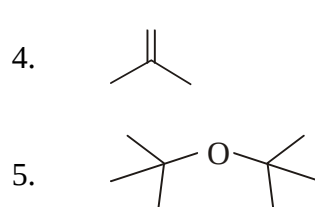
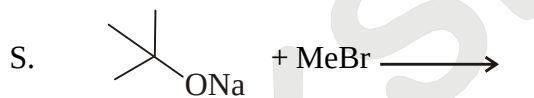
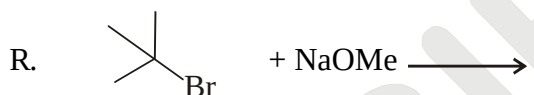
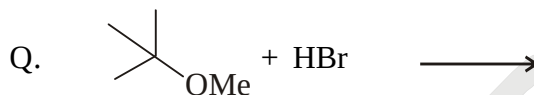
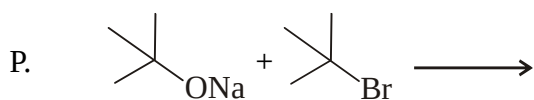
67. In the following reaction sequence, the correct structure(s) of X is (are): [JEE Advance 2018]



68. LIST-I contains reactions and LIST-II contains major products. [JEE Advance 2018]

LIST-I

LIST-II



Match each reaction in LIST-I with one or more products in LIST-II and choose the correct option.

(A) P → 1,5; Q → 2; R → 3; S → 4

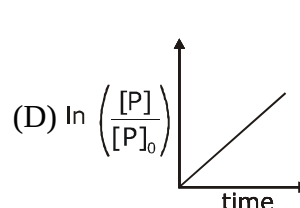
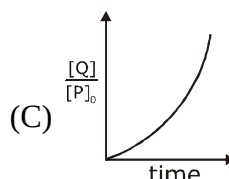
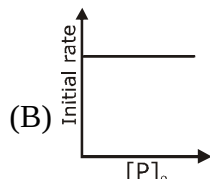
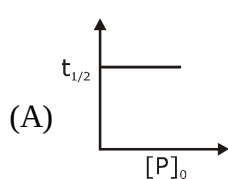
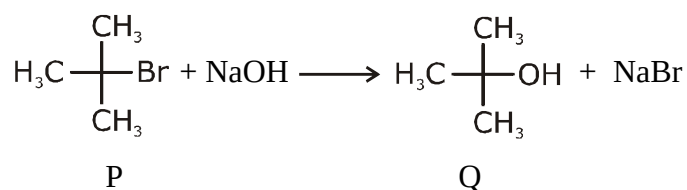
(B) P → 1,4; Q → 2; R → 4; S → 3

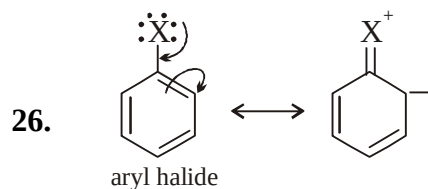
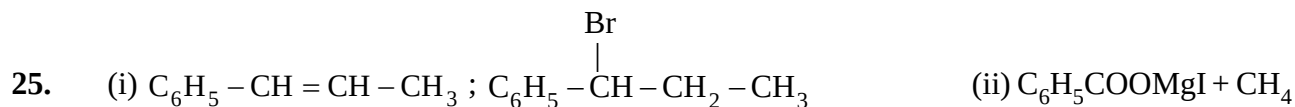
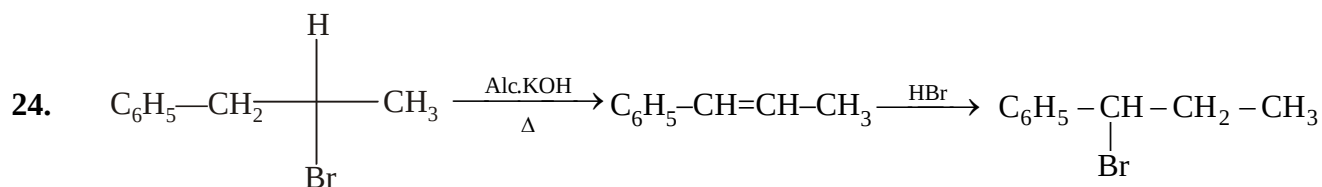
(C) P → 1,4; Q → 1,2; R → 3,4; S → 4

(D) P → 4,5; Q → 4; R → 4; S → 3,4

69. Which of the following plots is(are) correct for the given reaction? [JEE Advance 2020]

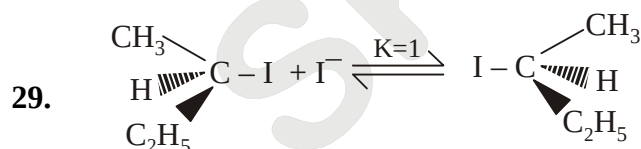
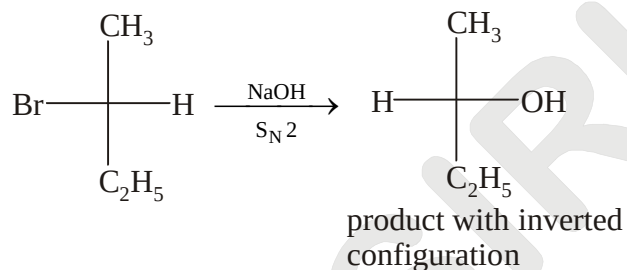
([P]₀ is the initial concentration of P)



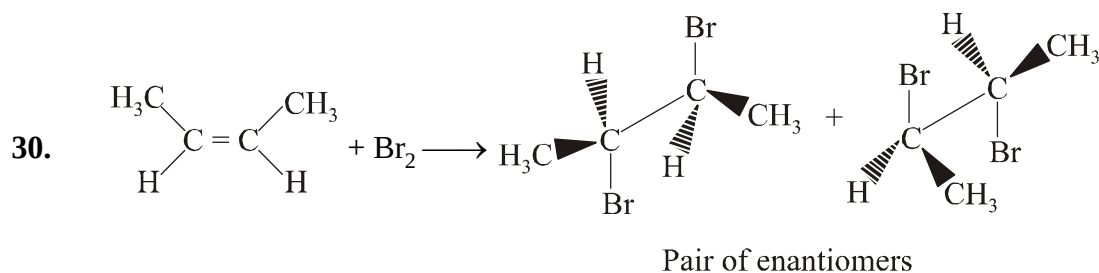


Due to the above resonance phenomena, C-X bond acquire partial double bond character and becomes difficult to break in the rate determining step of $\text{S}_{\text{N}}2$ reaction.

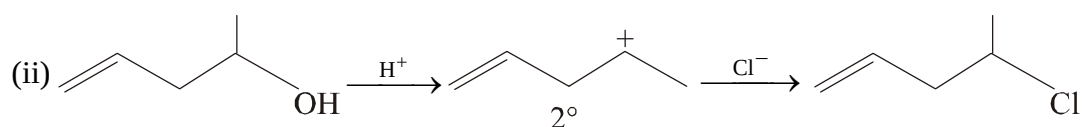
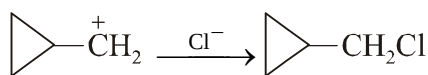
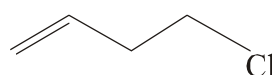
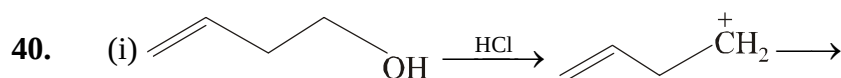
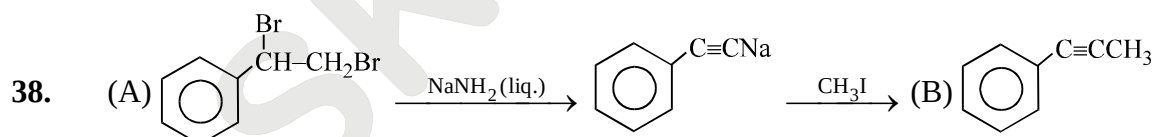
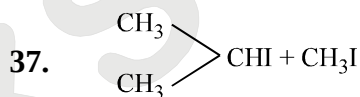
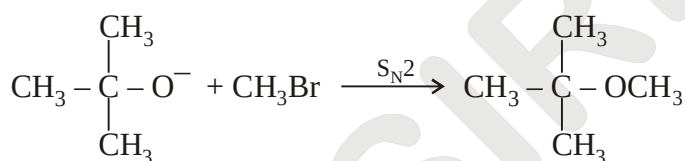
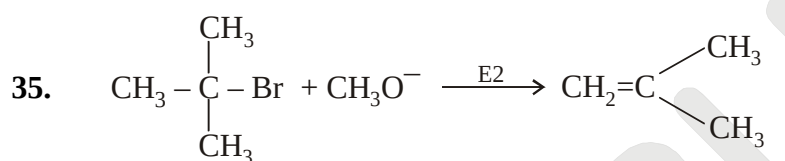
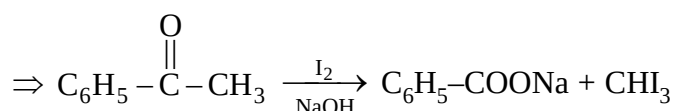
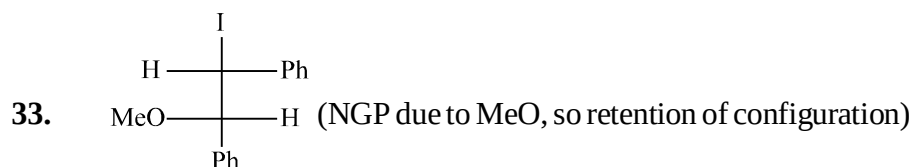
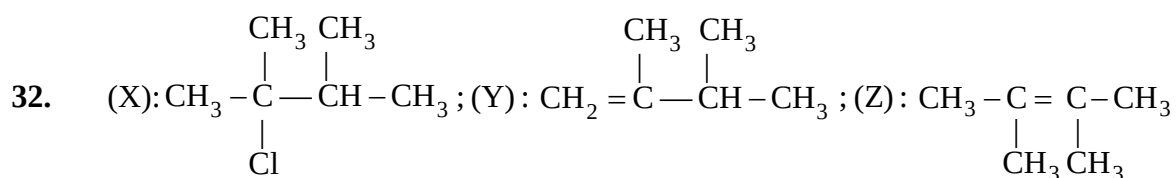
27. $\text{S}_{\text{N}}2$ reactions leads to inversion of configuration at $\alpha\text{-C}$.



Above equilibrium is established which has equilibrium constant equal to 1. Therefore, equilibrium mixture will have both the enantiomers in equal amount giving racemic mixture.



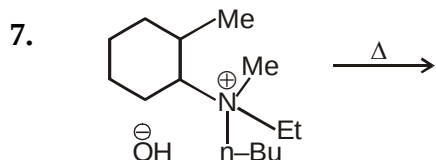
31. B



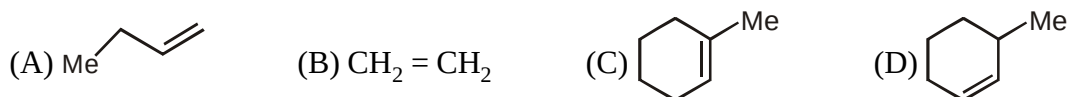
41. D 42. C 43. C 44. $\text{CH}_3 - \overset{\text{CH}_3}{\underset{|}{\text{C}}} = \text{CHCH}_3$
45. D 46. C 47. B 48. B 49. B
50. $\text{CH}_3\text{CH}_2 - \text{C} \equiv \text{CH} \xrightarrow[\text{(ii) CH}_3\text{CH}_2\text{Br}]{\text{(i) NaNH}_2} \text{CH}_3\text{CH}_2 - \text{C} \equiv \text{C} - \text{CH}_2\text{CH}_3$
51. B 52. B 53. A
54. [(A) Q; (B) Q; (C) R,S; (D) P,S] 55. B 56. B 57. A
58. D 59. 5 60. B 61. C 62. A
63. D 64. C 65. 5 66. ACD 67. B
68. B 69. A

[JEE-MAIN / AIEEE]

1. $\text{CH}_3\text{CH}_2\text{COOH} \xrightarrow[\text{red P}]{\text{Cl}_2} \text{A} \xrightarrow{\text{alc. KOH}} \text{B}$. What is B ? [AIEEE-2002 & IIT 1995]
- (A) $\text{CH}_3\text{CH}_2\text{COCl}$ (B) $\text{CH}_3\text{CH}_2\text{CHO}$
 (C) $\text{CH}_2 = \text{CHCOOH}$ (D) $\text{ClCH}_2\text{CH}_2\text{COOH}$
2. On vigorous oxidation by permanganate solution [AIEEE-2002]
 $(\text{CH}_3)_2\text{C} = \text{CH} - \text{CH}_2 - \text{CHO}$ gives
- (A) $\text{CH}_3 - \underset{\text{CH}_3}{\overset{\text{CH}_3\text{OH}}{\text{C}}} - \text{CH} - \text{CH}_2\text{CH}_3$ (B) $\text{CH}_3\text{COCH}_3 + \text{CH}_2 \begin{matrix} \text{CO}_2\text{H} \\ \text{CO}_2\text{H} \end{matrix}$
 (C) $\text{CH}_3 \diagup \text{CH} - \text{OH} + \text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (D) $\text{CH}_3 \diagup \text{C} = \text{O} + \text{CH}_3\text{CH}_2\text{CHO}$
 $\text{CH}_3 \diagdown$
3. An ether is more volatile than an alcohol having the same molecular formula. This is due to [AIEEE-2003]
 (A) dipolar character of ethers
 (B) alcohols having resonance structures
 (C) inter-molecular hydrogen bonding in ethers
 (D) inter-molecular hydrogen bonding in alcohols
4. Acid catalyzed hydration of alkenes except ethene leads to the formation of [AIEEE-2005]
 (A) primary alcohol (B) secondary or tertiary alcohol
 (C) mixture of primary and secondary alcohols (D) mixture of secondary and tertiary alcohols
5. Reaction of one molecule of HBr with one molecule of 1, 3-butadiene at 40°C gives predominantly [AIEEE-2005]
 (A) 3-bromobutene under kinetically controlled conditions
 (B) 1-bromo-2-butene under thermodynamically controlled conditions
 (C) 3-bromobutene under thermodynamically controlled conditions
 (D) 1-bromo-2-butene under kinetically controlled conditions
6. Among the following the one that gives positive iodoform test upon reaction with I_2 and NaOH is [AIEEE-2006]
 (A) $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ (B) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{OH}$
 (C) $\text{H}_3\text{C} - \underset{\text{OH}}{\overset{\text{CH}_3}{\text{C}}} - \text{CH}_3$ (D) PhCHOHCH_3



The alkene formed as a major product in the above elimination reaction is [AIEEE-2006]



8. The compound formed as a result of oxidation of ethyl benzene by KMnO_4 is: [AIEEE-2007]
 (A) benzophenone (B) acetophenone (C) benzoic acid (D) benzyl alcohol

9. In the following sequence of reactions, the alkene affords the compound 'B'



The compound B is

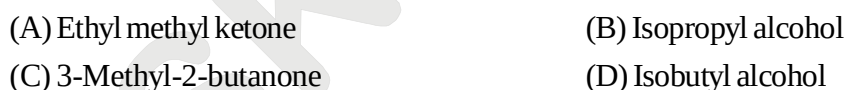


10. A liquid was mixed with ethanol and a drop of concentrated H_2SO_4 was added. A compound with a fruity smell was formed. The liquid was: [AIEEE-2009]



11. Iodoform can be prepared from all except:

[AIEEE-2012]



12. Rate of dehydration of alcohols follows the order:

[JEE MAINS ONLINE-2013]



13. An ether (A), $\text{C}_5\text{H}_{12}\text{O}$, when heated with excess of hot concentrated HI produced two alkyl halides which when treated with NaOH yielded compounds (B) and (C). Oxidation of (B) and (C) gave a propanone and an ethanoic acid respectively. The IUPAC name of the ether (A) is:

[JEE MAINS ONLINE-2013]



14. In the reaction, $\text{CH}_3\text{COOH} \xrightarrow{\text{LiAlH}_4} \text{A} \xrightarrow{\text{PCl}_5} \text{B} \xrightarrow{\text{Alc. KOH}} \text{C}$, the product C is

[JEE MAINS-2014]

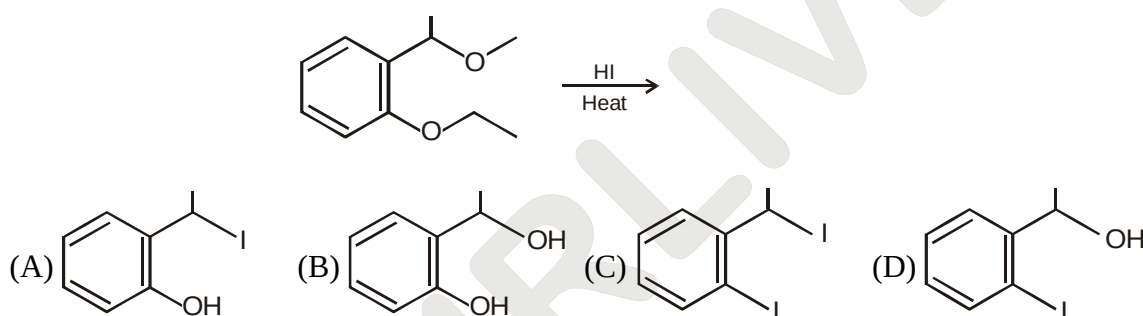


15. The most suitable reagent for the conversion of $R-CH_2-OH \rightarrow R-CHO$ is
 (A) $K_2Cr_2O_7$ (B) CrO_3 [JEE MAINS-2014]
 (C) PCC (Pyridinium Chlorochromate) (D) $KMnO_4$

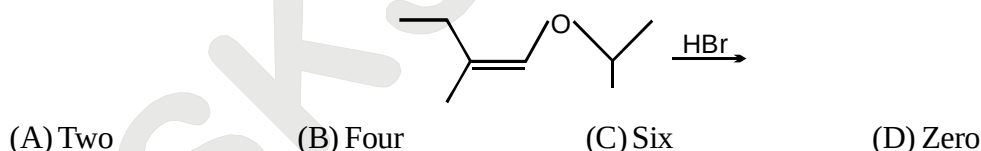
16. The gas evolved on heating CH_3MgBr in methanol is : [JEE MAINS ONLINE-2016]
 (A) Methane (B) Ethane (C) Propene (D) HBr

17. The distillation technique most suited for separating glycerol from spent-lye in the soap industry is :
 (A) Simple distillation (B) Fractional distillation [JEE MAINS-2016]
 (C) Steam distillation (D) Distillation under reduced pressure

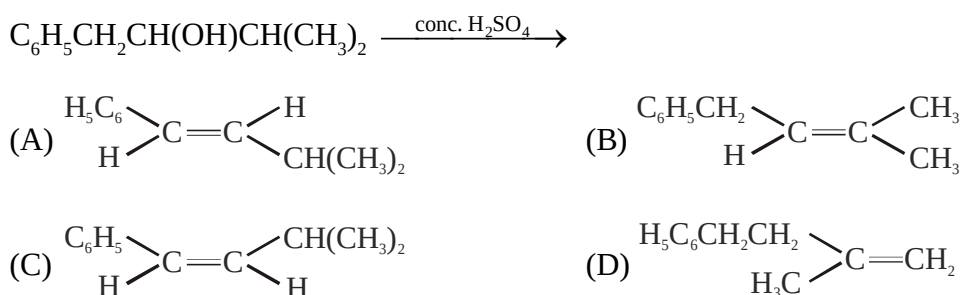
18. The major product formed in the following reaction is : [JEE MAINS (OFFLINE)-2018]



19. The total number of optically active compounds formed in the following reaction is : [JEE MAINS (ONLINE)-2018]



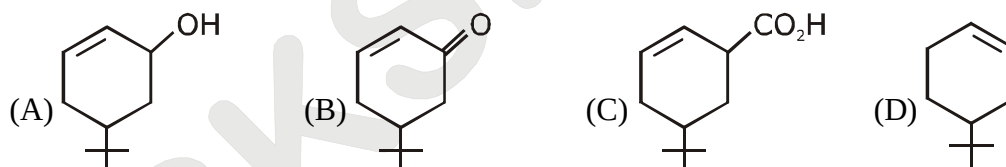
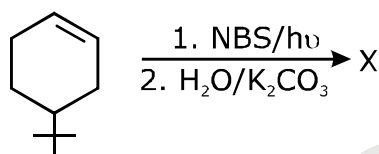
20. The main product of the following reaction is : [AIEEE-2010]



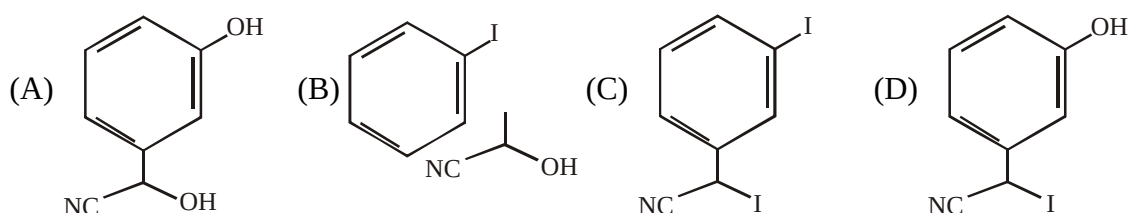
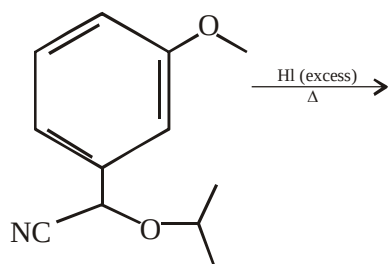
21. From amongst the following alcohols the one that would react fastest with conc. HCl and anhydrous $ZnCl_2$, is [AIEEE-2010]
 (A) 2-Butanol (B) 2-Methylpropan-2-ol
 (C) 2-Methylpropanol (D) 1-Butanol

22. Consider thiol anion ($\text{RS}^\ominus$) and alkoxy anion ($\text{RO}^\ominus$). Which of the following statements is correct ? [AIEEE-2011]
(A) $\text{RS}^\ominus$ is less basic but more nucleophilic than $\text{RO}^\ominus$.
(B) $\text{RS}^\ominus$ is more basic and more nucleophilic than $\text{RO}^\ominus$.
(C) $\text{RS}^\ominus$ is more basic but less nucleophilic than $\text{RO}^\ominus$.
(D) $\text{RS}^\ominus$ is less basic and less nucleophilic than $\text{RO}^\ominus$.
23. An unknown alcohol is treated with the "Lucas reagent" to determine whether the alcohol is primary, secondary or tertiary. Which alcohol reacts fastest and by what mechanism : [JEE (Mains) 2013]
(A) secondary alcohol by $\text{S}_{\text{N}}1$
(B) tertiary alcohol by $\text{S}_{\text{N}}1$
(C) secondary alcohol by $\text{S}_{\text{N}}2$
(D) tertiary alcohol by $\text{S}_{\text{N}}2$
24. The most suitable reagent for the conversion of $\text{R}-\text{CH}_2-\text{OH} \rightarrow \text{R}-\text{CHO}$ is : [JEE Main 2014]
(A) CrO_3
(B) PCC (Pyridinium Chlorochromate)
(C) KMnO_4
(D) $\text{K}_2\text{Cr}_2\text{O}_7$

25. The product of the reaction given below is : [JEE Main 2019]



26. The major product of the following reaction is [JEE Main 2019]



27. The synonym for water gas when used in the production of methanol is [JEE Main 2019]
(A) natural gas (B) laughing gas (C) syn gas (D) fuel gas

28. The major product of the following reaction is

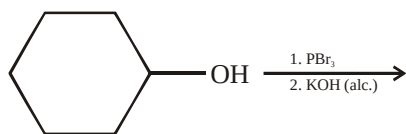


[JEE Main 2019]

- (A) $\text{CH}_3\text{CH}=\text{CHCH}_2\text{OH}$ (B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
(C) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{CH}_3$ (D) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$

29. The major product of the following reaction is

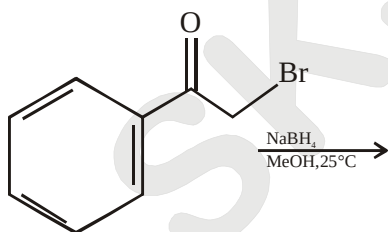
[JEE Main 2019]



- (A) (B) (C) (D)

30. The major product of the following reaction is

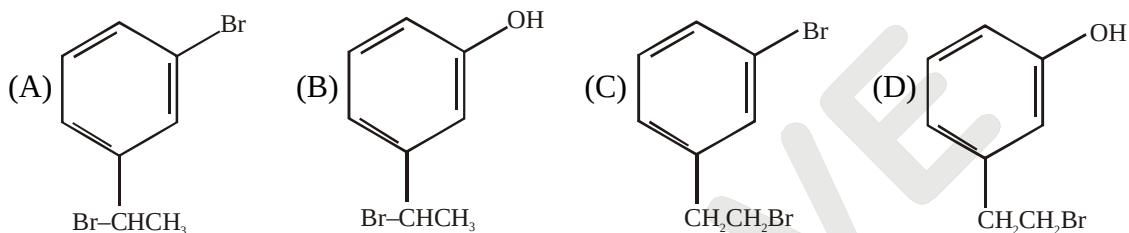
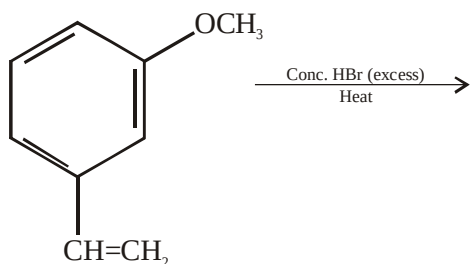
[JEE Main 2019]



- (A) (B)
(C) (D)

31. The major product of the following reaction is

[JEE Main 2019]



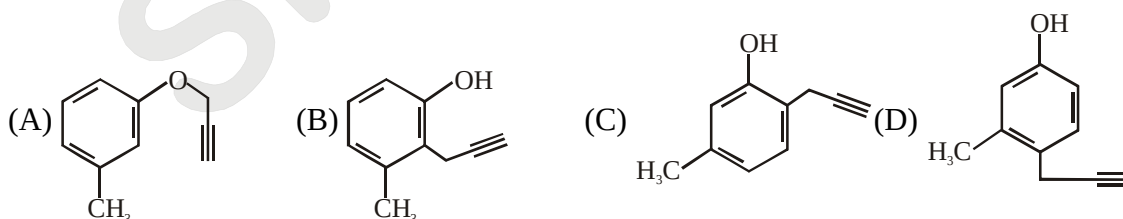
32. $\text{CH}_3\text{CH}_2-\text{C}(\text{OH})(\text{Ph})-\text{CH}_3$ cannot be prepared by

[JEE Main 2019]

- (A) $\text{CH}_3\text{CH}_2\text{COCH}_3 + \text{PhMgX}$ (B) $\text{PhCOCH}_3 + \text{CH}_3\text{CH}_2\text{MgX}$
 (C) $\text{PhCOCH}_2\text{CH}_3 + \text{CH}_3 + \text{CH}_3\text{MgX}$ (D) $\text{HCHO} + \text{PhCH}(\text{CH}_3)\text{CH}_2\text{MgX}$

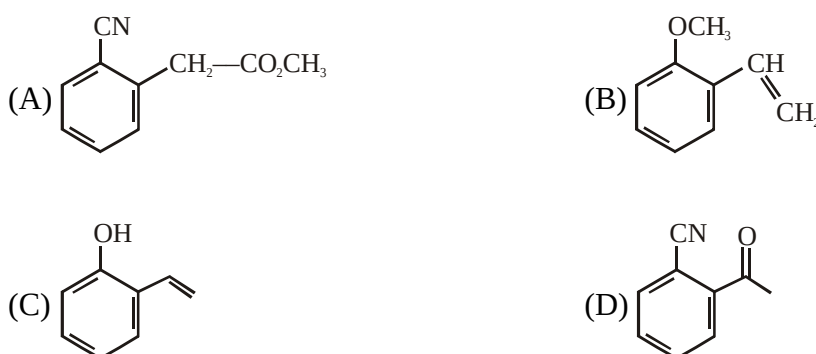
33. What will be the major product when m-cresol is reacted with propargyl bromide ($\text{HC}\equiv\text{C}-\text{CH}_2\text{Br}$) in presence of K_2CO_3 in acetone?

[JEE Main 2019]

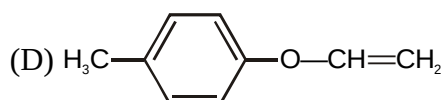
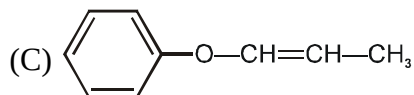
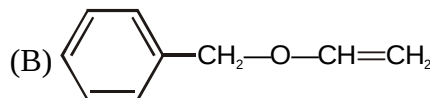
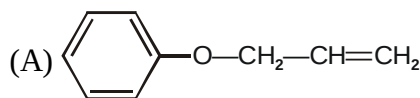


34. Which of the following compounds reacts with ethyl magnesium bromide and also decolourises bromine water solution

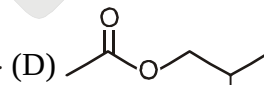
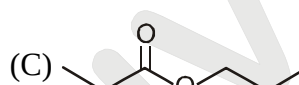
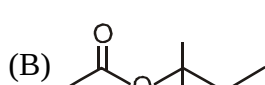
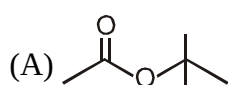
[JEE Main 2019]



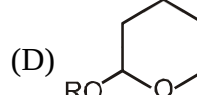
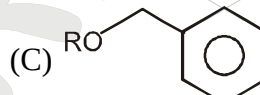
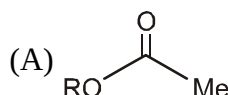
35. An organic compound 'A' ($C_9H_{10}O$) when treated with conc. HI undergoes cleavage to yield compound 'B' and 'C'. 'B' gives yellow precipitate with $AgNO_3$ where as 'C' tautomerizes to 'D'. 'D' gives positive iodoform test. 'A' could be: [JEE Main 2020]



36. An organic compound (A) (molecular formula $C_6H_{12}O_2$) was hydrolysed with dil. H_2SO_4 to give a carboxylic acid (B) and an alcohol (C). 'C' gives white turbidity immediately when treated with anhydrous $ZnCl_2$ and conc. HCl. The organic compound (A) is : [JEE Main 2020]



37. Which of the following derivatives of alcohols is unstable in an aqueous base ? [JEE Main 2020]

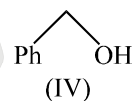
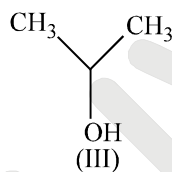
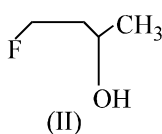
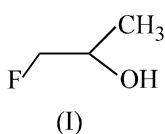


[JEE-ADVANCED]

1. An organic liquid 'A' containing C, H and O with boiling point $78^\circ C$, possessing a pleasant odour, on heating with concentrated sulphuric acid gives a gaseous product 'B' with the empirical formula, CH_2 . 'B' decolourises bromine water as well as alkaline permanganate solution and takes up one mole of H_2 (per mole of B) in presence of finely divided nickel at high temperature. Identify the substances A and B. [JEE 1979]
2. Which of the following is soluble in water? [JEE 1980]
(A) CS_2 (B) C_2H_5OH (C) CCl_4 (D) $CHCl_3$
3. Ethyl alcohol is heated with conc. H_2SO_4 . The product formed is [JEE 1980]
(A) $CH_3COOC_2H_5$ (B) C_2H_2 (C) C_2H_4 (D) C_2H_6
4. The compound which reacts fastest with Lucas reagent at room temperature is [JEE 1981]
(A) butan-2-ol (B) butan-1-ol
(C) 2-methyl propan-1-ol (D) 2-methyl propan-2-ol
5. A compound 'X' containing C, H and O is unreactive towards sodium. It does not add with bromine. It also does not react with Schiff's reagent. On refluxing with an excess of HI, 'X' yields only one organic product 'Y'. 'Y' on hydrolysis yields a new compound 'Z' which can be converted into 'Y' by reaction with red phosphorus and iodine. The compound 'Z' on oxidation with potassium permanganate gives a carboxylic acid. The equivalent weight of acid is 60. What are the compounds 'X', 'Y' and 'Z'? Write chemical leading to the conversion of 'X' to 'Y'. [JEE 1981]

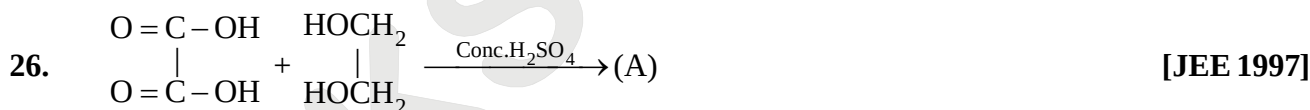
6. Diethyl ether on heating with conc. HI give two moles of [JEE 1983]
(A) ethanol (B) iodoform (C) ethyl iodide (D) methyl iodide
7. State the conditions under which the following preparations are carried out. Give necessary equations which need not be balanced. [JEE 1983]
(i) Ethanol from acetylene
(ii) Tetraethyl lead from sodium-lead alloy
(iii) Methyl chloride from aluminium carbide.
8. An alcohol A, when heated with conc. H_2SO_4 gives an alkene B. When B is bubbled through bromine water and the product obtained is dehydro halogenated with excess of sodamide, a new compound C is obtained. The compound C gives D when treated with warm dilute H_2SO_4 in presence of HgSO_4 . D can also be obtained either by oxidising A with KMnO_4 or from acetic acid through its calcium salt. Identify A, B, C and D. [JEE 1983]
9. Suggest a reason for the large difference between the boiling points of butanol and butanal, although they have almost the same solubility in water. [JEE 1985]
10. Give a chemical test to distinguish between methanol and ethanol. [JEE 1985]
11. Write down the main product of the following reaction: [JEE 1985]
$$\text{Ethanol} \xrightarrow{\text{I}_2 / \text{NaOH}}$$
12. HBr reacts fastest with [JEE 1986]
(A) 2-methyl propan-2-ol (B) propan-1-ol
(C) propan-2-ol (D) 2-methyl propan-1-ol
13. How may be the following transformation be carried out (in not more than six steps)? [JEE 1986]
"Ethyl alcohol to vinyl acetate."
14. Hydrogen bonding is maximum in [JEE 1987]
(A) ethanol (B) diethyl ether (C) ethyl chloride (D) triethyl amine
15. In $\text{CH}_3\text{CH}_2\text{OH}$, the bond that undergoes heterolytic cleavage most readily is [JEE 1988]
(A) C–C (B) C–O (C) C–H (D) O–H
16. Arrange the following in increasing order of boiling point: [JEE 1988]
n-butane, n-butanol, n-butylchloride, iso-butane
17. How will you bring about the following conversion? [JEE 1990]
"Ethanal to 2-hydroxy-3-butenic acid."
18. The products of combustion of an aliphatic thiol (RSH) at 298 K are: [JEE 1992]
(A) $\text{CO}_2(l)$, $\text{H}_2\text{O}(g)$ and $\text{SO}_2(g)$ (B) $\text{CO}_2(g)$, $\text{H}_2\text{O}(g)$ and $\text{SO}_2(g)$
(C) $\text{CO}_2(l)$, $\text{H}_2\text{O}(l)$ and $\text{SO}_2(g)$ (D) $\text{CO}_2(g)$, $\text{H}_2\text{O}(l)$ and $\text{SO}_2(l)$
19. Compound X (molecular formula, $\text{C}_5\text{H}_8\text{O}$) does not react appreciably with Lucas reagent at room temperature but gives a precipitate with ammoniacal silver nitrate with excess of MeMgBr , 0.42 g of X gives 224 mL of CH_4 at STP. Treatment of X with H_2 in presence of Pt catalyst followed by boiling with excess HI, gives n-pentane. Suggest structure for X and write the equation involved. [JEE 1992]

20. When t-butanol and n-butanol are separately treated with a few drops of dilute KMnO_4 in one case only, purple colour disappears and a brown precipitate is formed. Which of the two alcohols gives the above reaction and what is the brown precipitate? [JEE 1994]
21. The reaction products of $\text{C}_6\text{H}_5\text{OCH}_3 + \text{HI} \xrightarrow{\Delta}$ are: [JEE 1995]
 (A) $\text{C}_6\text{H}_5\text{OH} + \text{CH}_3\text{I}$ (B) $\text{C}_6\text{H}_5\text{I} + \text{CH}_3\text{OH}$
 (C) $\text{C}_6\text{H}_5\text{CH}_3 + \text{HOI}$ (D) $\text{C}_6\text{H}_6 + \text{CH}_3\text{OI}$
22. 3,3-Dimethylbutan-2-ol loses a molecule of water in the presence of a concentrated sulphuric acid to give tetramethyl ethylene as a major product. Suggest a suitable mechanism. [JEE 1996]
23. A compound D ($\text{C}_8\text{H}_{10}\text{O}$) upon treatment with alkaline solution of iodine gives a yellow precipitate. The filtrate on acidification gives a white solid (E) ($\text{C}_7\text{H}_6\text{O}_2$). Write the structures of (D) and (E) and explain the formation of (E). [JEE 1996]
24. The order of reactivity of the following alcohols: [JEE 1997]



towards conc. HCl is:

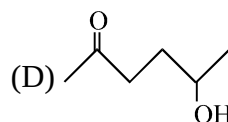
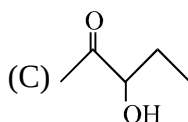
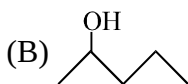
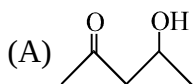
- (A) $\text{I} > \text{II} > \text{III} > \text{IV}$ (B) $\text{I} > \text{III} > \text{II} > \text{IV}$ (C) $\text{IV} > \text{III} > \text{II} > \text{I}$ (D) $\text{IV} > \text{II} > \text{III} > \text{I}$
25. Which of the following is the correct method for synthesising methyl-t-butyl ether and why?
 (i) $(\text{CH}_3)_3\text{CBr} + \text{NaOMe} \longrightarrow$ (ii) $\text{CH}_3\text{Br} + \text{tert-BuONa} \longrightarrow$ [JEE 1997]



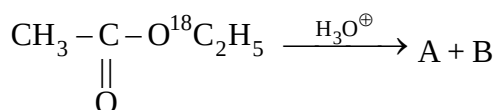
27. Among the following compounds, the strongest acid is: [JEE 1998]
 (A) $\text{HC}\equiv\text{CH}$ (B) C_6H_6 (C) C_2H_6 (D) CH_3OH

28. Discuss why o-hydroxy benzaldehyde is a liquid at room temperature while p-hydroxy benzaldehyde is a high melting solid? [JEE 1999]

29. Which one of the following will most readily be dehydrated in acidic condition: [JEE 2000]

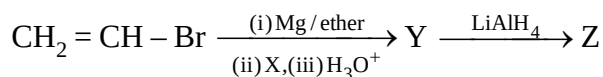
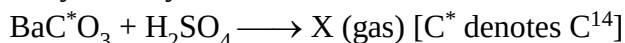


30. Write the structures of the product A & B [JEE 2000]



31. Cyclobutyl bromide on treatment with magnesium in dry ether forms an organometallic A. The organometallic reacts with ethanal to give an alcohol B after mild acidification. Prolonged treatment of alcohol B with an equivalent amount of HBr gives 1-bromo-1-methylcyclopentane (C). Write the structures of A, B and explain how C is obtained from B. [JEE 2001]

32. Identify X, Y and Z in the following synthetic scheme and write their structure. Explain the formation of labelled formaldehyde ($\text{H}_2\text{C}^*\text{O}$) as one of the products when compound Z is treated with HBr and subsequently ozonolysed. Mark the C^* carbon in the entire scheme. [JEE 2001]



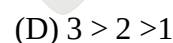
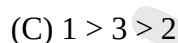
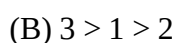
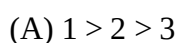
33. Identify the correct order of boiling point of the following compounds: [JEE 2002]

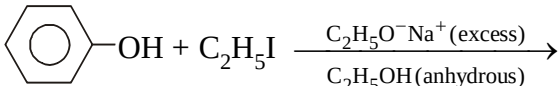


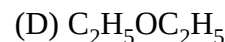
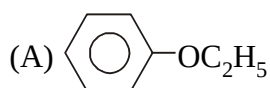
1

2

3



34.  [JEE 2003]



35. Reaction of enantiomerically pure acid with 1 chiral carbon and racemic alcohol with 1 chiral carbon gives an ester which is: [JEE 2003]

(A) Meso

(B) Optically active mixture

(C) Racemic mixture

(D) Enantiomerically pure

36. Mention two esters produced when a racemic mixture of 2-phenyl propanoic acid is treated with (+) 2-butanol. What is the stereochemical relationship between these esters? [JEE 2003]

37. On acid catalysed hydration, 2-phenyl propene gives: [JEE 2004]

(A) 3-phenyl-2-propanol

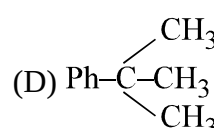
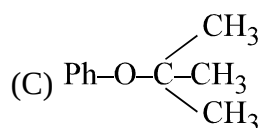
(B) 2-phenyl-1-propanol

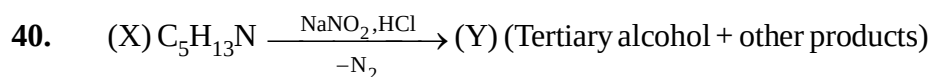
(C) 1-phenyl-3-propanol

(D) 2-phenyl-2-propanol

38. An organic compound $\text{P}(\text{C}_5\text{H}_{10}\text{O})$ reacts 10^{15} times faster than ethylene with dil. H_2SO_4 to give two products Q and R. Both Q and R give positive iodoform test. Identify P, Q and R and also give reason for very high reactivity of P. [JEE 2004]

39. Phenyl magnesium bromide reacting with t-Butyl alcohol gives [JEE 2005]

(A) $\text{Ph} - \text{OH}$ (B) $\text{Ph} - \text{H}$ 



(Optically active)

Find X and Y. Is Y optically active? Write the intermediate steps.

[JEE 2005]

41. **Statement-1:** p-Hydroxybenzoic acid has a lower boiling point than o-hydroxybenzoic acid. [JEE 2007]
because

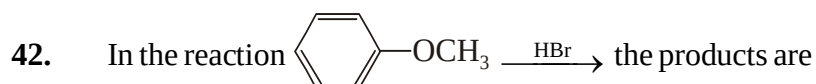
Statement-2: o-Hydroxybenzoic acid has intramolecular hydrogen bonding.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

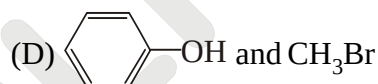
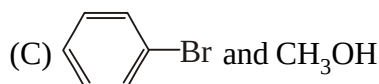
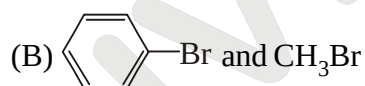
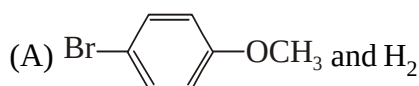
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True.

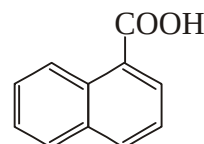
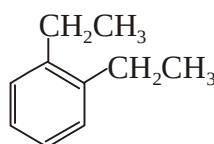
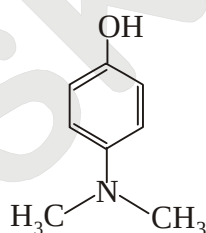
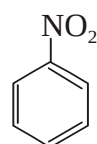
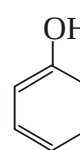
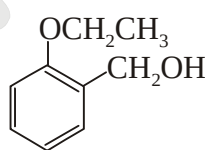
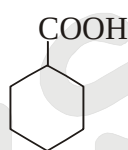
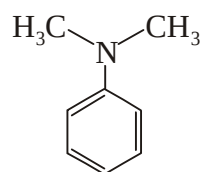


[JEE 2010]



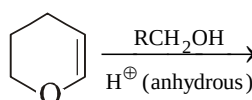
43. Amongst the following, the total number of compounds soluble in aqueous NaOH is

[JEE 2010]



44. The major product of the following reaction is

[JEE 2011]



(A) a hemiacetal

(B) an acetal

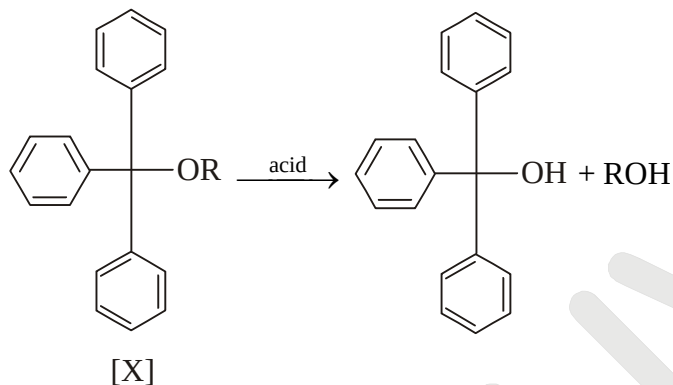
(C) an ether

(D) an ester

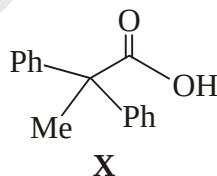
45. Consider all possible isomeric ketones, including stereoisomers of MW = 100. All these isomers are independently reacted with NaBH_4 (NOTE : Stereoisomers are also reacted separately). The total number of ketones that give a racemic product(s) is / are



46. The acidic hydrolysis of ether (X) shown below is fastest when [JEE Advance 2014]

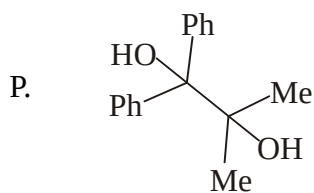


- (A) one phenyl group is replaced by a methyl group.
 (B) one phenyl group is replaced by a para-methoxyphenyl group.
 (C) two phenyl groups are replaced by two para-methoxyphenyl groups.
 (D) no structural change is made to X.
47. The desired product X can be prepared by reacting the major product of the reactions in LIST-I with one or more appropriate reagents in LIST-II.
 (given, order of migratory aptitude: aryl > alkyl > hydrogen) [JEE Advance 2018]



LIST-I

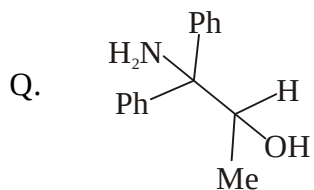
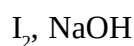
LIST-II



+



1.

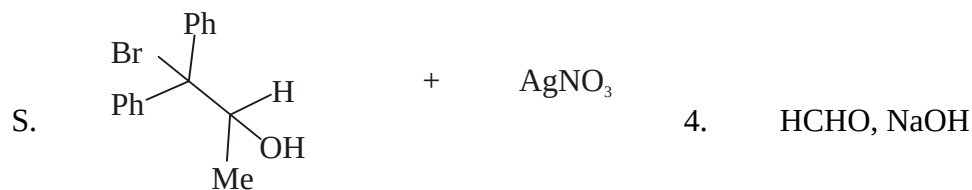
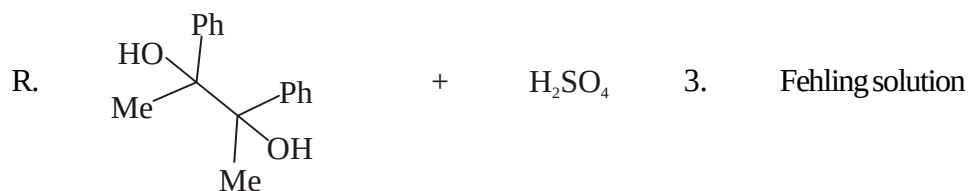


+



2.





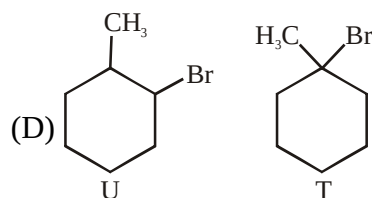
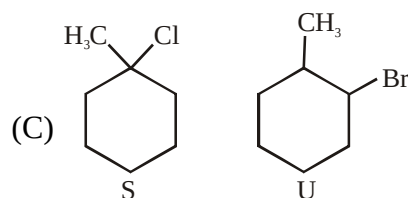
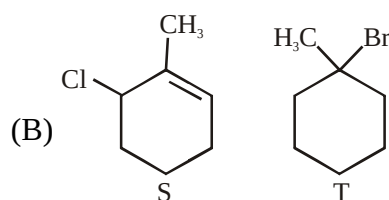
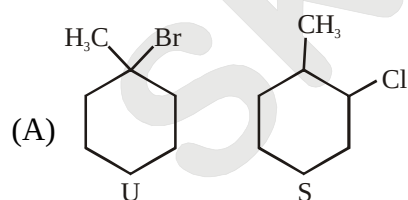
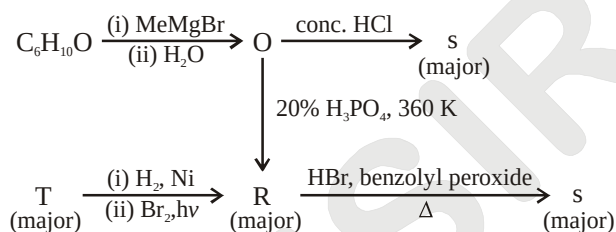
5. NaOBr

The Correct options is

- (A) $\text{P} \rightarrow 1$; $\text{Q} \rightarrow 2,3$; $\text{R} \rightarrow 1, 4$; $\text{S} \rightarrow 2,4$
 (B) $\text{P} \rightarrow 1,5$; $\text{Q} \rightarrow 3, 4$; $\text{R} \rightarrow 4, 5$; $\text{S} \rightarrow 3$
 (C) $\text{P} \rightarrow 1,5$; $\text{Q} \rightarrow 3,4$; $\text{R} \rightarrow 5$; $\text{S} \rightarrow 2, 4$
 (D) $\text{P} \rightarrow 1,5$; $\text{Q} \rightarrow 2,3$; $\text{R} \rightarrow 1, 5$; $\text{S} \rightarrow 2,3$

48. Choose the correct option(s) for the following set of reactions

[JEE Advanced 2019]

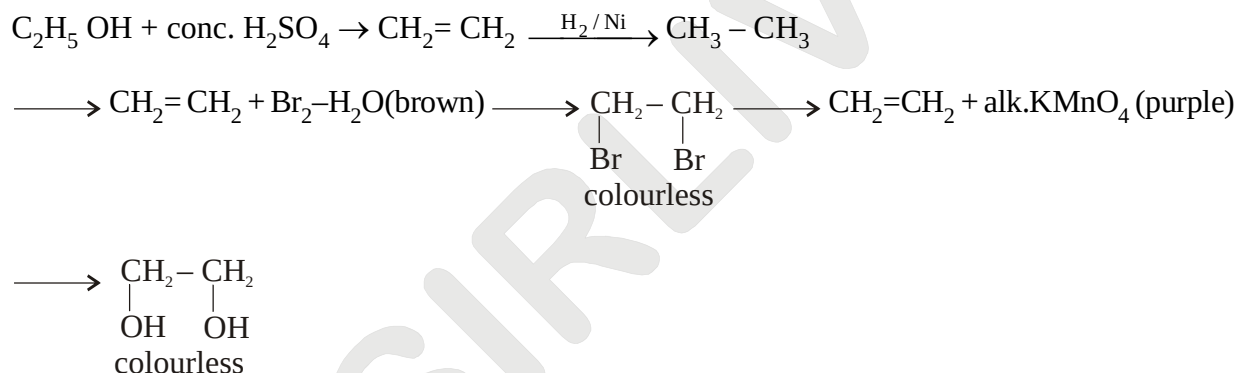


ANSWER KEY**[JEE-MAIN/AIEEE]**

1.	C	2.	B	3.	D	4.	B	5.	B	6.	D	7.	B
8.	C	9.	D	10.	B	11.	D	12.	A	13.	C	14.	B
15.	C	16.	A	17.	D	18.	A	19.	B	20.	A	21.	B
22.	A	23.	B	24.	B	25.	A	26.	D	27.	C	28.	A
29.	C	30.	D	31.	B	32.	D	33.	A	34.	C	35.	B
36.	A	37.	A										

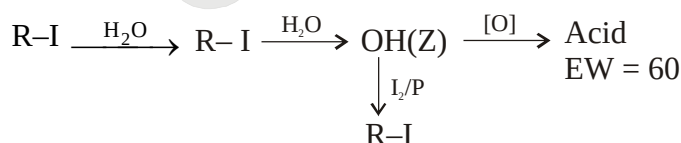
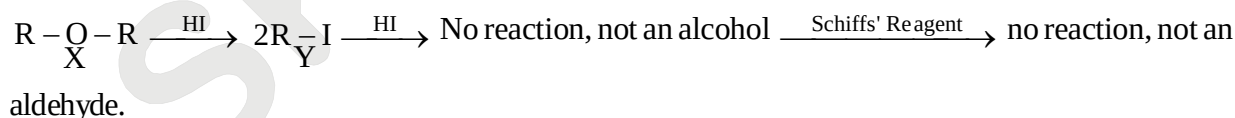
[JEE-ADVANCED]

1. A is ethanol because B is an alkene (Ethane)



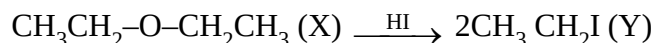
2. B 3. C 4. D

5. Compound X must be a symmetrical ether :

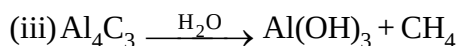
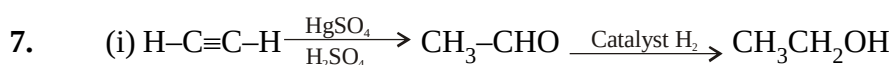


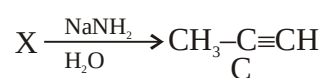
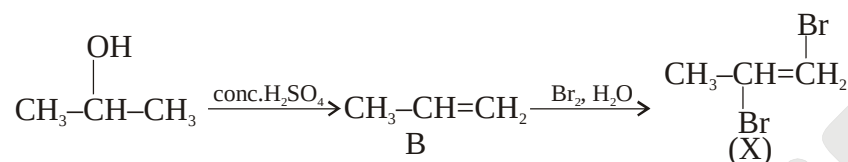
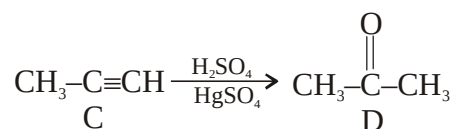
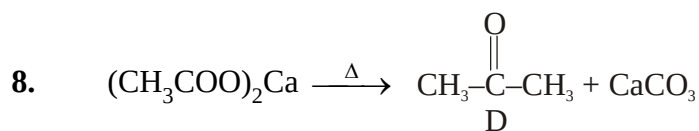
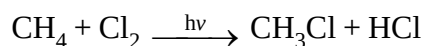
$\Rightarrow$ Acid is CH_3COOH and $\text{Z} = \text{CH}_3\text{CH}_2\text{OH}$

$\Rightarrow \text{X} = \text{C}_2\text{H}_5-\text{O}-\text{C}_2\text{H}_5$, $\text{Y} = \text{CH}_3-\text{CH}_2-\text{I}$

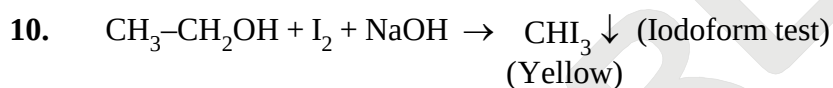


6. C

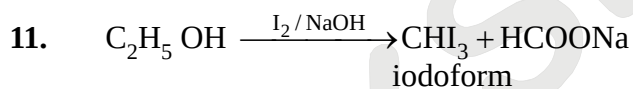




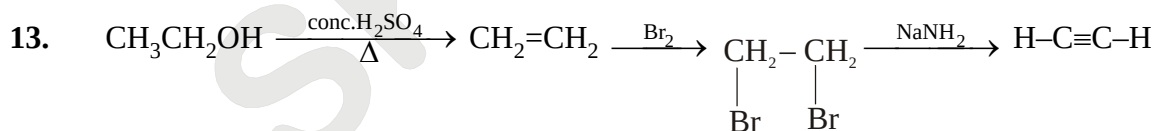
9. Butanol forms intermolecular H-bonds, has higher boiling point than butanal.



Iodoform test is not given by methanol.



12. A

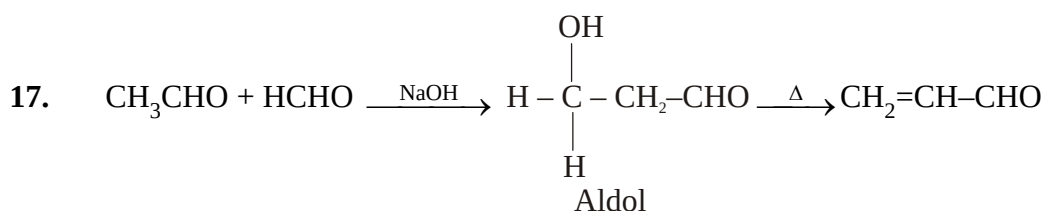


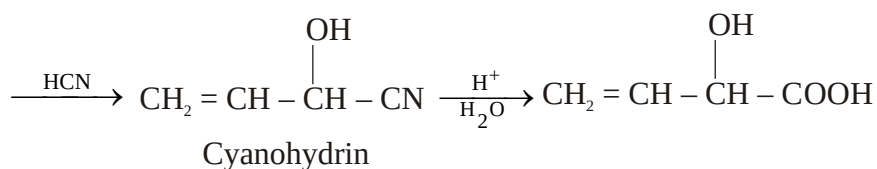
(Vinyl acetate)

14. A

15. D

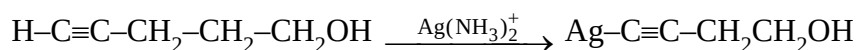
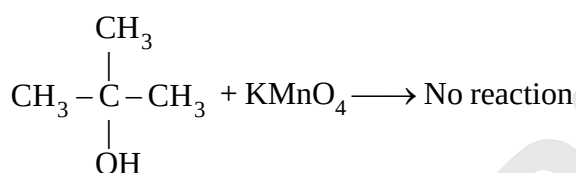
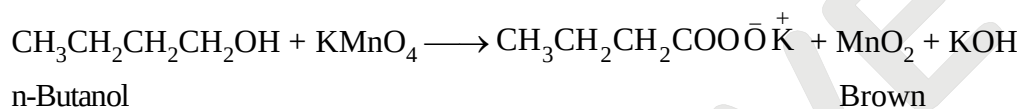
16. iso-butane < n-butane < n-butylchloride < n-butanol





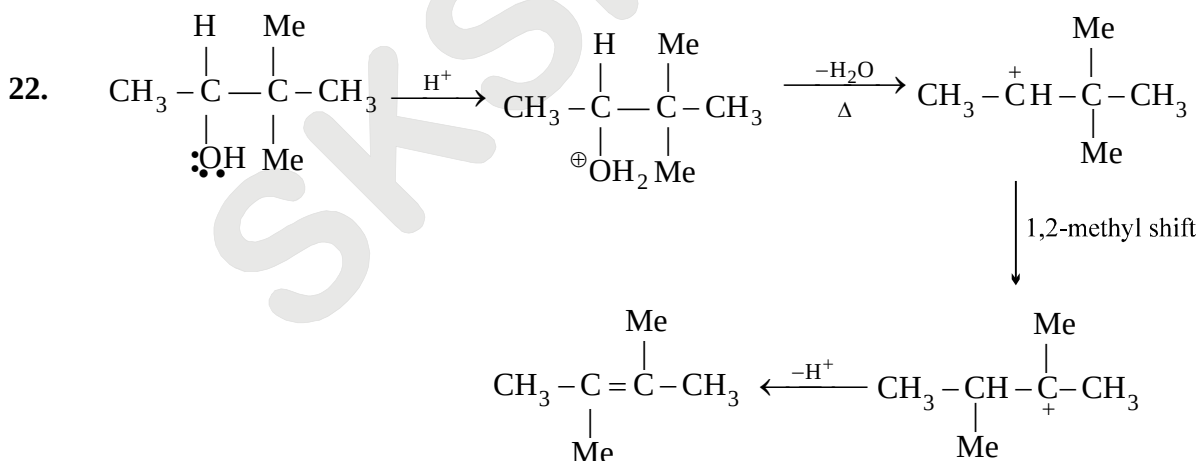
18. B

19. Above information suggest that X has a terminal triple bond and it contain primary –OH group.

20. n-butanol is oxidised by KMnO_4 and not t-butanol as the latter does not contain H atom attached to carbinol carbon atom.

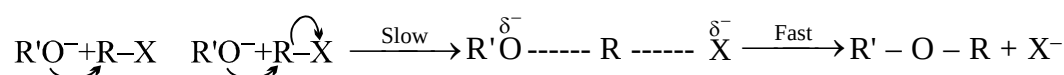
t-Butanol

21. A

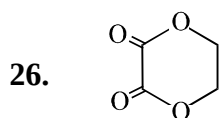
23. (D) $\text{Ph} - \underset{\text{OH}}{\text{CH}} - \text{CH}_3$ (E) $\text{C}_6\text{H}_5\text{COOH}$

24. C

25. The ether formation involves nucleophilic substitution of alkoxide ion for halide ion.



3° alkyl halide can also involve elimination of HX to give alkene in the presence of a base. So, it is better to start with 3° alkoxide and 1° alkyl halide, i.e., equation (b)

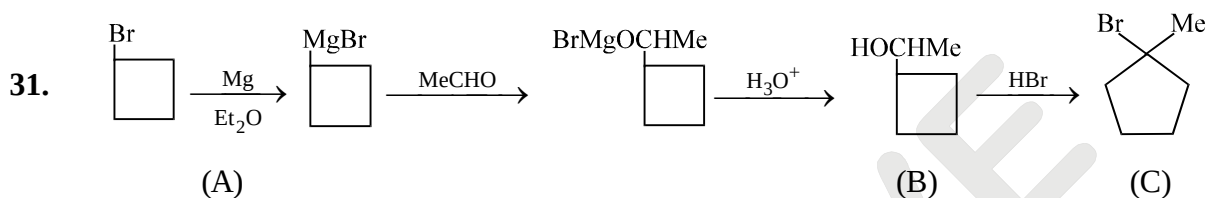


27. D

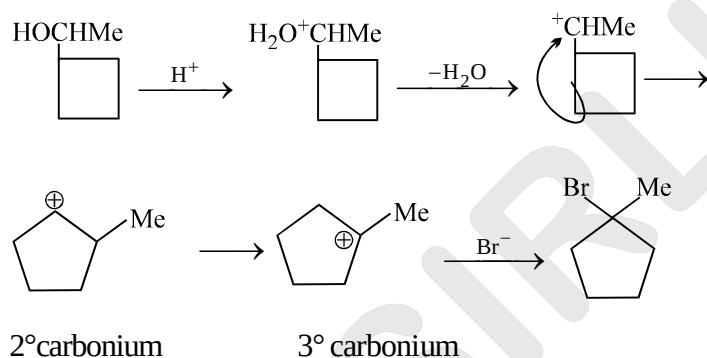
28. Due to intermolecular H-Bonding

29. A

30. $A = \text{CH}_3 - \underset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{OH}$, $B = \text{C}_2\text{H}_5\text{O}^{18}\text{H}$

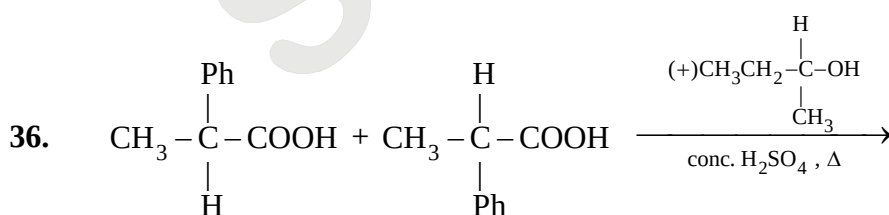


Mechanism

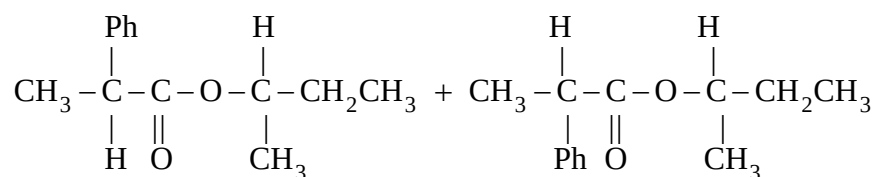


32. $X: \overset{*}{\text{CO}_2}$; $Y: \text{CH}_2 = \overset{*}{\text{CH}} - \text{COOH}$; $Z: \text{CH}_2 = \overset{*}{\text{CH}} \overset{*}{\text{CH}_2}\text{OH}$; $\overset{*}{\text{CH}_2}\text{O} \quad \text{R}'\text{O}^- + \text{R}-\text{X}$

33. B 34. D 35. B

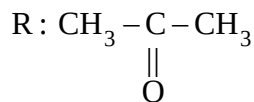
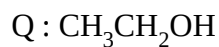
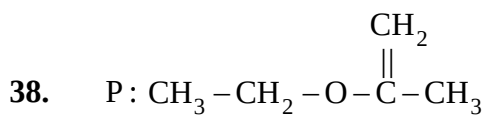


(racemic mixture)

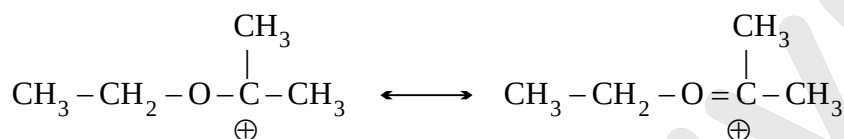


during esterification reaction only $-\text{COOH}$ and $-\text{OH}$ participates. There is no effect on structure of configuration of carbon adjacent of these groups. So when $(\pm)$ acid reacts with pure $(+)$ alcohol two esters are produced which are **diastereoisomers** of each other.

37. D



when ethylene reacts with dil. H_2SO_4 $\text{CH}_3\text{CH}_2^\oplus$ is produced during rate determining step, whereas P gives resonance stabilized intermediate.



due to extra stability of intermediate the rate of reaction is very fast

39. B



Y is optically inactive

41. D 42. D 43. 4 44. B 45. 5 46. C 47. D
48. CD

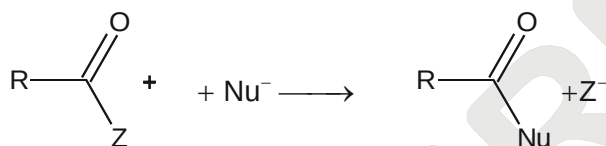
[JEE-MAIN / AIEEE]

1. Which of these will not react with acetylene? [AIEEE-2002]
 (A) NaOH (B) ammonical AgNO_3 (C) Na (D) HCl

2. $\text{CH}_3 - \text{Mg} - \text{Br}$ is an organo metallic compound due to [AIEEE-2002]
 (A) Mg – Br bond (B) C – Mg bond (C) C – Br bond (D) C – H bond

3. Ethyl isocyanide on hydrolysis in acidic medium generates [AIEEE-2003]
 (A) ethylamine salt and methanoic acid (B) propanoic acid and ammonium salt
 (C) ethanoic acid and ammonium salt (D) methylamine salt and ethanoic acid

4. Rate of the reaction [AIEEE-2004]

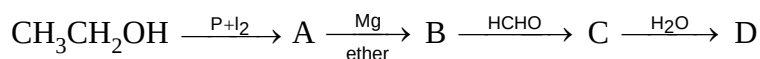


- is fastest when Z is [AIEEE-2004]
 (A) Cl (B) OCOCH_3 (C) OC_2H_5 (D) NH_2

5. Acetyl bromide reacts with excess of CH_3MgI followed by treatment with a saturated solution of NH_4Cl given [AIEEE-2004]
 (A) acetone (B) acetyl iodide
 (C) 2- methyl -2- propanol (D) acetamide

6. Phenyl magnesium bromide reacts with methanol to give [AIEEE-2006]
 (A) a mixture of anisole and $\text{Mg}(\text{OH})\text{Br}$ (B) a mixture of benzene and $\text{Mg}(\text{OMe})\text{Br}$
 (C) a mixture of toluene and $\text{Mg}(\text{OH})\text{Br}$ (D) a mixture of phenol and $\text{Mg}(\text{Me})\text{Br}$

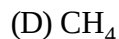
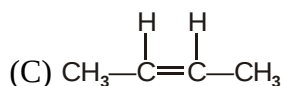
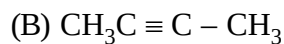
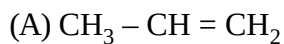
7. In the following sequence of reactions, [AIEEE-2007]



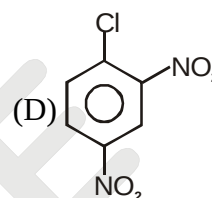
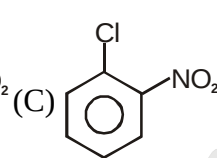
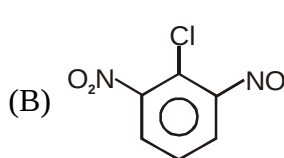
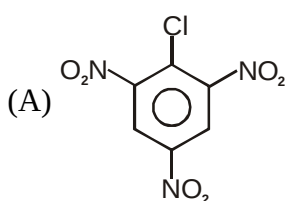
the compound 'D' is :

- (A) butanal (B) n-butyl alcohol (C) n-propyl alcohol (D) propanal
8. Which one of the following is the strongest base in aqueous solution? [AIEEE-2007]
 (A) Trimethylamine (B) Aniline (C) Dimethylamine (D) Methylamine

9. The treatment of CH_3MgX with $\text{CH}_3\text{C}\equiv\text{C-H}$ produces [AIEEE-2008]



10. A major component of Borsch reagent is obtained by reacting hydrazine hydrate with which of the following? [IIT Mains Online 2013]



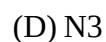
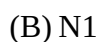
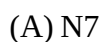
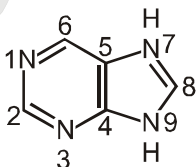
11. The hydrocarbon with seven carbon atoms containing a neopentyl and a vinyl group is :

[IIT-JEE Mains (online) 2016]



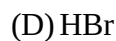
12. The "N" which does not contribute to the basicity for the compound is :

[IIT-JEE Mains (online) 2016]

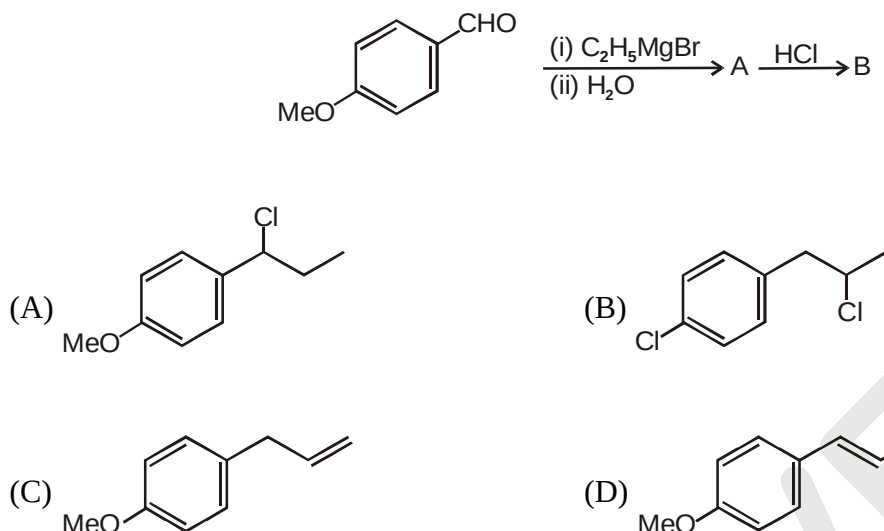


13. The gas evolved on heating CH_3MgBr in methanol is

[JEE Mains (online) 2016]



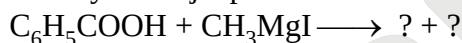
14. The major product B formed in the following reaction sequence is [JEE Main (online) 2018]



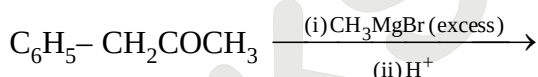
[JEE-ADVANCED]

1. Write the structural formula of main organic product formed when ethyl acetate is treated with double the molar quantity of methyl magnesium bromide and the reaction mixture is poured into water. [JEE 1981]

2. Identify the major product. [JEE 1993]



3. Predict the major product. [JEE 1994]



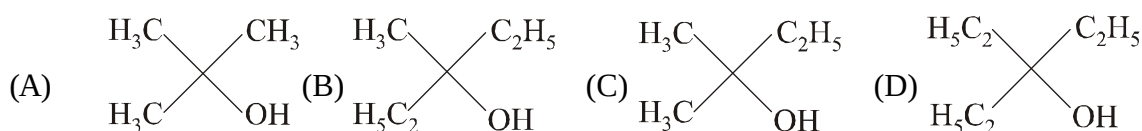
4. Which of the following is an organometallic compound? [JEE 1997]

- (A) Lithium methoxide (B) Lithium acetate
(C) Lithiumdimethylamide (D) Methyl lithium

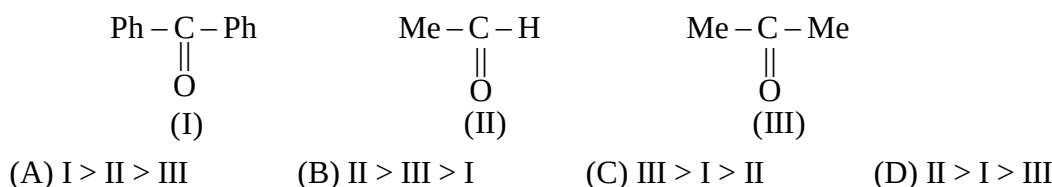
5. $(\text{CH}_3)_3\text{CMgCl}$ on treatment with D_2O produces [JEE 1997]

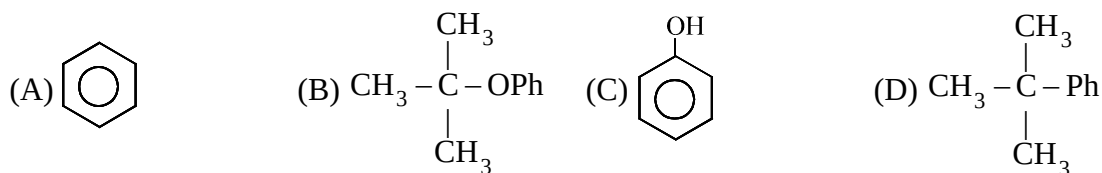
- (A) $(\text{CH}_3)_3\text{CD}$ (B) $(\text{CH}_3)_3\text{COD}$ (C) $(\text{CD})_3\text{CD}$ (D) $(\text{CD})_3\text{COD}$

6. $\text{C}_4\text{H}_8\text{O}_2$ $\xrightarrow[\text{(excess)}]{\text{CH}_3\text{MgBr}}$ P, the product 'P' will be [JEE 2003]

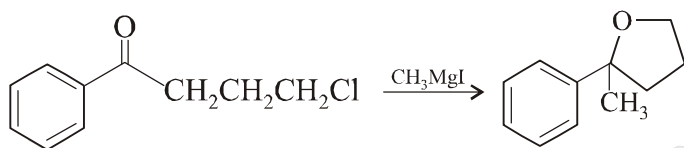


7. Order of rate of reaction of following compounds with phenyl magnesium bromide is: [JEE 2004]

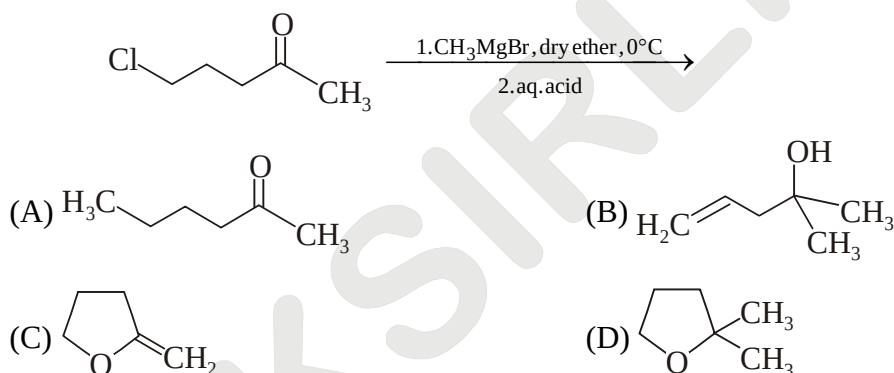




9. Identify the reaction mechanism : [JEE 2011]



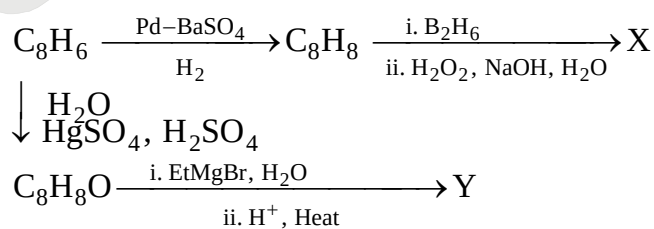
10. The major product in the following reaction is [JEE Advance 2014]



Paragraph for question nos. 11 & 12

In the following reactions :

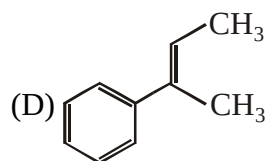
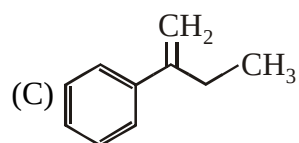
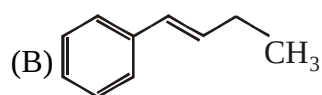
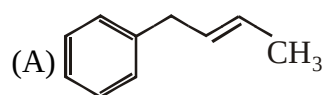
[JEE Advance 2015]



11. Compound X is



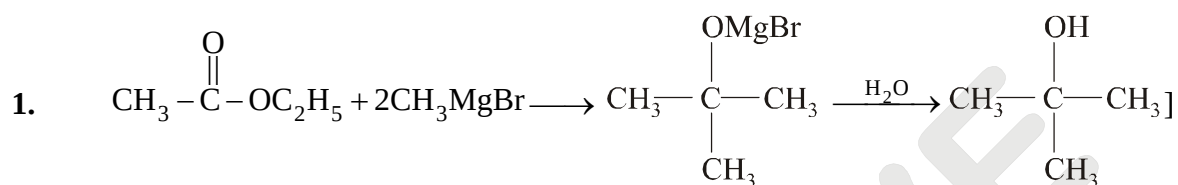
12. The major compound Y is



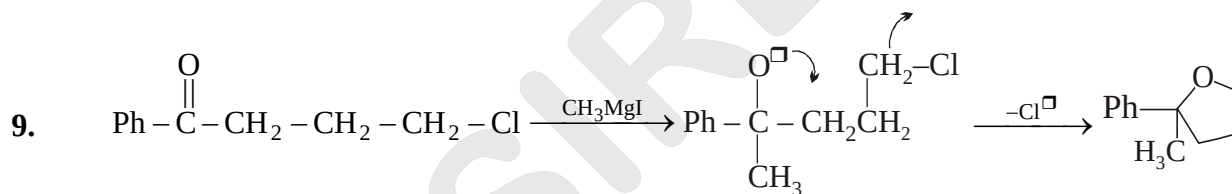
SKSIRLIVE

ANSWER KEY**[JEE-MAIN/AIEEE]**

- | | | | | | | | | | | | | | |
|----|---|----|---|-----|---|-----|---|-----|---|-----|---|-----|---|
| 1. | A | 2. | B | 3. | A | 4. | A | 5. | C | 6. | B | 7. | C |
| 8. | C | 9. | D | 10. | D | 11. | B | 12. | C | 13. | A | 14. | A |

[JEE-ADVANCED]

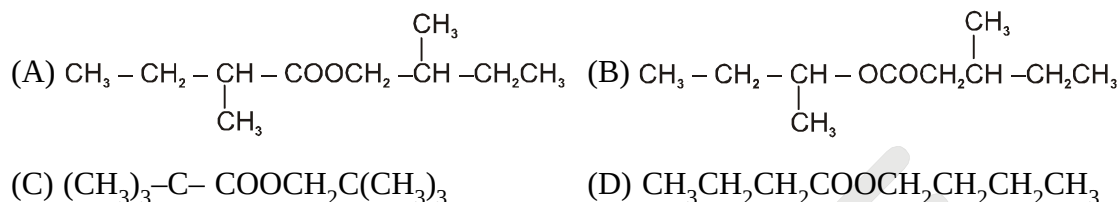
4. D 5. A 6. A 7. B 8. A



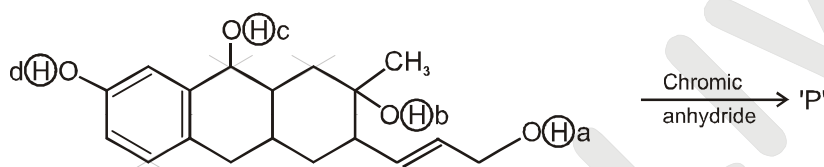
10. D 11. C 12. D

[JEE-MAIN]

1. An organic compound [A], molecular formula $C_{10}H_{20}O_2$ was hydrolysed with dilute sulphuric acid to give a carboxylic acid [B] and an alcohol [C]. Oxidation of [C] with $CrO_3 - H_2SO_4$ produced [B]. Which of the following structures are not possible for [A] [JEE-Main Online 2020]



2. Consider the following reaction : [JEE-Main Online 2020]



The product 'P' gives positive ceric ammonium nitrate test. This is because of the presence of which of these -OH group(s) ?

- (A) (b) only (B) (d) only (C) (b) and (d) (D) (c) and (d)

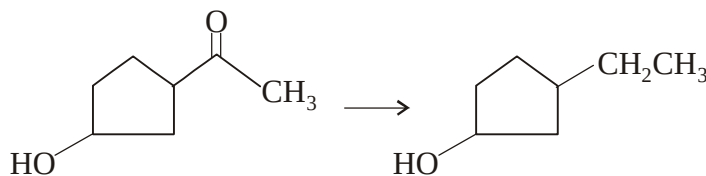
[JEE-ADVANCED]

OBJECTIVE

1. Which of the following will decolourise alkaline $KMnO_4$ solution? [JEE 1980]
 (A) C_3H_8 (B) CH_4 (C) CCl_4 (D) C_2H_4
2. The reagent with which both acetaldehyde and acetone react easily is [JEE 1981]
 (A) Tollen's reagent (B) Schiff's reagent (C) Grignard reagent (D) Fehling reagent
3. When acetaldehyde is treated with Fehling's solution, it gives a precipitate of [JEE 1982]
 (A) Cu (B) CuO (C) Cu_2O (D) $Cu + Cu_2O + CuO$
4. Baeyer's reagent is [JEE 1984]
 (A) alkaline permanganate solution (B) acidified permanganate solution
 (C) neutral permanganate solution (D) aqueous bromine solution
5. Hydrogenation of benzoyl chloride in the presence of Pd on $BaSO_4$ gives [JEE 1992]
 (A) benzyl alcohol (B) benzaldehyde (C) benzoic acid (D) phenol

6. The appropriate reagent for the following transformation:

[JEE 2000]



- (A) Zn(Hg), HCl (B) $\text{NH}_2\text{NH}_2, \text{OH}^-$ (C) H_2 / Ni (D) NaBH_4

7. **Statement-1:** Dimethyl sulphide is commonly used for the reduction of an ozonide of an alkene to get the carbonyl compound. [JEE 2001]

Statement-2: It reduces the ozonide giving water soluble dimethyl sulfoxide and excess of it evaporates.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

8. 1-propanol and 2-propanol can be best distinguished by [JEE 2001]

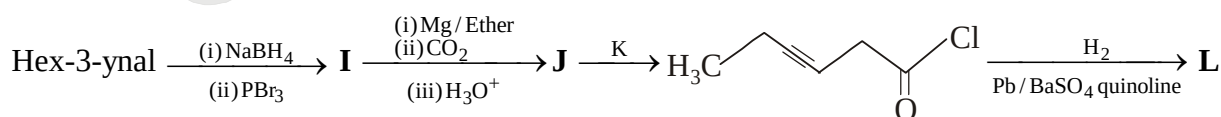
- (A) Oxidation with alkaline KMnO_4 followed by reaction with Fehling solution.
 (B) Oxidation with acidic dichromate followed by reaction with Fehling solution
 (C) Oxidation by heating with copper followed by reaction with Fehling solution
 (D) Oxidation with concentrated H_2SO_4 followed by reaction with Fehling solution.

9. Butan-2-one can be converted to propanoic acid by which of the following? [JEE 2006]

- (A) $\text{NaOH, NaI} / \text{H}^+$ (B) Fehling solution
 (C) $\text{NaOH, I}_2 / \text{H}^+$ (D) Tollen's reagent

COMPREHENSION :

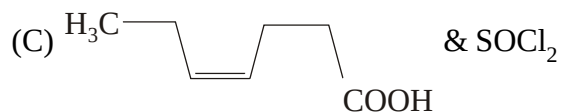
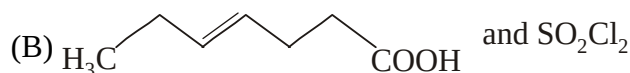
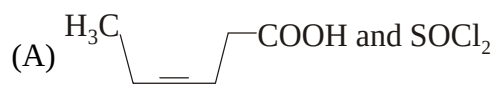
In the following sequence, product I, J and L are formed. K represents a reagent. [JEE 2008]



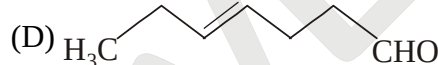
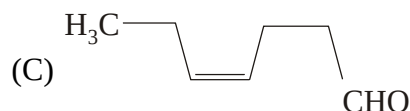
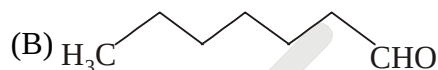
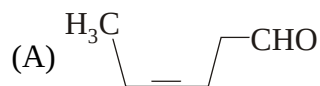
10. The structure of the product I is

- (A) (B)
 (C) (D)

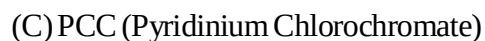
11. The structures of compound J and K, respectively, are



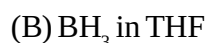
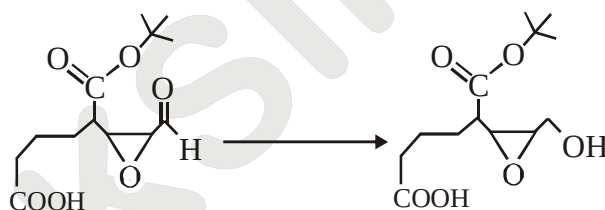
12. The structure of product L is



13. The most suitable reagent for the conversion of $\text{R}-\text{CH}_2-\text{OH} \rightarrow \text{R}-\text{CHO}$ is :[JEE Main 2014]



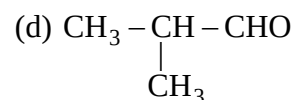
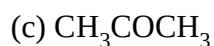
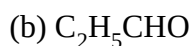
14. Reagent(s) which can be used to bring about the following transformation is(are) [JEE Advance 2016]



SUBJECTIVE

1. One mole of a hydrocarbon A reacts with 1 mol of bromine giving a dibromo compound, $\text{C}_5\text{H}_{10}\text{Br}_2$. Compound A on treatment with cold dilute alkaline potassium permanganate solution forms a compound, $\text{C}_5\text{H}_{12}\text{O}_2$. On ozonolysis, A gives equimolar quantities of propanone and ethanal. Deduce the structural formula of A. [JEE 1981]

2. A white precipitate was formed slowly when silver nitrate was added to compound A with the molecular formula $\text{C}_6\text{H}_{13}\text{Cl}$. Compound A on treatment with hot alcoholic potassium hydroxide gave a mixture of two isomeric alkenes B and C, having the formula C_6H_{12} . The mixture of b and C, on ozonolysis, furnished four compounds. [JEE 1986]



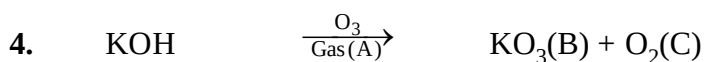
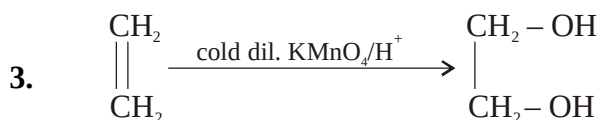
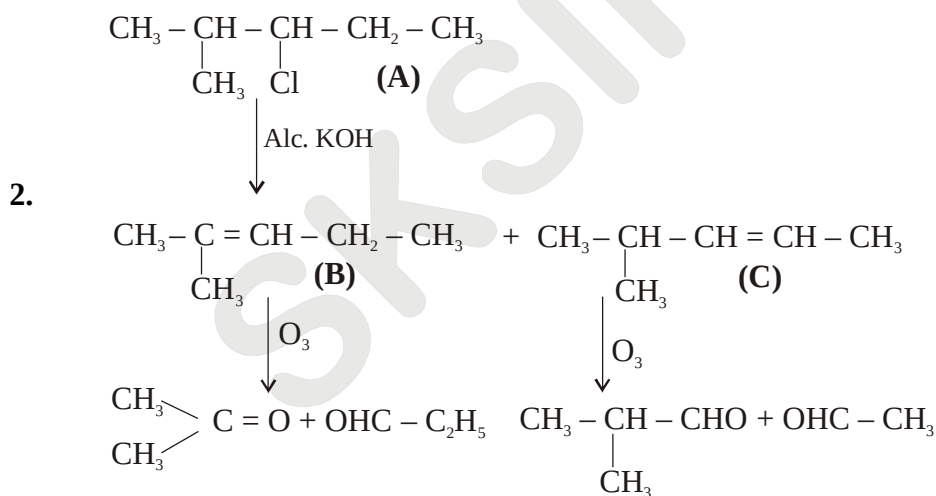
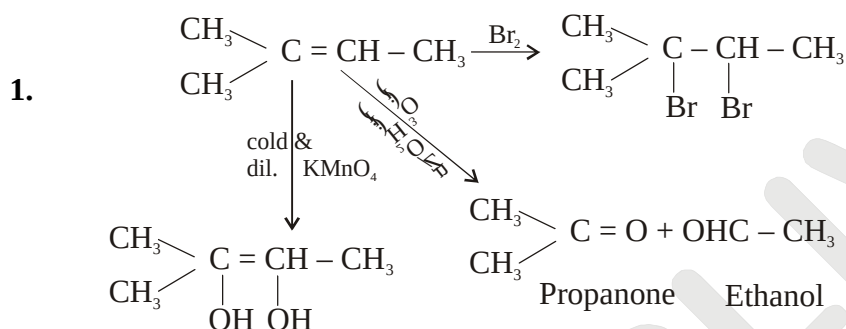
What are the structures of A, B and C?

3. Write the balanced chemical equations for the following: [JEE 1991]
"Ethylene glycol is obtained by the reaction of ethylene with potassium permanganate".
4. When gas A is passed through dry KOH at low temperature, a deep-red coloured compound B and a gas C are obtained. Gas A, on reaction with but-2-ene, followed by treatment with $\text{Zn}/\text{H}_2\text{O}$ yields acetaldehyde. Identify A, B and C. [JEE 1994]
5. An organic compound E (C_5H_8) on hydrogenation gives compound F (C_5H_{12}). Compound E on ozonolysis gives formaldehyde and 2-ketopropanal. Deduce the structure of compound E. [JEE 1995]
6. Hydrocarbon A adds 1 mol of hydrogen in the presence of a platinum catalyst to form n-hexane. When A is oxidised vigorously with KMnO_4 , a single carboxylic acid, containing three carbon atoms, is isolated. Give the structure of A and explain. [JEE 1997]
7. Monomer A of a polymer on ozonolysis yields 2 mol of HCHO and 1 mol of CH_3COCHO .
(a) Deduce the structure of A. [JEE 2005]
(b) Write the structure of all cis forms of polymer of compound A.

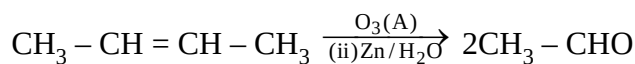
[JEE-MAIN]

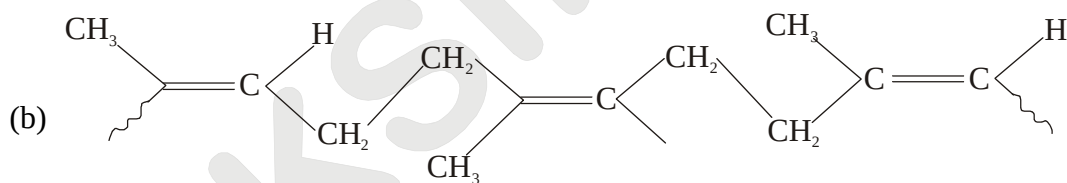
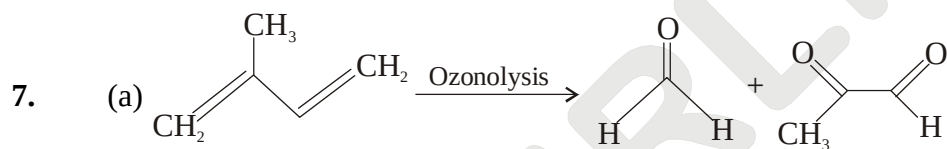
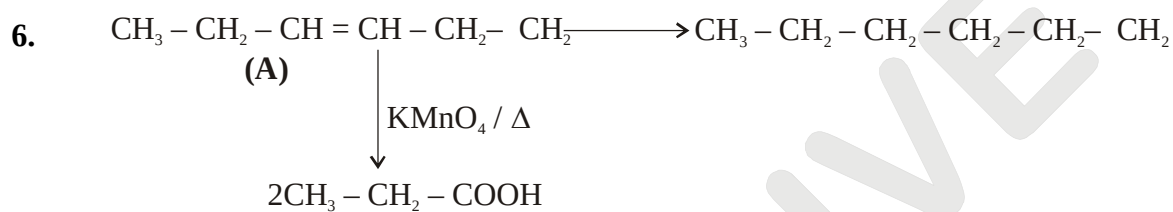
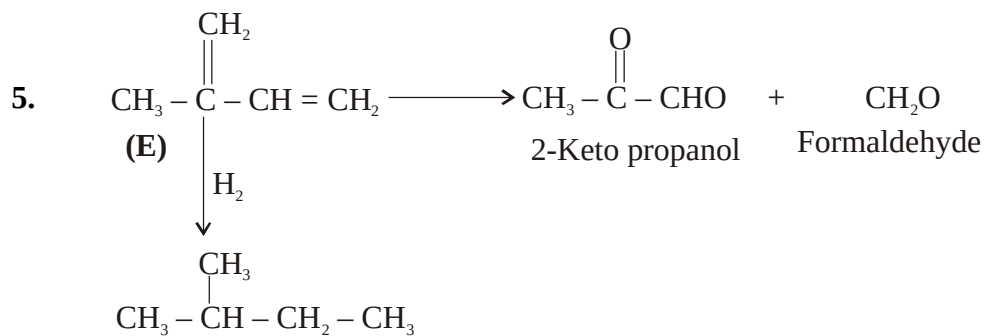
[JEE-ADVANCED]

1.	D	2.	C	3.	C	4.	A	5.	B	6.	B	7.	A
8.	C	9.	C	10.	D	11.	A	12.	C	13.	C	14.	CD



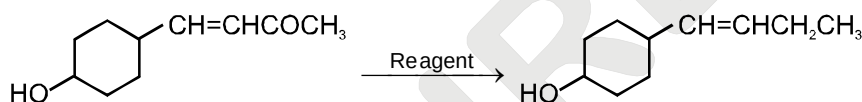
Pottasium ozonide (Deep Red Coloured)


$$(A) \rightarrow O_3; (B) \rightarrow KO_3; (C) \rightarrow O_3$$

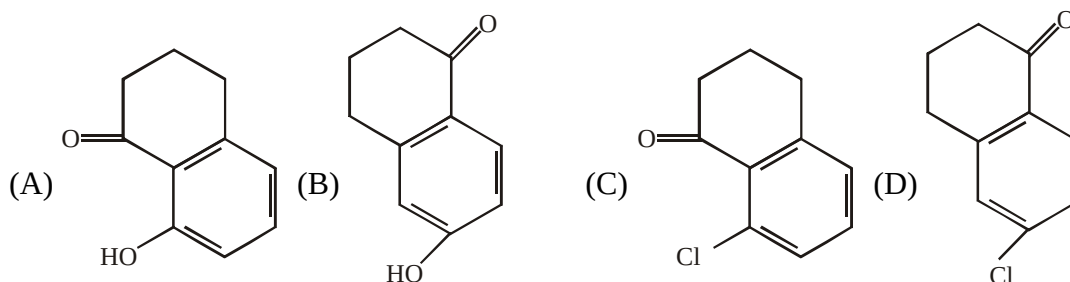
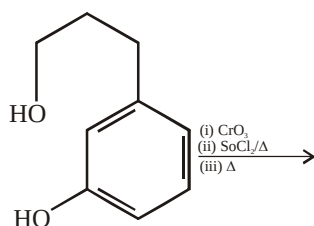


[JEE-MAIN / AIEEE]

- Trichloroacetaldehyde was subjected to Cannizzaro's reaction by using NaOH. The mixture of the products contains sodium trichloroacetate ion and another compound. The other compound is :
 (A) 2, 2, 2-Trichloroethanol (B) Trichloromethanol [AIEEE 2011]
 (C) 2, 2, 2-Trichloropropanol (D) Chloroform
- Ozonolysis of an organic compound 'A' produces acetone and propionaldehyde in equimolar mixture. Identify 'A' from the following compounds : [AIEEE 2011]
 (A) 1-Pentene (B) 2-Pentene
 (C) 2-Methyl-2-pentene (D) 2-Methyl-1-pentene
- Iodoform can be prepared from all except : [AIEEE 2012]
 (A) Ethyl methyl ketone (B) Isopropyl alcohol
 (C) 3-Methyl-2-butanone (D) Isobutyl alcohol
- In the given transformation, which of the following is the most appropriate reagent ? [AIEEE 2012]

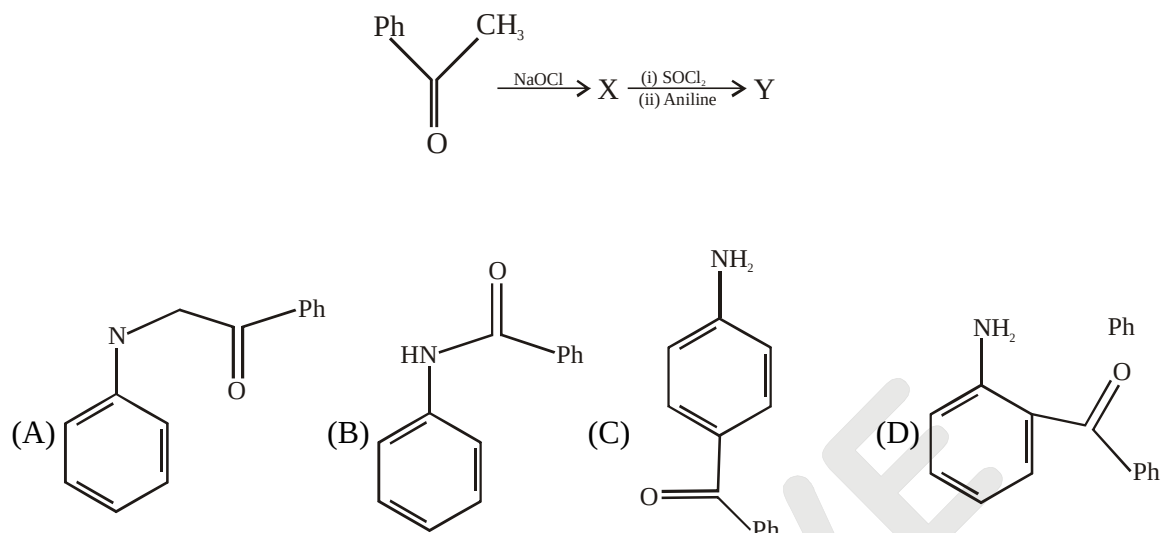


- (A) $\text{NH}_2\text{NH}_2, \text{OH}^-$ (B) Zn-Hg/HCl (C) Na, Liq, NH_3 (D) NaBH_4
- The correct sequence of reagents for the following conversion will be : [JEE Mains - 2017]
 (A) $[\text{Ag}(\text{NH}_3)_2]^+\text{OH}^-$, $\text{H}^+/\text{CH}_3\text{OH}$, CH_3MgBr (B) CH_3MgBr , $\text{H}^+/\text{CH}_3\text{OH}$, $[\text{Ag}(\text{NH}_3)_2]^+\text{OH}^-$
 (C) CH_3MgBr , $[\text{Ag}(\text{NH}_3)_2]^+\text{OH}^-$, $\text{H}^+/\text{CH}_3\text{OH}$ (D) $[\text{Ag}(\text{NH}_3)_2]^+\text{OH}^-$, CH_3MgBr , $\text{H}^+/\text{CH}_3\text{OH}$
 - The major product of the following reaction is [JEE-MAIN 2019]



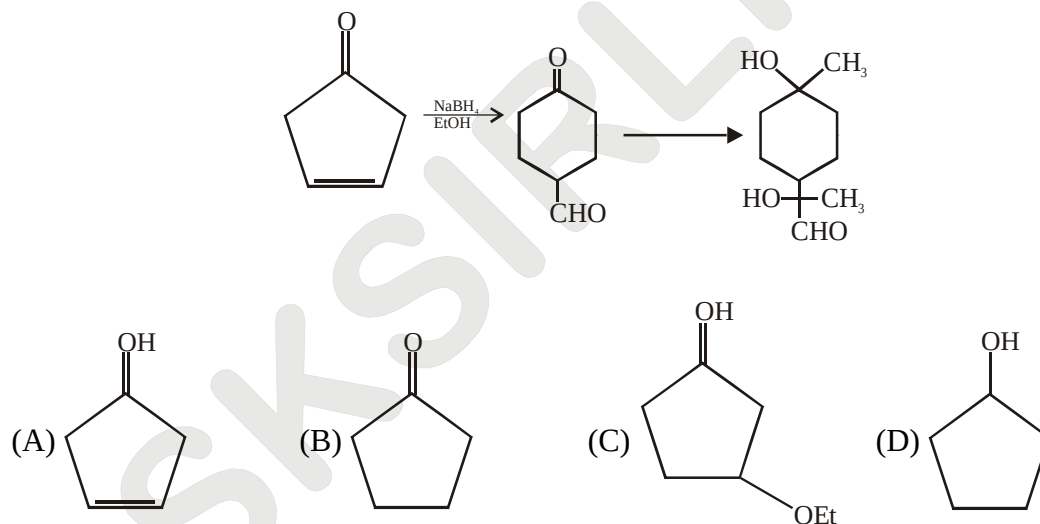
7. The major product Y in the following reaction is

[JEE-MAIN 2019]

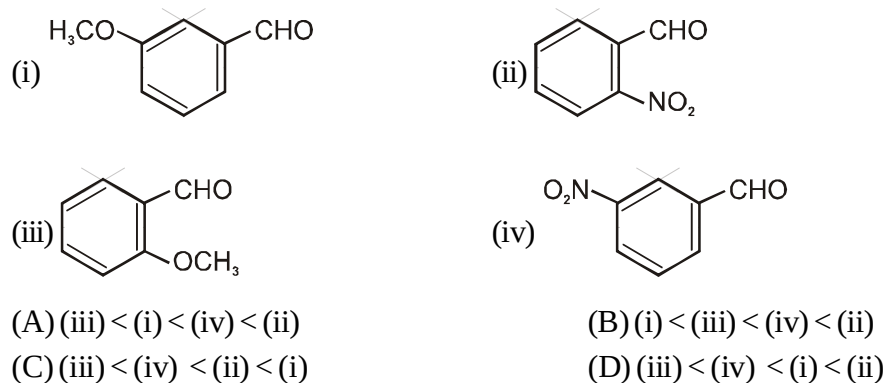


8. The major product of the following reaction is

[JEE-MAIN 2019]

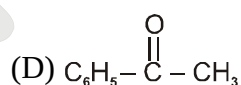
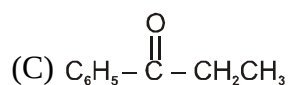
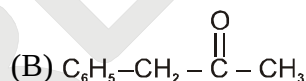
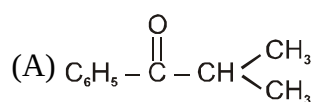
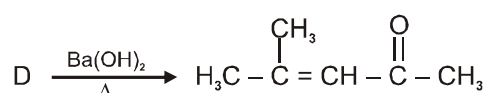
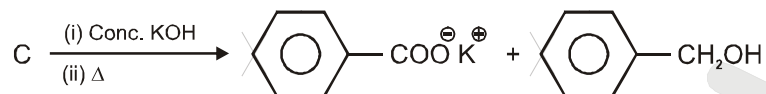
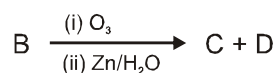
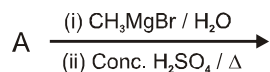


9. The increasing order of the following compounds towards HCN addition is : [JEE-MAIN 2020]

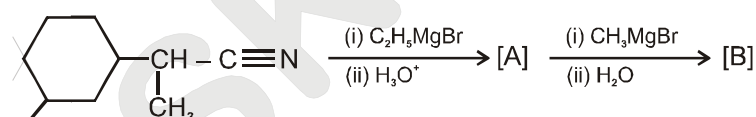


10. The increasing order of the reactivity of the following compounds in nucleophile addition reaction is:
 Propanal, Benzaldehyde, Propanone, Butanone [JEE-MAIN 2020]
 (A) Benzaldehyde > Butanone < Propanone < Propanal
 (B) Butanone < Propanone < Benzaldehyde < Propanal
 (C) Benzaldehyde < Propanal < Propanone < Butanone
 (D) Propanal < Propanone < Butanone < Benzaldehyde

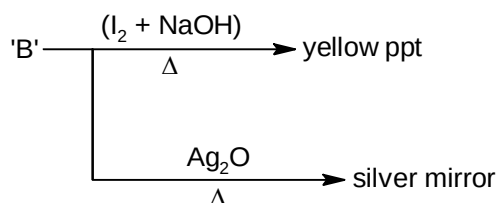
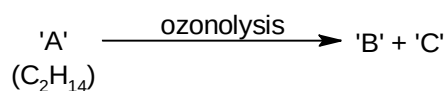
11. The compound A in the following reactions: [JEE-MAIN 2020]

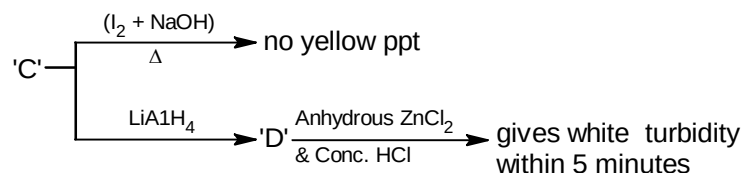


12. The number of chiral centres present in [B] is [JEE-MAIN 2020]

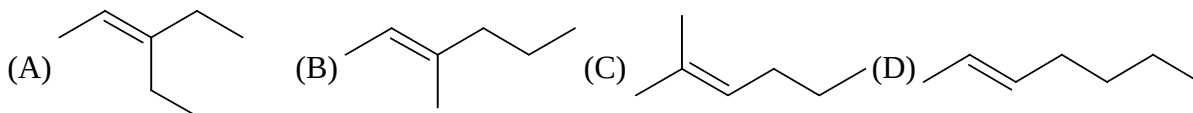


13. Consider the following reactions: [JEE-MAIN 2020]





'A' is



14. The correct match between Item – I (starting material) and Item-II (reagent) for the preparation of benzaldehyde is: [JEE-MAIN 2020]

Item-I	Item-II
(I) Benzene	(P) HCl and SnCl_2 , H_3O^+
(II) Benzonitrile	(Q) H_2 , Pd- BaSO_4 , S and quinoline
(III) Benzoyl Chloride	(R) Co, HCl and AlCl_3
(A) (I)-(R), (II)-(Q) and (III)-(P)	(B) (I)-(Q), (II)-(R) and (III)-(P)
(C) (I)-(P), (II)-(Q) and (III)-(R)	(D) (I)-(R), (II)-(P) and (III)-(Q)

[JEE-ADVANCED]

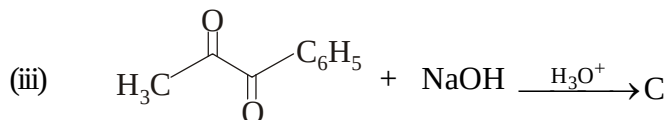
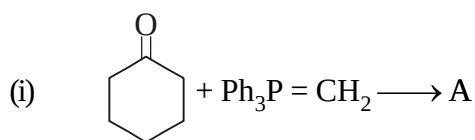
- Which reagent both acetaldehyde and acetone react easily is [JEE 1981]
 (A) Tollen's reagent (B) Schiff's reagent (C) Grignard reagent (D) Fehling reagent
- Write the structural formula of the main organic product formed when methanal reacts with ammonia. [JEE 1981]
- Outline the reaction sequence for the conversion of methanal to ethanal (the number of steps should not be more than three) [JEE 1981]
- A compound that gives a positive iodoform test is [JEE 1982]
 (A) 1-pentanol (B) 3-pentanone (C) 2-pentanone (D) pentanal
- An alkene 'A' on ozonolysis yields acetone and an aldehyde. The aldehyde is easily oxidised to an acid 'B'. When 'B' is treated with bromine in presence of phosphorus yields a compound 'C' which on hydrolysis gives a hydroxyl acid 'D'. This acid can also be obtained from acetone by the reaction with hydrogen cyanide followed by hydrolysis. Identify the compounds A,B,C and D. [JEE 1982]
- When acetaldehyde is treated with Fehling's solution, it gives a precipitate of [JEE 1983]
 (A) Cu (B) CuO (C) Cu_2O (D) $\text{Cu} + \text{Cu}_2\text{O} + \text{CuO}$
- The Cannizzaro's reaction is not given by [JEE 1983]
 (A) trimethyl acetaldehyde (B) acetaldehyde
 (C) benzaldehyde (D) formaldehyde

8. Base catalysed aldol condensation occurs with [JEE 1984]
(A) propionaldehyde (B) benzaldehyde
(C) 2-methyl propionaldehyde (D) 2,2-dimethyl propionaldehyde
9. Which of the following compounds will give a yellow precipitate with iodine and alkali? [JEE 1984]
(A) 2-hydroxy propane (B) Acetophenone
(C) Methyl acetate (D) Acetamide
10. Which of the following compounds will react with ethanolic KCN? [JEE 1984]
(A) Ethyl chloride (B) Acetyl chloride
(C) Chlorobenzene (D) Benzaldehyde
11. Show with balanced equation, what happens, when the following are mixed: [JEE 1984]
"Chloral is heated with aqueous hydroxide"
12. Which of the following will not react with alkali and iodine is [JEE 1985]
(A) acetone (B) ethanol (C) diethyl ketone (D) isopropyl alcohol
13. Arrange the following in order of their increasing reactivity towards HCN : [JEE 1985]
 CH_3CHO , CH_3COCH_3 , HCHO , $\text{C}_2\text{H}_5\text{COCH}_3$
14. Write down product of the following reaction. [JEE 1985]
$$\text{Propanal} \xrightarrow[\text{heat}]{\text{NaOH}}$$
15. Give reason in one or two sentences for the following [JEE 1986]
"Hydrazones of aldehydes and ketones are not prepared in highly acidic medium"
16. How may the following transformation be carried out (in not more than six steps) "Benzaldehyde to cyanobenzene"? [JEE 1986]
17. Complete the following with appropriate structures [JEE 1986]
$$? \xrightarrow{\text{NaOH}} \text{C}_6\text{H}_5\text{CH=CHCHO}$$
18. Which of the following compounds is oxidised to prepare methyl ethyl ketone? [JEE 1987]
(A) 2-propanol (B) 1-butanol (C) 2-butanol (D) t-butyl alcohol
19. Answer the followings with suitable equations wherever necessary [JEE 1987]
(i) Suggest a reagent to distinguish acetaldehyde from acetone.
(ii) What happens when excess chlorine is passed through boiling toluene in the presence of sunlight?

20. Which of the following is/are an example of aldol condensation? [JEE 1989]
- (A) $2\text{CH}_3\text{CHO} \xrightarrow{\text{dil. NaOH}} \text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CHO}$
- (B) $2\text{CH}_3\text{COCH}_3 \xrightarrow{\text{dil. NaOH}} \text{H}_3\text{C}-\overset{\text{OH}}{\underset{\text{CH}_3}{\text{C}}}-\text{CH}_2\text{COCH}_3$
- (C) $2\text{HCHO} \xrightarrow{\text{dil. NaOH}} \text{CH}_3\text{OH} + \text{HCOONa}$
- (D) $\text{C}_6\text{H}_5\text{CHO} + \text{HCHO} \xrightarrow{\text{dil. NaOH}} \text{C}_6\text{H}_5\text{CH}_2\text{OH} + \text{HCOONa}$
21. A ketone A, which undergoes haloform reaction, gives compound B on reduction. B on heating with sulphuric acid gives compound C, which forms mono-ozonide D. D on hydrolysis in presence of zinc dust gives only acetaldehyde. Identify A, B and C. Write down the reactions involved. [JEE 1989]
22. The formation of cyanohydrin from a ketone is an example of: [IIT 1990]
- (A) Electrophilic addition (B) Nucleophilic addition
(C) Nucleophilic substitution (D) Electrophilic substitution
23. The enolic form of acetone contains: [IIT 1990]
- (A) 9 sigma bonds, 1 pi bond and 2 lone pairs
(B) 8 sigma bonds, 2 pi bonds and 2 lone pairs
(C) 10 sigma bonds, 1 pi bond and 1 lone pair
(D) 9 sigma bonds, 2 pi bonds and 1 lone pair
24. m-chlorobenzaldehyde on reaction with conc. KOH at room temperature gives: [IIT 1991]
- (A) Potassium m-chlorobenzoate and m-hydroxybenzaldehyde
(B) m-hydroxybenzaldehyde and m-chlorobenzyl alcohol
(C) m-chlorobenzyl and m-hydroxybenzyl alcohol
(D) Potassium m-chlorobenzoate and m-chlorobenzyl alcohol
25. Give reason in one or two sentences: [JEE 1991]
- "Iodoform is obtained by the reaction of acetone with hypoiodite but not with iodide."
26. Hydrogenation of benzoyl chloride in the presence of Pd and BaSO_4 gives: [IIT 1992]
- (A) Benzyl alcohol (B) Benzaldehyde (C) Benzoic acid (D) Phenol
27. Arrange the following in increasing order of expected enol content. [JEE 1992]
- $\text{CH}_3\text{COCH}_2\text{CHO}$, CH_3COCH_3 , CH_3CHO , $\text{CH}_3\text{COCH}_2\text{COCH}_3$
28. Write the structure of the major organic product expected from the following reaction. [JEE 1992]
- $\text{MeO}-\text{C}_6\text{H}_4-\text{CHO} + \text{HCHO} \xrightarrow{\text{KOH}}$
29. An organic compound $\text{C}_3\text{H}_6\text{O}$ does not give a precipitate with 2,4-Dinitrophenyl hydrazine reagent and does not react with metallic sodium. It could be: [IIT 1993]
- (A) $\text{CH}_3\text{CH}_2\text{CHO}$ (B) CH_3COCH_3
(C) $\text{CH}_2=\text{CH}-\text{CH}_2\text{OH}$ (D) $\text{CH}_2=\text{CH}-\text{O}-\text{CH}_3$

30. Under Wolff Kishner reduction conditions, the conversions which may be brought about is?
 (A) Benzaldehyde into Benzyl alcohol (B) Cyclohexanol into Cyclohexane
 (C) Cyclohexanone into Cyclohexanol (D) Benzophenone into Diphenylmethane [IIT 1995]
31. In the reaction, P is [IIT 1995]
- $$\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{CO} \xrightarrow{\text{SeO}_2} \text{P} + \text{Se} + \text{H}_2\text{O} \\ \diagup \\ \text{CH}_3 \end{array}$$
- (A) CH_3COCHO (B) $\text{CH}_3\text{COOCH}_3$ (C) $\text{CH}_3\text{COCH}_2\text{OH}$ (D) None
32. $\text{C}_6\text{H}_5\text{-CHO} + \text{CH}_3\text{-COOC}_2\text{H}_5 \xrightarrow[\text{C}_2\text{H}_5\text{OH and heat}]{\text{NaOC}_2\text{H}_5 \text{ in absolute}} (\text{D}).$ [IIT 1995]
33. In the Cannizzaro reaction given below, $2\text{Ph-CHO} \xrightarrow{\text{OH}^-} \text{Ph-CH}_2\text{OH} + \text{PhCO}_2^-$ the slowest step is:
 (A) the attack of OH^- at the carbonyl group
 (B) the transfer of hydride to the carbonyl group
 (C) the abstraction of proton from the carboxylic acid
 (D) the deprotonation of $\text{Ph-CH}_2\text{OH}$ [IIT 1996]
34. $\text{CH}_3\text{CH}_2\text{-C(=O)-H} \xrightarrow[\text{(ii) LiAlH}_4]{\text{(i) KCN/H}_2\text{SO}_4} \text{D.}$ [IIT 1996]
35. Suggest appropriate structures for the missing compounds. (the number of carbon atoms remains the same throughout the reaction) [JEE 1996]
- $$\begin{array}{c} \text{CH}_3 \\ \diagup \\ \text{Cyclopentene} \\ \diagdown \\ \text{CH}_3 \end{array} \xrightarrow{\text{dil. KMnO}_4} \text{A} \xrightarrow{\text{HIO}_4} \text{B} \xrightarrow[\Delta]{\text{HO}^-} \text{C}$$
36. Among the given compounds, the most susceptible to nucleophilic attack at the carbonyl group is
 (A) MeCOCl (B) MeCHO (C) MeCOOMe (D) MeCOOCOMe [IIT 1997]
37. In a Cannizzaro reaction the intermediate which is the best hydride donor is: [IIT 1997]
- (A) $\text{C}_6\text{H}_5\text{-}\overset{\text{H}}{\underset{\text{OH}}{\text{C}}}\text{-O}^-$ (B) $\text{C}_6\text{H}_5\text{-}\overset{\text{H}}{\underset{\text{O}^-}{\text{C}}}\text{-O}^-$
- (C) $\text{O}_2\text{N-C}_6\text{H}_4\text{-}\overset{\text{H}}{\underset{\text{O}^-}{\text{C}}}\text{-O}^-$ (D) $\text{CH}_3\text{O-C}_6\text{H}_4\text{-}\overset{\text{H}}{\underset{\text{O}^-}{\text{C}}}\text{-O}^-$
38. $\text{CH}_3\text{CHO} + \text{H}_2\text{NOH} \longrightarrow \text{CH}_3\text{-CH=N-OH}$. The above reaction occurs at:
 (A) $\text{pH} = 1$ (B) $\text{pH} = 4.5$ (C) Any value of pH (D) $\text{pH} = 12$ [IIT 1997]

39. Complete the following, giving the structures of the principal organic products [JEE 1997]



40. Acetophenone on reaction with hydroxylamine-hydrochloride can produce two isomeric oximes. Write structures of the oximes. [IIT 1997]

41. Among the following compounds, which will react acetone to give a product containing $>\text{C}=\text{N}-$

- (A) $\text{C}_6\text{H}_5\text{NH}_2$ (B) $(\text{CH}_3)_3\text{N}$ (C) $\text{C}_6\text{H}_5\text{NHC}_6\text{H}_5$ (D) $\text{C}_6\text{H}_5\text{NHNH}_2$ [IIT 1998]

42. The product obtained via oxymercuration ($\text{HgSO}_4-\text{H}_2\text{SO}_4$) of 1-butyne would be

- (A) $\text{CH}_3\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$ (B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$
(C) $\text{CH}_3\text{CH}_2\text{CHO} + \text{HCHO}$ (D) $\text{CH}_3\text{CH}_2\text{COOH} + \text{HCOOH}$ [IIT 1998]

43. Which of the following will undergo aldol condensation: [IIT 1998]

- (A) Acetaldehyde (B) Propanaldehyde
(C) Benzaldehyde (D) Trideutero acetaldehyde

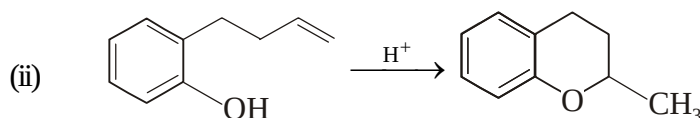
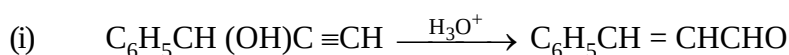
44. Which of the following will react with water: [IIT 1998]

- (A) CHCl_3 (B) Cl_3CCHO (C) CCl_4 (D) $\text{ClCH}_2\text{CH}_2\text{Cl}$

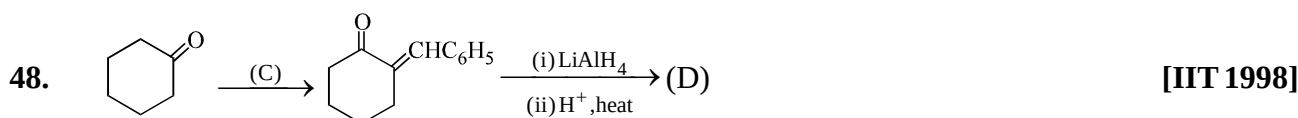
45. A new carbon-carbon bond formation is possible in [IIT 1998]

- (A) Cannizzaro reaction (B) Friedel-Crafts alkylation
(C) Clemmensen reduction (D) Reimer-Tiemann reaction

46. Write the intermediate steps for each of the following reactions. [JEE 1998]



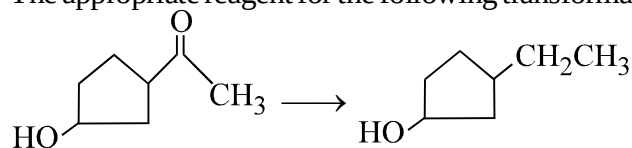
47. An aldehyde (A) ($\text{C}_{11}\text{H}_8\text{O}$), which does not undergo self aldol condensation, gives benzaldehyde and two mole of (B) on ozonolysis. Compound (B), on oxidation with silver ion, gives oxalic acid. Identify the compounds (A) and (B). [IIT 1998]



49. Which of the following has the most acidic hydrogen: [IIT 2000]

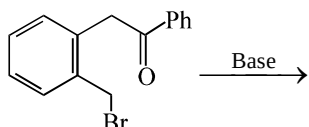
- (A) 3-hexanone (B) 2,4-hexanedione (C) 2,5-hexanedione (D) 2,3-hexanedione

50. The appropriate reagent for the following transformation: [IIT 2000]

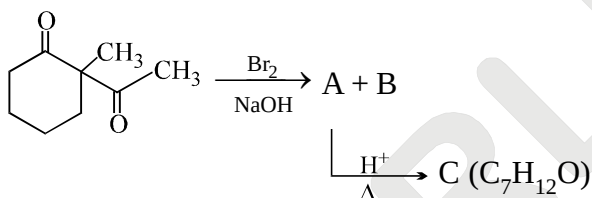


- (A) Zn(Hg), HCl (B) $\text{NH}_2\text{NH}_2, \text{OH}^-$ (C) H_2/Ni (D) NaBH_4

51. What would be the major product in each of the following reaction? [IIT 2000]



52. Identify (A), (B) and (C), and give their structures. [IIT 2000]



53. A mixture of benzaldehyde and formaldehyde on heating with aqueous NaOH solution gives:

- (A) benzyl alcohol and sodium formate (B) sodium benzoate and methyl alcohol
(C) sodium benzoate and sodium formate (D) benzyl alcohol and methyl alcohol [IIT 2001]

54. 1-propanol & 2-propanol can be best distinguished by: [JEE 2001]

- (A) Oxidation with alkaline KMnO_4 followed by reaction with Fehling solution
(B) Oxidation with acedid dichromate followed by reaction with Fehling solution
(C) Oxidation by heating with copper followed by reaction with Fehling solution
(D) Oxidation with concentrated H_2SO_4 followed by reaction with Fehling solution

55. An alkene (A) $\text{C}_{16}\text{H}_{16}$ on ozonolysis gives only one product (B) $\text{C}_8\text{H}_8\text{O}$. Compound (B) on reaction with NaOH / I_2 yields sodium benzoate. Compound (B) reacts with $\text{KOH} / \text{NH}_2\text{NH}_2$ yielding a hydrocarbon (C) C_8H_{10} . Write the structures of compound (B) and (C). Based on this information two isomeric structures can be proposed for alkene (A). Write their structures and identify the isomer which on catalytic hydrogenation ($\text{H}_2 / \text{Pd-C}$) gives a racemic mixture. [JEE 2001]

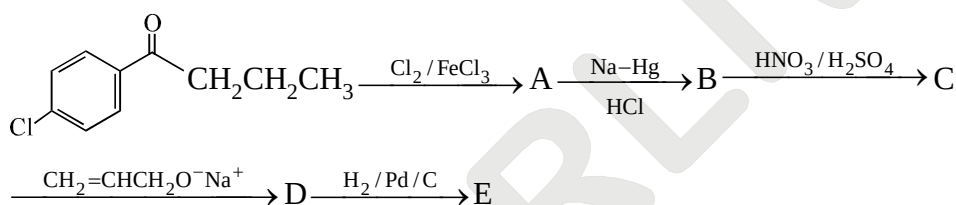
56. A compound $\text{C}_9\text{H}_7\text{O}_2\text{Cl}$ exists predominantly in enol form (A) and also in keto form (B). On oxidation with KMnO_4 it gives m-chlorobenzoic acid as one of the products. Identify the compounds (A) and (B). [JEE 2001]

57. Compound A (molecular formula $\text{C}_3\text{H}_8\text{O}$) is treated with acidified potassium dichromate to form a product B (molecular formula $\text{C}_3\text{H}_6\text{O}$). B forms a shining silver mirror on warming with ammonical silver nitrate. B when treated with an aqueous solution of $\text{H}_2\text{NCONHNH}_2$, HCl and sodium acetate gives a product C. Identify the structure of C.

- (A) $\text{CH}_3\text{CH}_2\text{CH}=\text{NNHCONH}_2$ (B) $\text{CH}_3-\overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}=\text{NNHCONH}_2$
- (C) $\text{CH}_3-\overset{\text{CH}_3}{\text{C}}=\text{NCONHNH}_2$ (D) $\text{CH}_3\text{CH}_2\text{CH}=\text{NCONHNH}_2$ [IIT 2002]

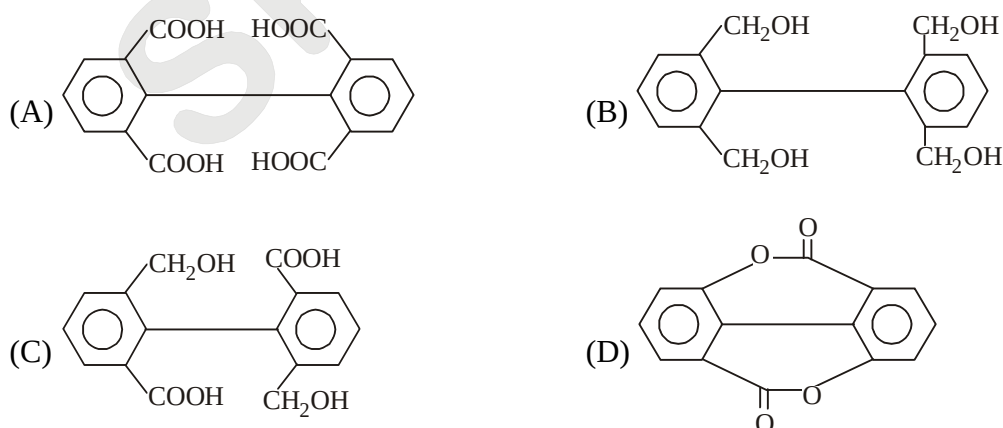
58. Five isomeric para-disubstituted aromatic compounds A to E with molecular formula $\text{C}_8\text{H}_8\text{O}_2$ were given for identification. Based on the following observations, give structure of the compounds.
- Both A and B form a silver mirror with Tollen's reagent; also, B gives a positive test with FeCl_3 solution
 - C gives positive iodoform test.
 - D is readily extracted in aqueous NaHCO_3 solution.
 - E on acid hydrolysis gives 1,4-dihydroxybenzene.
- [IIT 2002]

59. Write structures of the products A, B, C, D and E in the following scheme. [IIT 2002]



60. [IIT 2003]

any one of the products formed is :

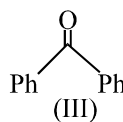
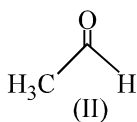
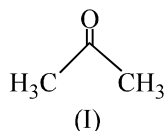


61. Products formed by P & Q can be differentiated by: [IIT 2003]

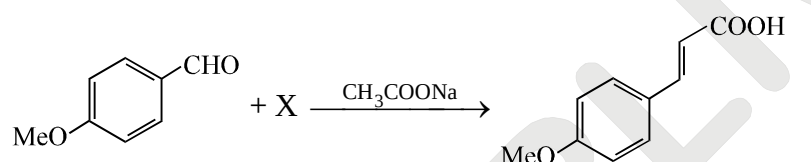
- (A) 2, 4 DNP (B) Lucas reagent (ZnCl_2) conc. HCl
- (C) NaHSO_3 (D) Fehlings solution

62. $C_6H_{12} \xrightarrow{HCl} C_6H_{13}Cl + (C)$
 (A) (B)
 $\xrightarrow{Alc. KOH} (D) \xrightarrow{ozonolysis} (E)$
 (A) $\xrightarrow{ozonolysis} (F) + (G) \xrightarrow{NaOH} HCOONa + 1^\circ \text{ alcohol}$
 (D) is isomer of A. E gives negative test with Fehling solution but gives iodoform test. F and G gives Tollen's test but do not give iodoform test. Identify A to G. [IIT 2003]

63. The order of reactivity of phenyl Magnesium Bromide with the following compounds is [IIT 2004]



- (A) II > III > I (B) I > III > II (C) II > I > III (D) All react with the same rate

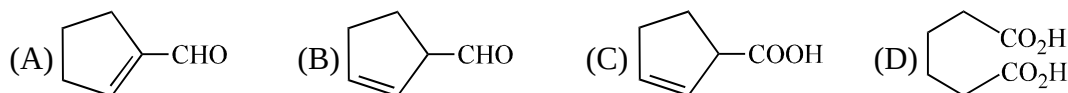
64.  [IIT 2005]

What is X?

- (A) CH_3COOH (B) $BrCH_2COOH$ (C) $(CH_3CO)_2O$ (D) $CHO-COOH$
65. The smallest ketone and its next homologue are reacted with NH_2OH to form oxime. Which statement is correct about this : [JEE 2006]

- (A) Two different oximes are formed (B) Three different oximes are formed
 (C) Two oximes are optically active (D) All oximes are optically active

66. Cyclohexene on ozonolysis followed by reaction with zinc dust and water gives compound E. Compound E on further treatment with aqueous KOH yields compound F. Compound F is [JEE 2007]



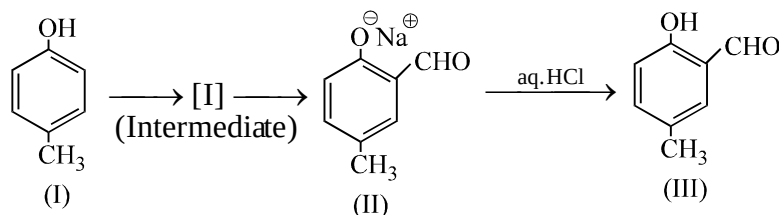
67. **Statement-1 :** Glucose gives a reddish-brown precipitate with Fehling's solution.
 because

Statement-2 : Reaction of glucose with Fehling's solution gives CuO and gluconic acid.

- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True. [JEE 2007]

Paragraph for Question Nos. 68 to 70 (3 questions)

Riemer-Tiemann reaction introduces an aldehyde group, on to the aromatic ring of phenol, ortho to the hydroxyl group. This reaction involves electrophilic aromatic substitution. This is a general method for the synthesis of substituted salicylaldehydes as depicted below.



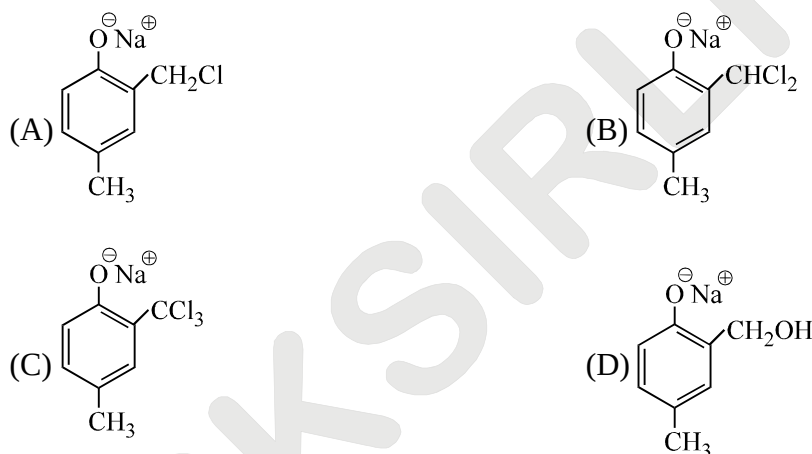
68. Which one of the following reagents is used in the above reaction? [JEE 2007]

- (A) aq. NaOH + CH₃Cl (B) aq. NaOH + CH₂Cl₂
 (C) aq. NaOH + CHCl₃ (D) aq. NaOH + CCl₄

69. The electrophile in this reaction is [JEE 2007]

- (A) :CHCl (B) $^+\text{CHCl}_2$ (C) :CCl_2 (D) $\cdot\text{CCl}_3$

70. The structure of the intermediate I is [JEE 2007]



71. Match the compounds/ion in column I with their properties/ reaction in Column II. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the OMR. [JEE 2007]

Column I

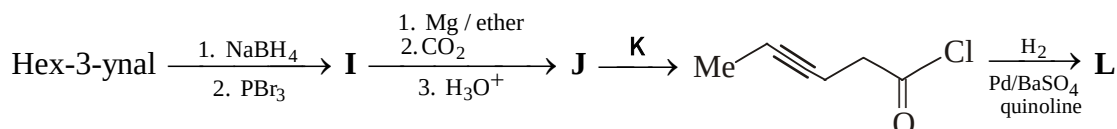
- (A) C₆H₅CHO
 (B) CH₃C≡CH
 (C) CN⁻
 (D) I⁻

Column II

- (P) gives precipitate with 2,4-dinitrophenylhydrazine
 (Q) gives precipitate with AgNO₃
 (R) is a nucleophile
 (S) is involved in cyanohydrin formation

Paragraph for Question No. 72 to 74

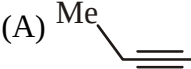
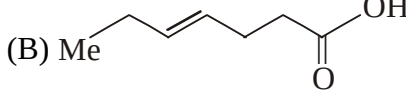
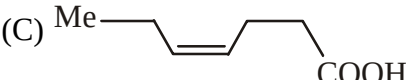
In the following reaction sequence, products I, J and L are formed. K represents a reagent.



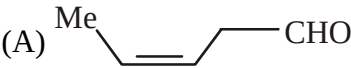
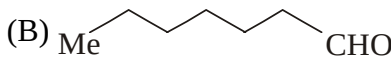
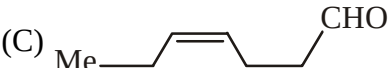
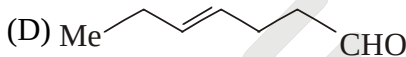
72. The structure of the product I is [JEE 2008]



73. The structures of compounds **J** and **K**, respectively are [JEE 2008]

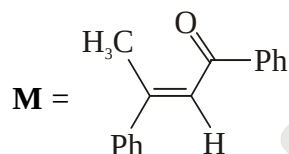
- (A)  and SOCl_2 (B)  and SO_2Cl_2
- (C)  and SOCl_2 (D)  and $\text{CH}_3\text{SO}_2\text{Cl}$

74. The structure of product **L** is [JEE 2008]

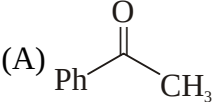
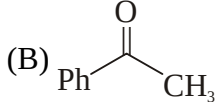
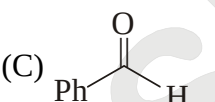
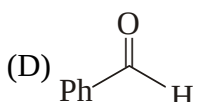
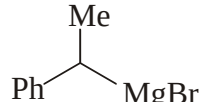
- (A)  (B) 
- (C)  (D) 

Paragraph for Question No. 75 to 77

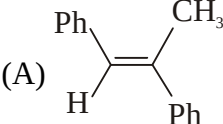
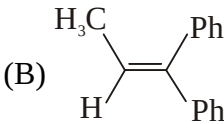
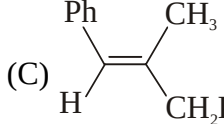
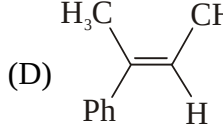
A tertiary alcohol **H** upon acid catalysed dehydration gives a product **I**. Ozonolysis of **I** leads to compounds **J** and **K**. Compound **J** upon reaction with KOH gives benzyl alcohol and a compound **L**, whereas **K** on reaction with KOH gives only **M**.



75. Compound **H** is formed by the reaction of [JEE 2008]

- (A)  + PhMgBr (B)  + PhCH_2MgBr
- (C)  + PhCH_2MgBr (D)  + 

76. The structure of compound **I** is [JEE 2008]

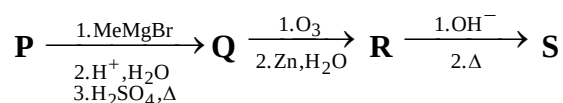
- (A)  (B) 
- (C)  (D) 

77. The structures of compounds **J**, **K** and **L**, respectively, are [JEE 2008]

- (A) PhCOCH_3 , $\text{PhCH}_2\text{COCH}_3$ and $\text{PhCH}_2\text{COO}^-\text{K}^+$
 (B) PhCHO , PhCH_2CHO and PhCOO^-K^+
 (C) PhCOCH_3 , PhCH_2CHO and $\text{CH}_3\text{COO}^-\text{K}^+$
 (D) PhCHO , PhCOCH_3 and PhCOO^-K^+

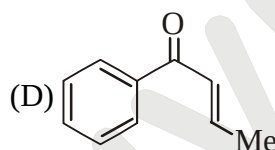
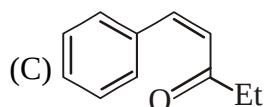
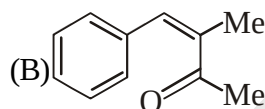
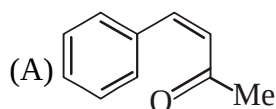
Paragraph for Question Nos. 78 to 80

A carbonyl compound **P**, which gives positive iodoform test, undergoes reaction with MeMgBr followed by dehydration to give an olefin **Q**. Ozonolysis of **Q** leads to a dicarbonyl compound **R**, which undergoes intramolecular aldol reaction to give predominantly **S**.



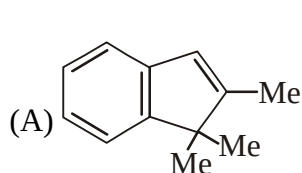
78. The structure of the carbonyl compound **P** is

[JEE 2009]

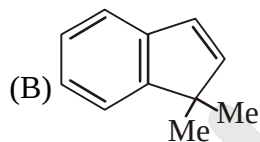
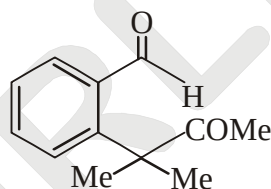


79. The structure of the products **Q** and **R**, respectively, are

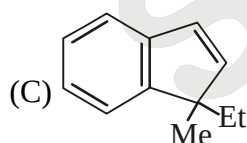
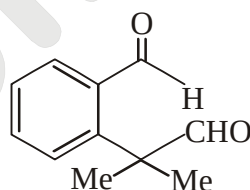
[JEE 2009]



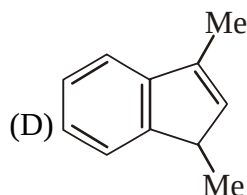
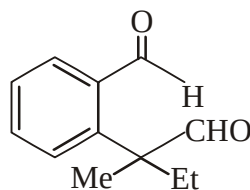
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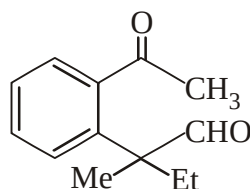
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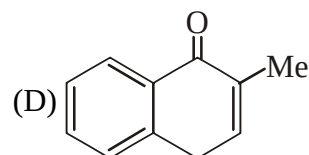
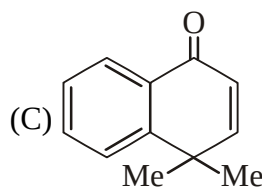
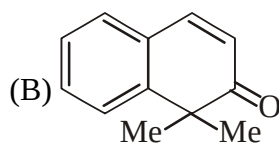
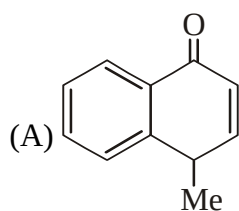


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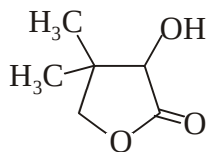
80. The structure of the product **S** is

[JEE 2009]



Paragraph for Questions 81 to 83

Two aliphatic aldehydes P and Q react in the presence of aqueous K_2CO_3 to give compound R, which upon treatment with HCN provides compound S. On acidification and heating, S gives the product shown below:



81. The compounds P and Q respectively are

[JEE 2010]

- (A) $\text{H}_3\text{C}-\text{CH}(\text{CH}_3)-\text{C}(=\text{O})\text{H}$ and $\text{H}_3\text{C}-\text{C}(=\text{O})\text{H}$
- (B) $\text{H}_3\text{C}-\text{CH}(\text{CH}_3)-\text{C}(=\text{O})\text{H}$ and $\text{H}-\text{C}(=\text{O})\text{H}$
- (C) $\text{H}_3\text{C}-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{C}(=\text{O})\text{H}$ and $\text{H}_3\text{C}-\text{C}(=\text{O})\text{H}$
- (D) $\text{H}_3\text{C}-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{C}(=\text{O})\text{H}$ and $\text{H}-\text{C}(=\text{O})\text{H}$

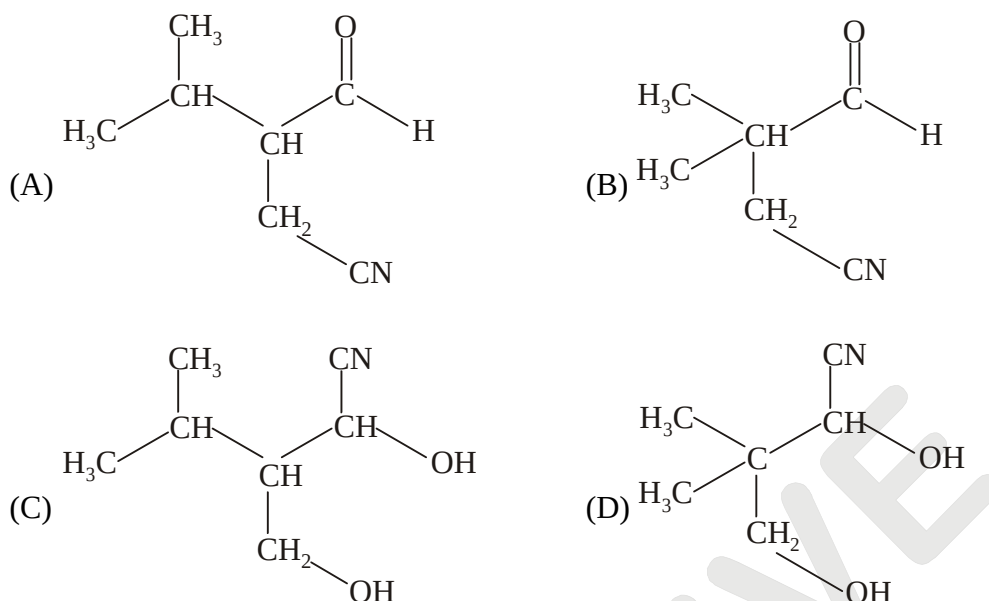
82. The compound R is

[JEE 2010]

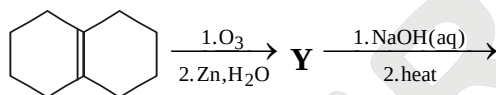
- (A) $\text{H}_3\text{C}-\text{C}(\text{H}_3)(\text{CH}_2\text{OH})-\text{C}(=\text{O})\text{H}$
- (B) $\text{H}_3\text{C}-\text{C}(\text{H}_3)(\text{CH}(\text{OH})\text{H})-\text{C}(=\text{O})\text{H}$
- (C) $\text{H}_3\text{C}-\text{CH}(\text{CH}_3)-\text{CH}(\text{CH}_2\text{OH})-\text{C}(=\text{O})\text{H}$
- (D) $\text{H}_3\text{C}-\text{CH}(\text{CH}_3)-\text{CH}(\text{CH}(\text{OH})\text{H})-\text{C}(=\text{O})\text{H}$

83. The compound S is

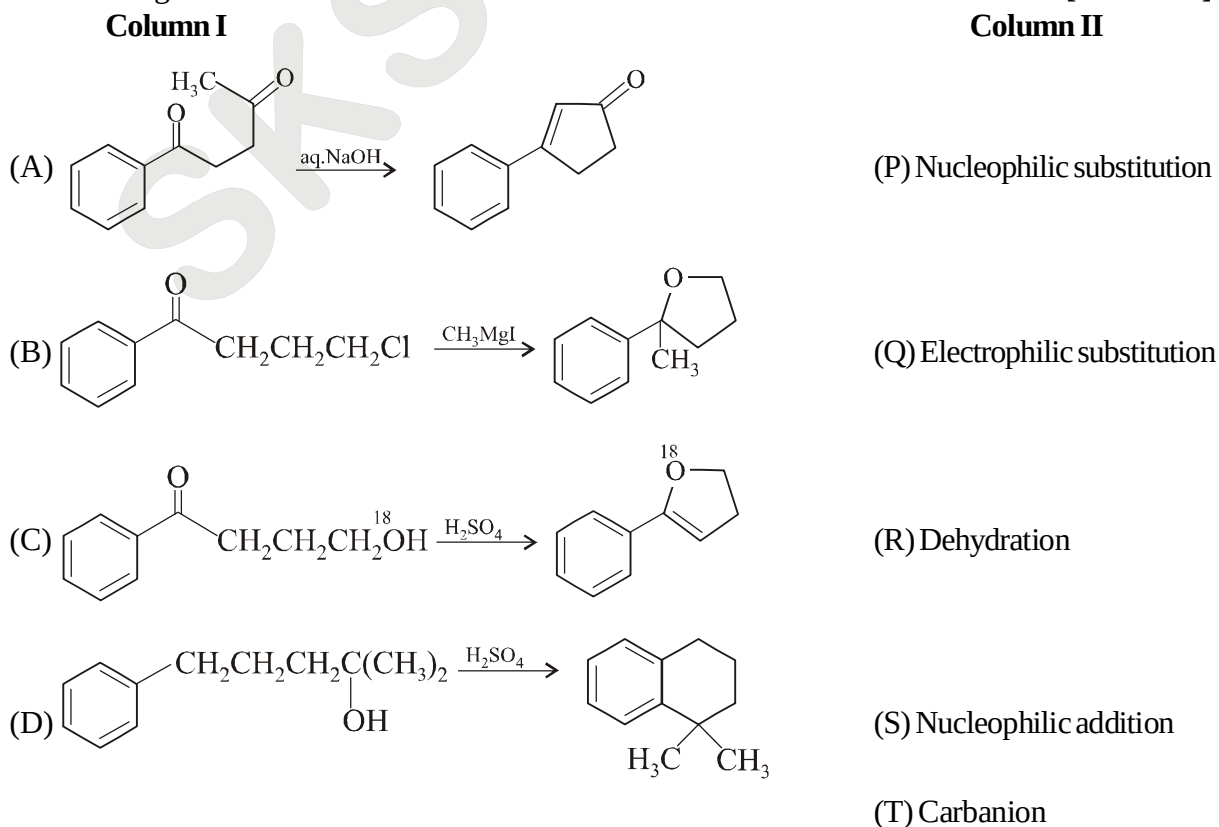
[JEE 2010]



84. In the scheme given below, the total number of intramolecular aldol condensation products formed from 'Y' is [IIT 2010]

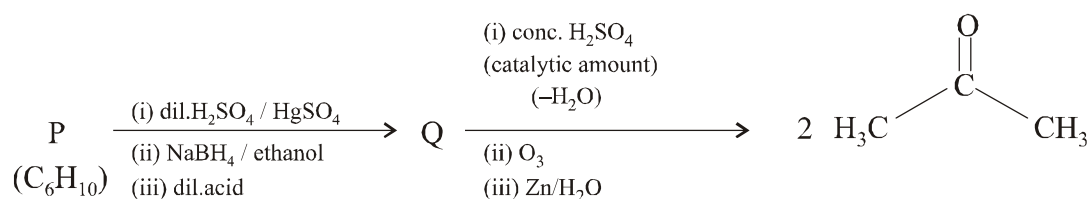


85. Match the reactions in column I with appropriate types of steps / reactive intermediate involved in these reactions as given in column II [JEE 2011]



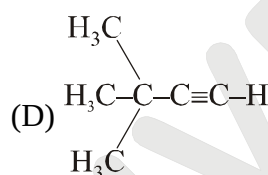
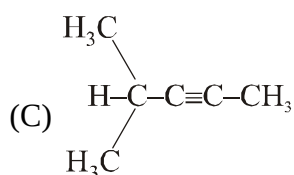
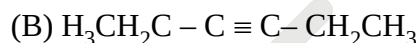
Paragraph for Questions 86 & 87

An acyclic hydrocarbon P, having molecular formula C_6H_{10} , gave acetone as the only organic product through the following sequence of reactions, in which Q is an intermediate organic compound.



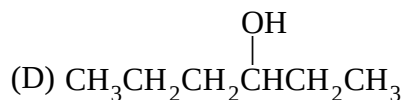
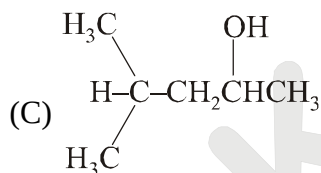
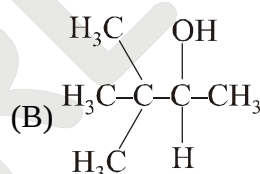
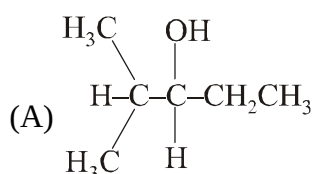
86. The structure of compound P is

[IIT 2011]

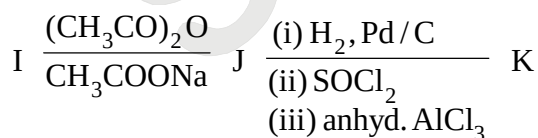


87. The structure of the compound Q is

[IIT 2011]

**Paragraph for Questions 88 to 89**

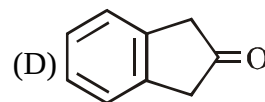
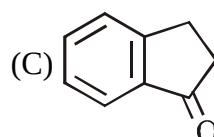
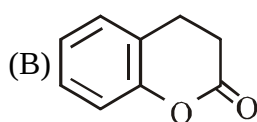
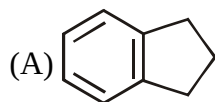
In the following reaction sequence, the compound J is an intermediate.



J ($C_9H_8O_2$) gives effervescence on treatment with $NaHCO_3$ and a positive Baeyer's test.

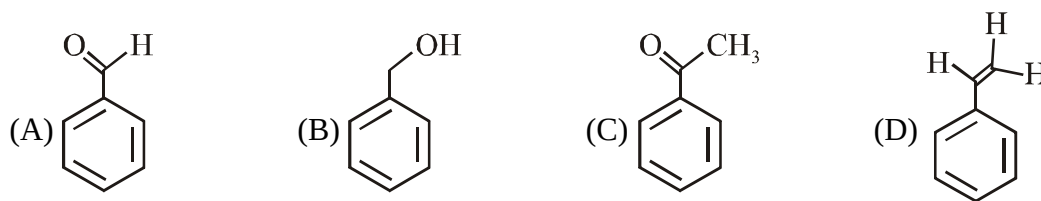
88. The compound 'K' is :

[JEE 2012]

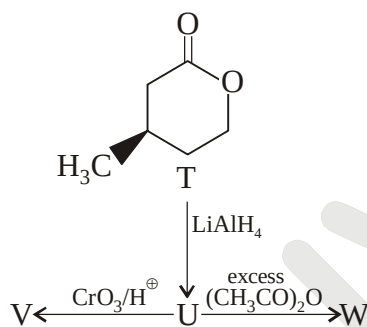


89. The compound 'T' is :

[JEE 2012]



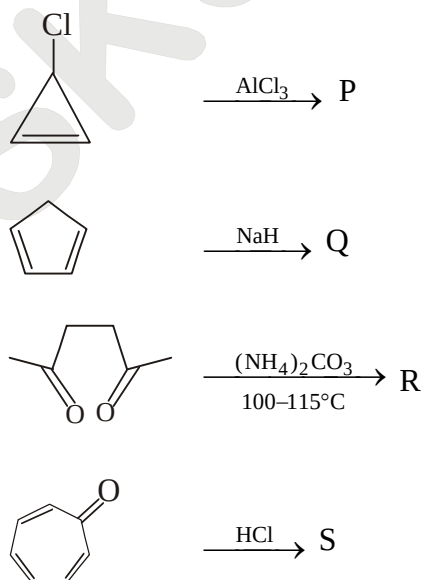
90. With reference to the scheme given, which of the given statement(s) about T, U, V and W is (are) correct? [JEE 2012]



- (A) T is soluble in hot aqueous NaOH
 (B) U is optically active
 (C) Molecular formula of W is $C_{10}H_{18}O_4$
 (D) V gives effervescence on treatment with aqueous $NaHCO_3$

91. Among P, Q, R and S, the aromatic compound(s) is / are

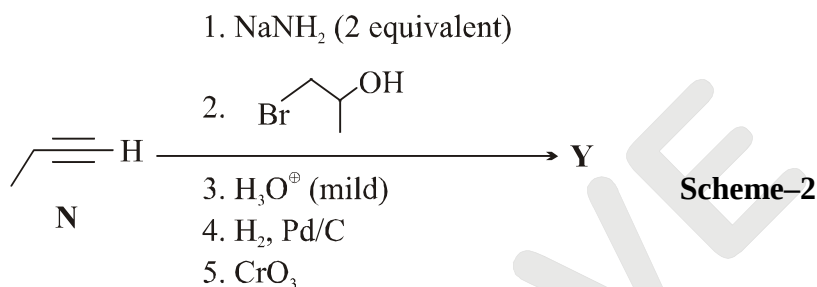
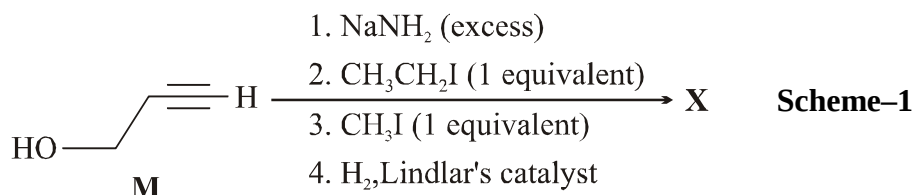
[JEE Advance 2013]



- (A) P (B) Q (C) R (D) S

Paragraph For Questions 92 to 93

Schemes 1 and 2 describe sequential transformation of alkynes M and N. Consider only the major products formed in each step for both the schemes.



92. The product X is

[JEE Advance 2014]



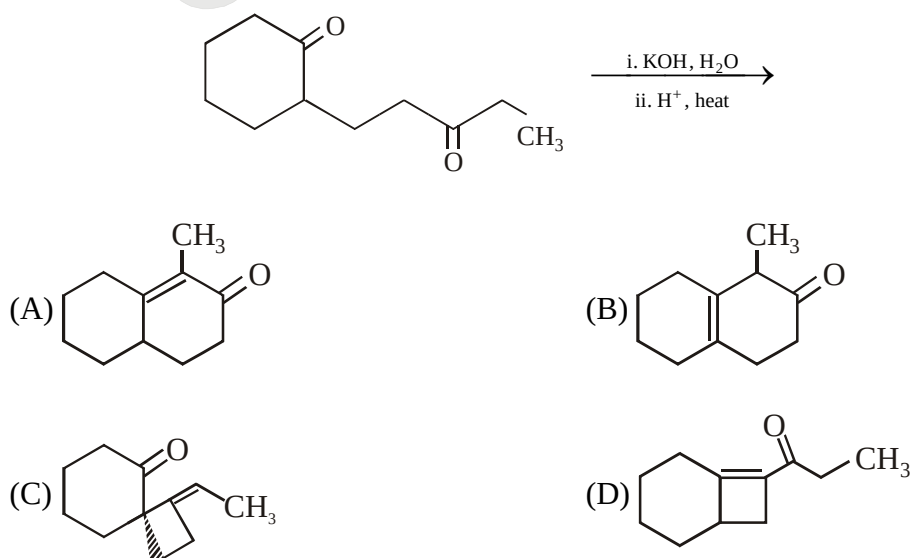
93. The correct statement with respect to product Y is

[JEE Advance 2014]

- (A) It gives a positive Tollens test and is functional isomer of X
 (B) It gives a positive Tollens test and is a geometrical isomer of X
 (C) It gives a positive iodoform test and is a functional isomer of X.
 (D) It gives a positive iodoform test and is a geometrical isomer of X.

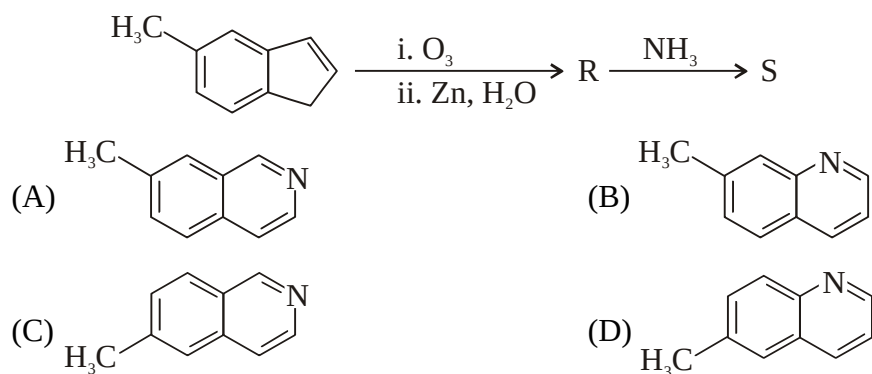
94. The major product of the following reaction is

[JEE Advance 2015]



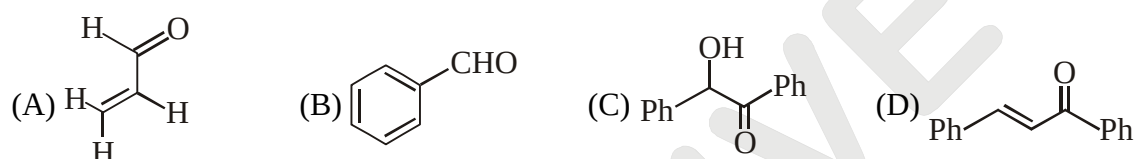
95. In the following reactions, the product S is

[JEE Advance 2015]



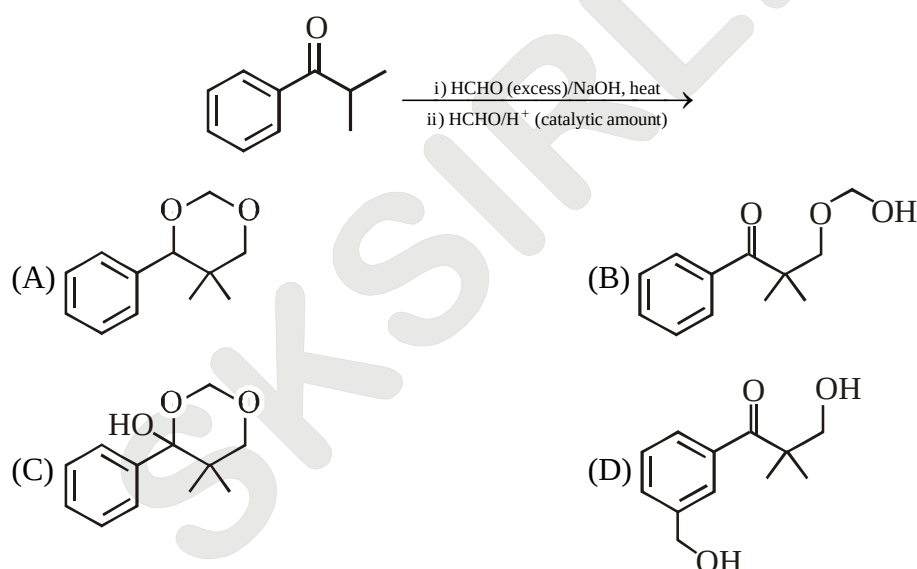
96. Positive Tollen's test is observed for

[JEE Advance 2016]



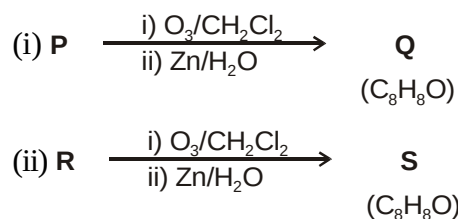
97. The major product of the following reaction sequence is

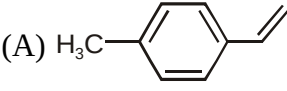
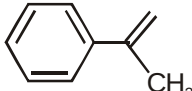
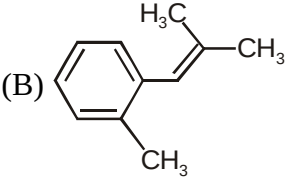
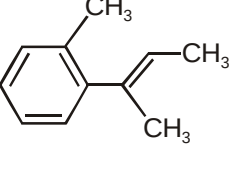
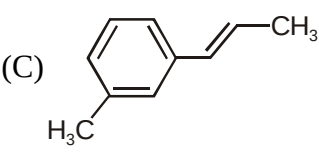
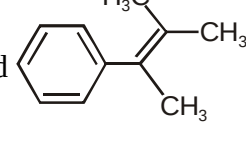
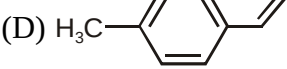
[JEE Advance 2016]



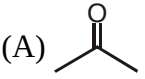
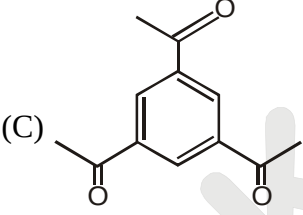
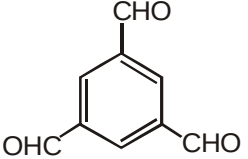
98. Compounds **P** and **R** upon ozonolysis produce **Q** and **S**, respectively. The molecular formula of **Q** and **S** is $\text{C}_8\text{H}_8\text{O}$. **Q** undergoes Cannizzaro reaction but not haloform reaction, whereas **S** undergoes haloform reaction but not Cannizzaro reaction.

[JEE ADVANCE 2017]

The option(s) with suitable combination of **P** and **R**, respectively, is(are)

- (A)  and 
- (B)  and 
- (C)  and 
- (D)  and 

99. The reaction(s) leading to the formation of 1,3,5-trimethylbenzene is(are) [JEE ADVANCE 2018]

- (A)  $\xrightarrow[\Delta]{\text{Conc. H}_2\text{SO}_4}$
- (B) $\text{Me}-\text{C}\equiv\text{H} \xrightarrow[873\text{ K}]{\text{heated iron tube}}$
- (C)  $\xrightarrow[\text{(3) sodalime, } \Delta]{\text{(1) Br}_2, \text{NaOH; (2) H}_3\text{O}^+}$
- (D)  $\xrightarrow{\text{Zn/Hg, HCl}}$

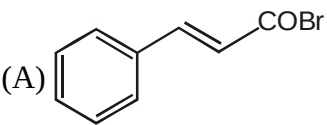
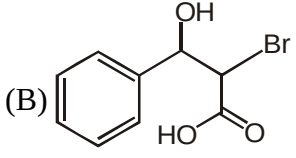
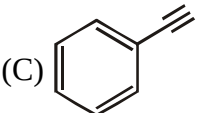
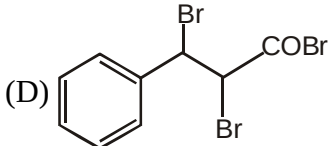
Paragraph for Question Nos. 100 and 101

[IIT-JEE Advance 2018]

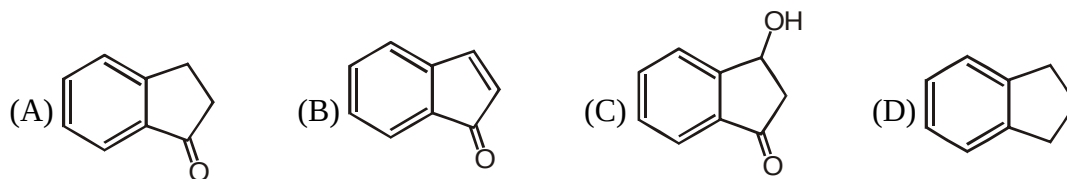
Treatment of benzene with CO/HCl in the presence of anhydrous $\text{AlCl}_3/\text{CuCl}$ followed by reaction with $\text{Ac}_2\text{O}/\text{NaOAc}$ gives compound **X** as the major product. Compound **X** upon reaction with $\text{Br}_2/\text{Na}_2\text{CO}_3$, followed by heating at 473 K with moist KOH furnishes **Y** as the major product. Reaction of **X** with $\text{H}_2/\text{Pd-C}$, followed by H_3PO_4 treatment gives **Z** as the major product.

(There are two questions based on Paragraph "X", the question given below is one of them)

100. The compound **Y** is

- (A) 
- (B) 
- (C) 
- (D) 

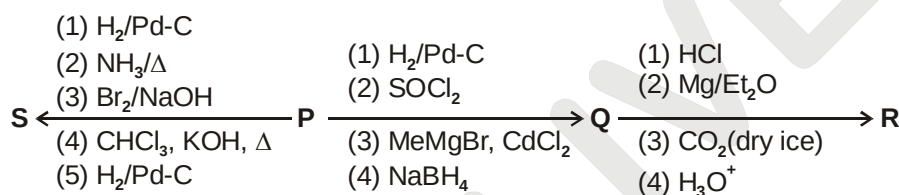
101. The compound **Z** is



Paragraph for Question Nos. 102 and 103

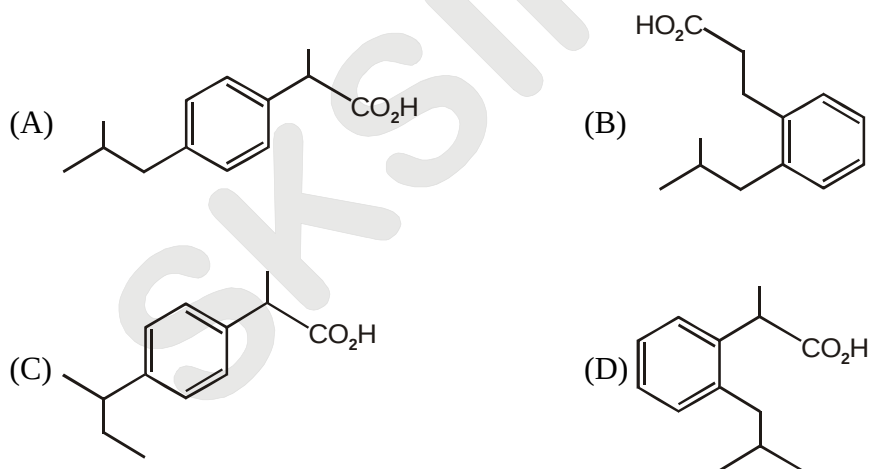
[IIT-JEE Advance 2018]

An organic acid **P** ($C_{11}H_{12}O_2$) can easily be oxidized to a dibasic acid which reacts with ethyleneglycol to produce a polymer dacron. Upon ozonolysis, **P** gives an aliphatic ketone as one of the products. **P** undergoes the following reaction sequences to furnish **R** via **Q**. The compound **P** also undergoes another set of reactions to produce **S**.

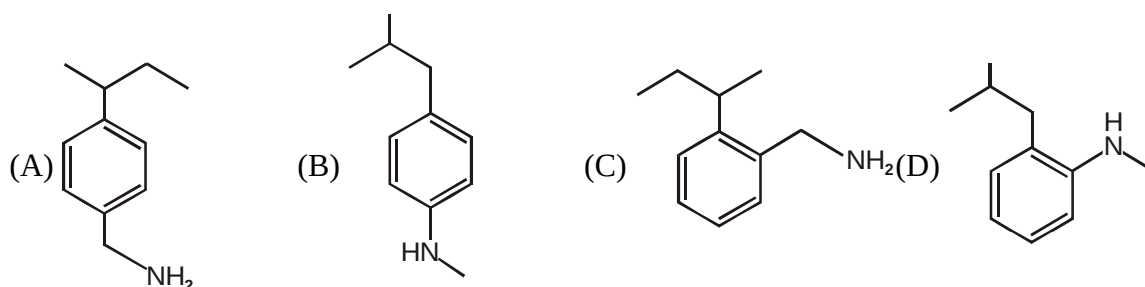


(There are two questions based on PARAGRAPH "A", the question given below is one of them)

102. The compound **R** is



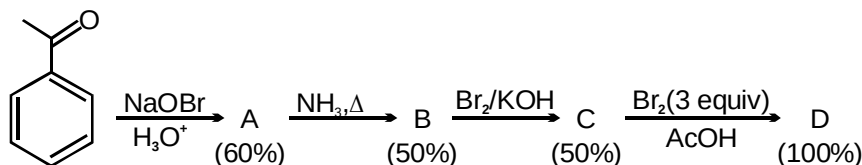
103. The compound **S** is



104. In the following reaction sequence, the amount of D(in g) formed from 10 moles of acetophenone is _____.

(Atomic weights in g mol^{-1} : H = 1, C = 12, N = 14, O = 16, Br = 80. The yield (%) corresponding to the product in each step is given in the parenthesis)

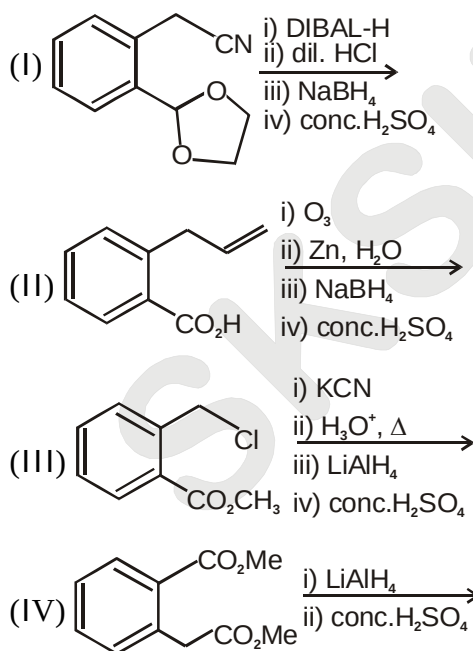
[JEE ADVANCE : 2018]



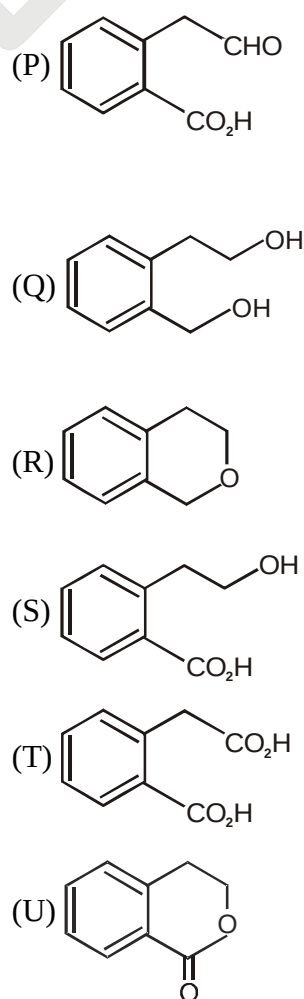
Answer the following by appropriately matching the lists based on the information given in the paragraph (Q.105 to Q.106) [IIT Adv. 2019]

List-I includes starting materials and reagents of selected chemical reactions. List-II gives structures of compound that may be formed as intermediate products and/or final products from the reactions of List-I.

List-I



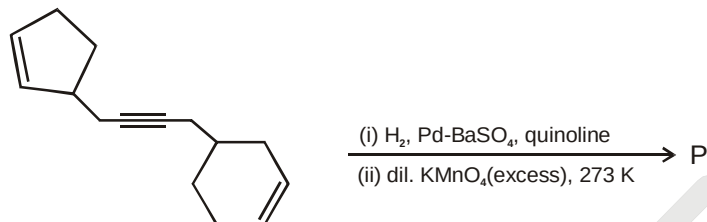
List-II



105. Which of the following options has correct combination considering List-I and List-II ?
 (A) (I), (Q), (T), (U) (B) (II), (P), (S), (U) (C) (II), (P), (S), (T) (D) (I), (S), (Q), (R)

106. Which of the following options has correct combination considering List-I and List-II ?
 (A) (III), (S), (R) (B) (IV), (Q), (U)
 (C) (III), (T), (U) (D) (IV), (Q), (R)
107. Total number of hydroxyl groups present in a molecule of the major product **P** is _____.

[IIT Adv. 2019]

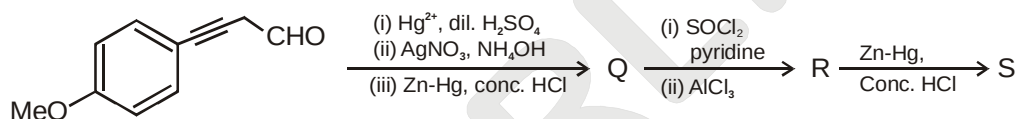


108. Total number of isomers, considering both structural and stereoisomers, of cyclic ethers with the molecular formula $\text{C}_4\text{H}_8\text{O}$ is _____.

[IIT Adv. 2019]

109. Choose the correct option(s) for the following reaction sequence :

[IIT Adv. 2019]

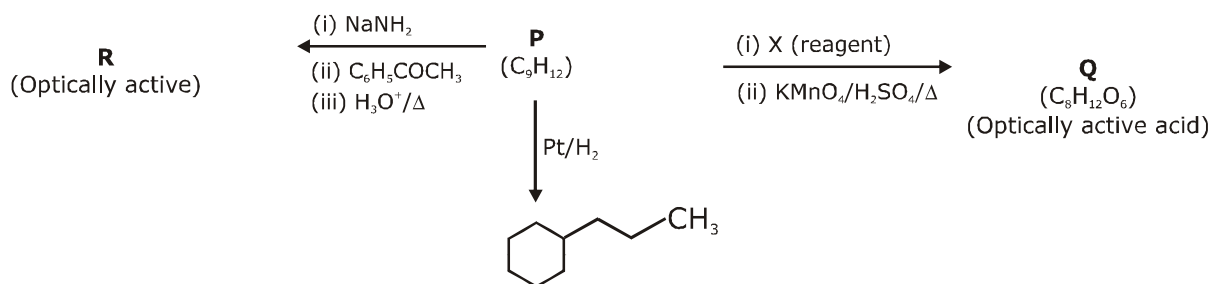


Consider Q, R and S as major products

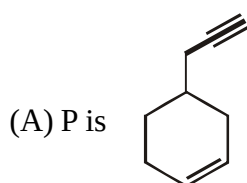
- (A) (Q) (R)
- (B) (Q) (S)
- (C) (R) (S)
- (D) (R) (S)

110. Consider the following transformations of a compound P.

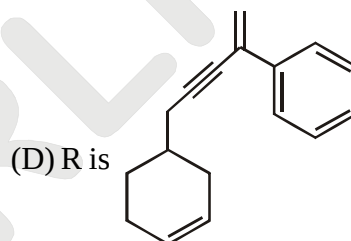
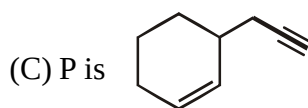
[IIT Adv. 2020]



Choose the correct option(s).



(B) X is $Pd-C/quinoline/H_2$

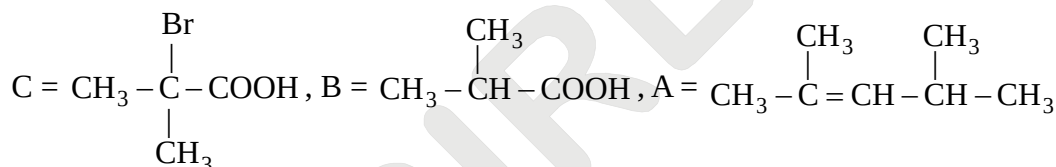
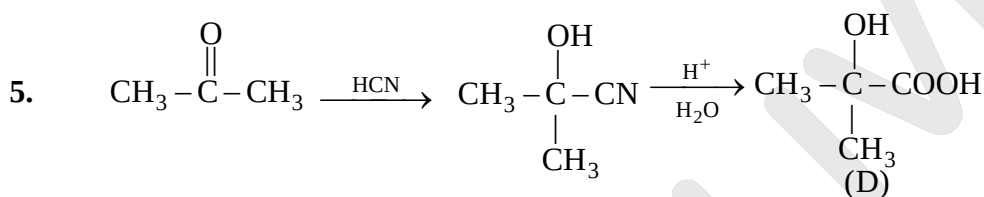
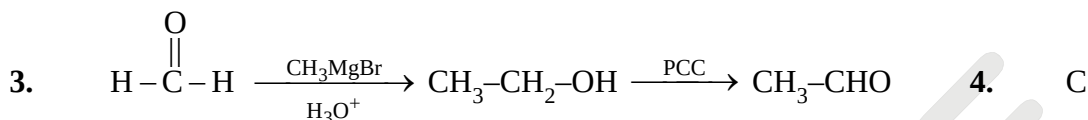


ANSWER KEY**[JEE-MAIN/AIEEE]**

1. A 2. C 3. D 4. A 5. A 6. B 7. B
8. A 9. A 10. B 11. B 12. D 13. A 14. D

[JEE-ADVANCED]

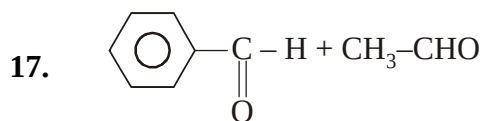
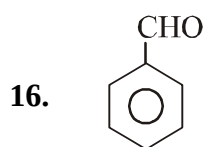
1. C 2. $(\text{CH}_2)_6\text{N}_4$



6. C 7. B 8. A 9. AB 10. ABD



15. Hydrazine form salt therefore no further reaction with aldehyde & ketones.



18. C

19. (i) Tollen's reagent (ii) Benzo chloride is obtained

20. AB

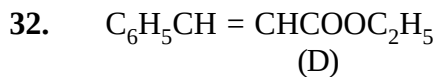


22. B 23. A 24. D

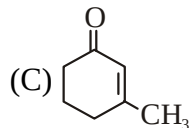
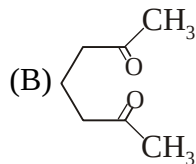
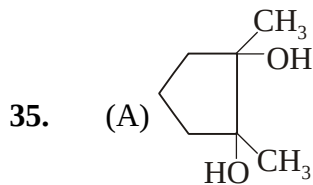
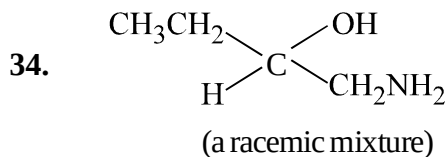
25. Hypoiodite is oxidising reagent whereas I^- is reducing agent.



29. D 30. D 31. A



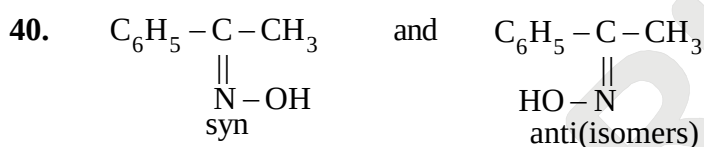
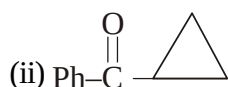
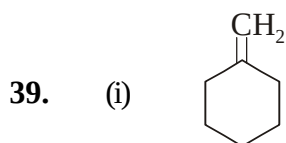
33. B



36. A

37. D

38. B



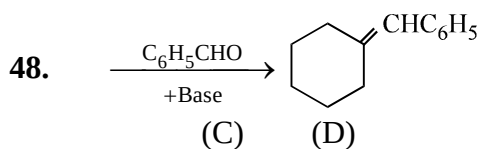
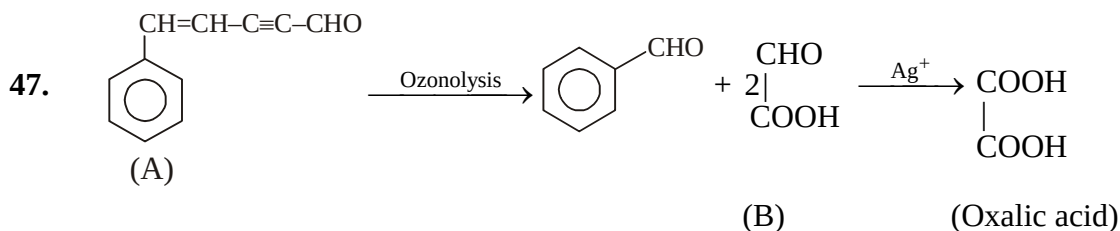
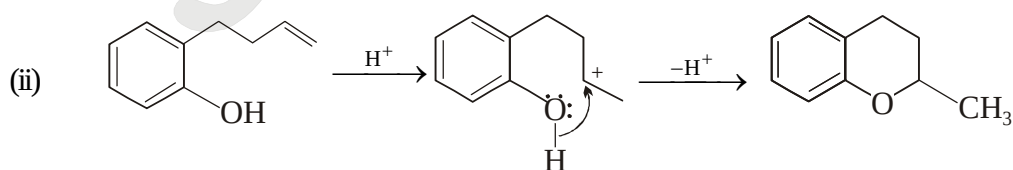
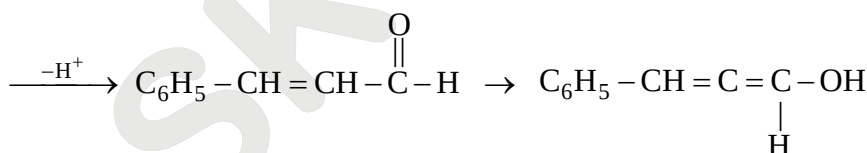
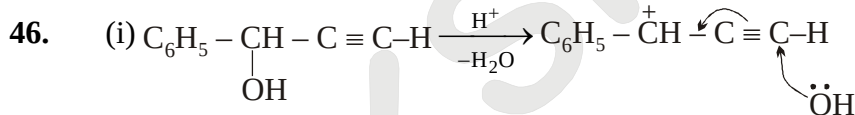
41. AD

42. A

43. ABD

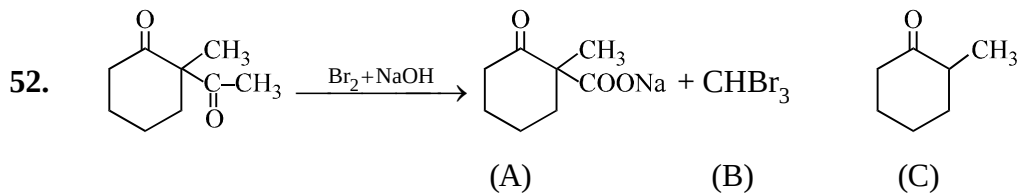
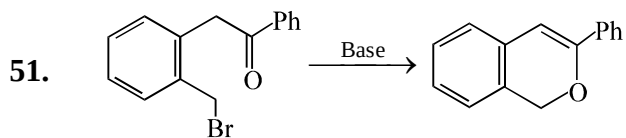
44. B

45. BD

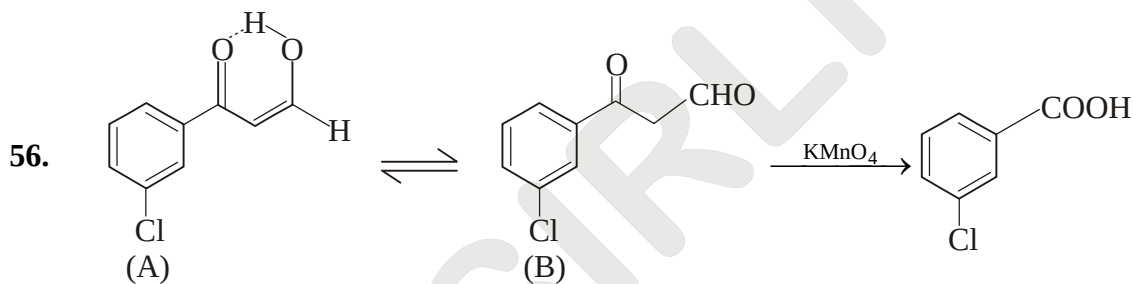
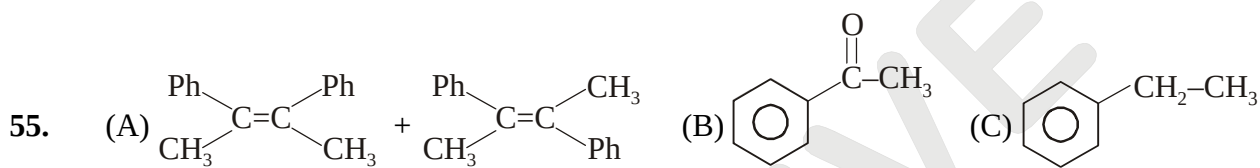


49. B

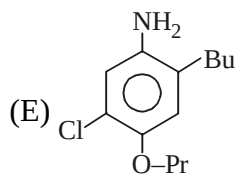
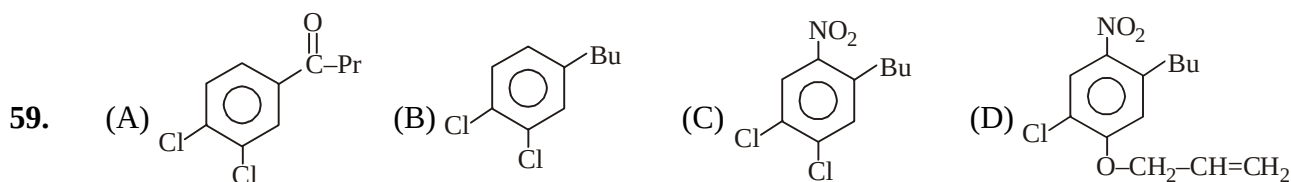
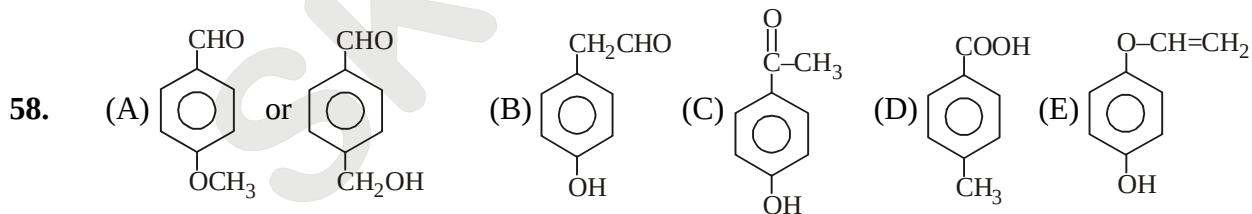
50. B



53. A 54. C

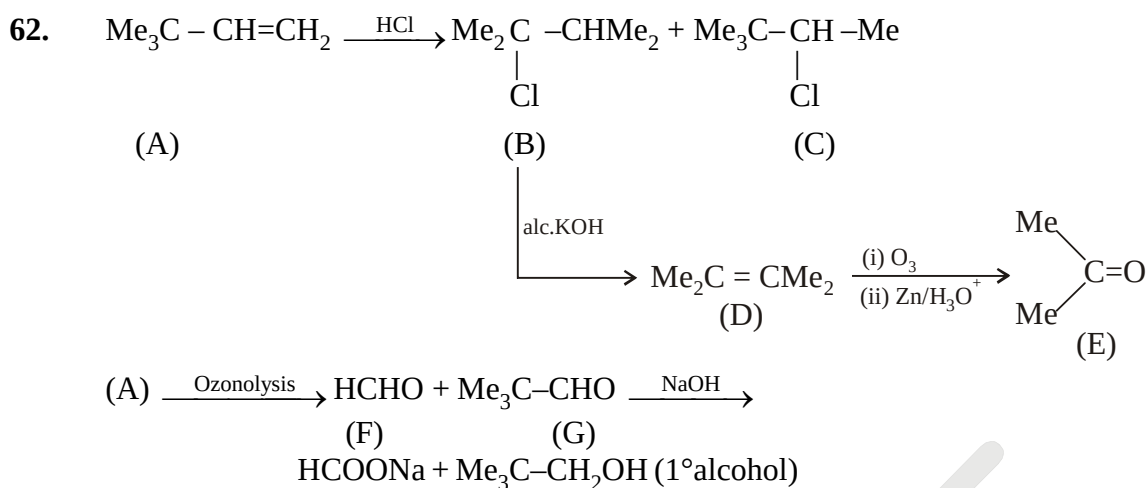


57. A



60. C

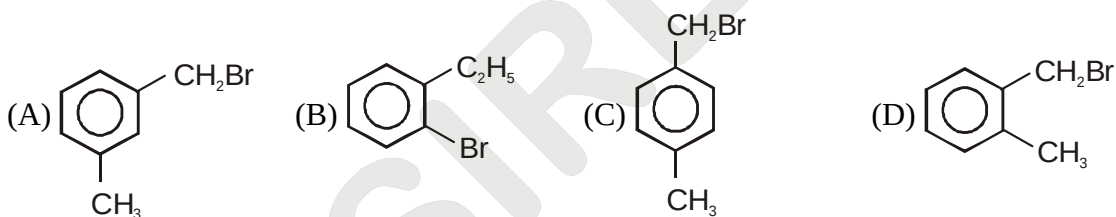
61. D



- | | | | | | | |
|---------|--|------------|---|---------|----------|--------|
| 63. C | 64. C | 65. B | 66. A | 67. C | 68. C | 69. C |
| 70. B | 71. (A) P, S; (B) Q; (C) Q, R, S; (D) Q, R | 72. D | 73. A | 74. C | | |
| 75. B | 76. A | 77. D | 78. B | 79. A | 80. B | 81. B |
| 82. A | 83. D | 84. 1 | 85. (A) R, S, T (B) P, S, T (C) R, S (D) Q, R | | | |
| 86. D | 87. B | 88. C | 89. A | 90. ACD | 91. ABCD | |
| 92. A | 93. C | 94. A | 95. A | 96. ABC | 97. A | 98. AC |
| 99. ABD | 100. C | 101. A | 102. A | 103. B | 104. 495 | 105. B |
| 106. D | 107. 6.00 | 108. 10.00 | 109. AC | 110. BC | | |

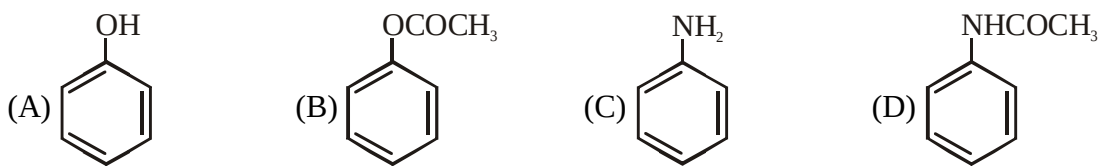
[JEE-MAIN / AIEEE]

1. Sodium ethoxide has reacted with ethanoyl chloride. The compound that is produced in the above reaction is : [AIEEE-2011]
 (A) Diethyl ether (B) 2-Butanone (C) Ethyl chloride (D) Ethyl ethanoate
2. A compound with molecular mass 180 is acylated with CH_3COCl to get a compound with molecular mass 390. The number of amino groups present per molecule of the former compound is : [JEE(Mains) 2013]
 (A) 2 (B) 5 (C) 4 (D) 6
3. Compound (A), $\text{C}_8\text{H}_9\text{Br}$, gives a white precipitate when warmed with alcoholic AgNO_3 . Oxidation of (A) gives an acid (B), $\text{C}_8\text{H}_6\text{O}_4$. (B) easily forms anhydride on heating. Identify the compound (A). [JEE (Mains) 2013]



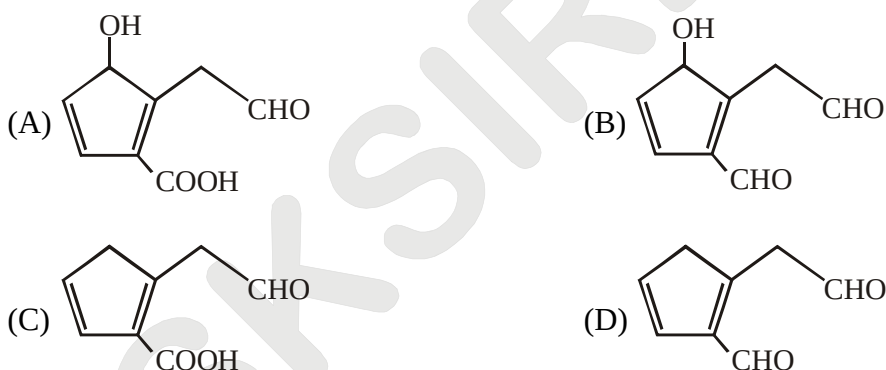
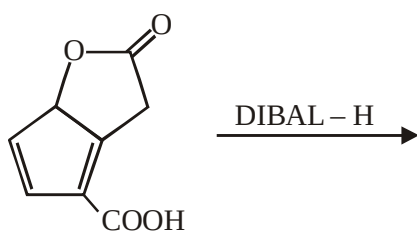
4. An organic compound A upon reacting with NH_3 gives B. On heating B gives C. C in presence of KOH reacts with Br_2 to give $\text{CH}_3\text{CH}_2\text{NH}_2$. A is [JEE (Mains) 2013]
 (A) CH_3COOH (B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
 (C) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{COOH}$ (D) $\text{CH}_3\text{CH}_2\text{COOH}$
5. In the Hofmann bromamide degradation reaction, the number of moles of NaOH and Br_2 used per mole of amine produced are : [JEE (Mains) 2016]
 (A) Four moles of NaOH and two moles of Br_2
 (B) Two moles of NaOH and two moles of Br_2
 (C) Four moles of NaOH and one mole of Br_2
 (D) One mole of NaOH and one mole of Br_2

6. Which of the following compounds will form significant amount of *meta* products during mono-nitration reaction? [JEE (Mains) 2017]

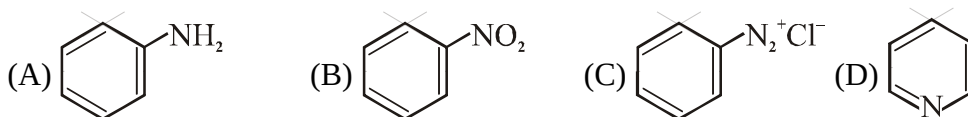


7. Sodium salt of an organic acid 'X' produces effervescence with conc. H_2SO_4 . 'X' reacts with the acidified aqueous CaCl_2 solution to give a white precipitate which decolourises acidic solution of KMnO_4 . 'X' is: [JEE (Mains) 2017]
- (A) $\text{C}_6\text{H}_5\text{COONa}$ (B) HCCONa (C) CH_3COONa (D) $\text{Na}_2\text{C}_2\text{O}_4$

8. The major product obtained in the following reaction is [JEE (Mains) 2017]



9. Which of the following compounds will be suitable for Kjeldahl's method for nitrogen estimation? [JEE (Mains) 2018]



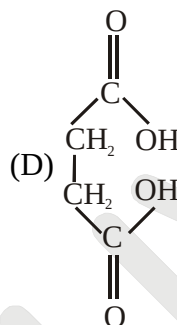
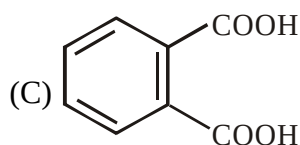
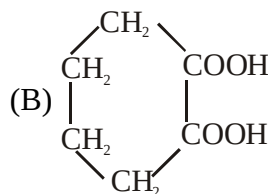
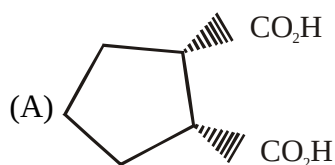
10. The increasing order of basicity of the following compounds is: [JEE (Mains) 2018]



- (A) (b) < (a) < (c) < (d) (B) (b) < (a) < (d) < (c)
 (C) (d) < (b) < (a) < (c) (D) (a) < (b) < (c) < (d)

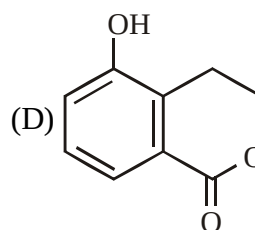
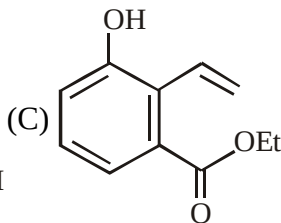
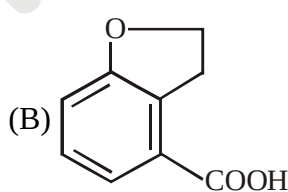
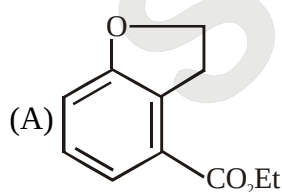
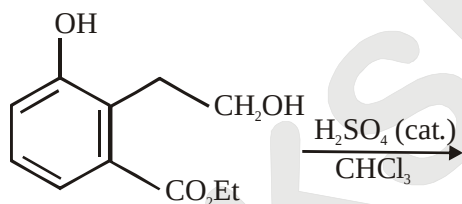
11. Which dicarboxylic acid in presence of a dehydrating agent is least reactive to give an anhydride ?

[2019 Mains]



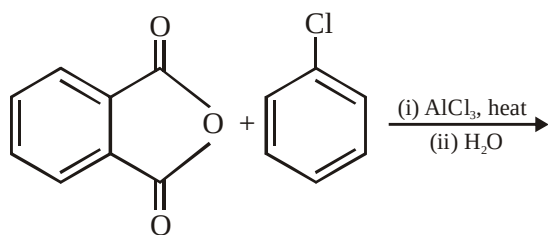
12. The major product of the following reaction is

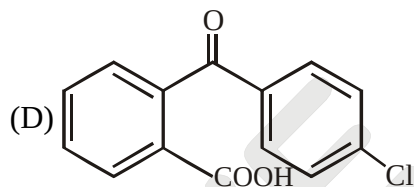
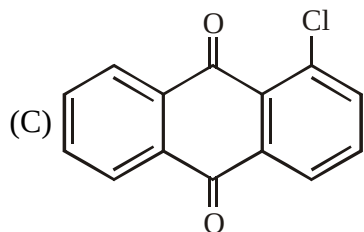
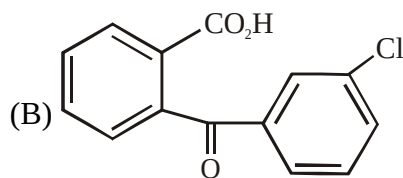
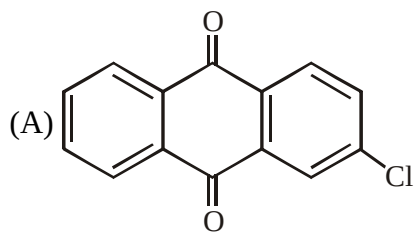
[2019 Mains]



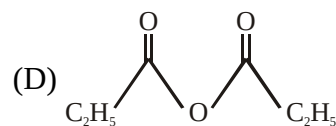
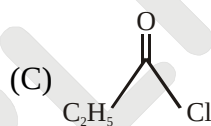
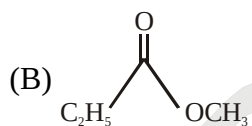
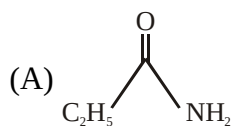
13. The major product of the following reaction is

[2019 Mains]





14. The increasing order of the reactivity of the following with LiAlH_4 is [2019 Mains]



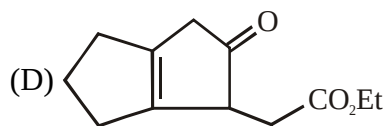
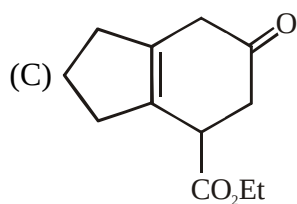
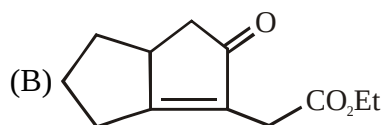
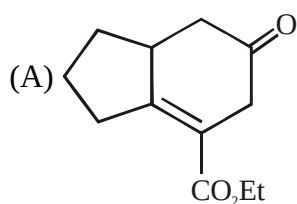
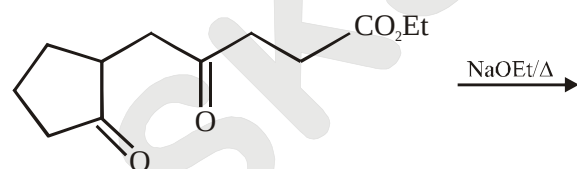
(A) (A) < (B) < (D) < (C)

(B) (A) < (B) < (C) < (D)

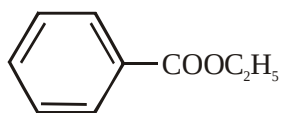
(C) (B) < (A) < (D) < (C)

(D) (B) < (A) < (C) < (D)

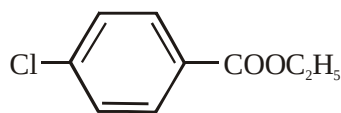
15. The major product obtained in the following reaction is [2019 Mains]



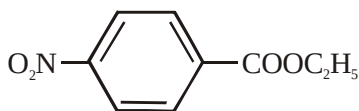
16. The decreasing order of ease of alkaline hydrolysis for the following esters is [2019 Mains]



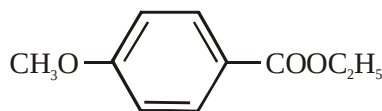
(I)



(II)

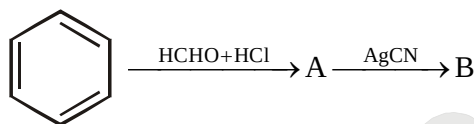


(III)

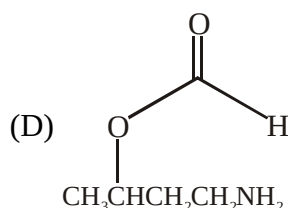
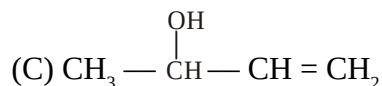
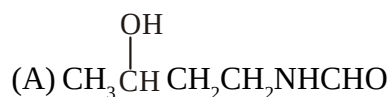
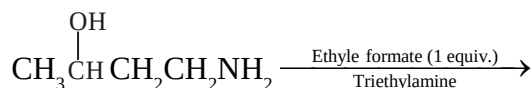


(IV)

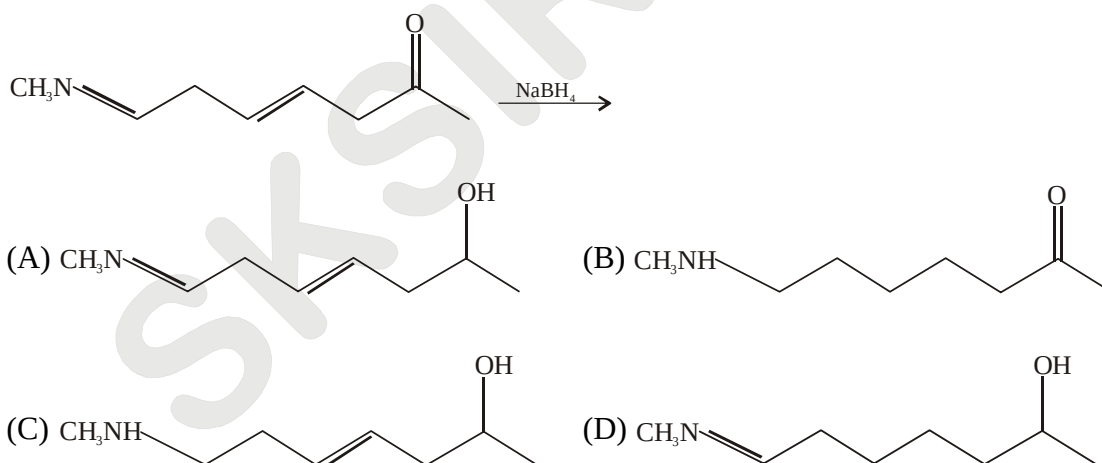
- (A) III > II > IV > I (B) III > II > I > IV (C) II > III > I > IV (D) IV > II > III > I
17. The compound A and B in the following reaction are, respectively [2019 Mains]



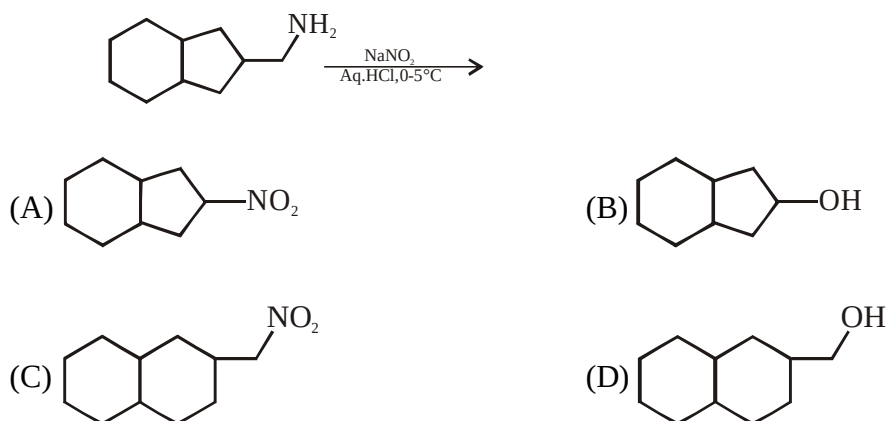
- (A) A = Benzyl alcohol, B = Benzyl isocyanide
 (B) A = Benzyl alcohol, B = Benzyl cyanide
 (C) A = Benzyl chloride, B = Benzyl isocyanide
 (D) A = Benzyl chloride, B = Benzyl cyanide
18. The major product of following reaction is [2019 Mains]
- $$R-C \equiv N \xrightarrow[(ii) H_2O]{(i) AlH(i-Bu)_2} \quad (A) RCHO \quad (B) RCONH_2 \quad (C) RCOOH \quad (D) RCH_2NH_2$$
19. The major product of the following reaction is [JEE-Main 2019]



20. Ethylamine ($C_2H_5NH_2$) can be obtained from N-ethylphthalimide on treatment with [JEE-Main 2019]
(A) $NaBH_4$ (B) NH_2NH_2 (C) H_2O (D) CaH_2
21. Hinsberg's reagent is [JEE-Main 2019]
(A) $SOCl_2$ (B) C_6H_5COCl (C) $C_6H_5SO_2Cl$ (D) $(COCl)_2$
22. Which of the following amines can be prepared by Gabriel phthalimide reaction? [JEE-Main 2019]
(A) n-butylamine (B) triethylamine (C) t-butylamine (D) neo-pentylamine
23. A compound 'X' on treatment with $Br_2/NaOH$, provided C_3H_9N , which gives positive carbylamine test. Compound 'X' is [JEE-Main 2019]
(A) $CH_3COCH_2NHCH_3$ (B) $CH_3CH_2CH_2CONH_2$
(C) $CH_3CON(CH_3)_2$ (D) $CH_3CH_2COCH_2NH_2$
24. The major product of the following reaction is [JEE-Main 2019]

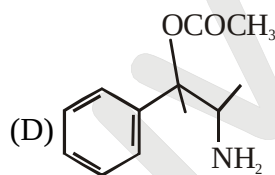
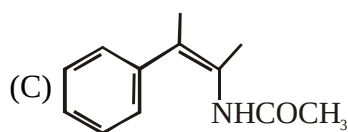
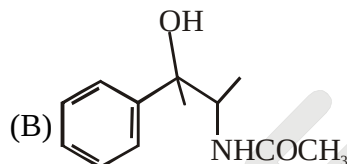
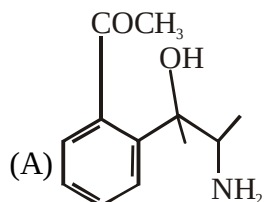
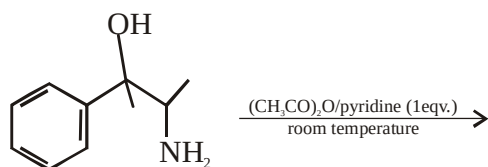


25. The major product formed in the reaction given below will be [JEE-Main 2019]



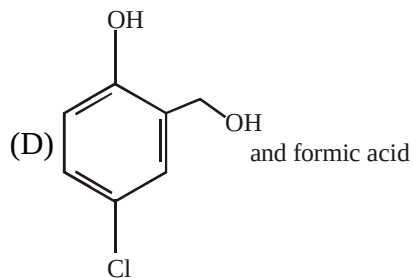
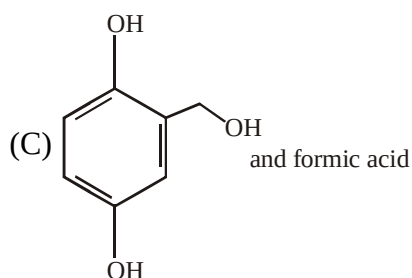
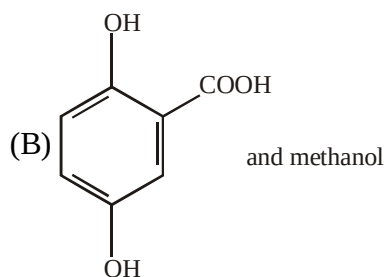
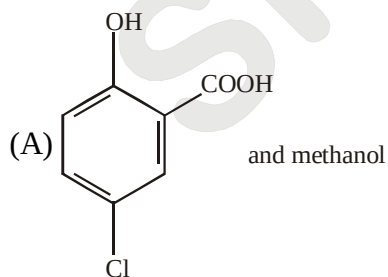
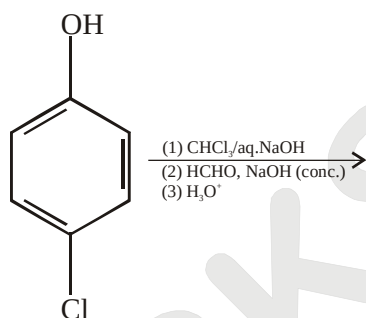
26. The major product obtained in the following reaction is

[JEE-Main 2019]



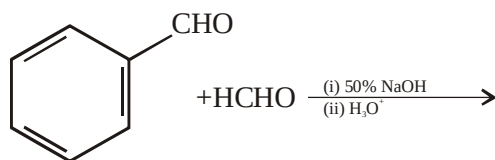
27. The major products of the following reaction are

[JEE-MAIN 2019]



28. Major products of the following reaction are

[JEE-MAIN 2019]

(A) CH₃OH and HCO₂H(B) CH₃OH and

(C) HCOOH and

(D) and

[JEE-ADVANCED]

1. **Statement-1** : Acetate ion is more basic than the methoxide ion.**Statement-2** : The acetate ion is resonance stabilized.

[JEE-2004]

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

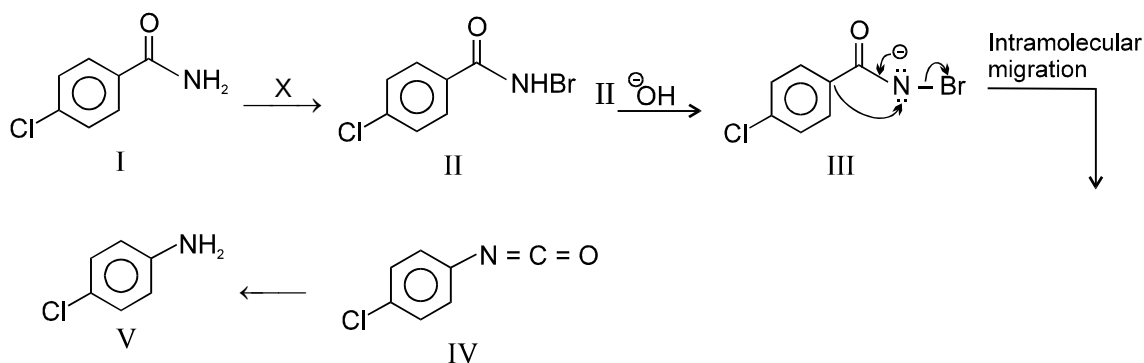
2. In conversion of 2-butanone to propanoic acid which reagent is used.

[JEE-2005]

(A) NaOH, NaI / H[⊕] (B) Fehling solution (C) NaOH, I₂ / H[⊕] (D) Tollen's reagent

Passage (3 and 4)

[JEE 2006]

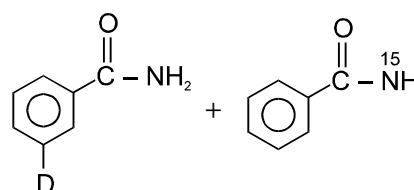


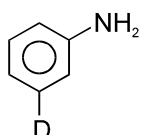
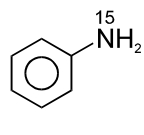
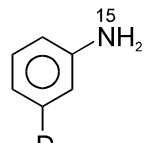
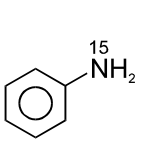
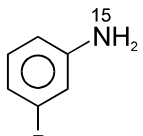
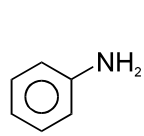
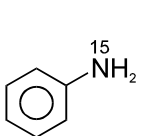
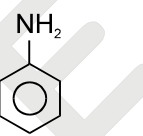
3. Which reagent (X) is used to convert I to II

(A) KBr / NaOH (B) Br₂ / NaOH (C) NaHCO₃ (D) N-Bromo succinimide

4. Which step is rate determining step [JEE 2006]

- (A) Formation of II (B) Formation of III (C) Formation of V (D) Formation of IV

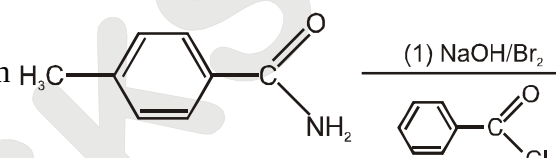
5.  under Hofmann conditions will give : [JEE 2006]

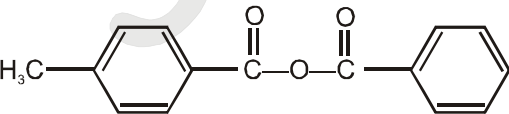
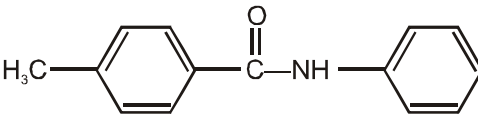
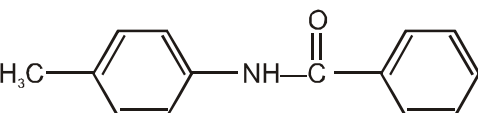
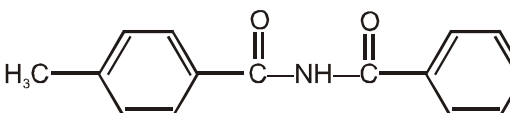
- (A)  +  (B)  + 
 (C)  +  (D)  + 

6. **Statement-1** : p-Hydroxybenzoic acid has a lower boiling point than o-hydroxybenzoic acid.

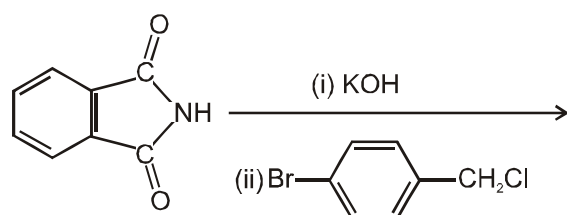
Statement-2 : o-Hydroxybenzoic acid has intramolecular hydrogen bonding. [JEE 2007]

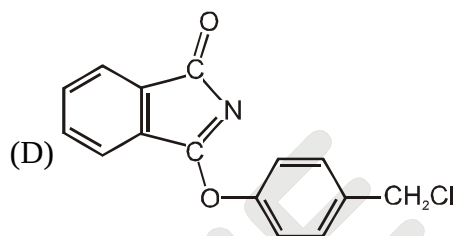
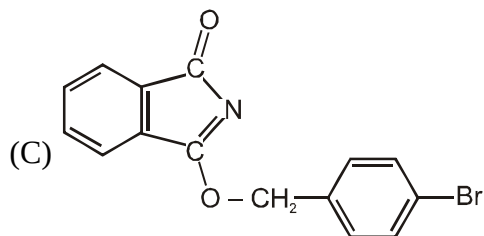
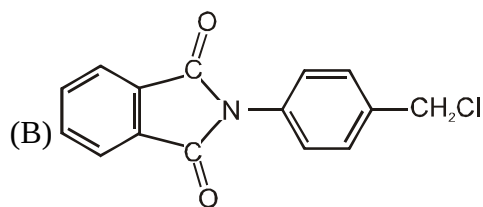
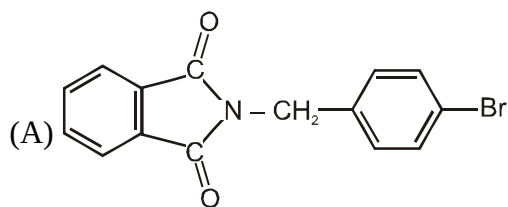
- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True.

7. In the reaction  T, the structure of the Product T is : [JEE-2010]

- (A)  (B) 
 (C)  (D) 

8. The major product of the following reaction is : [JEE-2011]

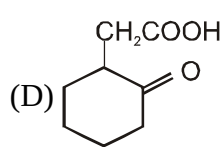
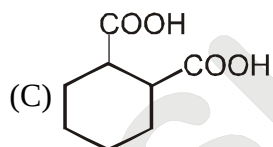
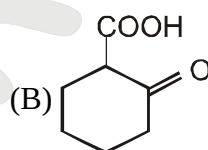
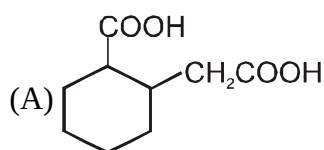




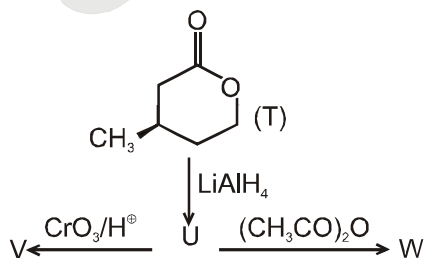
9. The carboxyl functional group ($-\text{COOH}$) is present in : [JEE-2012]

- (A) picric acid (B) barbituric acid (C) ascorbic acid (D) aspirin

10. The compound that undergoes decarboxylation most readily under mild condition is : [JEE-2012]

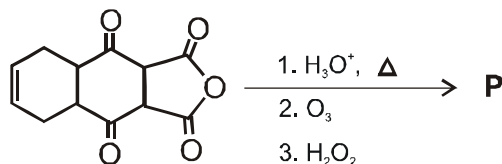


11. With reference to the scheme given, which of the given statements(s) about T, U, V and W is (are) correct? [JEE-2012]



- (A) T is soluble in hot aqueous NaOH
 (B) U is optically active
 (C) Molecular formula of W is $\text{C}_{10}\text{H}_{18}\text{O}_4$
 (D) V gives effervescence on treatment with aqueous NaHCO_3

12. The total number of carboxylic acid groups in the product **P** is [JEE(Advanced)-2013]

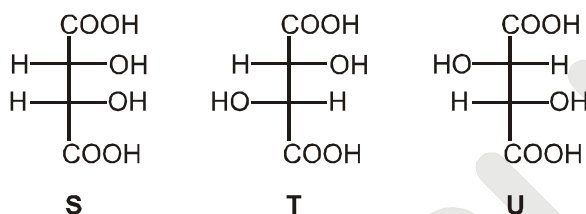


Paragraph for Question 13 and 14

[JEE(Advanced)-2013]

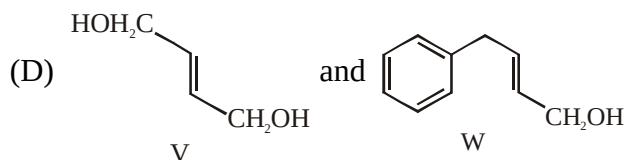
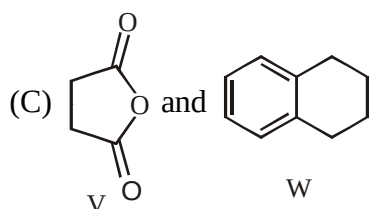
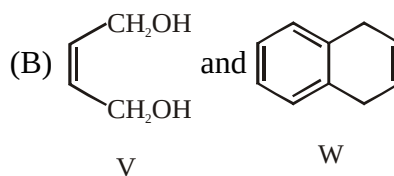
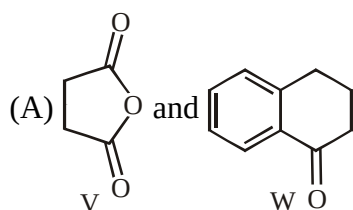
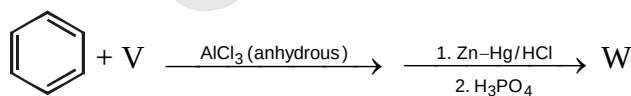
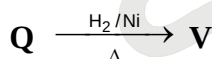
P and **Q** are isomers of dicarboxylic acid $C_4H_4O_4$. Both decolorize Br_2/H_2O . On heating, **P** forms the cyclic anhydride.

Upon treatment with dilute alkaline $KMnO_4$, **P** as well as **Q** could produce one or more than one from **S**, **T** and **U**.



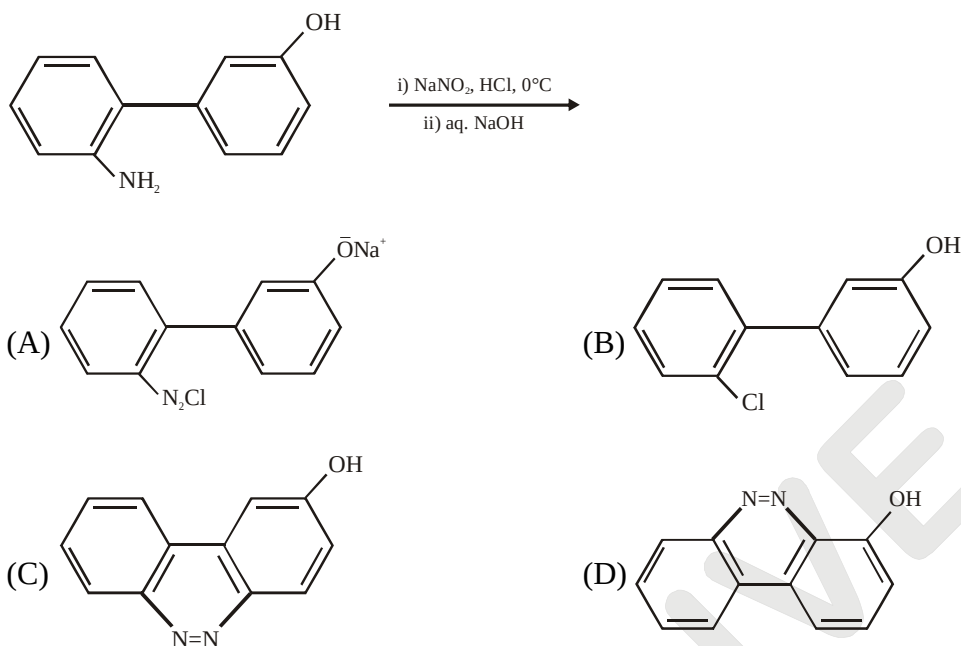
13. Compounds formed from **P** and **Q** are, respectively
 (A) Optically active **S** and optically active pair (**T**, **U**)
 (B) Optically inactive **S** and optically inactive pair (**T**, **U**)
 (C) Optically active pair (**T**, **U**) and optically active **S**
 (D) Optically inactive pair (**T**, **U**) and optically inactive **S**

14. In the following reaction sequences **V** and **W** are respectively : [JEE(Advanced)-2013]



15. The major product of the following reaction is

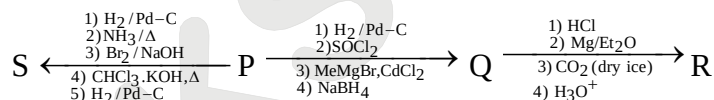
[JEE(Advanced)-2017]



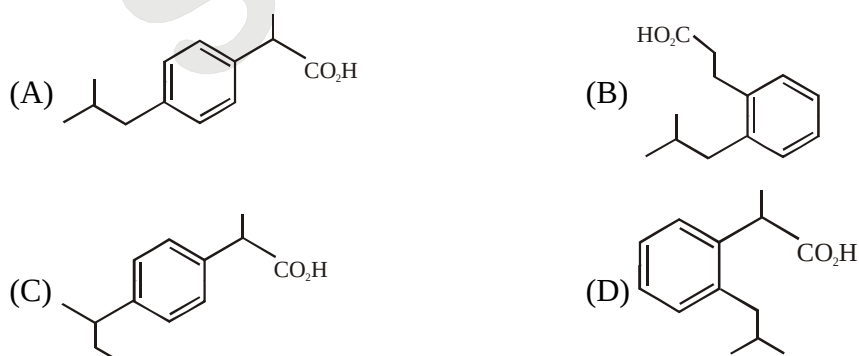
PARAGRAPH (16-17)

[JEE(Advanced)-2018]

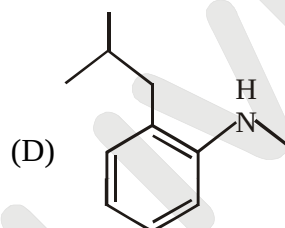
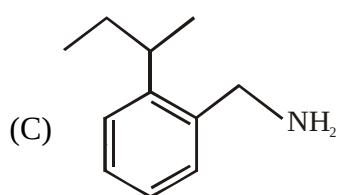
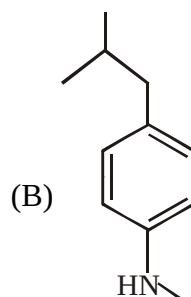
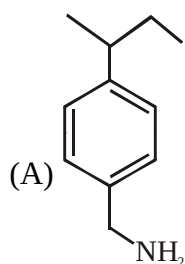
An organic acid P ($C_{11}H_{12}O_2$) can easily be oxidized to a dibasic acid which reacts with ethyleneglycol to produce a polymer decron. Upon ozonolysis, P gives an aliphatic ketone as one of the products. P undergoes the following reaction sequences to furnish R via Q. The compound P also undergoes another set of reactions to produce S.



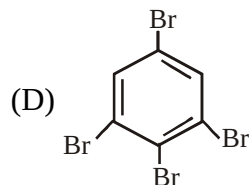
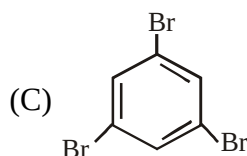
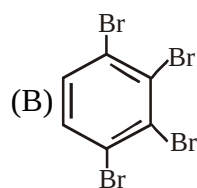
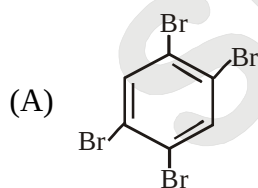
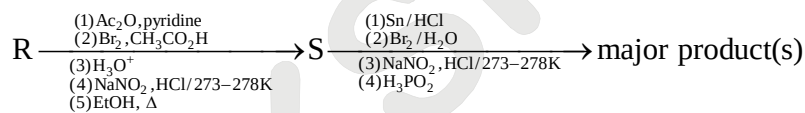
16. The compound R is



17. The compound S is

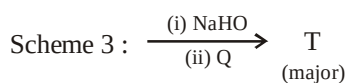
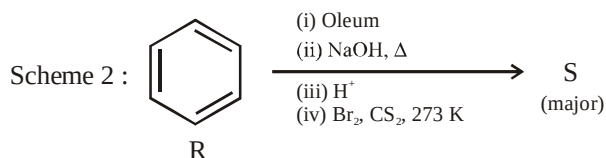
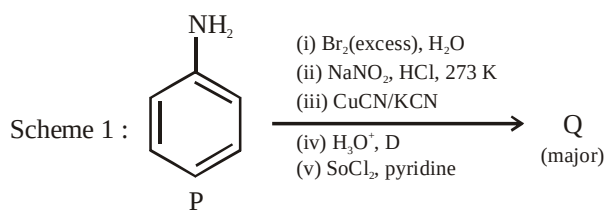


18. Aniline reacts with mixed acid (conc. HNO_3 and conc. H_2SO_4) at 288 K to give P (51%), Q (47%) and R (2%). The major product(s) of the following reaction sequence is/are
[JEE(Advanced)-2018]



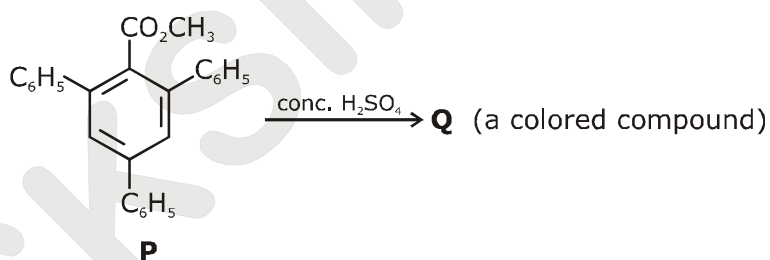
19. Schemes 1 and 2 describe the conversion of P to Q and R to S, respectively. Scheme 3 describes the synthesis of T from Q and S. The total number of Br atoms in a molecule of T is

[JEE(Advanced)-2019]



20. In the following reaction, compound Q is obtained from compound P via an ionic intermediate.

[JEE(Advanced)-2020]



What is the degree of unsaturation of Q?

ANSWER KEY

[JEE-MAIN/AIEEE]

1.	D	2.	B	3.	D	4.	D	5.	C	6.	C	7.	D
8.	B	9.	A	10.	B	11.	B	12.	D	13.	D	14.	A
15.	B	16.	B	17.	C	18.	A	19.	A	20.	B	21.	C
22.	A	23.	B	24.	C	25.	B	26.	B	27.	D	28.	C

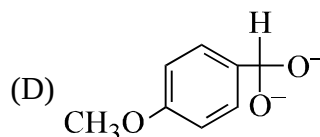
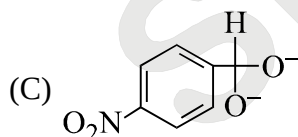
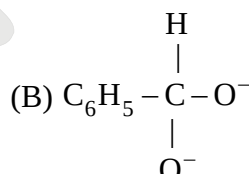
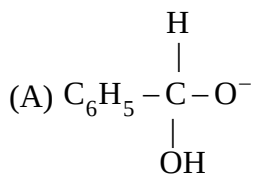
[JEE-ADVANCED]

1.	D	2.	C	3.	B	4.	D	5.	A	6.	D	7.	C
8.	A	9.	D	10.	B	11.	A,C,D	12.	2	13.	B	14.	A
15.	C	16.	A	17.	B	18.	D	19.	4.00	20.	18		

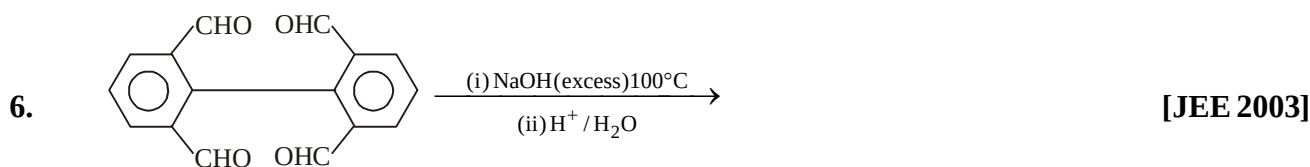
[JEE-ADVANCED]

(OBJECTIVE)

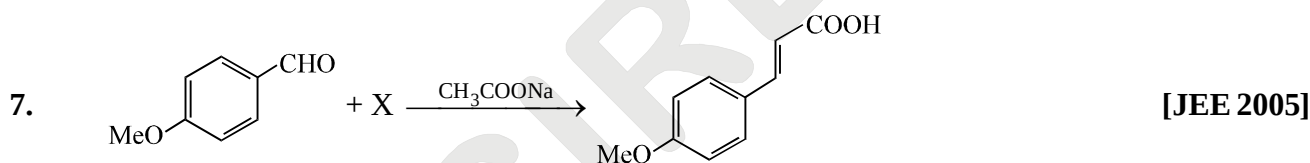
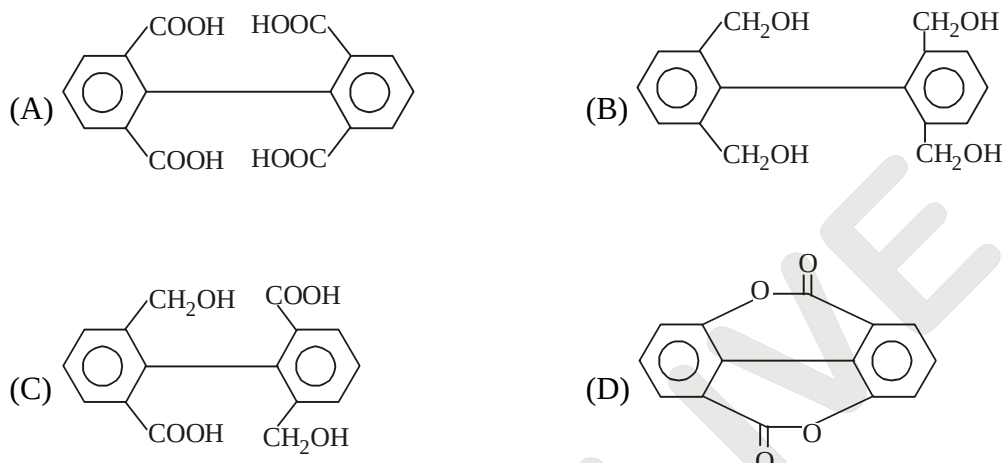
1. m-chlorobenzaldehyde on reaction with conc. KOH at room temperature gives: [JEE 1991]
 (A) Potassium m-chlorobenzoate and m-hydroxybenzaldehyde
 (B) m-hydroxybenzaldehyde and m-chlorobenzyl alcohol
 (C) m-chlorobenzyl and m-hydroxybenzyl alcohol
 (D) Potassium m-chlorobenzoate and m-chlorobenzyl alcohol
2. In the Cannizzaro reaction given below, $2\text{Ph-CHO} \xrightarrow{\text{OH}^-} \text{Ph-CH}_2\text{OH} + \text{PhCO}_2^-$ the slowest step is:
 (A) the attack of OH^- at the carbonyl group
 (B) the transfer of hydride to the carbonyl group
 (C) the abstraction of proton from the carboxylic acid
 (D) the deprotonation of $\text{Ph-CH}_2\text{OH}$ [JEE 1996]
3. In a Cannizzaro reaction the intermediate which is the best hydride donor is: [JEE 1997]



4. Which of the following will undergo aldol condensation: [JEE 1998]
 (A) Acetaldehyde (B) Propanaldehyde
 (C) Benzaldehyde (D) Trideutero acetaldehyde
5. A mixture of benzaldehyde and formaldehyde on heating with aqueous NaOH solution gives: [JEE 2001]
 (A) benzyl alcohol and sodium formate
 (B) sodium benzoate and methyl alcohol
 (C) sodium benzoate and sodium formate
 (D) benzyl alcohol and methyl alcohol



any one of the products formed is :



What is X?

- (A) CH_3COOH (B) BrCH_2COOH
 (C) $(\text{CH}_3\text{CO})_2\text{O}$ (D) $\text{CHO}-\text{COOH}$

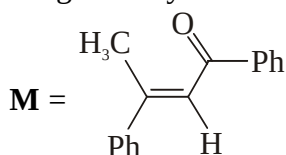
8. Cyclohexene on ozonolysis followed by reaction with zinc dust and water gives compound E. Compound E on further treatment with aqueous KOH yields compound F. Compound F is

[JEE 2007]

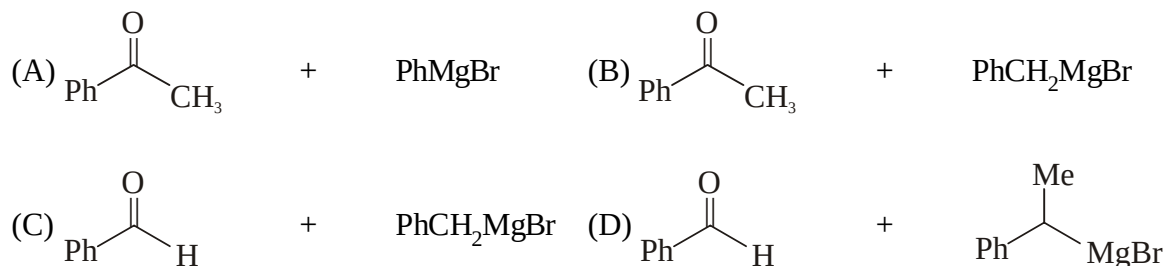


Paragraph for Question No. 9 to 11

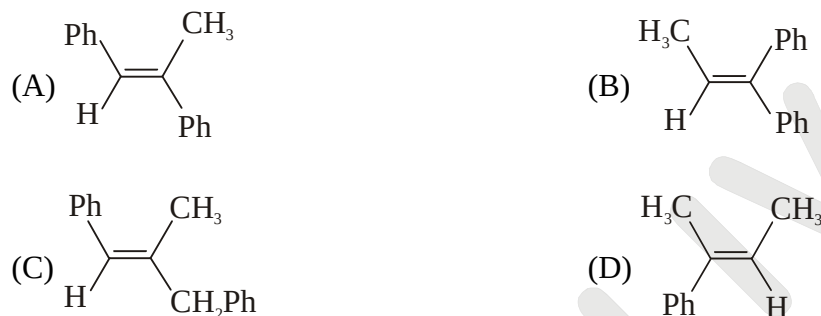
A tertiary alcohol **H** upon acid catalysed dehydration gives a product **I**. Ozonolysis of **I** leads to compounds **J** and **K**. Compound **J** upon reaction with KOH gives benzyl alcohol and a compound **L**, whereas **K** on reaction with KOH gives only **M**.



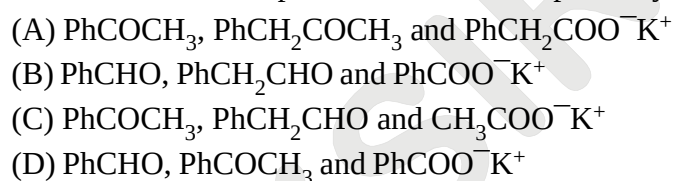
9. Compound **H** is formed by the reaction of [JEE 2008]



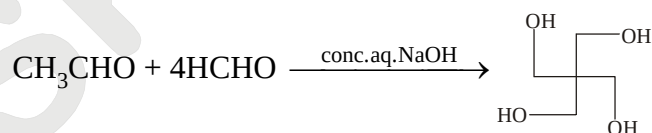
10. The structure of compound **I** is [JEE 2008]



11. The structures of compounds **J**, **K** and **L**, respectively, are [JEE 2008]



12. The number of aldol reaction(s) that occurs in the given transformation is [JEE 2012]

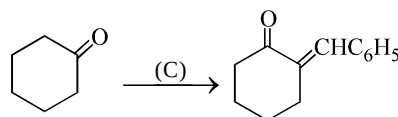


- (A) 1 (B) 2 (C) 3 (D) 4

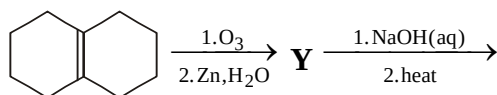
SUBJECTIVE

1. Write the chemical equation to show what happens when "Ethyl acetate is treated with sodium ethoxide in ethanol and the reaction mixture is acidified". [JEE 1981]

2. $\text{C}_6\text{H}_5-\text{CHO} + \text{CH}_3-\text{COOC}_2\text{H}_5 \xrightarrow[\text{C}_2\text{H}_5\text{OH and heat}]{\text{NaOC}_2\text{H}_5 \text{ in absolute}} (\text{D}).$ [JEE 1995]

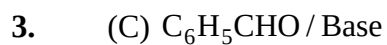
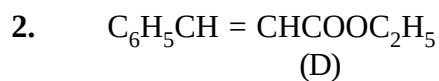
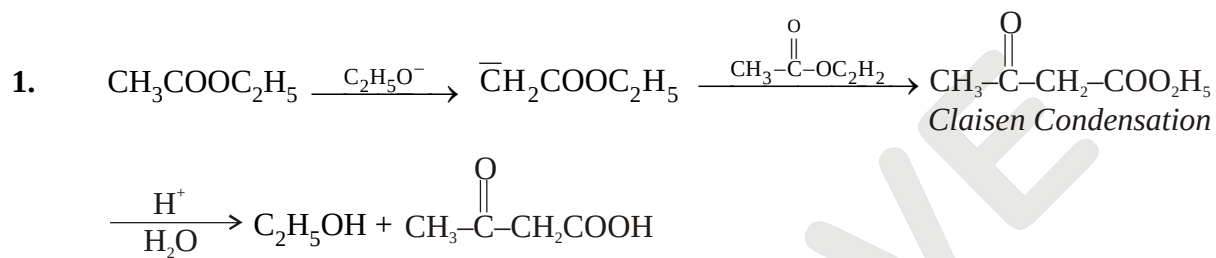
3.  [JEE 1998]

4. In the scheme given below, the total number of intramolecular aldol condensation products formed from 'Y' is [JEE 2010]



ANSWER KEY**[JEE-ADVANCED]**OBJECTIVE

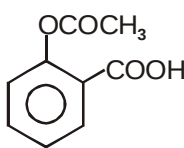
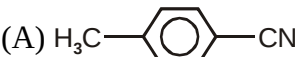
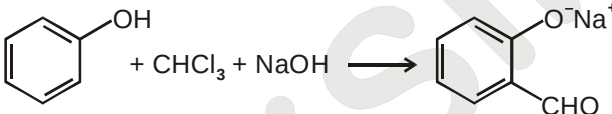
- | | | | | | | | | | | | | | |
|----|---|----|---|-----|---|-----|-----|-----|---|----|---|----|---|
| 1. | D | 2. | B | 3. | D | 4. | ABD | 5. | A | 6. | C | 7. | C |
| 8. | A | 9. | B | 10. | A | 11. | D | 12. | C | | | | |

SUBJECTIVE

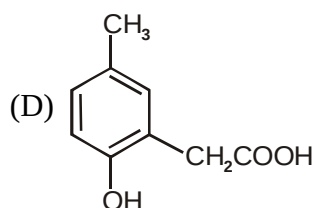
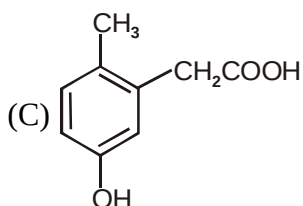
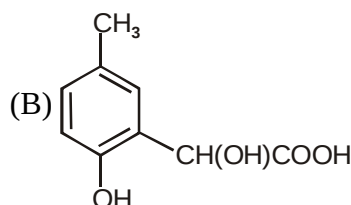
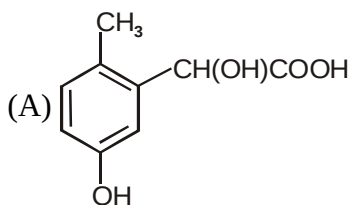
4. 1

(CARBENE & NITRENE MECHANISM BASED)

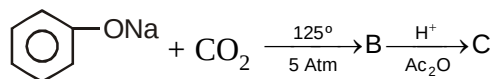
[JEE-MAIN / AIEEE]

1. The compound  is used as [AIEEE-2002]
- (A) antiseptic (B) antibiotic (C) analgesic (D) pesticide
2. When primary amine reacts with chloroform in ethanoic KOH then the product is [AIEEE-2002]
- (A) an isocyanide (B) an aldehyde (C) a cyanide (D) an alcohol
3. The reaction of chloroform with alcoholic KOH and p-toluidine form [AIEEE-2003]
- (A)  (B) 
 (C)  (D) 
4. 
 The electrophile involved in the above reaction is [AIEEE-2006]
- (A) dichloromethyl cation (CHCl_2^+) (B) dichlorocarbene ($:\text{CCl}_2$)
 (C) trichloromethyl anion (CCl_3^-) (D) formyl cation (CHO^+)
5. In the chemical reaction, [AIEEE-2007]
 $\text{CH}_3\text{CH}_2\text{NH}_2 + \text{CHCl}_3 + 3\text{KOH} \longrightarrow (\text{A}) + (\text{B}) + 3\text{H}_2\text{O}$, the compound (A) and (B) are respectively
- (A) $\text{C}_2\text{H}_5\text{CN}$ and 3KCl (B) $\text{CH}_3\text{CH}_2\text{CONH}_2$ and 3KCl
 (C) $\text{C}_2\text{H}_5\text{NC}$ and K_2CO_3 (D) $\text{C}_2\text{H}_5\text{NC}$ and 3KCl
6. The major product obtained on interaction of phenol with sodium hydroxide and carbon dioxide is : [AIEEE-2009]
- (A) salicylic acid (B) phthalic acid (C) benzoic acid (D) salicylaldehyde
7. Aspirin is known as : [AIEEE-2012]
- (A) Acetyl salicylic acid (B) Phenyl salicylate (C) Acetyl salicylate (D) Methyl salicylic acid
8. An organic compound A upon reacting with NH_3 gives B. On heating, B gives C. C in presence of KOH reacts with Br_2 to give $\text{CH}_3\text{CH}_2\text{NH}_2$. A is : [IIT-JEE Mains 2013]
- (A) $\text{CH}_3\text{CH}_2\text{COOH}$ (B) CH_3COOH
 (C) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$ (D) $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \text{COOH}$

9. The reaction of phenol with benzoyl chloride to give phenyl benzoate is known as :
[IIT-JEE Mains (Online)2013]
(A) Schotten - Baumann reaction (B) Reimer - Tiemann reaction
(C) Claisen reaction (D) Gatterman-Koch reaction
10. Phenol on heating with CHCl_3 and NaOH gives salicylaldehyde. The reaction is called :
[IIT-JEE Mains (Online)2013]
(A) Claisen reaction (B) Hell - Volhard - Zelinsky reaction
(C) Cannizzaro's reaction (D) Reimer - Tiemann reaction
11. Carbylamine forms from aliphatic or aromatic primary amine via which of the following intermediates ?
[IIT-JEE Mains (Online)2013]
(A) Carbene (B) Carbon radical (C) Carbo cation (D) Carbanion
12. Which one of the following reactions will not result in the formation of carbon bond ?
[IIT Mains Online 2014]
(A) Reimer-Tieman reaction (B) Friedel Craft's acylation
(C) Wurtz reaction (D) Cannizzaro reaction
13. In the Hofmann bromamide degradation reaction, the number of moles of NaOH and Br_2 used per mole of amine produced are :
[JEE MAIN 2016]
(A) One mole of NaOH and one mole of Br_2 . (B) Four moles of NaOH and two moles of Br_2 .
(C) Two moles of NaOH and two moles of Br_2 . (D) Four moles of NaOH and one mole of Br_2 .
14. In the Hofmann bromamide degradation reaction, the number of moles of NaOH and Br_2 used per mole of amine produced are :
[IIT-JEE Mains 2016]
(A) One mole of NaOH and one mole of Br_2 .
(B) Four moles of NaOH and two moles of Br_2 .
(C) Two moles of NaOH and two moles of Br_2 .
(D) Four moles of NaOH and one mole of Br_2 .
15. p-cresol reacts with chloroform in alkaline medium to give the compound A which adds hydrogen cyanide to form, the compound B. The latter on acidic hydrolysis gives chiral carboxylic acid. The structure of the carboxylic acid is
[AIEEE-2005]

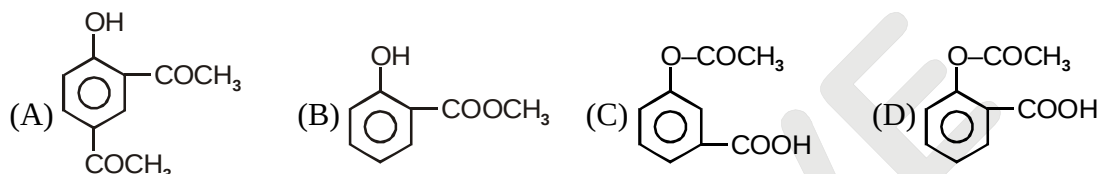


16. Which one of the following methods is neither meant for the synthesis nor for separation of amines?
(A) Hinsberg method (B) Hofmann method [AIEEE-2005]
(C) Wurtz reaction (D) Curtius reaction
17. Sodium phenoxide when heated with CO_2 under pressure at 125°C yields a product which on acetylation produces C.



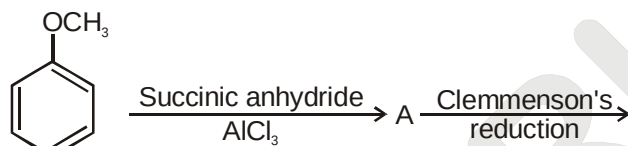
The major product C would be

[JEE MAINS-2014]

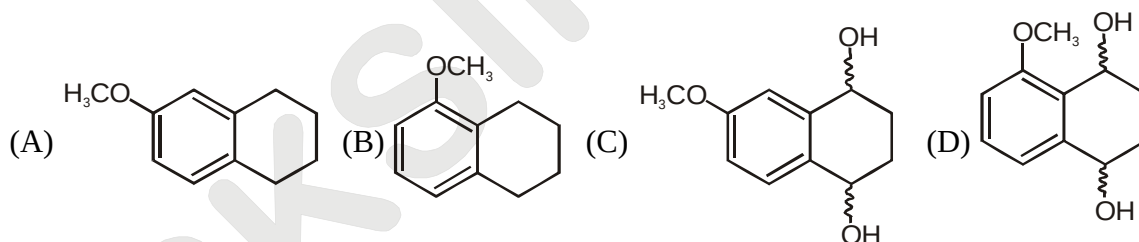


18. Consider the reaction sequence below :

[JEE MAINS ONLINE-2016]

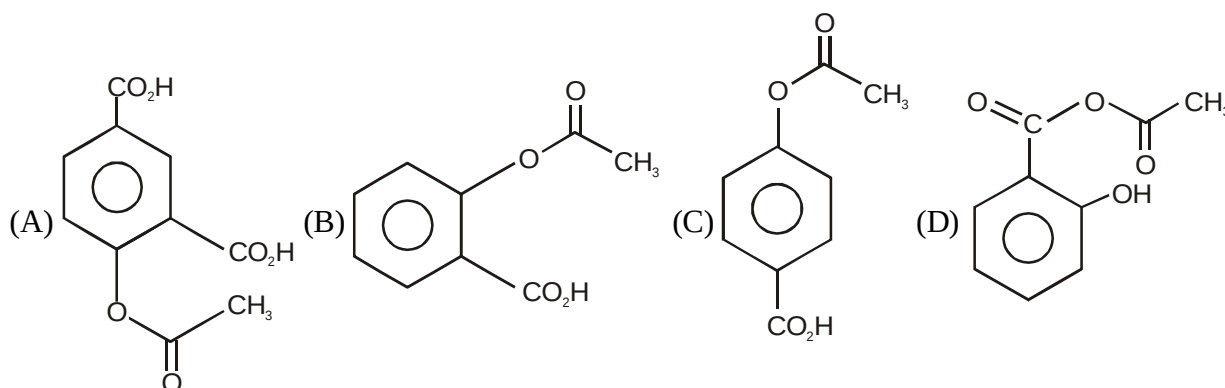


X is :



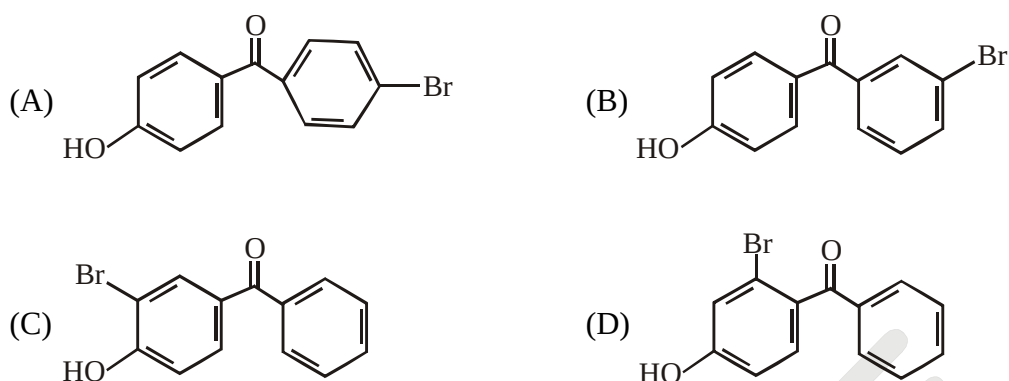
19. Phenol on treatment with CO_2 in the presence of NaOH followed by acidification produces compound X as the major product. X on treatment with $(\text{CH}_3\text{CO})_2\text{O}$ in the presence of catalytic amount of H_2SO_4 produces

[JEE MAINS OFFLINE-2018]



20. p-hydroxybenzophenone upon reaction with bromine in carbon tetrachloride gives

[JEE MAINS ONLINE-2019]



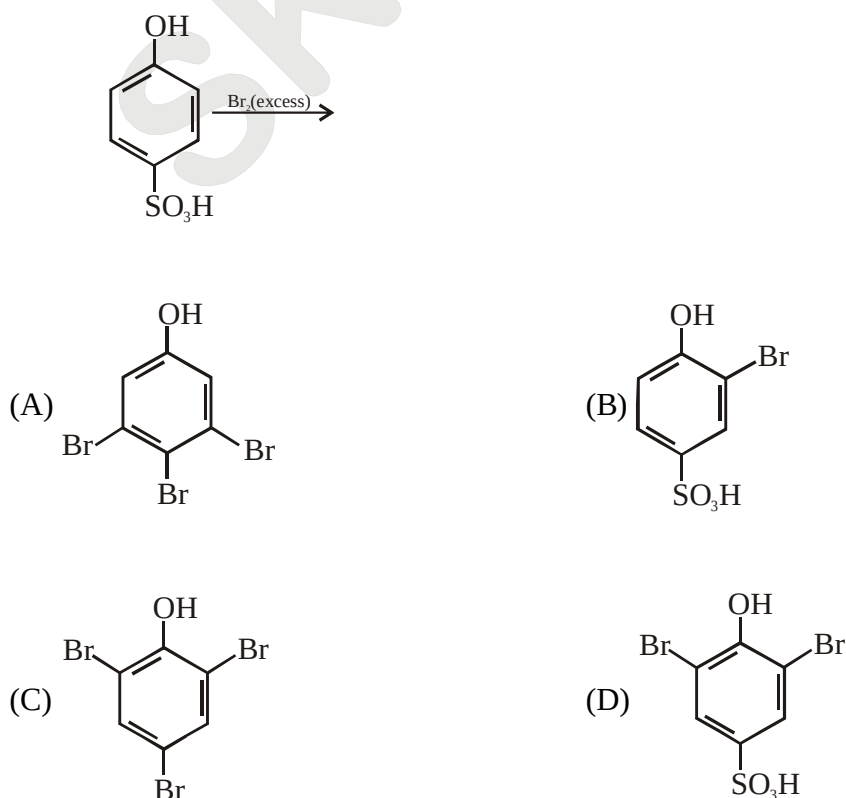
21. The organic compound that gives following qualitative analysis is [JEE MAINS ONLINE-2019]

Test	Inference
(i) Dil. HCl	Insoluble
(ii) NaOH solution	Soluble
(iii) Br ₂ /water	Decolourisation



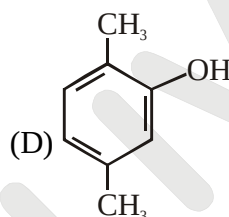
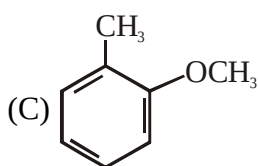
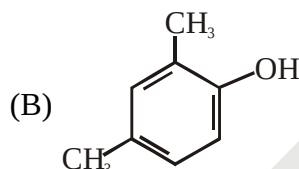
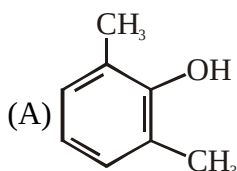
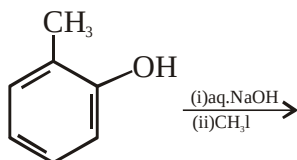
22. The major product of the following reaction is

[JEE MAINS ONLINE-2019]



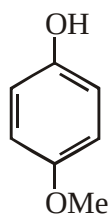
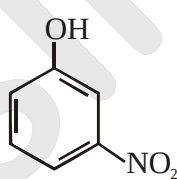
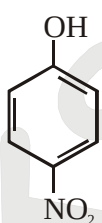
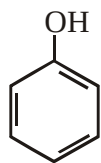
23. The major product of the following reaction is

[JEE MAINS ONLINE-2019]



24. The increasing order of the pK_a values of the following compound is

[JEE MAINS ONLINE-2019]



A

B

C

D

(A) D < A < C < B (B) B < C < A < D (C) C < B < A < D (D) B < C < D < A

25. A solution of phenol in chloroform when treated with aqueous NaOH gives compound P as a major product. The mass percentage of carbon in P is _____. (to the nearest integer)

(Atomic mass : C = 12; H = 1; O = 16)

[JEE MAINS ONLINE-2020]

[JEE-ADVANCED]

1. Acetamide is treated separately with the following reagents. Which of these would give methylamine?

[JEE 1983]

(A) PCl₅

(B) Sodalime

(C) NaOH + Br₂

(D) Hot, conc. H₂SO₄

2. Carbylamine test is performed in alc. KOH by heating a mixture of

[JEE 1984]

(A) chloroform and silver powder

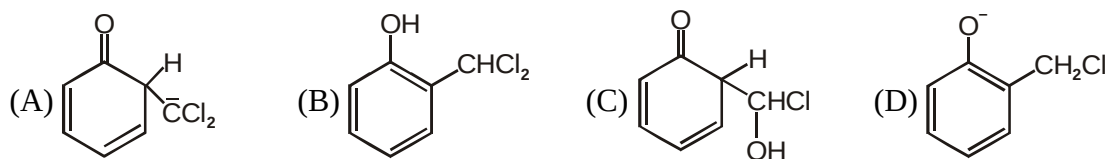
(B) trihalogenated methane and a primary amine

(C) an alkyl halide and a primary amine

(D) an alkyl cyanide and a primary amine

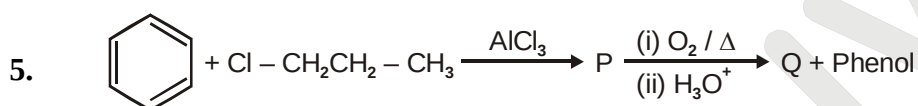
3. When phenol is reacted with CHCl_3 and NaOH followed by acidification, salicylaldehyde is formed. Which of the following species are involved in the above mentioned reaction as intermediates?

[JEE 1995]



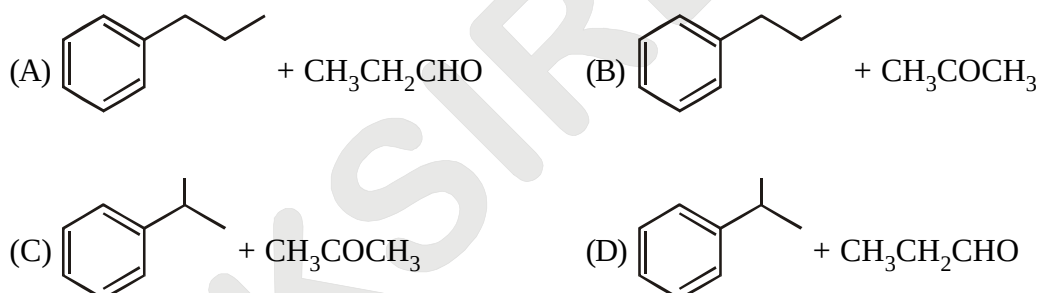
4. $\text{CH}_3\text{NH}_2 + \text{CHCl}_3 + \text{KOH} \rightarrow \text{Nitrogen containing compound} + \text{Cl} + \text{H}_2\text{O}$. Nitrogen containing compound is

[JEE 2006]



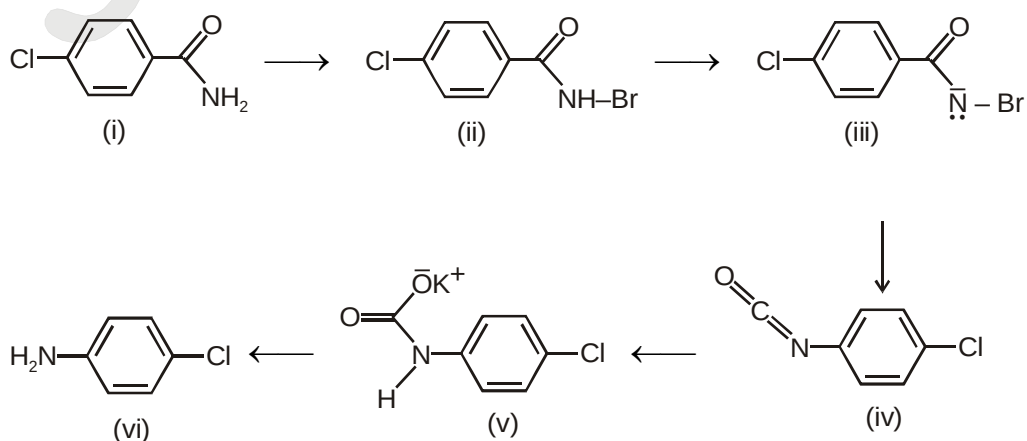
[JEE 2006]

P and Q are respectively

**Comprehension (Q.6 to Q.8)**

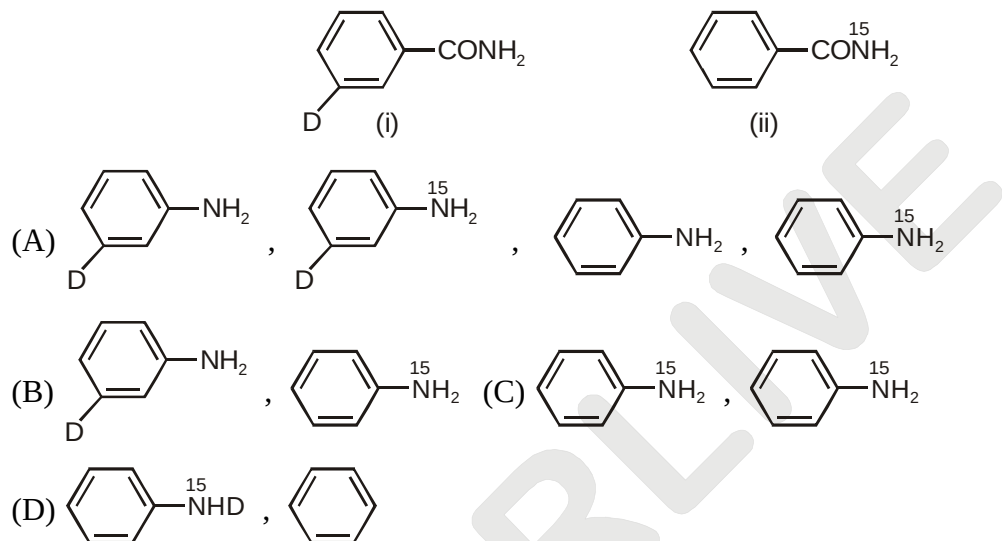
[JEE-2006]

RCONH_2 is converted into RNH_2 by means of Hofmann bromamide degradation.



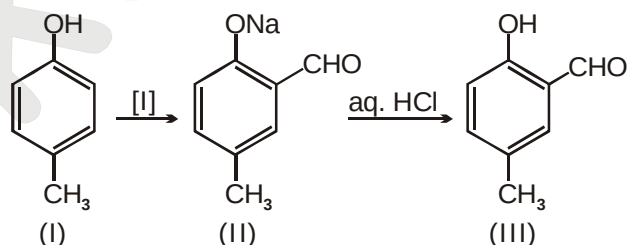
In this reaction, RCONHBr is formed from which this reaction has derived its name. Electron donating group at phenyl activates the reaction. Hofmann degradation reaction is an intramolecular reaction.

6. How can the conversion of (i) to (ii) be brought about?
 (A) KBr (B) KBr + CH₃ONa (C) KBr + KOH (D) Br₂ + KOH
7. Which is the rate determining step in Hofmann bromamide degradation?
 (A) Formation of (i) (B) Formation of (ii) (C) Formation of (iii) (D) Formation of (iv)
8. What are the constituent amines formed when the mixture of (i) and (ii) undergoes Hofmann bromamide degradation?



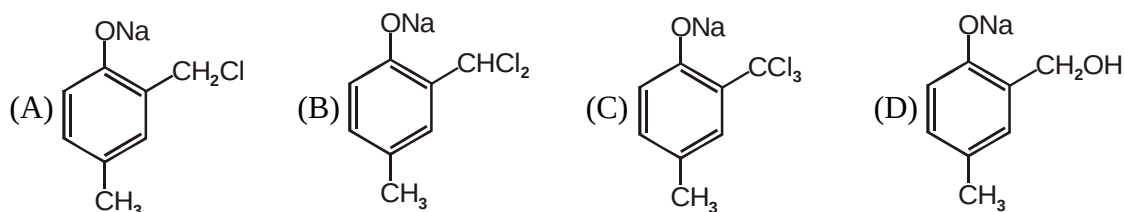
Question No.9 to 11 (3 questions) :

Reimer-Tiemann reaction introduces an aldehyde group, on to the aromatic ring of phenol, ortho to the hydroxyl group. This reaction involves electrophilic aromatic substitution. This is a general method for the synthesis of substituted salicylaldehydes as depicted below.



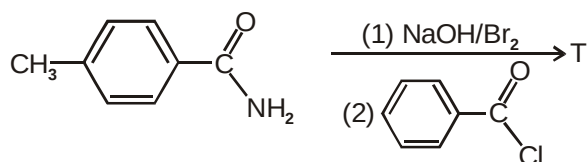
[JEE 2007]

9. Which one of the following reagents is used in the above reaction ?
 (A) aq NaOH + CH₃Cl (B) aq NaOH + CH₂Cl₂
 (C) aq NaOH + CHCl₃ (D) aq NaOH + CCl₄
10. The electrophile in this reaction is
 (A) :CHCl (B) ⁺CHCl₂ (C) :CCl₂ (D) ⁺CCl₃
11. The structure of the intermediate I is

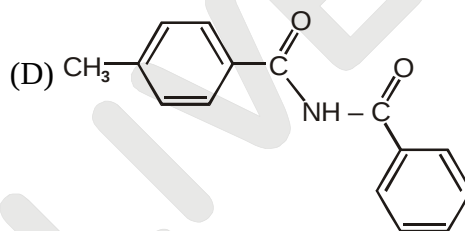
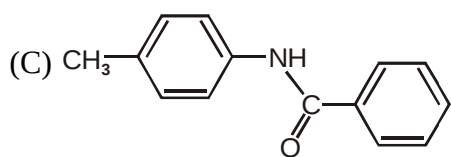
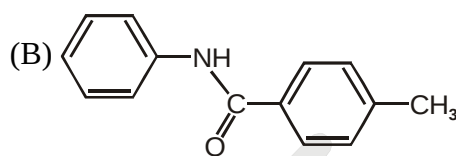
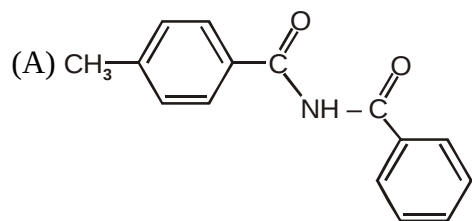


12. In the reaction

[JEE 2010]



the structure of the product T is

**Multiple Choice Questions :**

13. A new carbon-carbon bond formation is possible in

[JEE 1998]

- (A) wolf-kishner reduction (B) Friedel-Craft alkylation
(C) Clemmensen reduction (D) Reimer-Tiemann reaction

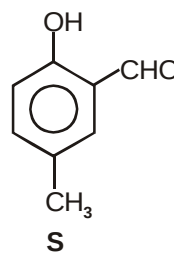
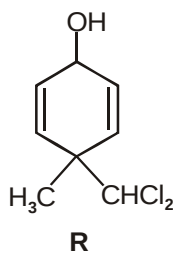
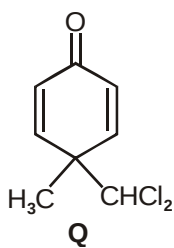
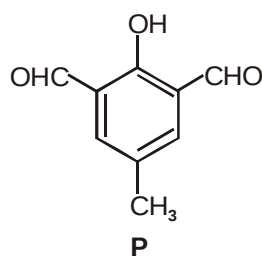
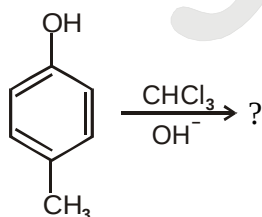
14. A positive carbylamine test is given by

[JEE 1999]

- (A) N, N-dimethylaniline (B) 2,4-dimethylaniline
(C) N-methyl-o-methylaniline (D) p-methylbenzylamine

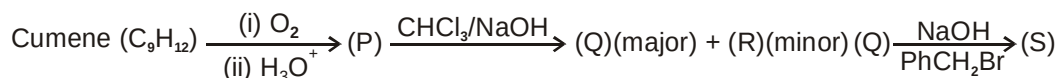
15. In the following reaction, the product(s) formed is(are)

[IIT-JEE Advance 2013]

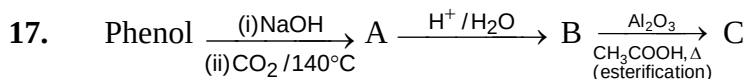


- (A) **P** (major) (B) **Q** (minor) (C) **R** (minor) (D) **S** (major)

16. The correct statement(s) about the following reaction sequence is(are) [JEE ADVANCE 2016]



- (A) R is steam volatile
 (B) Q gives dark violet coloration with 1% aqueous FeCl_3 solution
 (C) S gives yellow precipitate with 2,4-dinitrophenylhydrazine.
 (D) S gives dark violet coloration with 1% aqueous FeCl_3 solution.



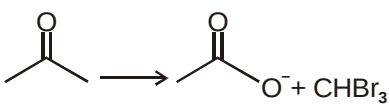
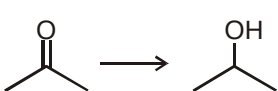
In this reaction, the end product C is:

- (A) salicylaldehyde
 (B) salicylic acid
 (C) phenyl acetate
 (D) aspirin

18.

Column I

Column II

- | | |
|---|---------------------------------|
| (A)  | (P) $\text{NaOH} + \text{Br}_2$ |
| (B)  | (Q) KOH |
| (C) $\text{R}-\text{C}(=\text{O})-\text{NH}_2 \longrightarrow \text{R}-\text{NH}_2$ | (R) LiAlH_4 |
| (D) $\text{R}-\text{C}(=\text{O})-\text{NH}_2 \longrightarrow \text{R}-\text{CH}_2-\text{NH}_2$ | (S) DIBAL-H |

Match the Column :

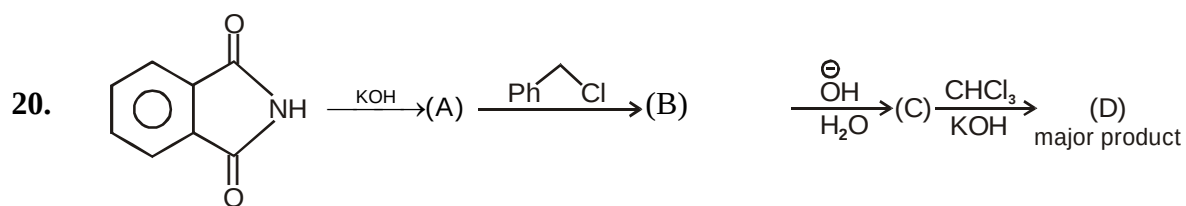
19.

Column-I

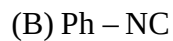
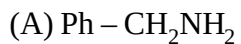
Column-II

[IIT 2009]

- | | |
|---|---|
| (A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CN}$ | (P) Reduction with $\text{Pd}-\text{C}/\text{H}_2$ |
| (B) $\text{CH}_3\text{CH}_2\text{OCOCH}_3$ | (Q) Reduction with SnCl_2/HCl |
| (C) $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_2\text{OH}$ | (R) Development of foul smell on treatment with chloroform and alcoholic KOH |
| (D) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$ (s) | (S) Reduction with diisobutylaluminium hydride (DIBAL-H) |
| | (T) Alkaline hydrolysis |

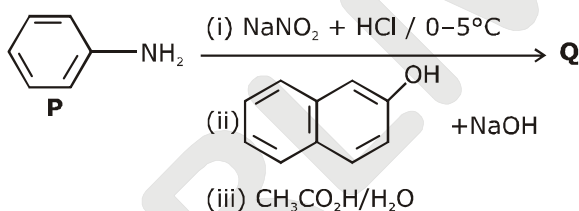


Structure of (D) is :



21. Consider the reaction sequence from P to Q shown below. The overall yield of the major product Q from P is 75%. What is the amount in grams of Q obtained from 9.3 mL of P ? (Use density of P = 1.00 g mL^{-1} ; Molar mass of C = 12.0, H = 1.0, O = 16.0 and N = 14.0 g mol^{-1})

[JEE-Advanced 2020]



ANSWER KEY**[JEE-MAIN/AIEEE]**

- | | | | | | | | | | | | | | |
|-----|---|-----|---|-----|---|-----|----|-----|---|-----|---|-----|---|
| 1. | C | 2. | A | 3. | D | 4. | B | 5. | D | 6. | A | 7. | A |
| 8. | A | 9. | A | 10. | D | 11. | A | 12. | D | 13. | D | 14. | D |
| 15. | B | 16. | C | 17. | D | 18. | B | 19. | B | 20. | C | 21. | B |
| 22. | C | 23. | C | 24. | B | 25. | 69 | | | | | | |

[JEE-ADVANCED]

- | | | | | | | | | | | | | | |
|-----|---------------------------------------|-----|-----------|-----|---|-----|---|-----|---|-----|------|-----|----|
| 1. | C | 2. | B | 3. | A | 4. | D | 5. | C | 6. | D | 7. | D |
| 8. | B | 9. | C | 10. | C | 11. | B | 12. | C | 13. | BD | 14. | BD |
| 15. | BD | 16. | BC or BCD | 17. | D | | | | | | | | |
| 18. | A → PQ ; B → R ; C → PQ ; D → R; | | | | | | | | | | | | |
| 19. | (A) → QST, (B) → ST, (C) → P, (D) → R | | | | | | | 20. | B | 21. | 18.6 | | |

[JEE-MAIN / AIEEE]

1. Match the catalysts to the correct processes:

[JEE Main 2015]

Catalyst**Process**(A) TiCl_3

(i) Wacker process

(B) PdCl_2

(ii) Ziegler-Natta Polymerization

(C) CuCl_2

(iii) Contact process

(D) V_2O_5

(iv) Deacon's process

(A) (A) - (ii), (B) - (iii), (C) - (iv), (D) - (i)

(B) (A) - (iii), (B) - (i), (C) - (ii), (D) - (iv)

(C) (A) - (iii), (B) - (ii), (C) - (iv), (D) - (i)

(D) (A) - (ii), (B) - (i), (C) - (iv), (D) - (iii)

2. The distillation technique most suited for separating glycerol from spent-lye in the soap industry is :

(A) Distillation under reduced pressure

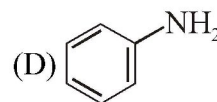
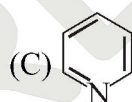
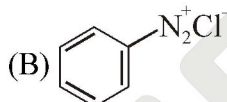
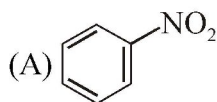
(B) Simple distillation

[JEE Main 2016]

(C) Fractional distillation

(D) Steam distillation

3. Which of the following compounds will be suitable for Kjeldahl's method for nitrogen estimation?



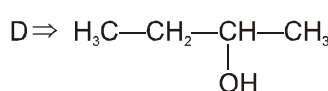
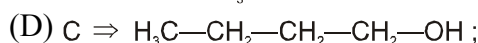
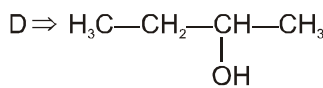
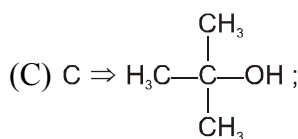
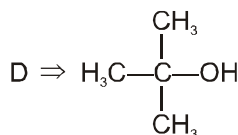
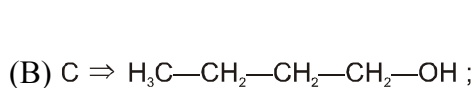
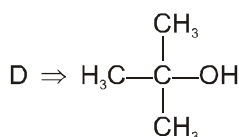
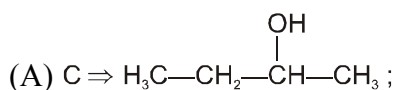
[JEE Main 2018]

4. Two compounds A and B with same molecular formula (
- $\text{C}_3\text{H}_6\text{O}$
-) undergo Grignard reaction with methylmagnesium bromide to give products C and D. Products C and D show following chemical tests.

Test	C	D
Ceric ammonium nitrate Test	Positive	Positive
Lucas Test	Turbidity obtained after five minutes	Turbidity obtained immediately
Iodoform Test	Positive	Negative

C and D respectively are:

[JEE Main (Online) 2020]



5. In an estimation of bromine by Carius method, 1.6 g of an organic compound gave 1.88 g of AgBr. The mass percentage of bromine in the compound is..... [JEE Main (Online) 2020]
(Atomic mass, Ag = 108, Br = 80 g mol⁻¹)

6. Match the following : [JEE Main (Online) 2020]

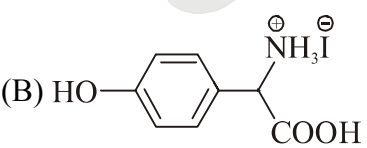
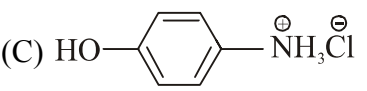
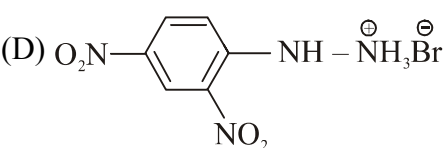
Test / Method	Reagent
(i) Lucas Test	(a) C ₆ H ₅ SO ₂ Cl / aq. KOH
(ii) Dumas method	(b) HNO ₃ / AgNO ₃
(iii) Kjeldahl's method	(c) CuO / CO ₃
(iv) Hinsberg Test	(d) Conc. HCl and ZnCl ₂
	(e) H ₂ SO ₄
(A) (i)-(d), (ii)-(c), (iii)-(b), (iv)-(e)	(B) (i)-(d), (ii)-(c), (iii)-(e), (iv)-(a)
(C) (i)-(b), (ii)-(a), (iii)-(c), (iv)-(d)	(D) (i)-(b), (ii)-(d), (iii)-(e), (iv)-(a)

[JEE-ADVANCED]

1. Match the compounds/ion in column I with their properties/ reaction in Column II. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the OMR. [JEE 2007]

Column I	Column II
(A) C ₆ H ₅ CHO	(P) gives precipitate with 2,4-dinitrophenylhydrazine
(B) CH ₃ C≡CH	(Q) gives precipitate with AgNO ₃
(C) CN ⁻	(R) is a nucleophile
(D) I ⁻	(S) is involved in cyanohydrin formation

2. Match the compounds in Column I with their characteristic test(s)/reaction(s) given in Column II. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the OMR.

Column I	Column II
(A) $\text{H}_2\text{N}-\overset{\oplus}{\text{N}}\text{H}_3^+\text{Cl}^-$	(P) sodium fusion extract of the compound gives Prussian blue colour with FeSO ₄
(B) 	(Q) gives positive FeCl ₃ test
(C) 	(R) gives white precipitate with AgNO ₃
(D) 	(S) reacts with aldehydes to form the corresponding hydrazone derivative

3. An organic compound (C₈H₁₀O₂) rotates plane-polarized light. It produces pink color with neutral FeCl₃ solution. What is the total number of all the possible isomers for this compound ?

[JEE Advanced 2020]

ANSWER KEY**[JEE-MAIN/AIEEE]**

1. D 2. A 3. D 4. A 5. 50.00 6. B

[JEE-ADVANCED]

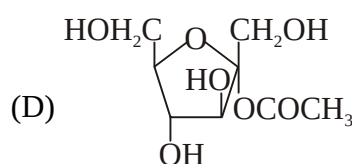
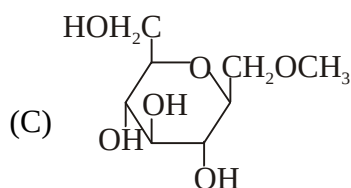
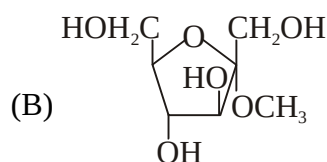
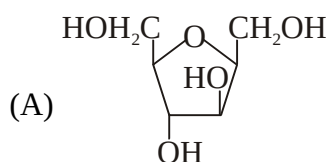
1. (A) P, S; (B) Q; (C) Q, R, S; (D) Q, R 2. (A) R, S (B) P, Q (C) P, Q, R (D) P, S
3. 6

SKSIRLIVE

[JEE-MAIN / AIEEE]

1. RNA contains - [AIEEE-2002]
 (A) Urasil, Ribose (B) Thiamine, Ribose
 (C) Cytocine, Deoxyribose (D) Adenine, Deoxyribose
2. Complete hydrolysis of cellulose gives - [AIEEE-2003]
 (A) D-glucose (B) L-glucose (C) D-fructose (D) D-ribose
3. The reason for double helical structure of DNA is operation of - [AIEEE-2003]
 (A) Hydrogen bonding (B) Electrostatic attractions
 (C) vander Waal's forces (D) Dipole-dipole interaction
4. Coordination compounds have great importance in biological systems, In this context which of the following statements is incorrect ? [AIEEE-2004]
 (A) Chlorophylls are green pigments in plants and contain calcium
 (B) haemoglobin is the red pigment of blood and contains iron
 (C) Cyanocobalamin is B₁₂ and contains cobalt
 (D) Carboxypeptidase-A is an enzyme and contains zinc
5. Which base is present in RNA but not in DNA ? [AIEEE-2004]
 (A) Uracil (B) Cytosine (C) Guanine (D) Thymine
6. Insulin production and its action in human body are responsible for the level of diabetes. This compound belongs to which of the following categories ? [AIEEE-2004]
 (A) A co-enzyme (B) A hormone (C) An enzyme (D) An antibiotic
7. Identify the correct statement regarding enzymes : [AIEEE-2004]
 (A) Enzymes are specific biological catalysts that can normally function at very high temperatures (T~ 1000 K)
 (B) Enzymes are normally heterogeneous catalysts that are very specific in their action
 (C) Enzymes are specific biological catalysts that cannot be poisoned
 (D) Enzymes are specific biological catalysts that possess well-defined active sites
8. In both DNA and RNA, heterocyclic base and phosphate ester linkages are at - [AIEEE-2005]
 (A) C₂' and C₅' respectively of the sugar molecule
 (B) C₅' and C₂' respectively of the sugar molecule
 (C) C₅' and C₁' respectively of the sugar molecule
 (D) C₁' and C₅' respectively of the sugar molecule
9. The pyrimidine bases present in DNA are - [AIEEE 2006]
 (A) cytosine and guanine (B) cytosine and thymine
 (C) cytosine and uracil (D) cytosine and adenine
10. The term anomers of glucose refers to - [AIEEE 2006]
 (A) a mixture of (D)-glucose and (L)-glucose
 (B) enantiomers of glucose
 (C) isomers of glucose that differ in configuration at carbon one (C-1)
 (D) isomers of glucose that differ in configurations at carbons one and four (C-1 and C-4)

11. The secondary structure of a protein refers to – [AIEEE 2007]
(A) α -helical backbone (B) hydrophobic interactions
(C) sequence of α -amino acids (D) fixed configuration of the polypeptide backbone
12. α -D-(+)-glucose and β -D-(+)-glucose are [AIEEE 2008]
(A) epimers (B) anomers (C) enantiomers (D) conformers
13. The two functional groups present in a typical carbohydrate are : [AIEEE 2009]
(A) – OH and –COOH (B) – CHO and –COOH
(C) $>C=O$ and –OH (D) – OH and –CHO
14. Which of the following compounds can be detected by Molisch's test ? [AIEEE 2012]
(A) Amines (B) Primary alcohols (C) Nitro compounds (D) Sugars
15. Which one of the following statement is correct? [AIEEE 2012]
(A) All amino acids except glycine are optically active.
(B) All amino acids except glutamic acid are optically active.
(C) All amino acids except lysine are optically active.
(D) All amino acids are optically active.
16. Synthesis of each molecule of glucose in photosynthesis involves: [JEE Main 2013]
(A) 10 molecules of ATP (B) 8 molecules of ATP
(C) 6 molecules of ATP (D) 18 molecules of ATP
17. Which one of the following bases is not present in DNA? [JEE Main 2014]
(A) Adenine (B) Cytosine (C) Thymine (D) Quinoline
18. Thiol group is present in : [JEE Main 2016]
(A) Methionine (B) Cytosine (C) Cystine (D) Cysteine
19. Which of the following compounds will behave as a reducing sugar in an aqueous KOH solution ? [JEE Main 2017]



20. Glucose on prolonged heating with HI gives: [JEE Main 2018]
(A) Hexanoic acid (B) 6-iodohexanal (C) n-Hexane (D) 1-Hexene

21. Which of the given statements is incorrect about glycogen ? [JEE Mains 2019]
(A) It is straight chain polymer similar to amylose
(B) Only α -linkages are present in the molecule
(C) It is present in animal cells
(D) It is present in some yeast and fungi
22. Which of the following statement is not true about RNA? [JEE Mains 2019]
(A) It controls the synthesis of protein
(B) It has always double stranded α -helix structure
(C) It usually does not replicate
(D) It is present in the nucleus of the cell
23. Number of stereo-centres present in linear and cyclic structures of glucose are respectively [JEE Mains 2019]
(A) 4 and 5 (B) 4 and 4 (C) 5 and 4 (D) 5 and 5
24. Amylopectin is composed of [JEE Mains 2019]
(A) β -D-glucose, C_1 - C_4 and C_2 - C_6 linkages (B) α -D-glucose, C_1 - C_4 and C_2 - C_6 linkages
(C) β -D-glucose, C_1 - C_4 and C_1 - C_6 linkages (D) α -D-glucose, C_1 - C_4 and C_1 - C_6 linkages
25. The peptide that gives positive ceric ammonium nitrate and carbylamine tests is [JEE Mains 2019]
(A) Lys-Asp (B) Ser-Lys (C) Gln-Asp (D) Asp-Gln
26. Which of the following statement is not true about sucrose ? [JEE Mains 2019]
(A) It is also named as invert sugar
(B) The glycosidic linkage is present between C_1 of α -glucose and C_1 of β -fructose
(C) It is a non-reducing sugar
(D) On hydrolysis, it produces glucose and fructose
27. Fructose and glucose can be distinguished by [JEE Mains 2019]
(A) Fehling's test (B) Barfoed's test (C) Benedict's test (D) Seliwanoff's test
28. Maltose on treatment with dilute HCl gives [JEE Mains 2019]
(A) D-glucose and D-fructose (B) D-fructose
(C) D-galactose (D) D-glucose
29. Which of the following tests cannot be used for identifying amino acids ? [JEE Mains 2019]
(A) Barfoed test (B) Ninhydrin test (C) Xanthoproteic test (D) Biuret test
30. The increasing order of pK_a of the following amino acids in aqueous solution is Gly, Asp, Lys, Arg [JEE Mains 2019]
(A) Asp < Gly < Arg < Lys (B) Arg < Lys < Gly < Asp
(C) Gly < Asp < Arg < Lys (D) Asp < Gly < Lys < Arg

31. Consider the following reactions : [JEE Mains 2020]
- (i) $\text{Glucose} + \text{ROH} \xrightarrow{\text{dry HCl}} \text{Acetal} \xrightarrow[\text{(CH}_3\text{CO)}_2\text{O}]{\text{x.eq.of}} \text{acetyl derivative}$
- (ii) $\text{Glucose} \xrightarrow{\text{Ni/H}_2} \text{A} \xrightarrow[\text{(CH}_3\text{CO)}_2\text{O}]{\text{y.eq.of}} \text{acetyl derivative}$
- (iii) $\text{Glucose} \xrightarrow[\text{(CH}_3\text{CO)}_2\text{O}]{\text{z.eq.of}} \text{acetyl derivative}$
- 'x', 'y' and 'z' in these reactions are respectively :
- (A) 5, 6 & 6 (B) 4, 5 & 5 (C) 4, 6 & 5 (D) 5, 4 & 5
32. The correct observation in the following reaction is: [JEE Mains 2020]
- $$\text{Sucrose} \xrightarrow[\text{Cleavage (Hydrolysis)}]{\text{Glycosidic bond}} \text{A} + \text{B} \xrightarrow[\text{Reagent}]{\text{Seliwanoff}} ?$$
- (A) Formation of blue colour (B) Formation of violet colour
(C) Formation of red colour (D) Gives no colour
33. The number of >C=O groups present in a tripeptide Asp–Glu–Lys is [JEE Mains 2020]
34. What are the functional groups present in the structure of maltose ? [JEE Mains 2020]
(A) Two acetals (B) One acetal and one ketal
(C) One ketal and one hemiketal (D) One acetal and one hemiacetal
35. Which of the following is not an essential amino acid ? [JEE Mains 2020]
(A) Tyrosine (B) Valine (C) Leucine (D) Lysine
36. The number of chiral carbon(s) present in peptide, Ile–Arg–Pro, is [JEE Mains 2020]
37. The number of chiral carbons present in sucrose is..... [JEE Mains 2020]
38. Which one of the following statement is not true ? [JEE Mains 2020]
(A) Lactose contains α -glycosidic linkage between C_1 of galactose and C_4 of glucose
(B) On acid hydrolysis, lactose gives one molecule of D(+) - glucose and one molecule of D(+)-galactose.
(C) Lactose is a reducing sugar and it gives Fehling's test.
(D) Lactose ($\text{C}_{11}\text{H}_{22}\text{O}_{11}$) is a disaccharide and it contains 8 hydroxyl groups.

[JEE-ADVANCED]

1. Give the structures of the products in each of the following reactions. [JEE 2000]

$$\text{Sucrose} \xrightarrow{\text{H}^+} \text{A} + \text{B}$$
2. Aspartame, an artificial sweetener, is a peptide and has the following structure: [JEE 2001]

$$\begin{array}{c} \text{CH}_2 - \text{C}_6\text{H}_5 \\ | \\ \text{H}_2\text{N} - \text{CH} - \text{CONH} - \text{CH} - \text{COOCH}_3 \\ | \\ \text{CH}_2 - \text{COOH} \end{array}$$
 - Identify the four functional groups.
 - Write the zwitterionic structure.
 - Write the structures of the amino acids obtained from the hydrolysis of aspartame.
 - Which of the two amino acids is more hydrophobic?
3. Following two amino acids lysine and glutamine form dipeptide linkage. What are two possible dipeptides?

$$\text{H}_2\text{N} - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}(\text{NH}_2) - \text{COOH} + \text{HOOC} - \text{CH}(\text{NH}_2) - \text{CH}_2 - \text{CH}_2 - \text{COOH}$$
 [JEE 2003]
4. The structure of D-Glucose is as follows- [JEE 2004]

$$\begin{array}{c} \text{CHO} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{HO} \end{array}$$
 - Draw the structure of L -Glucose
 - Give the reaction of L - Glucose with Tollen's reagent.
5. Which of the following pairs give positive Tollen's test? [JEE 2004]
 (A) Glucose, sucrose (B) Glucose, fructose
 (C) Hexanal, Acetophenone (D) Fructose, sucrose
6. Two forms of D – glucopyranose, are called. [JEE 2005]
 (A) Enantiomers (B) Anomers (C) Epimers (D) Diastereomers
7. When benzene sulphonic acid and p-nitrophenol are treated with NaHCO_3 , the gases released respectively are [JEE 2006]
 (A) SO_2 , NO_2 (B) SO_2 , NO (C) SO_2 , CO_2 (D) CO_2 , CO_2
8. **Statement-1** : Glucose gives a reddish-brown precipitate with Fehling's solution. [JEE 2007]
because
Statement-2 : Reaction of glucose with Fehling's solution gives Cu_2O and gluconic acid.
 (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.

9. **Statement-1** : p-Hydroxybenzoic acid has a lower boiling point than o-hydroxybenzoic acid.
because [JEE 2007]
Statement-2 : o-Hydroxybenzoic acid has intramolecular hydrogen bonding.
 (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.

10. Match the chemical substances in Column I with type of polymers / type of bonds in Column II. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS. [JEE 2007]

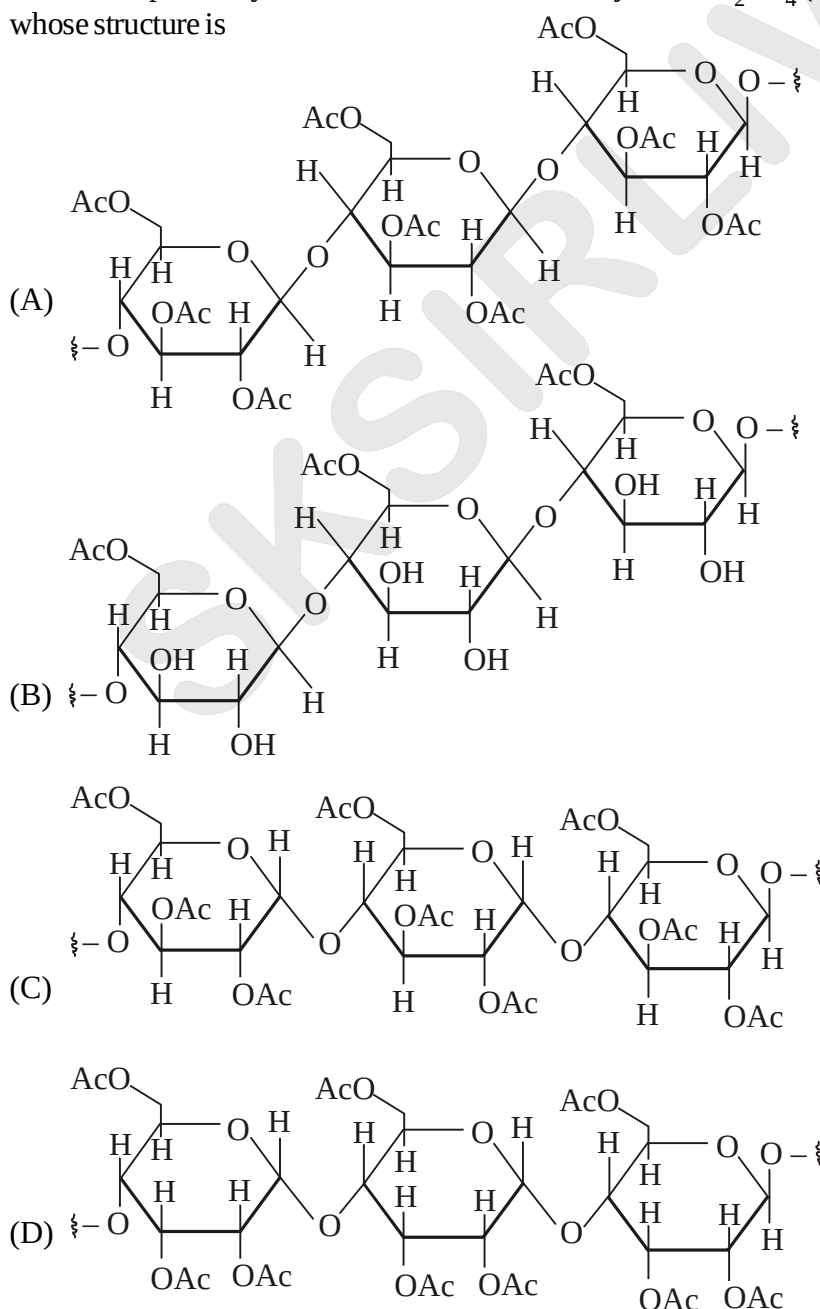
Column I

- (A) Cellulose
 (B) Nylon-6, 6
 (C) Protein
 (D) Sucrose

Column II

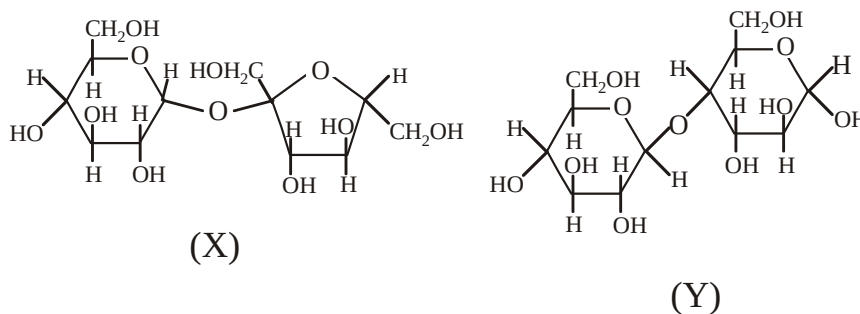
- (P) Natural polymer
 (Q) Synthetic polymer
 (R) amide linkage
 (S) Glycoside linkage

11. Cellulose upon acetylation with excess acetic anhydride / H_2SO_4 (catalytic) gives cellulose triacetate whose structure is [JEE 2008]



12. The correct statement(s) about the following sugars X and Y is(are)

[JEE 2009]



- (A) X is a reducing sugar and Y is a non-reducing sugar
 (B) X is a non-reducing sugar and Y is a reducing sugar
 (C) The glucosidic linkages in X and Y are α and β , respectively.
 (D) The glucosidic linkages in X and Y are β and α , respectively.

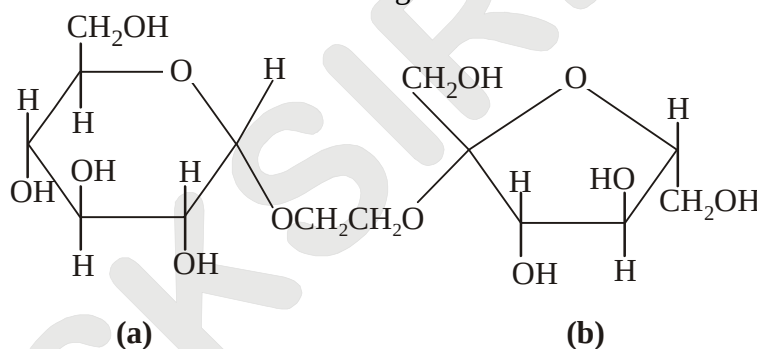
13. Among cellulose, poly vinyl chloride, nylon and natural rubber, the polymer in which the intermolecular force of attraction is weakest in

[JEE 2009]

- (A) Nylon (B) Poly vinyl chloride
 (C) Cellulose (D) Natural Rubber

14. The correct statement about the following disaccharide is

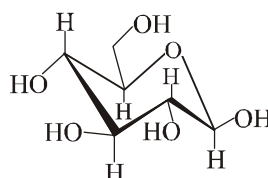
[JEE 2010]



- (A) Ring (a) is pyranose with α -glycosidic link.
 (B) Ring (a) is furanose with α -glycosidic link
 (C) Ring (b) is furanose with α -glycosidic link
 (D) Ring (b) is pyranose with β -glycosidic link

15. The following carbohydrate is

[JEE 2011]

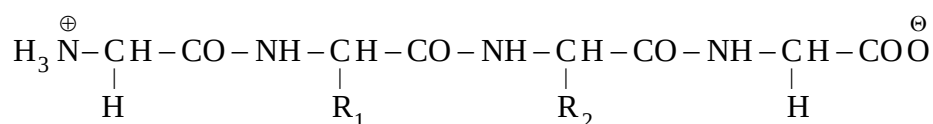


- (A) a ketohexose (B) an aldohexose (C) an α -furanose (D) an α -pyranose

16. A decapeptide (Mol. Wt. 796) on complete hydrolysis gives glycine (Mol. Wt. 75), alanine and phenylalanine. Glycine contributes 47.0 % to the total weight of the hydrolysed products. The number of glycine units present in the decapeptide is

[JEE 2011]

17. The substituents R_1 and R_2 for nine peptides are listed in the table given below. How many of these peptides are positively charged at pH = 7.0? [JEE 2012]

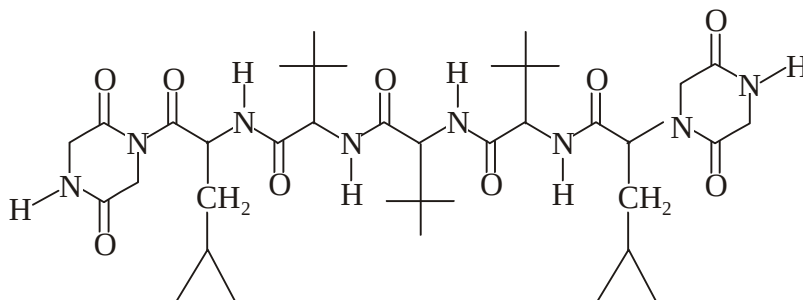


Peptide	R_1	R_2
I	H	H
II	H	CH_3
III	CH_2COOH	H
IV	CH_2CONH_2	$(\text{CH}_2)_4\text{NH}_2$
V	CH_2CONH_2	CH_2CONH_2
VI	$(\text{CH}_2)_4\text{NH}_2$	$(\text{CH}_2)_4\text{NH}_2$
VII	CH_2COOH	CH_2CONH_2
VIII	CH_2OH	$(\text{CH}_2)_4\text{NH}_2$
IX	$(\text{CH}_2)_4\text{NH}_2$	CH_3

18. When the following aldohexose exists in its D-configuration, the total number of stereoisomers in its pyranose form is [JEE 2012]

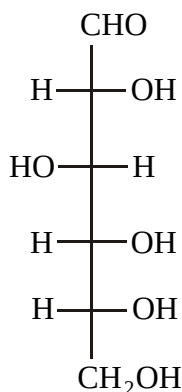


19. A tetrapeptide has $-\text{COOH}$ group on alanine. This produces glycine (Gly), Valine (Val), Phenyl Alanine (Phe) and Alanine (Ala), on complete hydrolysis. For this tetrapeptide, the number of possible sequences (primary structures) with $-\text{NH}_2$ group attached to a chiral center is [JEE Advance 2013]
20. The total number of distinct naturally occurring amino acids obtained by complete acidic hydrolysis of the peptide shown below is [JEE Advance 2014]

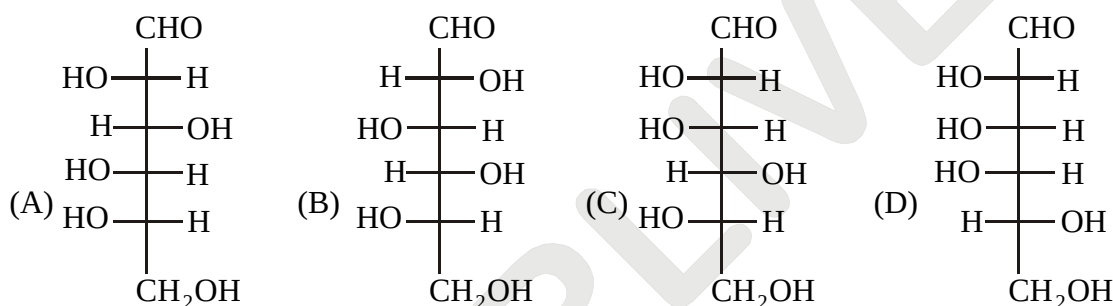


21. The structure of D-(+) glucose is :

[JEE Advance 2015]



The structure of L-(−) glucose is :



22. For 'invert sugar', the correct statement(s) is (are)

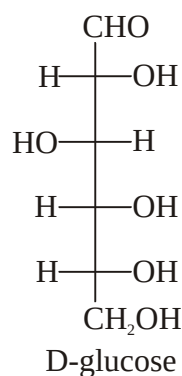
[JEE Advance 2016]

[Given : specific rotations of (+)-sucrose, (+)-maltose, L-(−)-glucose and L-(+)-fructose in aqueous solution are +66°, +140°, −52° and +92°, respectively]

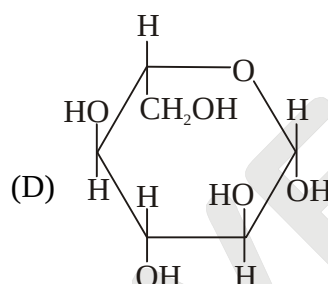
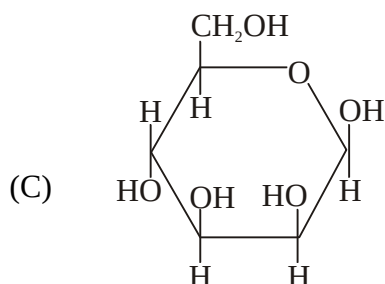
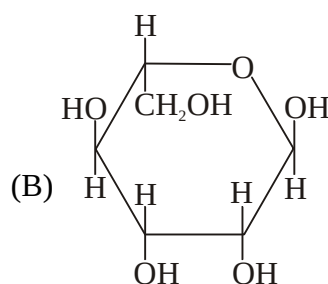
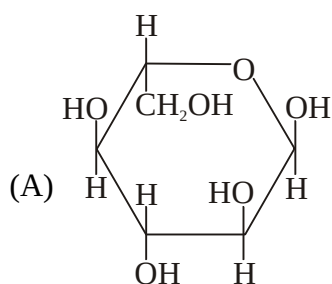
- (A) 'invert sugar' is prepared by acid catalysed hydrolysis of maltose
- (B) 'invert sugar' is an equimolar mixture of D-(+)-glucose and D-(−)-fructose
- (C) specific rotation of 'invert sugar' is −20°
- (D) on reaction with Br₂ water, 'invert sugar' forms saccharic acid as one of the products

23. The Fischer presentation of D-glucose is given below.

[JEE Advance 2018]



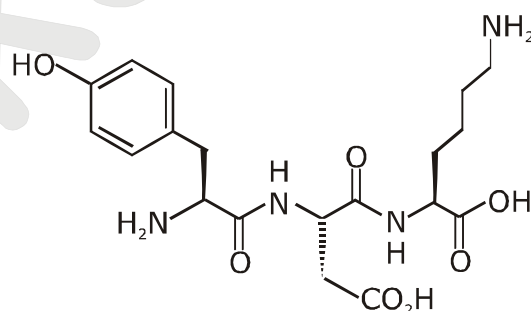
The correct structure(s) of β-L-glucopyranose is (are)



24. Which of the following statements(s) is (are) true ? [JEE Advanced 2019]

- (A) Oxidation of glucose with bromine water gives glutamic acid
- (B) The two six-membered cyclic hemiacetal forms of D-(+)-glucose are called anomers
- (C) Monosaccharides cannot be hydrolysed to give polyhydroxy aldehydes and ketones
- (D) Hydrolysis of sucrose gives dextrorotatory glucose and laevorotatory fructose

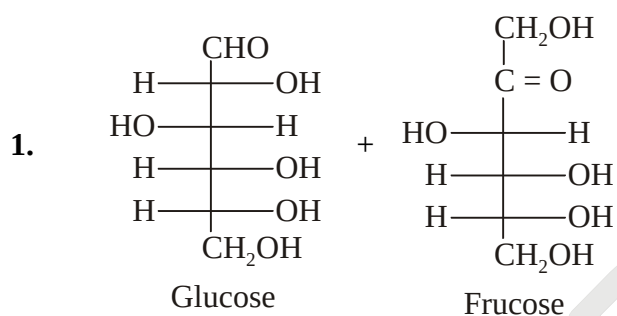
25. The structure of a peptide is given below : [JEE Advanced 2020]



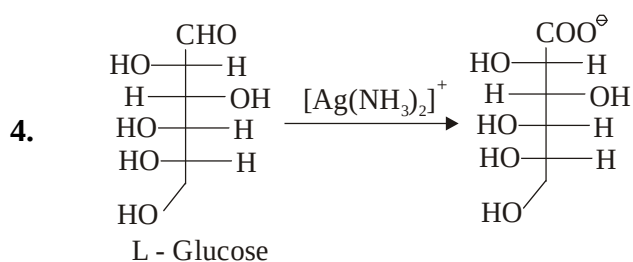
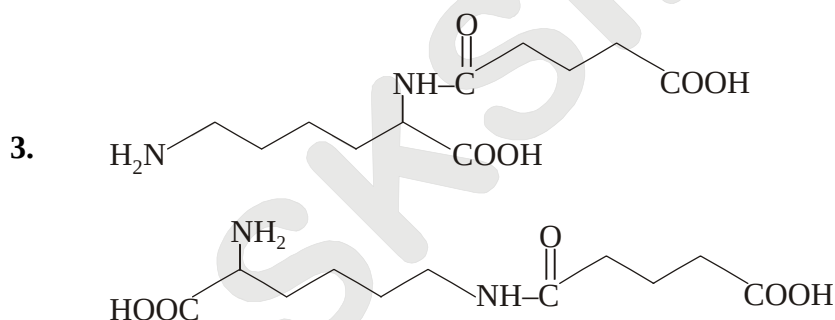
If the absolute values of the net charge of the peptide at pH = 2, pH = 6, and pH = 11 are $|z_1|$, $|z_2|$, and $|z_3|$, respectively, then what is $|z_1| + |z_2| + |z_3|$?

ANSWER KEY**[JEE-MAIN/AIEEE]**

1.	A	2.	A	3.	A	4.	A	5.	A	6.	B	7.	D
8.	D	9.	B	10.	C	11.	A	12.	B	13.	C	14.	D
15.	A	16.	D	17.	D	18.	D	19.	D	20.	C	21.	A
22.	B	23.	A	24.	D	25.	B	26.	B	27.	D	28.	D
29.	A	30.	D	31.	C	32.	C	33.	5	34.	4	35.	A
36.	4	37.	9	38.	A								

[JEE-ADVANCED]

2. (i) Amine, carboxylic acid, Amide, Ester



5.	B	6.	B	7.	D	8.	A	9.	D				
10.	(A) P,S; (B) Q, R; (C) P, R; (D) PRS	11.	A	12.	BC	13.	D	14.	A				
15.	B	16.	6	17.	4	18.	8	19.	4	20.	1	21.	A
22.	BC	23.	D	24.	BCD	25.	5						

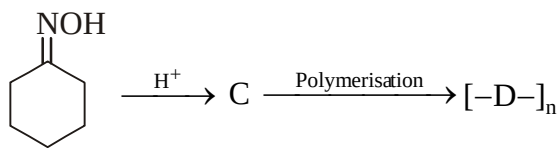
[JEE-MAIN / AIEEE]

1. Monomers are converted to polymer by – [AIEEE-2003]
 (A) Hydrolysis of monomers
 (B) Condensation reaction between monomers
 (C) Protonation of monomers
 (D) None is correct
2. Nylon threads are made of – [AIEEE-2003]
 (A) Polyamide polymer (B) Polyethylene polymer
 (C) Polyvinyl polymer (D) Polyester polymer
3. Which of the following is a polyamide ? [AIEEE-2005]
 (A) Nylon-66 (B) Teflon (C) Bakelite (D) Terylene
4. Which of the following is fully fluorinated polymer- [AIEEE-2005]
 (A) Teflon (B) Neoprene (C) PVC (D) Thiokol
5. Bakelite is obtained from phenol by reacting with [AIEEE 2008]
 (A) CH_3CHO (B) CH_3COCH_3 (C) HCHO (D) $(\text{CH}_2\text{OH})_2$
6. Buna-N synthetic rubber is a copolymer of - [AIEEE 2009]
 (A) $\text{H}_2\text{C} = \text{CH} - \overset{\text{Cl}}{\underset{|}{\text{C}}} = \text{CH}_2$ and $\text{H}_2\text{C} = \text{CH} - \text{CH} = \text{CH}_2$
 (B) $\text{H}_2\text{C} = \text{CH} - \text{CH} = \text{CH}_2$ and $\text{H}_5\text{C}_6 - \text{CH} = \text{CH}_2$
 (C) $\text{H}_2\text{C} = \text{CH} - \text{CN}$ and $\text{H}_2\text{C} = \text{CH} - \text{CH} = \text{CH}_2$
 (D) $\text{H}_2\text{C} = \text{CH} - \text{CN}$ and $\text{H}_2\text{C} = \text{CH} - \overset{\text{CH}_3}{\underset{|}{\text{C}}} = \text{CH}_2$
7. The species which can best serve as an initiator for the cationic polymerization is [AIEEE 2012]
 (A) AlCl_3 (B) BuLi (C) LiAlH_4 (D) HNO_3
8. Which one is classified as a condensation polymer? [JEE Main 2014]
 (A) Neoprene (B) Teflon (C) Acrylonitrile (D) Dacron
9. Which polymer is used in the manufacture of paints and lacquers ? [JEE Main 2015]
 (A) Polypropene (B) Poly vinyl chloride
 (C) Bakelite (D) Glyptal

10. Which of the following statements about low density polythene is FALSE? [JEE Main 2016]
(A) It is used in the manufacture of buckets, dust-bins etc.
(B) Its synthesis requires high pressure.
(C) It is a poor conductor of electricity.
(D) Its synthesis requires dioxygen or a peroxide initiator as a catalyst.
11. The formation of which of the following polymers involves hydrolysis reaction ? [JEE Main 2017]
(A) Bakelite (B) Nylon 6, 6 (C) Terylene (D) Nylon 6
12. Which of the following polymers is not obtained by condensation polymerisation? [JEE Main 2020]
(A) Nylon 6, 6 (B) Buna – N (C) Bakelite (D) Nylon 6
13. Consider the Assertion and Reason given below : [JEE Main 2020]
Assertion (A) : Ethene polymerized in the presence of Ziegler Natta Catalyst at high temperature and pressure is used to make buckets and dustbins.
Reason (R) : High density polymers are closely packed and are chemically inert. Choose the correct answer from the following :
(A) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(B) Both (A) and (R) are correct and (R) is the correct explanation of (A).
(C) (A) is correct but (R) is wrong.
(D) (A) and (R) both are wrong.
14. The correct match between Item-I and Item-II is : [JEE Main 2020]
- | Item-I | Item-II |
|--|---|
| (a) Natural rubber | (I) 1, 3-butadiene + styrene |
| (b) Neoprene | (II) 1, 3-butadiene |
| (c) Buna-N | (III) Chloroprene |
| (d) Buna-S | (IV) Isoprene |
| (A) (a)-(IV), (b)-(III), (c)-(II), (d)-(I) | (B) (a)-(III), (b)-(IV), (c)-(III), (d)-(I) |
| (C) (a)-(IV), (b)-(III), (c)-(I), (d)-(II) | (D) (a)-(III), (b)-(IV), (c)-(I), (d)-(II) |

[JEE-ADVANCED]

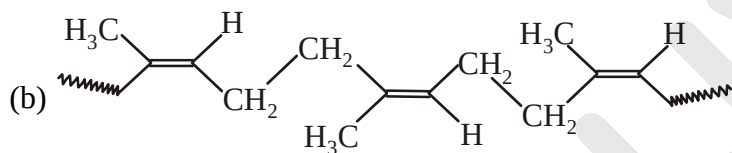
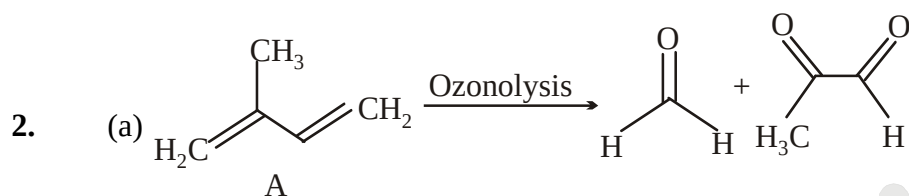
1. Give the structures of the products in each of the following reactions :

[JEE 2000]

2. Monomer A of a polymer on ozonolysis yields two moles of $HCHO$ and one mole of CH_3COCHO .
(a) Deduce the structure of A. **[JEE 2005]**
(b) Write the structure of “all cis” – form of polymer of compound A.
3. On complete hydrogenation, natural rubber produces **[JEE Advance 2016]**
(A) ethylene-propylene copolymer (B) vulcanised rubber
(C) polypropylene (D) polybutylene
4. Choose the correct option(s) from the following: **[JEE Advance 2019]**
(A) Cellulose has only α -D-glucose units that are joined by glycosidic linkages
(B) Teflon is prepared by heating tetrafluoroethene in presence of a persulphate catalyst at high pressure
(C) Natural rubber is polyisoprene containing *trans* alkene units
(D) Nylon-6 has amide linkages

ANSWER KEY**[JEE-MAIN/AIEEE]**

- | | | | | | | | | | | | | | |
|----|---|----|---|-----|---|-----|---|-----|---|-----|---|-----|---|
| 1. | B | 2. | A | 3. | A | 4. | A | 5. | C | 6. | C | 7. | A |
| 8. | D | 9. | D | 10. | A | 11. | D | 12. | B | 13. | B | 14. | A |

[JEE-ADVANCED]

3. A
4. BD

[JEE-MAIN / AIEEE]

- The compound formed in the positive test for nitrogen with the Lassaigne solution of an organic compound is - **[AIEEE - 2004]**
 (A) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ (B) $\text{Na}_3[\text{Fe}(\text{CN})_6]$
 (C) $\text{Fe}(\text{CN})_3$ (D) $\text{Na}_4[\text{Fe}(\text{CN})_5\text{NOS}]$
- Which one of the following method is neither meant for the synthesis nor for separation of amines ? (A) Hinsberg method (B) Hofmann method **[AIEEE-2005]**
 (C) Wurtz reaction (D) Curtius reaction
- Which one of the following types of drugs reduces fever ? **[AIEEE - 2005]**
 (A) Tranquiliser (B) Antibiotic (C) Antipyretic (D) Analgesic
- The hydrocarbon which can react with sodium in liquid ammonia is : **[AIEEE-2008]**
 (A) $\text{CH}_3\text{CH}_2\text{C} \equiv \text{CH}$ (B) $\text{CH}_3\text{CH} = \text{CHCH}_3$
 (C) $\text{CH}_3\text{CH}_2\text{C} \equiv \text{CCH}_2\text{CH}_3$ (D) $\text{CH}_3\text{CH}_2\text{CH}_2\text{C} \equiv \text{CCH}_2\text{CH}_2\text{CH}_3$
- Which of the following reagents may be used to distinguish between phenol and benzoic acid ? **[AIEEE-2011]**
 (A) Aqueous NaOH (B) Tollen's reagent (C) Molisch reagent (D) Neutral FeCl_3
- Aspirin is known as : **[AIEEE-2012]**
 (A) Acetyl salicylic acid (B) Phenyl salicylate
 (C) Acetyl salicylate (D) Methyl salicylic acid
- What is DDT among the following : **[AIEEE-2012]**
 (A) Greenhouse gas (B) A fertilizer
 (C) Biodegradable pollutant (D) Non-biodegradable pollutant
- The gas leaked from a storage tank of the Union Carbide plant in Bhopal gas tragedy was : **[JEE(Mains)-2013]**
 (A) Methylisocyanate (B) Methylamine
 (C) Ammonia (D) Phosgene
- Which of the following is an anionic detergent ? **[JEE(Mains)-2016]**
 (A) Sodium lauryl sulphate (B) Cetyltrimethyl ammonium bromide
 (C) Glyceryl oleate (D) Sodium stearate
- Which of the following is a thermosetting polymer **[JEE(Mains)-2019]**
 (A) Bakelite (B) Buna-N (C) Nylon-6 (D) PVC

11. The correct match between item-I and Item-II is [JEE(Mains)-2019]

Item-I	Item-II
A. High density polythene	I. Peroxide catalyst
B. Polyacrylonitrile	II. Condensation at high temp. and Pressure
C. Novolac	III. Ziegler-Natta catalyst
D. Nylon-6	IV. Acid or base catalyst

Codes

	A	B	C	D		A	B	C	D
(A)	III	I	IV	II	(B)	IV	II	I	III
(C)	II	IV	I	III	(D)	III	I	II	IV

12. Which of the following is a condensation polymer ? [JEE(Mains)-2019]

(A) Nylon-6, 6 (B) Neoprene (C) Teflon (D) Buna-S

13. Noradrenaline is a/an [JEE(Mains)-2019]

(A) antidepressant (B) antithistamine (C) neurotransmitter (D) antacid

14. Which of the following compounds is a constituent of the polymer $\text{--}[\text{HN} - \overset{\text{O}}{\parallel}{\text{C}} - \text{NHCH}_2]_n\text{--}$?

(A) N-methyl urea (B) Methylamine (C) Ammonia (D) Formaldehyde

15. The two monomers for the synthesis of nylon 6, 6 are [JEE(Mains)-2019]

(A) $\text{HOOC}(\text{CH}_2)_4\text{COOH}$, $\text{H}_2\text{N}(\text{CH}_2)_4\text{NH}_2$ (B) $\text{HOOC}(\text{CH}_2)_6\text{COOH}$, $\text{H}_2\text{N}(\text{CH}_2)_4\text{NH}_2$
 (C) $\text{HOOC}(\text{CH}_2)_4\text{COOH}$, $\text{H}_2\text{N}(\text{CH}_2)_6\text{NH}_2$ (D) $\text{HOOC}(\text{CH}_2)_6\text{COOH}$, $\text{H}_2\text{N}(\text{CH}_2)_6\text{NH}_2$

16. Poly- β -hydroxybutyrate-Co- β -hydroxyvalerate (PHBV) is a copolymer of

(A) 3-hydroxybutanoic acid and 2-hydroxypentanoic acid [JEE(Mains)-2019]
 (B) 2-hydroxybutanoic acid and 3-hydroxypentanoic acid
 (C) 3-hydroxybutanoic acid and 4-hydroxypentanoic acid
 (D) 3-hydroxybutanoic acid and 3-hydroxypentanoic acid

17. The antifertility drug "Novestrol" can react with : [JEE(Mains)-2020]

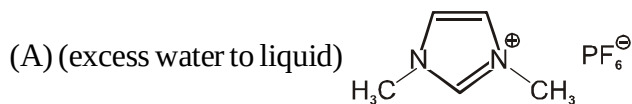
(A) $\text{ZnCl}_2 / \text{HCl}$; FeCl_3 ; Alcoholic HCN (B) Alcoholic HCN ; NaOCl ; $\text{ZnCl}_2 / \text{HCl}$
 (C) $\text{Br}_2 / \text{water}$, $\text{ZnCl}_2 / \text{HCl}$; NaOCl (D) $\text{Br}_2 / \text{water}$, $\text{ZnCl}_2 / \text{HCl}$; FeCl_3

18. Glycerol is separated in soap industries by : [JEE(Mains)-2020]

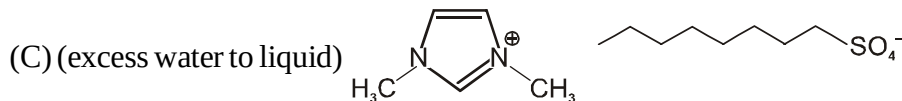
(A) Steam distillation (B) Differential distillation
 (C) Fractional distillation (D) Distillation under reduced pressure

19. An ionic micelle is formed on the addition of :

[JEE(Mains)-2020]



(B) liquid diethyl ether to aqueous NaCl solution.



(D) Sodium stearate to pure toluene

20. Match the following drugs with their therapeutic actions :

[JEE(Mains)-2020]

- | | |
|---------------------------------|--------------------|
| (i) Ranitidine | (a) Antidepressant |
| (ii) Nardil (Phenelzine) | (b) Antibiotic |
| (iii) Chloramphenicol | (c) Antihistamine |
| (iv) Dimetane (Brompheniramine) | (d) Antacid |
| | (e) Analgesic |

(A) (i)-(a) ; (ii)-(c) ; (iii)-(b) ; (iv)-(e)

(B) (i)-(d) ; (ii)-(c) ; (iii)-(a) ; (iv)-(e)

(C) (i)-(d) ; (ii)-(a) ; (iii)-(b) ; (iv)-(c)

(D) (i)-(e) ; (ii)-(a) ; (iii)-(c) ; (iv)-(d)

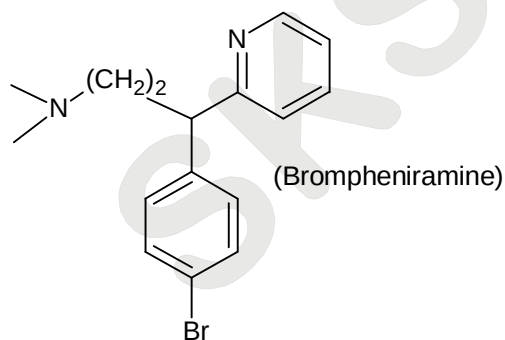
21. If a person is suffering from the deficiency of nor-adrenaline, what kind of drug can be suggested ?

[JEE(Mains)-2020]

- (A) Antihistamine (B) Analgesic (C) Anti-inflammatory (D) Antidepressant

22. The following molecule acts as an :

[JEE(Mains)-2020]



- (A) Anti-depressant (B) Antiseptic (C) Anti-histamine (D) Anti-bacterial

[JEE-ADVANCED]

1. **Statement-1 :** Aniline on reaction with $\text{NaNO}_2 / \text{HCl}$ at 0°C followed by coupling with β -naphthol gives a dark blue precipitate. [IIT-JEE 2008]

Statement-2 : The colour of the compound formed in the reaction of aniline with $\text{NaNO}_2 / \text{HCl}$ at 0°C followed by coupling with β -naphthol is due to the extended conjugation.

- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True

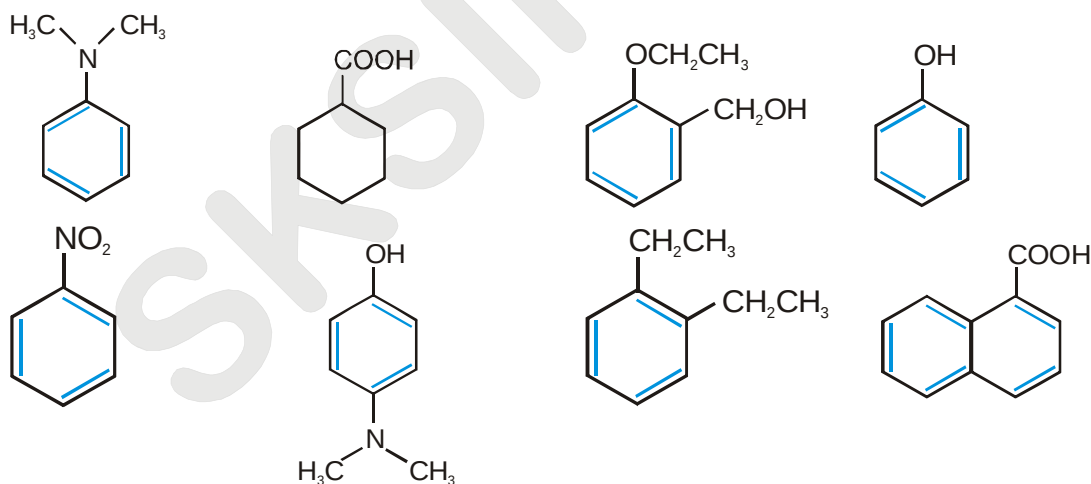
2. Match the entries in **Column I** with the correctly related quantum number(s) in **Column II**.

[IIT-JEE 2008]

Column I	Column II
(A) $\text{H}_2\text{N}-\text{NH}_3^+\text{Cl}^-$	(p) sodium fusion extract of the compound gives Prussian blue colour with FeSO_4
(B)	(q) gives positive FeCl_3 test
(C)	(r) gives white precipitate with AgNO_3
(D)	(s) reacts with aldehydes to form the corresponding hydrazone derivative

3. Amongst the following, the total number of compounds soluble in aqueous NaOH is :

[IIT-JEE 2010]



(A) 5

(B) 3

(C) 6

(D) 4

4. The compound that does **not** liberate CO_2 , on treatment with aqueous sodium bicarbonate solution, is

[JEE(Advanced) 2013]

(A) Benzoic acid
(C) Salicylic acid

(B) Benzenesulphonic acid
(D) Carboic acid (Phenol)

ANSWER KEY**[JEE-MAIN/AIEEE]**

- | | | | | | | | | | | | | | |
|-----|---|-----|---|-----|---|-----|---|-----|---|-----|---|-----|---|
| 1. | A | 2. | C | 3. | C | 4. | A | 5. | D | 6. | A | 7. | D |
| 8. | A | 9. | A | 10. | A | 11. | A | 12. | A | 13. | C | 14. | D |
| 15. | C | 16. | D | 17. | D | 18. | D | 19. | C | 20. | C | 21. | D |
| 22. | C | | | | | | | | | | | | |

[JEE-ADVANCED]

- | | | | | | | | |
|----|---|----|---|----|---|----|---|
| 1. | D | 2. | $A \rightarrow r, s ; B \rightarrow p, q, r ; C \rightarrow p, q, r, ; D \rightarrow p, r, s$ | 3. | D | 4. | D |
|----|---|----|---|----|---|----|---|